

Conditioning and first-passage fibres in the natural-density argument

Collatz workbench

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Abstract

Allikvere's natural-density manuscript replaces logarithmically distributed initial values in Tao's first-passage argument by uniformly distributed ones. This requires control of the Syracuse offset jointly with the total 2-valuation, and an endpoint calculation that is absent from the logarithmic argument. We reconstruct the sum-conditioned Fourier step, including its odd-length case, and give an exact arithmetic description of the first-passage fibres. We also extend an existing uniform harmonic-weight estimate to the time range used by the uniform-measure calculation. Keeping the offset and total valuation together gives a joint mixing estimate at every residue scale. Its proof uses Tao's positive majorant, the retained v7 stopping reserve, and exact convolution in the total coordinate. It recovers Allikvere's conditioned mixing theorem and extends its stated scale range; a separate calculation compares the conditioned coarse law with the unconditioned one. An integer-separation argument removes the logarithmic-form estimate from endpoint replacement. Together with a Gaussian kernel upper bound, it controls the absolute remainder in the uniform integer-fibre formula. A summation-by-parts estimate then evaluates its leading kernel with constant $1/\log(4/3)$, uniformly across the return band. Using Rhin's published exponent, this gives every kernel error exponent $c < 1/17.232$, improving the block estimate in the source. An explicit probability reduction carries that rate to the first-passage laws for the two uniform input intervals. The subsequent timed target-set iteration preserves it: outside a proportion $O_c((\log N_0)^{-c})$, positive odd inputs reach N_0 within $\log N/\log(4/3) + O((\log(N+2))^{4/5})$ odd returns. A disjoint scale cover removes the separate finite-cutoff error in the source's stated bound. Exact valuation telescoping gives a single raw Collatz clock for every diverging threshold, and thus the stated natural-density conclusion. These are written deductions from the cited analytic inputs; they are not inferred from finite orbit checks. A companion comparison retains the square-root scales in Mazur's terminal error and improves that scalar estimate from $(\log B)^{-1/20}$ to $O((\log B)^{-1/5} \log \log B)$. The exact leading coefficient and its propagation through the terminal scalar bounds are proved; the whole Mazur package is not thereby certified. The finite terminal comparison is reconstructed through its original integer inverses, time-window injectivity, boundary shells and two-row phase selector. Retaining the exact absorption denominator halves the leading coefficient of its terminal error majorant asymptotically. Fibre cancellation and signed tail comparison give a further refinement: the leading terminal coefficient becomes A_C in place of $(86/25)A_C$, with the same exponent and the exact exponential retained. Shaik's complementary parity/timeout construction is reconstructed through its critical shell estimate, with smaller deterministic transport charges, an integer timeout refinement, and explicit square-root errors about the shortcut, ordinary and odd-return drift clocks. For its fixed target exponent the exceptional count retains the endpoint power together with its logarithmic factor. At the two critical iterated-logarithm targets, the shell estimate retains the boundary exceptional powers, and an arbitrary diverging final factor admits an exact minorant profile. The fixed-barrier construction also retains the full stretched-exponential power with a common time and height witness. For a fixed power target, an explicit rank-sensitive clock gives a power-saving exceptional count

and simultaneous shortcut, raw and odd-return time bounds. Retaining the integer envelope horizon removes an additional startup payment from the fixed-depth clock and admits its closed power exponent on the original retained sets, with a joint exceptional-count bound. An explicit one-bit auxiliary barrier also retains the full binary-entropy rate for the original envelope set. Its propagation strengthens the fixed-depth and fixed-barrier exceptional exponents without changing their orbit witnesses or clocks. These refinements do not lower its critical target exponent or certify its complete formal source package.

Provenance. This note was developed in the PolyClank Collatz workbench through human–LLM collaboration. The present reconstruction and refinements were written with GPT-6 Astra, Ultra mode in Codex; the earlier coefficient reconstruction cited below was developed with GPT-5.6 Sol, Ultra mode in Codex. Literature results are credited where used. The written proofs and finite checks are available for independent review; the finite checks are not formal certification of the analysis.

1 Which change of measure has to be justified?

For a positive odd integer N , set

$$\text{Syr}(N) = \frac{3N+1}{2^{v_2(3N+1)}}.$$

Tao’s theorem concerns logarithmic density; its proof uses first entry below a threshold and fine-scale mixing of a random affine offset [8]. Allikvere proposes a natural-density theorem and a quantitative time bound [9]. The latter claims require additional arguments: uniform mass on $[y, y^\alpha]$ concentrates near y^α , whereas the logarithmic measure gives comparable weight to intervals of comparable logarithmic length.

In particular, for fixed $\alpha > 1$, $h \rightarrow \infty$ and $h = o(\log y)$, the upper layer $[y^\alpha e^{-h}, y^\alpha]$ has uniform probability $1 - o(1)$ among the odd integers of $[y, y^\alpha]$. Its logarithmic probability is $h/((\alpha - 1) \log y) + o(1) = o(1)$. This follows by counting odd integers and, respectively, summing $1/N$ along the odd progression. A discarded upper layer cannot therefore be transferred from one measure to the other.

Allikvere addresses this at the level of the first-passage formula rather than merely replacing the word “logarithmic” by “natural.” The components below identify what the conditioning argument proves and which arithmetic estimates can be reused over the extended interval.

2 A positive majorant survives sum-conditioning

Let $a_1, \dots, a_n$ be independent positive integers with $\mathbb{P}(a_i = r) = 2^{-r}$ for $r \geq 1$. Put

$$S_j = \sum_{i=1}^j a_i, \quad X_n = \sum_{i=1}^n 3^{i-1} 2^{-S_i} \pmod{3^n}.$$

Powers of 2^{-1} are taken in the ring $\mathbb{Z}/3^n\mathbb{Z}$. Fix a character $\chi(z) = \exp(-2\pi i \xi z/3^n)$ with $3 \nmid \xi$. For $1 \leq j \leq \lfloor n/2 \rfloor$ set $b_j = a_{2j-1} + a_{2j}$ and $B_j = b_1 + \dots + b_j$. Let $\mathcal{F}$ be generated by all b_j and, when n is odd, also by a_n .

The following identity is exact in $\mathbb{Z}[1/2]$, before reduction:

$$X_n = \sum_{j=1}^{\lfloor n/2 \rfloor} 3^{2j-2} 2^{-B_j} (2^{a_{2j}} + 3) + \mathbf{1}_{n \text{ odd}} 3^{n-1} 2^{-B_{\lfloor n/2 \rfloor} - a_n}. \quad (1)$$

Here and below an expression for X_n in $\mathbb{Z}[1/2]$ denotes the indicated representative before applying the reduction map. Given $\mathcal{F}$, the variables a_{2j} are independent and uniform on $\{1, \dots, b_j - 1\}$. Define

$$f_\chi(c, b) = \frac{1}{b-1} \sum_{r=1}^{b-1} \chi(c(2^r + 3)).$$

Taking the conditional expectation in (1) gives

$$\mathbb{E}(\chi(X_n) \mid \mathcal{F}) = \prod_j f_\chi(3^{2j-2} 2^{-B_j}, b_j) \chi(\mathbf{1}_{n \text{ odd}} 3^{n-1} 2^{-B_{\lfloor n/2 \rfloor} - a_n}). \quad (2)$$

For even n the last factor is 1. For odd n , a_n is already fixed by $\mathcal{F}$, so its factor has modulus exactly 1; it is not an independently averaged factor after conditioning on S_n .

Tao's white set $W_{n,\xi}$ consists of the pairs (j, l) for which the signed fractional part of $\xi 3^{2j-2} 2^{-l+1} / 3^n$ has absolute value greater than a fixed sufficiently small $\varepsilon > 0$ [8, §7.1]. For $b = 3$,

$$f_\chi(c, 3) = \frac{1}{2} \chi(5c) + \frac{1}{2} \chi(7c), \quad |f_\chi(c, 3)| = \frac{1}{2} |1 + \chi(2c)|.$$

Thus a white pair with $b_j = 3$ contributes a factor at most $e^{-\varepsilon^3}$; all other factors have modulus at most 1. Consequently

$$|\mathbb{E}(\chi(X_n) \mid \mathcal{F})| \leq Z_{n,\xi}, \quad Z_{n,\xi} = \exp(-\varepsilon^3 \#\{j : b_j = 3, (j, B_j) \in W_{n,\xi}\}). \quad (3)$$

The literature input is Tao's Proposition 7.3: $\mathbb{E} Z_{n,\xi} \ll_D n^{-D}$ for every $D > 0$, uniformly in primitive ξ . We use this stated result; its complete renewal-process proof is not re-certified in this note.

Proposition 1 (Weighted and sum-conditioned Fourier bounds). *For every bounded complex $\mathcal{F}$ -measurable random variable H ,*

$$|\mathbb{E}(H \chi(X_n))| \leq \mathbb{E}(|H| Z_{n,\xi}) \leq \|H\|_\infty \mathbb{E} Z_{n,\xi}. \quad (4)$$

Furthermore, for every $D > 0$,

$$\sum_{s \geq n} |\mathbb{E}(\chi(X_n) \mathbf{1}_{S_n=s})| \leq \mathbb{E} Z_{n,\xi} \ll_D n^{-D}. \quad (5)$$

For each fixed $A, C > 0$ and all sufficiently large n ,

$$|\mathbb{E}(\chi(X_n) \mid S_n = s)| \ll_{A,C} n^{-A} \quad \text{if } |s - 2n| \leq C \sqrt{n \log n}. \quad (6)$$

Proof. The tower property and (3) give (4). The total S_n is $\mathcal{F}$ -measurable, including in the odd case. Apply the first inequality to $H = \mathbf{1}_{S_n=s}$ and sum. These events partition the probability space, so monotone convergence gives (5) with no factor counting the possible sums.

The probability of a fixed sum is exactly

$$p_n(s) = \binom{s-1}{n-1} 2^{-s} \quad (s \geq n).$$

Stirling's formula, uniformly for $d = s - 2n$ in the displayed moderate window, gives

$$p_n(s) = \frac{e^{-d^2/(4n)}}{\sqrt{4\pi n}} \left(1 + O_C \left(n^{-1/2} (\log n)^{3/2} \right) \right).$$

For example, write $p_n(s) = \frac{n}{2n+d} \binom{2n+d}{n} 2^{-2n-d}$; the entropy term has logarithm $-d^2/(4n) + O(|d|^3/n^2)$ and the prefactor has logarithm $-\frac{1}{2} \log(4\pi n) + O(|d|/n + 1/n)$. The relative errors tend to zero uniformly. In particular $p_n(s) \geq c_C n^{-1/2-C^2/4}$ once n is large enough. Divide the single- s bound from (5) by $p_n(s)$ and use Tao's estimate with $D = A + 1 + C^2/4$. This proves (6). $\square$

Equation (6) reconstructs Allikvere's Theorem 3.1 (source label `thm:B`). Equation (5) records its weighted precursor explicitly. It is a direct consequence of the same positive-majorant argument, not a claim of independent novelty. An unconditional bound on $|\mathbb{E}\chi(X_n)|$ alone would not justify conditioning: cancellation between distinct values of S_n could hide large conditional expectations. The pointwise bound (3) is what prevents that error here.

3 Exact arithmetic fibres and first passage

For a word $\mathbf{a} = (a_1, \dots, a_r)$ of positive integers let

$$A_j = a_1 + \dots + a_j, \quad B_j = \sum_{i=1}^j 3^{j-i} 2^{A_{i-1}}, \quad A_0 = B_0 = 0.$$

Composition of $z \mapsto (3z + 1)/2^{a_j}$ gives

$$\text{Aff}_{\mathbf{a}}(N) = \frac{3^r N + B_r}{2^{A_r}} = 3^r 2^{-A_r} N + F_r(\mathbf{a}), \quad F_r(\mathbf{a}) = 2^{-A_r} B_r. \quad (7)$$

This is the forward-order offset; the reversed word gives the X_r -orientation of the preceding section.

Proposition 2 (Valuation cylinder and inverse fibre). *For $r \geq 1$ there is a unique odd residue*

$$u_{\mathbf{a}} \equiv 3^{-r} (2^{A_r} - B_r) \pmod{2^{A_r+1}}$$

such that the actual first r Syracuse valuations of a positive odd N equal $\mathbf{a}$ if and only if $N \equiv u_{\mathbf{a}} \pmod{2^{A_r+1}}$. For a specified positive odd terminal value M , the only possible initial value is

$$N = \frac{2^{A_r} M - B_r}{3^r}. \quad (8)$$

This is a positive odd integer with the specified valuation word precisely when $M > F_r(\mathbf{a})$ and $M \equiv F_r(\mathbf{a}) \pmod{3^r}$.

Proof. The terminal value in (7) is an odd integer exactly when $3^r N + B_r \equiv 2^{A_r} \pmod{2^{A_r+1}}$. Since 3^r is a unit modulo 2^{A_r+1} , this defines one residue; since B_r is odd, this residue is odd. Necessity for an actual valuation word is immediate.

For sufficiency, the congruence first implies $2^{a_1} \mid 3N + 1$: $B_r \equiv 3^{r-1} \pmod{2^{a_1}}$ and $A_r \geq a_1$. Put $N_1 = (3N + 1)/2^{a_1}$, a positive integer. If $r = 1$, the terminal congruence says directly that N_1 is odd. If $r > 1$, divide the congruence by 2^{a_1} and use $B_r = 3^{r-1} + 2^{a_1} B_{r-1}(a_2, \dots, a_r)$. The resulting congruence for the tail word forces N_1 odd modulo 2, and supplies the same terminal condition for that tail. Induction proves all remaining valuations; oddness of N_1 gives $v_2(3N + 1) = a_1$ exactly.

Solving (7) gives (8). Its integrality is equivalent to the stated congruence modulo 3^r , since 2 is invertible there. Its numerator is odd, so an integral quotient by 3^r is odd. Positivity is equivalent to $M > F_r$. Substitution into the terminal congruence and the first part of the proof completes the converse. $\square$

For example, the word $(1, 2)$ has $A_2 = 3$, $B_2 = 5$. The exact map is

$$\begin{array}{ccc} 11 + 16j & \xrightarrow{(3N+1)/2} & 17 + 24j \xrightarrow{(3N+1)/4} 13 + 18j, \quad j \geq 0, \\ N \equiv 11 \pmod{16} & & M \equiv 13 \pmod{18}. \end{array}$$

This displays actual positive integer trajectories, not a drawing of a random affine process. The last congruence is the combination of oddness and $M \equiv 5/8 \equiv 4 \pmod{9}$.

For a real $x \geq 1$, put $T_x(N) = \inf\{j \geq 0 : \text{Syr}^j(N) \leq x\}$. On one valuation cylinder the condition $T_x(N) = r$ is the exact interval condition

$$\max_{0 \leq j < r} \frac{2^{A_j}x - B_j}{3^j} < N \leq \frac{2^{A_r}x - B_r}{3^r}. \quad (9)$$

It follows by imposing $\text{Syr}^j(N) > x$ at every $j < r$ and $\text{Syr}^r(N) \leq x$ in (7), including $j = 0$. The strict and non-strict endpoints are retained.

Proposition 3 (Finite-word residue discrepancy). *Let $\mathcal{A}$ be a finite collection of distinct length- r valuation words. In each cylinder select its first-passage interval (9), intersected with an arbitrary further interval. Let E be the union of the resulting sets of positive odd integers and let P be the number of nonempty selected cylinders. For every positive odd integer q , every residue $z \pmod{q}$ and every interval W ,*

$$\left| \#(E \cap W \cap (z + q\mathbb{Z})) - \frac{\#(E \cap W)}{q} \right| \leq P.$$

Proof. One selected cylinder intersected with W is an interval in the progression $u_{\mathbf{a}} + 2^{A_r+1}\mathbb{Z}$. Its elements correspond to consecutive integers in the progression parameter. Because q is odd, the map from that parameter to the residue modulo q is bijective on each block of q consecutive parameters. A count of L consecutive parameters puts either $\lfloor L/q \rfloor$ or $\lceil L/q \rceil$ in each residue, giving error at most 1. Distinct valuation words have disjoint cylinders, so summing gives the bound. Empty pieces contribute zero. $\square$

Allikvere's residue-flattening lemma (source label `lem:flat`) uses exactly this mechanism with $q = 3^k$ and a count of the contributing prefixes. No independence between the terminal event and the residue is being assumed. Nor does the bound assert relative equidistribution in a class whose expected population is less than its additive error.

4 The uniform harmonic bound covers the extended time range

Set $L = \log x$, $\alpha = 1.001$, $m = \lfloor L/100000 \rfloor$, $n_0 = \lfloor L/(10 \log 2) \rfloor$. For an arbitrary set of terminal values $E \subseteq 2\mathbb{N}+1 \cap [1, x]$, let E' consist of the positive odd M such that $T_x(M) = m$, $\text{Syr}^m(M) \in E$, and

$$e^{-L^{7/10}}(4/3)^m x \leq M \leq e^{L^{7/10}}(4/3)^m x.$$

For every integer $m \leq n \leq n_0$, define

$$c_n(z) = 3^{n-m} \sum_{\substack{M \in E' \\ M \equiv z \pmod{3^{n-m}}}} \frac{1}{M}.$$

The local reconstruction of Tao's Lemma 5.3 proves

$$\sup_{m \leq n \leq n_0} \sup_z c_n(z) \ll 1. \quad (10)$$

Its statement previously restricted n to Tao's trimmed interval; the proof in fact uses only $m \leq n \leq n_0$. That wider range is now stated in the local manuscript [10].

Here is the summation that makes the range extension valid. Partition by the length- m valuation word, put $A = \sum_i a_i$, $b = a_m$, and keep the two wordwise bounds

$$c_{n,\mathbf{a}}(z) \ll \begin{cases} b2^{-A}, & 2^A \leq x^{1/2}, \\ b2^{-A} + 3^n 2^{-A}, & 2^A > x^{1/2}. \end{cases}$$

They follow by summing $1/M$ along the exact progression modulo $2^{A+1}3^{n-m}$ between the first-crossing bounds $3^{-m}2^{A-b}x \leq M \leq 3^{-m}2^A x$; the high- A case also uses $M \gg 3^{-m}2^A$. All constants are uniform over the indicated range of n , since $3^n \leq x^{4/25}$. The two sums are

$$\sum_{\mathbf{a} \in \mathbb{N}_{>0}^m} b2^{-A} = 2, \quad 3^n \sum_{2^A > x^{1/2}} 2^{-A} \leq 3^n x^{-1/4} (1 + \sqrt{2})^m \leq x^{-8999/100000}.$$

These establish (10); the complete progression estimates remain in [10]. The weaker summation $\sum_{\mathbf{a}} 2^{-A/2} = (1 + \sqrt{2})^m$ is not uniformly bounded as m grows and cannot replace them.

Allikvere's extended interval is

$$\tilde{I}_y = \left[\frac{\log(y/x)}{\log(4/3)} + L^{4/5}, \frac{\log(y^\alpha/x)}{\log(4/3)} + L^{4/5} \right], \quad y \in \{x^\alpha, x^{\alpha^2}\}.$$

For sufficiently large x , this lies in $[m, n_0]$. The upper endpoint is at most $((\alpha^3 - 1)/\log(4/3))L + L^{4/5} < 0.011L < n_0$; the lower endpoint exceeds m . Thus (10) applies throughout this extended interval, without importing Tao's discarded upper layer. Summing over residues gives $\sum_{M \in E'} 1/M \ll 1$; summing only over the refinements of a class modulo 3^j gives

$$\sum_{\substack{M \in E' \\ M \equiv z \pmod{3^j}}} \frac{1}{M} \ll 3^{-j} \quad (0 \leq j \leq n - m).$$

These are precisely the harmonic bounds consumed at source lines 1229–1236, 1302–1320 and 1411–1416 of Allikvere's TeX.

5 A constant that must remain attached to its use

Allikvere's fine-scale argument invokes Tao's 3-adic separation of offsets. It notes that the coefficient ρ in the stopping-sum reserve can be $1/2$ in an older edition or 0.99 in v7, and says its value is immaterial because the number of possible sums is $O(C^2 \log n)$. That counting statement is correct, but it does not account for the subsequent separation estimate.

In the size argument, write $a = \log(4/3)$ and $b = \log 2$. Completing the square gives the exact bound

$$\sup_{t \geq 0} \{-at + bC\sqrt{t \log n}\} = \frac{b^2}{4a} C^2 \log n.$$

A reserve $2^l \leq 3^n n^{-\rho b C^2}$ therefore leaves the leading exponent

$$\left(-\rho b + \frac{b^2}{4a}\right) C^2 \log n.$$

To retain a negative exponent and a small exponential tail in t , this argument needs

$$\rho > \frac{\log 2}{4 \log(4/3)} = 0.60235 \dots \quad (11)$$

The strict inequality allows replacing a by $a - \delta$ for a sufficiently small $\delta > 0$ and retaining $e^{-\delta t}$ in the sum. Terms of size $O(C \log n)$ are then absorbed by taking C sufficiently large. The value 0.99 satisfies (11); $1/2$ does not. This is a restriction of this particular displayed proof, not a counterexample to the separation statement with some other hypotheses. The present comparison fixes v_7 and keeps the 0.99 reserve at the separation step. Allikvere's later use of separation cannot be justified solely by counting how many l occur.

6 Mixing with the total valuation retained

The conditional Fourier estimate is not itself a total-variation estimate: there are 3^n frequencies, and multiplying a polynomial bound by their number would lose all useful decay. Tao's stopping split supplies the missing control. The early part of the valuation word has small collision probability modulo 3^n , while the remaining part supplies Fourier decay. Allikvere adapts this split to a fixed total in Theorem 4.1 and then compares scales in Proposition 4.2 [9]. We retain the total as an additional coordinate throughout the split. This gives the following joint version, from which the fixed-total statement follows by division only at the end.

Write $G_n = \mathbb{Z}/3^n \mathbb{Z}$ and extend every total-indexed measure by zero to all of $\mathbb{Z}$. Define

$$\mu_n(y, s) = \mathbb{P}(X_n = y, S_n = s), \quad p_j(r) = \mathbb{P}(S_j = r), \quad p_0(r) = \mathbf{1}_{r=0}.$$

For $1 \leq m \leq n$, the reduction and uniform lift on signed, absolutely summable measures are

$$\begin{aligned} R_{n,m} : \ell^1(G_n \times \mathbb{Z}) &\longrightarrow \ell^1(G_m \times \mathbb{Z}), & (R_{n,m}h)(z, s) &= \sum_{y \equiv z \pmod{3^m}} h(y, s), \\ U_{m,n} : \ell^1(G_m \times \mathbb{Z}) &\longrightarrow \ell^1(G_n \times \mathbb{Z}), & (U_{m,n}h)(y, s) &= 3^{m-n} h(y \bmod 3^m, s). \end{aligned}$$

Thus $R_{n,m}$ is a contraction, $U_{m,n}$ is an isometry, and $R_{n,m}U_{m,n} = I$. The lift preserves the total coordinate and spreads each residue mass equally over its 3^{n-m} refinements. Set

$$Q_{m,n} = U_{m,n}R_{n,m}, \quad J_{m,n} = \|\mu_n - Q_{m,n}\mu_n\|_1.$$

The norm here is the sum of absolute values, without a factor $1/2$. In particular,

$$J_{m,n} = \sum_{s \geq n} p_n(s) \|\text{Law}(X_n \mid S_n = s) - U_{m,n} \text{Law}(X_n \bmod 3^m \mid S_n = s)\|_1, \quad (12)$$

where on the right $U_{m,n}$ acts only on the displayed residue variable. This is an average of conditional errors, not independence of X_n and S_n .

Proposition 4 (Joint offset–total mixing). *For every $A > 0$ there is $K_A < \infty$ such that*

$$J_{m,n} \leq K_A m^{-A} \quad (1 \leq m \leq n). \quad (13)$$

The input from the literature is Tao’s positive-majorant estimate (3), with arbitrary polynomial decay. No first-passage or natural-density conclusion is used in this proof.

Proof. First suppose $0.9n \leq m \leq n$. Fix A , then choose a large constant C ; all ensuing lower bounds on n may depend on A, C . Put $L = \log n$, $b = \log 2$, $a = \log(4/3)$ and

$$H = n \frac{\log 3}{\log 2} - C^2 L, \quad \gamma = \frac{\log 3}{2 \log 2}.$$

Let E be the event

$$\left| \sum_{t=i}^j a_t - 2(j-i) \right| \leq C(\sqrt{(j-i)L} + L) \quad (1 \leq i \leq j \leq n).$$

The inclusive interval has $j-i+1$ entries; the centre $2(j-i)$ is kept exactly as in Tao’s source. The two-unit difference from its mean is absorbed in the CL term, not deleted from subsequent prefix estimates. The geometric moment-generating function $\mathbb{E}e^{u(a_i-2)} = e^{-u}/(2-e^u)$, for $u < \log 2$, has logarithm $O(u^2)$ near zero. Markov’s inequality with $|u| = \min(1/4, t/(Kr))$ for a sum of r entries gives $\mathbb{P}(|S_r - 2r| > t) \leq 2e^{-c \min(t^2/r, t)}$. Applying this bound to each interval (and adjusting the fixed two) gives $\mathbb{P}(E^c) \ll n^{2-cC}$. Choose C so that this is $O_A(n^{-A-2})$.

On E , there is a unique $k < n$ with $S_k \leq H < S_{k+1}$, and $|k - \gamma n| \leq 3C\sqrt{nL}$ for all sufficiently large n . Indeed the inequalities for S_k, S_{k+1} sandwich H between $2k + O(C\sqrt{nL} + CL + 2)$ and $2k + 2 + O(C\sqrt{nL} + CL + 2)$; the term C^2L is $o(C\sqrt{nL})$. Let $B_k = \{S_k \leq H < S_{k+1}\}$ and let E_k retain only the inequalities for $1 \leq i < j \leq k+1$. On $E_k \cap B_k$, the inequality with $(i, j) = (k, k+1)$ and $a_k \geq 1$ gives

$$a_{k+1} \leq 1 + C(\sqrt{L} + L), \quad H < S_{k+1} \leq n \frac{\log 3}{\log 2} - 0.99C^2L$$

once $C \geq 600$. Here $1 + C(\sqrt{L} + L) \leq 6CL \leq 0.01C^2L$ for $n \geq 2$. For admissible k and integer l in this last interval set $D_{k,l} = E_k \cap B_k \cap \{S_{k+1} = l\}$. These events are disjoint, since the sums S_j strictly increase. Their union contains E ; any additional points belong to E^c . Replacing the original law by its restriction to this union therefore changes its joint ℓ^1 norm by at most $\mathbb{P}(E^c)$, and changes $J_{m,n}$ by at most $2\mathbb{P}(E^c)$. There are at most $O_C(n^2)$ admissible pairs (k, l) .

Fix one such pair and put $K = k + 1$, $t = n - K$. On $D_{k,l}$ the exact split is

$$X_n = X_K + 3^K 2^{-l} X_t^{\text{tail}} \pmod{3^n}, \quad S_n = l + S_t^{\text{tail}},$$

where the tail consists of $a_{K+1}, \dots, a_n$, independently of the prefix. The same displayed formula for X_K is evaluated in G_n here, not reduced prematurely to G_K . Write $g_s(y) = \mathbb{P}(X_n = y, S_n = s, D_{k,l})$ and

$$\beta(\xi) = \mathbb{E}(e^{-2\pi i \xi X_K / 3^n} \mathbf{1}_{D_{k,l}}).$$

Frequencies surviving $I - Q_{m,n}$ have $v = v_3(\xi) < n - m$. Define the unit $\xi' = 2^{-l}(\xi/3^v) \pmod{3^{n-v}}$ and set $q = t - v$. Then $q \geq (0.9 - \gamma)n - O_C(\sqrt{nL}) \gg n$. Reduction of the tail offset modulo 3^q

deletes precisely its terms with indices greater than q . Independence gives, with no conditional probabilities,

$$\widehat{g}_s(\xi) = \beta(\xi)\Gamma_\xi(s-l), \quad \Gamma_\xi(r) = \sum_{u \in \mathbb{Z}} h_{q,\xi'}(u)p_v(r-u), \quad h_{q,\xi'}(u) = \mathbb{E}(e^{-2\pi i \xi' X_q/3^q} \mathbf{1}_{S_q=u}). \quad (14)$$

In this expression ξ' is reduced further to $G_q^\times$ when evaluating the character. Equation (5) and $\sum_r p_v(r) = 1$ show that for every $D > 0$

$$\sum_r |\Gamma_\xi(r)| \leq \sum_u |h_{q,\xi'}(u)| \ll_D q^{-D} \ll_{D,C} n^{-D}. \quad (15)$$

This includes $v = 0$, when convolution with p_0 changes nothing. In particular the pointwise bound holds for every r ; no moderate window for a tail sum, and no division by its probability, is needed.

We next bound the prefix factor. On E_k , for $1 \leq j \leq K$,

$$S_j \geq 2j - 2 - C(\sqrt{jL} + L).$$

For $j \geq 2$ this follows from the $(1, j)$ inequality, and for $j = 1$ it follows from $a_1 \geq 1$. The positive integer $V = 2^l X_K = \sum_{j=1}^K 3^{j-1} 2^{l-S_j}$ satisfies

$$\begin{aligned} \frac{V}{3^n} &\leq \frac{4}{3} n^{-0.99bC^2+bC} \sum_{j=1}^K e^{-a_j+bC\sqrt{jL}} \\ &\leq \frac{4}{3} n^{1-(0.99b-b^2/(4a))C^2+bC} < 1. \end{aligned}$$

The second line uses $K \leq n$ and completes the square in $\sqrt{j}$. The final inequality holds after enlarging C , because $0.99b - b^2/(4a) > 0$. Two such prefixes with the same l and the same residue $X_K \pmod{3^n}$ therefore have identical integers V , hence identical rational offsets. Their words coincide: after subtracting $3^{K-1}2^{-l}$ from each offset, the remaining rational has 2-valuation $-S_{K-1}$, since its last term is the unique term with least 2-valuation. Continue backwards to recover every S_j , hence every $a_j = S_j - S_{j-1}$. The case $K = 1$ is immediate.

Thus each residue in the restricted prefix law has mass at most 2^{-l} . Unnormalized Fourier transform and Plancherel give

$$\sum_{\xi \in G_n} |\beta(\xi)|^2 \leq 3^n 2^{-l} \leq n^{bC^2}. \quad (16)$$

Cauchy–Schwarz in G_n , followed by Plancherel, now yields for each s

$$\|g_s - Q_{m,n}g_s\|_1^2 \leq \sum_{v_3(\xi) < n-m} |\widehat{g}_s(\xi)|^2 \ll_{D,C} n^{-2D+bC^2}.$$

There are at most $4n$ possible integers s with $n \leq s \leq 4n$. The remaining total mass is exponentially small: using $\mathbb{E}(4/3)^{a_i} = 2$,

$$\mathbb{P}(S_n > 4n) \leq 2^n (3/4)^{4n} = (81/128)^n.$$

For every nonnegative submeasure its oscillation is at most twice its mass. Sum the displayed bound over $s \leq 4n$ and the $O_C(n^2)$ disjoint prefix events, and bound their combined $s > 4n$ part by twice the last tail probability. The result is

$$J_{m,n} \ll_{A,C,D} n^{-A-2} + n^{3-D+bC^2/2} + (81/128)^n.$$

Choose $D > A + 4 + bC^2/2$. This proves the assertion for $m \geq 0.9n$. The order of choices is A , then C , then D , then sufficiently large n ; no concentration constant is required to be large relative to a decay exponent that it determines in return.

To pass to all scales, define convolution in the total coordinate by

$$(\mathcal{C}_j h)(y, s) = \sum_u h(y, u) p_j(s - u).$$

It is an ℓ^1 contraction and commutes with every residue reduction and lift. The deterministic reduction $X_n \bmod 3^\nu = X_\nu$ and independence of the remaining valuations give

$$R_{n,\nu} \mu_n = \mathcal{C}_{n-\nu} \mu_\nu. \quad (17)$$

The exact maps used in this identity can be displayed as

$$\begin{array}{ccc} (X_\nu, S_\nu, S_n - S_\nu) & \xrightarrow{(y,u,r) \mapsto (y,u+r)} & (X_\nu, S_n) \\ \text{law } \mu_\nu \otimes p_{n-\nu} & & \text{law } \mathcal{C}_{n-\nu} \mu_\nu = R_{n,\nu} \mu_n \end{array}$$

Choose $n = \nu_0 > \dots > \nu_J = m$ by $\nu_{i+1} = \max(m, \lceil 0.9\nu_i \rceil)$. For $m \geq 20$ this terminates; apart from the final step, $\nu_{i+1} \leq 0.95\nu_i$. Telescoping the lifted projected laws, using the isometry of U , (17), and the contraction of $\mathcal{C}$, gives

$$J_{m,n} \leq \sum_{i=0}^{J-1} J_{\nu_{i+1}, \nu_i} \ll_A \sum_{i=0}^{J-1} \nu_i^{-A} \leq \frac{K'_A}{1 - 0.95^A} m^{-A}.$$

For $m < 20$, the bound $J_{m,n} \leq 2$ suffices after increasing K_A . All finitely many small- n cases in the earlier estimates are absorbed in the same way. This completes the proof for every $1 \leq m \leq n$. $\square$

Corollary 5 (Fixed totals and the scale range). *For every $A, C > 0$ and $0 < \varepsilon \leq 1$, uniformly for $n^\varepsilon \leq m \leq n$ and $|s - 2n| \leq C\sqrt{n \log n}$,*

$$\|\text{Law}(X_n \mid S_n = s) - U_{m,n} \text{Law}(X_n \bmod 3^m \mid S_n = s)\|_1 \ll_{A,C,\varepsilon} m^{-A}.$$

For an arbitrary possible total s , without a moderate-window condition, the same error is at most $\min(2, K_B m^{-B}/p_n(s))$ for every $B > 0$.

Proof. Each summand in (12) is nonnegative. Divide (13) by $p_n(s)$. In the moderate window, the Stirling estimate in Proposition 1 gives $p_n(s)^{-1} \ll_C n^{1/2+C^2/4} \leq m^{(1/2+C^2/4)/\varepsilon}$. Use $B = A + (1/2 + C^2/4)/\varepsilon$. The trivial bound is 2. $\square$

Taking $\varepsilon = 1/4$ recovers the scale range in Allikvere's Theorem 4.1 and Proposition 4.2. Equation (13) is stronger as an averaged statement and yields every fixed positive power scale. It does not assert a uniformly small conditional error at every possible total; the factor $p_n(s)^{-1}$ states exactly what is lost. For any bounded function $w(s)$ it also gives

$$\left\| \sum_s w(s) (\mu_n(\cdot, s) - Q_{m,n} \mu_n(\cdot, s)) \right\|_1 \leq \|w\|_\infty K_A m^{-A}.$$

Weights depending on both the residue and the total require their own calculation. No assertion about those weights is implicit here.

7 Removing the coarse conditioning

Uniform lifting does not make the remaining coarse residue independent of the total. The following calculation addresses that distinct question. It reconstructs Allikvere's Lemma 4.3 [9] with the ratio of probabilities, its exceptional set, and its error terms retained.

Lemma 6 (Effect of a fixed total on an initial block). *Fix $C > 0$. If $\log^4 n \leq m \leq n^{9/10}$ and $|s - 2n| \leq C\sqrt{n \log n}$, then*

$$\|\text{Law}(a_1, \dots, a_m \mid S_n = s) - \text{Law}(a_1, \dots, a_m)\|_1 \ll_C \sqrt{\frac{m \log n}{n}}.$$

The same bound holds after applying the exact map $(a_1, \dots, a_m) \mapsto X_m \in G_m$.

Proof. Relative to the product geometric law, the density at a word with sum σ is $w = p_{n-m}(s - \sigma)/p_n(s)$, with zero numerator outside its support. Write $d = s - 2n$, $e = \sigma - 2m$ and $\mathcal{E} = \{|e| > \sqrt{m \log n}\}$. The Chernoff estimate proved above gives $\mathbb{P}(\mathcal{E}) \leq 2 \exp(-c \log^2 n)$ because $m \geq \log^4 n$. Also

$$\mathbb{E}(|w - 1| \mathbf{1}_{\mathcal{E}}) \leq \mathbb{P}(\mathcal{E} \mid S_n = s) + \mathbb{P}(\mathcal{E}) \leq (1 + p_n(s)^{-1}) \mathbb{P}(\mathcal{E}) \ll_C n^{-10}.$$

Outside $\mathcal{E}$, $|d - e| \ll_C \sqrt{n \log n}$ since $m \leq n^{9/10}$, and $n - m \asymp n$. The same Stirling expansion used above therefore gives

$$\begin{aligned} \log w &= \frac{1}{2} \log \frac{n}{n - m} + \frac{d^2}{4n} - \frac{(d - e)^2}{4(n - m)} + O_C(n^{-1/2} \log^{3/2} n) \\ &= \frac{1}{2} \log \frac{n}{n - m} + \frac{2de - e^2}{4(n - m)} - \frac{d^2 m}{4n(n - m)} + O_C(n^{-1/2} \log^{3/2} n). \end{aligned}$$

Every term tends uniformly to zero on this complement. For example, $|de|/n \leq C\sqrt{m/n} \log^{3/2} n \leq Cn^{-1/20} \log^{3/2} n$, and $e^2/n \leq (m/n) \log^2 n$. Thus $|w - 1| \ll |\log w|$ there. Since $\mathbb{E}e^2 = 2m$ and $\mathbb{E}|e| \leq \sqrt{2m}$, integration gives

$$\mathbb{E}(|w - 1| \mathbf{1}_{\mathcal{E}^c}) \ll_C \frac{m}{n} + \frac{|d|\sqrt{m}}{n} + \frac{d^2 m}{n^2} + n^{-1/2} \log^{3/2} n.$$

The middle two terms are bounded by a constant times $\sqrt{m \log n/n}$: the third is $O_C(m \log n/n)$ and this ratio tends to zero. The final term is absorbed because $m \geq \log^4 n$. This proves the first assertion; summing over fibres of the deterministic offset map is an ℓ^1 contraction and proves the second. $\square$

Let $\rho_m(z) = \mathbb{P}(X_m = z)$. Combining the two proved comparisons gives

Corollary 7 (Product approximation). *For fixed $A, C > 0$ and $0 < \varepsilon < 9/10$, uniformly for $n^\varepsilon \leq m \leq n^{9/10}$ and $|s - 2n| \leq C\sqrt{n \log n}$, for sufficiently large n ,*

$$\sum_{y \in G_n} |\mathbb{P}(X_n = y \mid S_n = s) - 3^{m-n} \rho_m(y \bmod 3^m)| \ll_{A, C, \varepsilon} m^{-A} + \sqrt{\frac{m \log n}{n}}.$$

Proof. Insert the uniform lift of the actual conditioned coarse law. The first error is Corollary 5; the second is exactly the coarse-law error, because $U_{m,n}$ is an ℓ^1 isometry. Apply Lemma 6; $n^\varepsilon \geq \log^4 n$ eventually. $\square$

This includes Allikvere's Corollary 4.4, whose stated range is $n^{1/4} \leq m \leq n^{1/2}$. The additional range follows from the joint mixing argument and the range already present in the source's tilt lemma. The approximation concerns the random offset and total valuation. Passing from it to counts of actual integer trajectories still requires the first-passage fibre and endpoint calculations.

8 Endpoint replacement without a logarithmic-form estimate

We now return to actual integer fibres. In this section E' means the *full* set of positive odd M with $T_x(M) = m_0$ in the band

$$M_- := xe^{dm_0 - L^{7/10}} \leq M \leq M_+ := xe^{dm_0 + L^{7/10}},$$

$$L = \log x, \quad d = \log(4/3), \quad m_0 = \lfloor L/100000 \rfloor.$$

There is no additional restriction on the final value at this point. For a selected terminal set E , its corresponding $E'(E)$ is a subset of this full set. We will bound absolute errors over the full set, which then controls every such subset. This distinction matters: selecting arbitrary terminal values need not leave an interval in each valuation cylinder.

Put $\lambda = \log_2 3$ and let $\mathcal{I}_y = (\tilde{I}_y - m_0) \cap \mathbb{Z}$ be the set of prefix lengths $r = n - m_0$, for $y = x^\alpha$ or x^{α^2} , $\alpha = 1.001$. Uniformly in these choices, for sufficiently large x ,

$$c_\alpha L \leq r \leq C_\alpha L \quad (r \in \mathcal{I}_y), \quad 3^R \leq x^{0.0116}, \quad M_- > 4 \cdot 3^{2R}, \quad R := \max \mathcal{I}_y. \quad (18)$$

Indeed $\sup \tilde{I}_y \leq ((\alpha^3 - 1)/d)L + L^{4/5}$, and $(\alpha^3 - 1) \log 3/d < 0.0116$. For example the elementary bounds $\log 3 < 1.1$ and $\log(4/3) > 0.287$ already give a strict margin. Meanwhile $\log M_- = L + dm_0 - L^{7/10} = L + o(L) + dm_0$. The lower bound for r/L tends to at least $(\alpha - 1)/d - 1/100000 > 0$.

Lemma 8 (Number of cylinders in the full return band). *For every interval W , every odd modulus q and every residue $z \pmod{q}$,*

$$|\#(E' \cap W \cap (z + q\mathbb{Z})) - q^{-1} \#(E' \cap W)| \leq P_x, \quad P_x \ll L^{m_0} e^{O(L^{7/10})} \ll x^{10^{-4}}.$$

Proof. Partition by the actual length- m_0 word $\mathbf{b}$ of M . Each part is an interval in its exact odd cylinder by Proposition 2 and (9); Proposition 3 bounds the discrepancy by the number of nonempty parts.

Write $B_- = b_1 + \dots + b_{m_0-1}$. At the preceding iterate, $\text{Syr}^{m_0-1}(M) > x$ and the offset is at most $3^{m_0-1} < x/2$. The affine identity therefore gives

$$2^{B_-} < 2 \cdot 3^{m_0-1} M/x \leq (2/3) 4^{m_0} e^{L^{7/10}}, \quad B_- \leq 2m_0 + O(L^{7/10}).$$

The number of positive words of length $m_0 - 1$ with sum at most an integer B is $\binom{B}{m_0-1} \leq 2^B$ (the empty-word case is harmless). There are consequently at most $4^{m_0} e^{O(L^{7/10})}$ such prefixes. For the last valuation, the elementary iterate bound $\text{Syr}^j(M) \leq 3^j(M+1)$ gives $2^{b_{m_0}} \leq 3 \text{Syr}^{m_0-1}(M) + 1 \leq 3^{m_0+1}(M+1)$. Thus $b_{m_0} = O(L)$ uniformly in the band. Multiply these two counts. Since $(\log 4)/100000 < 10^{-4}$, the claimed final power follows. $\square$

Allikvere's Lemma 6.3 (source label `lem:flat`, lines 1126–1207) uses this mechanism for $q = 3^k$. The proof above also records why the discrepancy bound is independent of the odd modulus and why it is applied to the full return band.

Lemma 9 (Separated dyadic–triadic endpoint intervals). *Let $R \geq 1$, $M > 4 \cdot 3^{2R}$ and $v > 0$. There is at most one pair $(r, s) \in \{1, \dots, R\} \times \mathbb{Z}$ such that*

$$M - 3^r \leq v 3^r 2^{-s} \leq M. \quad (19)$$

Proof. Set $Q = 3^R$. Every displayed value lies in $[M - Q, M]$, whose endpoint ratio is less than 2. At fixed r , changing s multiplies the value by an integral power of 2, so there is at most one s . Suppose two distinct indices $r_2 > r_1$ occurred. Put $q = r_2 - r_1$, $p = s_2 - s_1$ and $z_i = v 3^{r_i} 2^{-s_i}$. Their ratio is $z_2/z_1 = 3^q/2^p$. Since $z_2/z_1 < 2$ and $3^q \geq 3$, necessarily $p \geq 1$. Both 3^q and 2^p are positive integers and they are unequal, since one is odd and the other even. However,

$$|3^q - 2^p| = 2^p \frac{|z_2 - z_1|}{z_1} \leq \frac{Q^2 M}{(M - Q)^2} \leq \frac{4Q^2}{M} < 1.$$

Here $|z_2 - z_1| \leq Q$, $z_1 \geq M - Q$, and $2^p = 3^q z_1/z_2 \leq QM/(M - Q)$; also $M - Q > M/2$. This contradicts the integer separation $|3^q - 2^p| \geq 1$. $\square$

Equivalently, the following intervals in the endpoint variable v are pairwise disjoint under the lemma's hypotheses:

$$(r, s) \mapsto \left[(M - 3^r) \frac{2^s}{3^r}, M \frac{2^s}{3^r} \right].$$

The two labelled endpoints retain the full possible offset 3^r . This is the interval geometry used in the proof, not an illustration of actual orbit convergence.

For a word $\mathbf{a}$ of length r and total s , write $F = F_r(\mathbf{a})$ and $t = s - r\lambda$. The true fibre point and its zero-offset comparison are

$$N = 2^t(M - F), \quad N^0 = 2^t M, \quad 0 < F \leq 3^r.$$

The true window tests $y \leq N \leq y^\alpha$, whereas the source's replaced window tests $y < N^0 \leq y^\alpha$. These are different endpoint conventions, and both are retained below.

Proposition 10 (Absolute cost of replacing the fibre window). *With the parameters and full band above, let*

$$\mathcal{D}_x = \frac{2}{y^\alpha} \sum_{r \in \mathcal{I}_y} \sum_{M \in E'} \sum_{\substack{\mathbf{a} \in \mathbb{N}_{>0}^r \\ F_r(\mathbf{a}) \equiv M \pmod{3^r}}} \left| \mathbf{1}_{y \leq 2^{|\mathbf{a}| - r\lambda} (M - F_r(\mathbf{a})) \leq y^\alpha} - \mathbf{1}_{y < 2^{|\mathbf{a}| - r\lambda} M \leq y^\alpha} \right|.$$

Then $\mathcal{D}_x \ll_\alpha L^{-1/2}$. This bound includes all words, not only words in the central valuation event, and uses no irrationality measure for $\log_2 3$.

Proof. If the indicators differ, one of $v = y, y^\alpha$ lies between N and N^0 , inclusively. Hence

$$M \in [v 3^r 2^{-s}, v 3^r 2^{-s} + 3^r].$$

For fixed (r, s) these two boundary intervals have at most two disjoint components. Let $N_{r,s}^\partial$ count their intersection with E' , and let $\nu_{r,s}^\partial(z)$ impose the additional residue $z \pmod{3^r}$. Lemma 8 gives

$$\nu_{r,s}^\partial(z) = 3^{-r} N_{r,s}^\partial + O(P_x).$$

There are $\binom{s-1}{r-1}$ words with total s . Reversal is a bijection between the forward offset F_r and the reversed offset X_r , preserving this total. Thus their conditioned residue histograms agree. Summing those histograms against $\nu_{r,s}^\partial$ bounds the discrepancy by

$$\mathcal{D}_x \leq \frac{2}{y^\alpha} \sum_{r,s: N_{r,s}^\partial > 0} \binom{s-1}{r-1} \left(3^{-r} N_{r,s}^\partial + O(P_x) \right), \quad (20)$$

where the error means a nonnegative upper bound of absolute size CP_x .

Use the exact identity $\binom{s-1}{r-1} = 3^r 2^t p_r(s)$. If a boundary interval meets E' , then $2^t \leq y^\alpha / (M_- - 3^R) \leq 2y^\alpha / M_-$. The discrepancy part of (20) is therefore at most

$$CP_x 3^R |\mathcal{I}_y| / M_- \ll x^{10^{-4} + 0.0116 - 1 + o(1)} \ll x^{-0.98}.$$

Here $\sum_s p_r(s) = 1$ was used before summing over the $O_\alpha(L)$ prefix lengths; no unweighted count of infinitely many totals occurs.

For the main part, interchange the nonnegative sums. Fix M and one endpoint v . By (18) and Lemma 9, at most one pair (r, s) contributes. At that pair $2^t \leq 2v/M$. Moreover

$$\sup_s p_r(s) \ll r^{-1/2} \ll_\alpha L^{-1/2}.$$

For completeness, $p_r(s+1)/p_r(s) = s/[2(s-r+1)]$ for $s \geq r$; its maximum occurs at the central mode or the two neighbouring central modes, where Stirling's formula bounds it by $C/\sqrt{r}$. The case $r = 1$ follows directly from $p_1(s) = 2^{-s}$. The main contribution is consequently bounded by

$$\frac{C_\alpha}{\sqrt{L}} (1 + y^{1-\alpha}) \sum_{M \in E'} \frac{1}{M} \ll_\alpha L^{-1/2},$$

using (10). This dominates the discrepancy part and proves the claim with the stated closed and half-open endpoints. $\square$

The source's window-replacement lemma, label `lem:winrep` (lines 1326–1425), obtains $L^{-1/2+o(1)}$ using Rhin's lower bound for a linear form in $\log 2, \log 3$. The integer argument above gives $L^{-1/2}$ and at most one incidence per endpoint across the whole prefix-length range, not only across its central Gaussian window. It removes that dependency from this step only. The averaged kernel asymptotic below remains a separate equidistribution question.

9 A kernel bound before equidistribution

For $M \in E'$ set $u = \log_2(y^\alpha/M)$ and $u_0 = \log_2(y/M)$. The source's nonnegative counting kernel is

$$D_y(M) = \sum_{r \in \mathcal{I}_y} 3^{-r} \sum_{r\lambda + u_0 < s \leq r\lambda + u} \binom{s-1}{r-1}, \quad (21)$$

where binomial coefficients with $s < r$ contribute zero. Its exact asymptotic constant requires an average over the fractional parts of $r\lambda + u$. An upper bound sufficient for transferring the mixing errors does not.

Proposition 11 (Uniform upper bound for the fibre kernel). *Uniformly for the full band and the two choices of y , $D_y(M) \ll_\alpha y^\alpha/M$.*

Proof. Put $s_r = \lfloor r\lambda + u \rfloor$ and $r_* = u/(2 - \lambda)$. Here $u \asymp_\alpha L$ and $r_* \asymp_\alpha L$, uniformly in M ; this follows by writing $M = xe^{dm_0 + \sigma}$, $|\sigma| \leq L^{7/10}$. Removing the lower cutoff increases the sum, and the exact hockey-stick identity gives

$$3^{-r} \sum_{s \leq s_r} \binom{s-1}{r-1} = 3^{-r} \binom{s_r}{r} = 2^{s_r - r\lambda} \frac{s_r}{r} p_r(s_r) \leq 2^u \frac{s_r}{r} p_r(s_r).$$

For all sufficiently large x , s_r/r lies in a fixed compact interval $[b_0, b_1] \subset (1, \infty)$, independent of M, y, x . Stirling's inequalities, uniformly on this interval, imply

$$p_r(s) \leq C_\alpha r^{-1/2} \exp(-c_\alpha(s - 2r)^2/r).$$

To see the exponent explicitly, for $b = s/r$ the exponential part is $rH(b)$ with $H(b) = b \log b - (b-1) \log(b-1) - b \log 2$. It satisfies $H(2) = H'(2) = 0$ and $H''(b) = -1/[b(b-1)]$. Enlarge the compact interval to contain 2; Taylor's theorem then gives $H(b) \leq -c_\alpha(b-2)^2$ throughout it. The factorial prefactors give the stated $r^{-1/2}$ factor. Since $s_r - 2r = -(2-\lambda)(r-r_*) - \{r\lambda + u\}$ and $r \asymp_\alpha L$, it follows that

$$\sum_{r \in \mathcal{I}_y} \frac{s_r}{r} p_r(s_r) \leq \frac{C_\alpha}{\sqrt{L}} \sum_{r \in \mathbb{Z}} \exp(-c_\alpha(r-r_*)^2/L) \leq C'_\alpha.$$

The last bound follows by separating the two sides of the maximum and comparing with the Gaussian integral; it is uniform in the real centre r_* . Multiply by $2^u = y^\alpha/M$. $\square$

10 Transferring the mixing error through integer fibres

Let $\mathcal{A}_r$ consist of positive words satisfying $|a_1 + \dots + a_j - 2j| < L^{3/5}$ for every $0 \leq j \leq r$. For $M \in E'$ define the actual uniform-input weight

$$W_y(M) = \sum_{r \in \mathcal{I}_y} \sum_{\mathbf{a} \in \mathcal{A}_r} \mathbb{P}(\text{Aff}_{\mathbf{a}}(\tilde{N}_y) = M), \quad \tilde{N}_y \sim \text{Unif}((2\mathbb{N} + 1) \cap [y, y^\alpha]).$$

This definition counts real integer inputs, not draws of the independent geometric model. Let $m_1 = \lfloor L^{2/5} \rfloor$ and $\rho_{m_1}(z) = \mathbb{P}(X_{m_1} = z)$.

Proposition 12 (Absolute error in the uniform fibre formula). *With the parameters above,*

$$W_y(M) = \frac{2}{y^\alpha} 3^{m_1} \rho_{m_1}(M \bmod 3^{m_1}) D_y(M) + R_y(M), \quad \sum_{M \in E'} |R_y(M)| \ll_\alpha L^{-1/4}.$$

Proof. For fixed $r, \mathbf{a}, M$, Proposition 2 gives precisely one possible positive odd input, $2^{|\mathbf{a}|}(M - F_r(\mathbf{a}))/3^r$, provided the residue condition holds. It has uniform mass H_y^{-1} when it lies in $[y, y^\alpha]$, where $H_y = \#((2\mathbb{N} + 1) \cap [y, y^\alpha])$. Counting odd integers gives $H_y^{-1} = (2/y^\alpha)(1 + O_\alpha(x^{-c_\alpha}))$. For each r the sum over actual words and M is at most 1, because an input has only one actual word and one terminal value. Hence replacing this normalization costs $O_\alpha(Lx^{-c_\alpha})$ over the full band. Replacing the true fibre window by $y < 2^{s-r\lambda}M \leq y^\alpha$ costs $O_\alpha(L^{-1/2})$ by Proposition 10, even after enlarging the word range to all words.

For each (r, s) the replaced window is the exact interval

$$W_{r,s} = (3^r 2^{-s} y, 3^r 2^{-s} y^\alpha]$$

in the variable M . Put $N_{r,s} = \#(E' \cap W_{r,s})$ and $\nu_{r,s}(z) = \#(E' \cap W_{r,s} \cap (z + 3^r \mathbb{Z}))$. Lemma 8 gives $\nu_{r,s}(z) = 3^{-r} N_{r,s} + O(P_x)$ uniformly. Whenever $N_{r,s} > 0$, $2^{s-r\lambda} \leq y^\alpha / M_-$, so

$$\frac{2P_x}{y^\alpha} \sum_{r,s: N_{r,s} > 0} \binom{s-1}{r-1} \leq \frac{2P_x 3^R |\mathcal{I}_y|}{M_-} \ll x^{-0.98}. \quad (22)$$

Also nonnegative summation and Proposition 11 give

$$\frac{2}{y^\alpha} \sum_{r,s} \binom{s-1}{r-1} 3^{-r} N_{r,s} = \sum_{M \in E'} \frac{2D_y(M)}{y^\alpha} \ll \sum_{M \in E'} \frac{1}{M} \ll 1. \quad (23)$$

Call a pair central when $|s - 2r| \leq C\sqrt{r \log r}$, where C is a fixed sufficiently large constant. On those pairs, reversal of a word preserves its total, so the unrestricted conditioned law of the forward offset F_r agrees with that of X_r . Corollary 7 therefore gives an error $h_{r,s}$ with

$$\sum_{z \in G_r} |h_{r,s}(z)| \leq \varepsilon_x, \quad \varepsilon_x \ll_A m_1^{-A} + \sqrt{m_1 \log L/L} \ll L^{-1/4}$$

from the uniform lift of ρ_{m_1} , uniformly in central (r, s) . Pair this absolute error with $\nu_{r,s}$ and use (22)–(23). The total contribution is at most $O(\varepsilon_x)$, with no factor equal to the length of the full band and no factor counting all residues.

Removing the restriction $\mathbf{a} \in \mathcal{A}_r$ also has to be justified with these residue weights. The Chernoff estimate and a union bound give $\mathbb{P}(\mathbf{a} \notin \mathcal{A}_r) \ll L \exp(-cL^{1/5})$. On central pairs, division by $p_r(s)$ changes this by only a fixed power of L , so the conditional bad fraction $\delta_{r,s}$ is $O(\exp(-c'L^{1/5}))$. The number of bad words is exactly $\binom{s-1}{r-1} \delta_{r,s}$. Summing their residue weights using the same two budgets bounds their contribution by $O(e^{-c'L^{1/5}}) + O(x^{-0.98})$. No invariance of $\mathcal{A}_r$ under word reversal is asserted: reversal sends it to a suffix-concentration event. The bad-word count uses the original event; the unrestricted histogram is the only place where the reversal identity is used.

For noncentral pairs, positivity and Chernoff give, for any prescribed $B > 0$ after choosing C ,

$$\sum_{|s-2r| > C\sqrt{r \log r}} p_r(s) \ll L^{-B-1}.$$

For a fixed M , the identity $3^{-r} \binom{s-1}{r-1} = 2^{s-r\lambda} p_r(s)$ and the window bound $2^{s-r\lambda} \leq y^\alpha / M$ show that all noncentral kernel terms, summed over r , are $O(y^\alpha L^{-B} / M)$. The actual noncentral words are bounded by this estimate and (22). The same estimate extends the lifted main term from central pairs to all pairs, since

$$3^{m_1} \sum_{M \in E'} \frac{\rho_{m_1}(M \bmod 3^{m_1})}{M} \ll 1.$$

The last inequality follows from (10): group M by its residue modulo 3^{m_1} , sum the bound over its refinements modulo 3^r for any $r \in \mathcal{I}_y$ (where $m_1 \leq r$), and then use $\sum_z \rho_{m_1}(z) = 1$.

The unrestricted, all-total main count is exactly $(2/y^\alpha) 3^{m_1} \rho_{m_1}(M) D_y(M)$. Combining the absolute bounds just proved gives the displayed remainder. The probability and counting normalizations, the endpoint discrepancy and the exceptional words have all been accounted for before taking a subset of the full return band. $\square$

This reconstructs the source's equation `eq:master` and its summed remainder estimate (lines 1079–1425), without using its equidistributed kernel asymptotic as an input to the error bound. It does not yet identify a common limiting passage measure: that requires the leading constant and its uniformity in M in the next kernel calculation.

11 The phase average and the leading constant

The remaining kernel has two scales: a Gaussian envelope varies over about $\sqrt{\log x}$ consecutive times, while the last included integer valuation introduces the factor $2^{-\{r\lambda+u\}}$. Its average, not its value at a chosen phase, supplies the constant. We first give the weighted discrepancy argument with all endpoint terms retained.

Lemma 13 (Summation by parts for the actual logarithmic rotation). *Set $\lambda = \log_2 3$, $\mu = 8.616$ and $\beta = 1 - 1/\mu$. For every real periodic function f of bounded variation V_f , every $u \in \mathbb{R}$, and every real weight w supported on $\{A, \dots, B\}$ of cardinality $N \geq 1$,*

$$\left| \sum_n w(n) f(\{n\lambda + u\}) - \left(\sum_n w(n) \right) \int_0^1 f(t) dt \right| \leq C_\lambda V_f N^\beta \sum_{n \in \mathbb{Z}} |w(n+1) - w(n)|. \quad (24)$$

The weight is extended by zero outside its support. The constant is independent of the interval, its location, the phase, and the weight.

Proof. Rhin's proposition, equation (8) on p. 160 [7], gives $|u_0 + u_1 \log 2 + u_2 \log 3| > H^{-7.616}$ for sufficiently large $H = \max(|u_1|, |u_2|)$. For an integer $q \geq 1$, take p nearest to $q\lambda$. Then $1 \leq p \leq 2q$ and $q\lambda - p = (q \log 3 - p \log 2)/\log 2$. The linear form is nonzero: otherwise $3^q = 2^p$. Thus the finitely many small q can be absorbed in a positive constant, yielding

$$\|q\lambda\| \geq c_\lambda q^{-(\mu-1)} \quad (q \geq 1).$$

This is an application of Rhin's published theorem, not a new verification of his auxiliary-polynomial computation.

For a block of k consecutive indices, the geometric-series identity gives $|\sum e(h(n\lambda + u))| \leq \min(k, (2\|h\lambda\|)^{-1})$. The Erdős–Turán inequality [6, Ch. 2, Theorem 2.5, pp. 112–114], in unnormalized counting form, consequently gives

$$kD_k \leq C \left(\frac{k}{H} + c_\lambda^{-1} \sum_{h=1}^H h^{\mu-2} \right) \leq C_\lambda \left(\frac{k}{H} + H^{\mu-1} \right).$$

Choose $H = \max(1, \lfloor k^{1/\mu} \rfloor)$. Koksma's inequality [6, Ch. 2, Theorem 5.1, p. 143] then bounds the centered sum on every block of length k by $C_\lambda V_f k^\beta$. The book's discrepancy is normalized by k ; the factor k above is essential. Star discrepancy is bounded by the interval discrepancy used here.

Put $a_n = f(\{n\lambda + u\}) - \int_0^1 f$ and $S_j = \sum_{n=A}^j a_n$, so $|S_j| \leq C_\lambda V_f N^\beta$. The exact finite identity

$$\sum_{n=A}^B w(n) a_n = w(B) S_B + \sum_{n=A}^{B-1} (w(n) - w(n+1)) S_n$$

proves the assertion, because $|w(B)|$ is included in the variation of the zero extension. There is no independence hypothesis between the deterministic phase and the deterministic weights. $\square$

In particular, a block-free estimate is available; no term of the form $\ell_0 V$ is incurred by freezing the weight. Allikvere's `lem:E` instead balances that term against block discrepancy. Passing to a smooth envelope remains useful because it gives a small variation, not because a deterministic weight must be independent of the phase.

Lemma 14 (Exact mass and variation of the smooth envelope). *Let $a = 2 - \lambda > 0$, $z > 0$, and*

$$g_z(t) = \frac{1}{\sqrt{\pi t}} \exp\left(-\frac{a^2(t-z)^2}{4t}\right) \quad (t > 0), \quad g_z(0) = 0.$$

Then $\int_0^\infty g_z(t) dt = 2/a$. For $z \asymp_\alpha L \rightarrow \infty$ and $J_z = \{r \in \mathbb{Z} : |r - z| \leq \sqrt{z} \log L\}$, the zero-extended weight $w_z(r) = g_z(r) \mathbf{1}_{J_z}(r)$ satisfies

$$\sum_{r \in \mathbb{Z}} |w_z(r+1) - w_z(r)| \ll_\alpha L^{-1/2}, \quad \sum_{r \in \mathbb{Z}} w_z(r) = \frac{2}{a} + O_\alpha(L^{-1/2}).$$

Both estimates are uniform in the real, not necessarily integral, centre.

Proof. Make the bijective changes $t = s^2$, $s > 0$, and $v = s - z/s \in \mathbb{R}$. Their inverse and derivative are

$$s = \frac{v + \sqrt{v^2 + 4z}}{2}, \quad \frac{ds}{dv} = \frac{1}{2} \left(1 + \frac{v}{\sqrt{v^2 + 4z}}\right) > 0.$$

Thus the integral is

$$\frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-a^2 v^2/4} \left(1 + \frac{v}{\sqrt{v^2 + 4z}}\right) dv = \frac{2}{a}.$$

The second summand is odd and absolutely integrable; the first is the Gaussian integral. No limiting replacement of t by z is used.

Moreover

$$4t^2(\log g_z)'(t) = a^2 z^2 - 2t - a^2 t^2.$$

The right side is strictly decreasing for $t > 0$, with its unique zero $t_c = \sqrt{z^2 + a^{-4}} - a^{-2}$. For sufficiently large z , $t_c \geq z/2$, so $\max g_z \leq \sqrt{2/(\pi z)}$. The function increases and then decreases, with zero limits at both ends. Its sampled sequence and its interval truncation are therefore unimodal; the variation of each zero-extended sequence is twice its maximum. This proves the first bound, including both boundary jumps.

For each integer $r \geq 0$, $|g_z(r) - \int_r^{r+1} g_z(t) dt| \leq \int_r^{r+1} |g'_z(t)| dt$. Summation gives an error at most $2 \max g_z$ between the full discrete sum and its exact integral. To truncate the sum, on $z/2 \leq r \leq 2z$ use $g_z(r) \leq C z^{-1/2} e^{-c(r-z)^2/z}$. Outside this interval, the exponent is at most $-cz^2/r$ on the left and $-cr$ on the right. Comparing the resulting positive sums with integrals shows that the discarded mass is $O_A(L^{-A})$ for every fixed $A > 0$: $e^{-c(\log L)^2}$ is smaller than every fixed negative power of L . This proves the second bound and its stated uniformity. $\square$

Proposition 15 (Uniform leading kernel and a stronger error exponent). *For both prescribed values of y , uniformly for every real $M \in [M_-, M_+]$, and for every fixed $0 < c < 1/17.232$,*

$$D_y(M) = \frac{y^\alpha}{M \log(4/3)} (1 + O_{\alpha,c}(L^{-c})). \quad (25)$$

In particular $c = 1/18$ is admissible. The definition of the kernel is (21); first passage is not needed for this pointwise estimate on the band.

Proof. Put $z = u/a$, $s_r = \lfloor r\lambda + u \rfloor$, $k_r = \lfloor r\lambda + u_0 \rfloor$ and $\theta_r = \{r\lambda + u\}$. Exact hockey-stick summation, with $\binom{k}{r} = 0$ for $k < r$, gives

$$D_y(M) = \sum_{r \in \mathcal{I}_y} 3^{-r} \left(\binom{s_r}{r} - \binom{k_r}{r} \right).$$

The upper term is $2^u 2^{-\theta_r} (s_r/r) p_r(s_r)$. The lower term is at most $C_\alpha 2^{u_0}$: when $k_r \geq r$ use $3^{-r} \binom{k_r}{r} = 2^{k_r - r\lambda} (k_r/r) p_r(k_r)$, $p_r(k_r) \leq 1$ and the uniform bound $k_r/r = O_\alpha(1)$. Since $u - u_0 \asymp_\alpha L$, the total relative bottom error is $O_\alpha(L 2^{-(u-u_0)})$, exponentially small in L .

Writing $M = x e^{dm_0 + \sigma}$, $|\sigma| \leq L^{7/10}$, gives

$$z = \frac{\log(y^\alpha/x)}{d} - m_0 - \frac{\sigma}{d}.$$

Its distance below the upper endpoint of $\tilde{I}_y - m_0$ is $L^{4/5} + \sigma/d \geq \frac{1}{2} L^{4/5}$; its distance above the lower endpoint is $(\alpha - 1) \log y/d - L^{4/5} - \sigma/d \gg_\alpha L$. Consequently $J_z \subset \mathcal{I}_y$ for sufficiently large x . The uniform Gaussian upper bound in Proposition 11 shows that the upper terms outside J_z , divided by 2^u , total $O_A(L^{-A})$ for every fixed A .

For $s = 2r + \delta$ in the remaining range, Stirling's formula gives

$$p_r(s) = \frac{e^{-\delta^2/(4r)}}{\sqrt{4\pi r}} \left(1 + O\left(r^{-1} + \frac{|\delta|}{r} + \frac{|\delta|^3}{r^2}\right) \right).$$

Indeed its logarithmic exponential term is $rH(s/r)$ with the function H computed above; $H''(2) = -1/2$ and H''' is bounded near 2. The factorial prefactor is $(2\pi r b(b-1))^{-1/2}$, $b = s/r$, and its relative deviation at $b = 2$ is $O(|\delta|/r)$; Stirling's remainder is $O(1/r)$. Here $\delta = -a(r-z) - \theta_r$, so $|\delta| = O_\alpha(\sqrt{L} \log L)$. Removing θ_r from the Gaussian exponent changes it by $O_\alpha(L^{-1/2} \log L)$, and $s_r/r = 2 + O_\alpha(L^{-1/2} \log L)$. It follows uniformly on J_z that

$$(s_r/r) p_r(s_r) = g_z(r) \left(1 + O_\alpha(L^{-1/2} (\log L)^3) \right).$$

Lemma 14 bounds its summed error by the same quantity. Thus

$$2^{-u} D_y(M) = \sum_{r \in J_z} g_z(r) 2^{-\theta_r} + O_\alpha(L^{-1/2} (\log L)^3).$$

Apply Lemma 13 to the zero extension of g_z on J_z , whose cardinality is $O_\alpha(\sqrt{L} \log L)$ and variation is $O_\alpha(L^{-1/2})$. For $f(\theta) = 2^{-\theta}$, periodically continued with $f(0) = 1$, the circle variation is 1, including the jump, and $\int_0^1 f = 1/(2 \log 2)$. We obtain the explicit error shape

$$2^{-u} D_y(M) = \frac{1}{a \log 2} + O_\alpha \left(L^{-1/(2\mu)} (\log L)^{1-1/\mu} + L^{-1/2} (\log L)^3 \right). \quad (26)$$

Since $a \log 2 = \log(4/3)$ and $2\mu = 17.232$, this proves the claim. The constants retain the fixed Diophantine constant for λ ; they are not uniform over rotations with unspecified lower-bound constants. $\square$

Corollary 16 (Common profile for the actual integer-fibre weights). *Define the finite measure on E' by*

$$\Pi_x(M) = \frac{2 \cdot 3^{m_1}}{\log(4/3)} \frac{\rho_{m_1}(M \bmod 3^{m_1})}{M}.$$

Then for every $0 < c < 1/17.232$ and both choices of y , $\sum_{M \in E'} |W_y(M) - \Pi_x(M)| \ll_{\alpha, c} L^{-c}$. Every pushforward under a map on E' satisfies the same bound.

Proof. Insert (25) in Proposition 12. Its error sums to $O(L^{-1/4})$. The kernel error sums to $O(L^{-c})$ by the weighted harmonic bound $3^{m_1} \sum_{E'} \rho_{m_1}(M)/M \ll 1$ proved there. Pushforward is an ℓ^1 contraction by the triangle inequality on each fibre. $\square$

This gives a common *finite measure*. The next section proves its probability identification using the actual-input reduction preceding the source's counting argument. The kernel bound improves the source's exponent ceiling $1/(2(\mu + 1))$ to $1/(2\mu)$ using the same Rhin input; it is not an independent novelty assertion about discrepancy theory or a completed upgrade to the source's final natural-density theorem.

12 From the integer fibres to the first-passage law

We now justify the probability identification rather than assuming it from the shape of the kernel. Let $O_x = (2\mathbb{N} + 1) \cap [1, x]$, $n_0 = \lfloor L/(10 \log 2) \rfloor$, and let $\tilde{N}_y$ be uniform on the positive odd integers in $[y, y^\alpha]$. We use $\text{Pass}_x(N) = \text{Syr}^{T_x(N)}(N)$ when the first-passage time is finite and $\text{Pass}_x(N) = 1$ otherwise. The latter convention defines a probability law without assuming Collatz convergence; the probability of using it will be bounded separately. Throughout, $\|\cdot\|_1$ denotes the sum of absolute atom masses, not half that sum.

Lemma 17 (Actual input, concentration, and the extended window). *For both values of y ,*

$$\|\text{Law}(\vec{a}^{(n_0)}(\tilde{N}_y)) - \text{Geom}(2)^{n_0}\|_1 \ll x^{-\eta}$$

for some $\eta > 0$. Put $G_y = \{\vec{a}^{(n_0)}(\tilde{N}_y) \in \mathcal{A}^{(n_0)}\}$, where every prefix satisfies $|a_{[1,j]} - 2j| < L^{3/5}$. There is $\kappa > 0$ such that

$$\mathbb{P}(G_y^c) \ll e^{-\kappa L^{1/5}}, \quad \mathbb{P}(T_x(\tilde{N}_y) \notin \tilde{I}_y) \ll e^{-\kappa L^{1/5}}, \quad \mathbb{P}(T_x(\tilde{N}_y) > n_0) \ll x^{-\eta}.$$

The first prefix bound is made explicit in Corollary 48. Constants may be decreased between the displayed coarse bounds. On G_y , uniformly for $0 \leq j \leq n_0$,

$$\log \text{Syr}^j(N) = \log N - dj - (\log 2)(a_{[1,j]} - 2j) + O(x^{-3/4}). \quad (27)$$

In particular $T_x(N) = \log(N/x)/d + O_\alpha(L^{3/5})$ on this event.

Proof. Let $K_y = \#((2\mathbb{N} + 1) \cap [y, y^\alpha])$ and $q = 3n_0$. An odd residue class modulo 2^q contains $K_y/2^{q-1} + O(1)$ of these consecutive odd integers. Thus its full ℓ^1 discrepancy is $O(2^q/K_y) \ll 2^{-q}$, since $K_y \asymp y^\alpha$ and $4^{3n_0} \leq x^{3/5} \leq y$ for large x . This verifies the precise hypothesis of Tao's valuation theorem [8, Proposition 1.9], with $c_0 = 1$. Its conclusion gives the first assertion.

Here is also the finite-cylinder argument behind that application, to make explicit which distribution is transported. For a word of length n_0 and total $s < q$, its cylinder is one odd class modulo 2^{s+1} , so its probability is $2^{-s} + O(2^{-q})$. The number of such words is $\binom{q-1}{n_0}$. For a word whose total reaches q , stop just before that happens: if the preceding length is $k < n_0$, its positive prefix has total less than q , with $\binom{q-1}{k}$ possibilities. The divisibility of the next affine numerator by 2^q fixes one odd residue modulo 2^q , independently of the size of the crossing valuation. Each such event has probability $O(2^{-q})$. Hence the actual tail is at most

$$C2^{-q} \sum_{k=0}^{n_0-1} \binom{q-1}{k} = O(e^{-cn_0}).$$

The binomial lower-tail Chernoff bound applies because $n_0 - 1 < (q - 1)/3$. The geometric tail is also $O(e^{-cn_0})$. For sufficiently large n_0 , also $n_0/(q - 1) < 2/5$; the same binomial lower-tail bound

controls $2^{-q} \binom{q-1}{n_0}$. Adding the low-total atom differences and the two tails proves the stated ℓ^1 estimate.

For completeness, if A has mass $\mathbb{P}(A = k) = 2^{-k}$ on $k \geq 1$,

$$\mathbb{E} e^{t(A-2)} = \frac{e^{-t}}{2 - e^t} \quad (t < \log 2).$$

The logarithm has value and derivative zero at 0, and bounded second derivative on any sufficiently small fixed interval about 0. Independence and exponential Markov bounds, with either sign of t , therefore give

$$\mathbb{P}(|A_1 + \dots + A_j - 2j| \geq h) \leq 2 \exp[-c \min(h^2/j, h)] \quad (j \geq 1).$$

Take $h = L^{3/5}$, sum over $j \leq n_0$, and add the valuation-law error once. The factor $n_0 = O(L)$ is absorbed by decreasing κ , giving the first concentration bound.

The exact affine formula has linear term $N e^{-dj} 2^{-(a_{[1,j]} - 2j)}$ and offset between 0 and 3^j . On G_y the linear term exceeds $x^{0.95}$ for large x : indeed $\log_2 3 > 3/2$ gives $\alpha - d/(10 \log 2) > \alpha - 1/20 = 0.951$. Also $\log_2 3 < 8/5$ gives $3^{n_0} < x^{0.16}$. These strict logarithmic inequalities follow from $3^2 > 2^3$ and $3^5 < 2^8$, respectively. Dividing the offset by the linear term proves (27), with room in the exponent. Its upper bound at n_0 is less than x , since $\alpha^3 - d/(10 \log 2) < \alpha^3 - 1/25 < 1$. The crossing exists by that time. At $T_x(N) - 1$ the orbit is above x and at $T_x(N)$ it is at most x ; applying (27) to both indices gives the asserted time estimate.

The upper endpoint of $\tilde{I}_y$ is respected on G_y because $N \leq y^\alpha$ and $L^{4/5}$ dominates $L^{3/5}$. Failure of the lower endpoint implies $N \leq y e^{2dL^{4/5}}$ for large x . By direct counting of odd integers this has probability $O(y^{1-\alpha} e^{2dL^{4/5}} + 1/K_y) = O(x^{-\eta})$. This proves the window concentration. Notice that no upper layer of uniform mass has been discarded.

Finally the same geometric exponential bound at $h = n_0/10$ gives $\mathbb{P}(a_{[1,n_0]} \leq 1.9n_0) \ll x^{-\eta}$. On the complementary event,

$$\text{Syr}^{n_0}(N) \leq C x^{\alpha^3 + (\log_2 3)/10 - 0.19} + x^{(\log_2 3)/10} < x$$

for large x : the first exponent is less than $\alpha^3 - 0.03 = 0.973003001 < 1$, and the second is less than 0.16. This proves the stronger power bound for $T_x > n_0$ without trying to obtain it from the weaker prefix-concentration event. $\square$

Proposition 18 (Probability reduction with its full mass error). *Let $q_x : E' \rightarrow O_x$ be the deterministic map $q_x(M) = \text{Syr}^{m_0}(M)$. With W_y defined in Proposition 12, for every fixed $B > 0$,*

$$\| \text{Law}(\text{Pass}_x(\tilde{N}_y)) - (q_x)_* W_y \|_1 \ll_{\alpha, B} L^{-B}. \quad (28)$$

The measure W_y is initially a finite counting-weight measure, not assumed to have mass one.

Proof. For a fixed input N form the finite measure on O_x

$$H_N = \sum_{n \in \tilde{I}_y \cap \mathbb{Z}} \mathbf{1}_{\tilde{a}^{(n-m_0)}(N) \in \mathcal{A}^{(n-m_0)}} \mathbf{1}_{\text{Syr}^{n-m_0}(N) \in E'} \delta_{q_x(\text{Syr}^{n-m_0}(N))}.$$

Its expected value is exactly $(q_x)_* W_y$, by the valuation-fibre identity of Proposition 2. The formula retains the true input-window indicator: an affine inverse outside $[y, y^\alpha]$ has probability zero.

For $N \in G_y$ and $n \in \tilde{I}_y \cap \mathbb{Z}$ set $r = n - m_0$. We prove

$$T_x(N) = n \iff \text{Syr}^r(N) \in E'.$$

In the forward direction, $M = \text{Syr}^r(N)$ has $T_x(M) = m_0$. Equation (27) and the passage-time estimate give $\log M = \log x + dm_0 + O_\alpha(L^{3/5})$, inside the band $[M_-, M_+]$ because $L^{7/10}$ dominates $L^{3/5}$. Conversely, $M = \text{Syr}^r(N) \in E'$ controls indices from r to n . For $j < r$, subtracting the two drift formulas gives

$$\log \text{Syr}^j(N) \geq \log M + d(r - j) - C_\alpha L^{3/5} \geq \log x + dm_0 - L^{7/10} - C_\alpha L^{3/5} > \log x.$$

Here dm_0 is a fixed positive multiple of L , up to a bounded floor error. There is thus no earlier crossing. At time n the crossing occurs and its value is $q_x(M)$, proving the equivalence. The only interval property used here is $\tilde{I}_y \subset [m_0, n_0]$, already checked in the source and in (18).

It follows that on $G_y \cap \{T_x(N) \in \tilde{I}_y\}$, $H_N = \delta_{\text{Pass}_x(N)}$; on G_y outside that window, $H_N = 0$. In all cases H_N has at most $J_y = \#(\tilde{I}_y \cap \mathbb{Z}) = O_\alpha(L)$ atoms counted with multiplicity. Consequently

$$\|\mathbb{E}H_N - \text{Law}(\text{Pass}_x(N))\|_1 \leq \mathbb{P}(T_x(N) \notin \tilde{I}_y, G_y) + (J_y + 1)\mathbb{P}(G_y^c) \ll_\alpha L e^{-\kappa L^{1/5}} \ll_{\alpha, B} L^{-B}.$$

This argument controls the entire measure at once, including the possible failure convention, rather than a separate estimate for each terminal atom. $\square$

Corollary 19 (Uniform first-passage stabilisation). *For sufficiently large x , $\Pi_x(E') > 0$, and*

$$Q_x = \frac{(q_x)_* \Pi_x}{\Pi_x(E')}$$

is a probability measure on O_x , independent of y . For every fixed $0 < c < 1/17.232$,

$$\|\text{Law}(\text{Pass}_x(\tilde{N}_y)) - Q_x\|_1 \ll_{\alpha, c} L^{-c}, \quad \Pi_x(E') = 1 + O_{\alpha, c}(L^{-c}).$$

In particular the two first-passage laws, for $y = x^\alpha$ and $y = x^{\alpha^2}$, have ℓ^1 distance $O_{\alpha, c}(L^{-c})$. Also $\mathbb{P}(T_x(\tilde{N}_y) = \infty) \ll x^{-\eta}$.

Proof. Proposition 18, Corollary 16, and contraction of pushforward give $\|\text{Law}(\text{Pass}_x(\tilde{N}_y)) - (q_x)_* \Pi_x\|_1 \ll L^{-c}$. Taking total masses proves $|\Pi_x(E') - 1| \ll L^{-c}$ and positivity for large x . Division by that positive mass changes the measure in ℓ^1 by exactly $|\Pi_x(E') - 1|$. The triangle inequality proves both probability comparisons. The last assertion follows from $T_x = \infty$ implying $T_x > n_0$ in Lemma 17. $\square$

This reconstructs Allikvere's uniform first-passage theorem (source label `thm:D`) and its probability reduction `prop:approx`, with the stronger component rate propagated to the first-passage law. The deterministic passage mechanism is Tao's Proposition 5.2; the change of initial measure and extended time window are Allikvere's. The common profile is divided by its *proved* mass, without replacing the original sampling measure or suppressing a factor. The following sections supply the subsequent target-set iteration and exact raw-clock comparison.

13 The scale iteration and the fixed-target estimate

Write U_y for the uniform probability on the positive odd integers in $[y, y^\alpha]$, whenever this set is nonempty. All probabilistic applications below have y larger than a fixed constant. The final counting statements hold for every real cutoff $X \geq 2$; no uniform probability on an empty short interval is needed. Fix $0 < c < 1/17.232$, and retain $\alpha = 1001/1000$ and $d = \log(4/3)$.

Lemma 20 (Nested passage, with failure kept separate). *For $1 \leq z \leq w$ and a positive odd integer N with $T_z(N) < \infty$, put $M = \text{Pass}_w(N)$. Then*

$$T_w(N) \leq T_z(N), \quad T_z(N) = T_w(N) + T_z(M), \quad \text{Pass}_z(N) = \text{Pass}_z(M).$$

For any subset E of the positive odd integers,

$$\mathbb{P}(\text{Pass}_z(N) \in E) - \mathbb{P}(T_z(N) < \infty, \text{Pass}_z(N) \in E) = \mathbf{1}_{1 \in E} \mathbb{P}(T_z(N) = \infty).$$

The second identity holds for any random positive odd input.

Proof. Before the first entry below w , every iterate exceeds $w \geq z$. Starting at that entry, the first entry below z is exactly the remaining part of the same orbit. This proves the three deterministic identities, including the case $M \leq z$, when the remaining time is zero. On failed passage the defined value of Pass_z is 1; splitting the probability according to finite or infinite passage proves the last equality. $\square$

Proposition 21 (The error survives the scale ladder). *For every integer $N_0 \geq 2$ and every sufficiently large y ,*

$$U_y\{N : \text{Syr}^k(N) > N_0 \text{ for all } k \geq 0\} \ll_c (\log N_0)^{-c}.$$

Proof. Put $E = \{M : \exists k \geq 0, \text{Syr}^k(M) \leq N_0\}$ and

$$b(z) = U_{z^\alpha}\{N : T_z(N) < \infty, \text{Pass}_z(N) \in E\}.$$

An orbit whose finite passage below z belongs to E has its earlier passage below z^α in E as well. Apply this inclusion to U_{z^α} , then use Corollary 19 to compare the two laws at the same target z . Lemma 20 gives

$$b(z^\alpha) \geq b(z) - C_c(\log z)^{-c} - Cz^{-\eta} \geq b(z) - C'_c(\log z)^{-c} \quad (*)$$

for sufficiently large z . In particular the failure convention is not being used to count an orbit as successful. Allikvere's final equality with $b(z)$ in the display following **sec:assembly** absorbs this failure mass into its O -term, which is legitimate. The inequality here makes that bookkeeping explicit; it is not a counterexample to the source argument. Tao's corresponding display in [8, §3] already uses that inequality.

Bounded N_0 are covered by increasing the constant. Assume henceforth that N_0 is large enough that N_0^{1/α^3} exceeds every scale threshold used above. If $y^\alpha \leq N_0$, the assertion has witness $k = 0$ for every input. Otherwise let $J \geq 1$ be least such that

$$v = y^{\alpha^{-J}} < N_0^{1/\alpha}.$$

Then $N_0^{1/\alpha^2} \leq v < N_0^{1/\alpha}$ and $y = v^{\alpha^J}$. Set $z_j = v^{\alpha^{j-1}}$ for $0 \leq j \leq J$. Thus $z_0 = v^{1/\alpha} \geq N_0^{1/\alpha^3}$, $z_J = y^{1/\alpha}$ and $z_{j+1} = z_j^\alpha$. At z_0 a finite passage lands at most $z_0 < N_0$, so $b(z_0) \geq 1 - Cz_0^{-\eta}$. Applying (*) successively gives

$$1 - b(z_J) \leq Cz_0^{-\eta} + C'_c \sum_{j=0}^{J-1} (\log z_j)^{-c} \leq Cz_0^{-\eta} + \frac{C'_c (\log z_0)^{-c}}{1 - \alpha^{-c}} \ll_c (\log N_0)^{-c}.$$

The source of $b(z_J)$ is exactly U_y . Every input counted by $b(z_J)$ has an actual iterate at most N_0 , proving the assertion. The factor $(1 - \alpha^{-c})^{-1}$ is retained in the constant; no uniformity as $c \downarrow 0$ or $\alpha \downarrow 1$ is asserted. $\square$

14 Keeping the time budget through the same iteration

The distribution comparison concerns passage *values*, not the joint distribution of values and passage times. To carry time, the target set itself must contain a subsequent time-bounded orbit statement. This is Allikvere's timed-target construction [9, source labels `sec:time`, `prop:timed`]; the deterministic inclusion below explains why value-distribution comparison suffices.

Put $p = 4/5$ and

$$\Delta_0 = (1 - \alpha^{-3}) \frac{\log N_0}{d},$$

$$E_z^\bullet = \left\{ M \in O_z : \exists k \in \mathbb{N}, k \leq \frac{(\log(M/N_0))_+}{d} + \Delta_0 + C_1(\log z)^p, \text{Syr}^k(M) \leq N_0 \right\}.$$

Here $(t)_+ = \max(t, 0)$; the nonnegative integer k includes zero. Let $B_z^\bullet$ be the event, on input $N \sim U_{z^\alpha}$, that

$$T_z(N) \leq \frac{\log(N/z)}{d} + C_1(\log z)^p, \quad \text{Pass}_z(N) \in E_z^\bullet.$$

Its first inequality requires finite passage.

Lemma 22 (The timed inclusion). *There are constants $C_1, z_* > 0$, independent of N_0 and c , for which*

$$\mathbb{P}(B_z^\bullet) \geq 1 - O((\log z)^{-10}) \quad (z_* \leq z \leq N_0),$$

and, for $z \geq \max(z_*, N_0^{1/\alpha^3})$,

$$\mathbb{P}(B_{z^\alpha}^\bullet) \geq \mathbb{P}(B_z^\bullet) - O_c((\log z)^{-c}).$$

Proof. The first estimate follows from Lemma 17: on its concentration event the time inequality holds, and every passage value is at most $z \leq N_0$, with subsequent witness zero.

For the second estimate take the common input $N \sim U_{z^\alpha}$. Intersect its concentration events at targets z and z^α , and call the intersection G . Both are instances of Lemma 17, so $\mathbb{P}(G^c) \ll e^{-\kappa(\log z)^{1/5}} \ll (\log z)^{-10}$. On G both passages exist, and, with $M = \text{Pass}_{z^\alpha}(N)$ and $M' = \text{Pass}_z(N)$, the nested-passage lemma gives

$$t := T_z(M) = T_z(N) - T_{z^\alpha}(N).$$

Subtracting the two passage-time estimates and applying the drift estimate at $T_{z^\alpha}(N)$ gives respectively

$$t \leq \frac{\log(z^\alpha/z)}{d} + C(\log z)^{3/5}, \quad \log M \geq \log z^\alpha - C(\log z)^{3/5}.$$

The drift estimate is applicable at that iterate because each passage lies within its corresponding length- n_0 valuation window on G . It follows that

$$t \leq \frac{\log(M/z)}{d} + C'(\log z)^{3/5}. \quad (**)$$

In particular this estimate concerns the same orbit, not a new uniform sample at M .

Suppose now that G holds and $M' \in E_z^\bullet$. If $M \leq N_0$, then $M \in E_{z^\alpha}^\bullet$ with witness zero. If $M > N_0$ and $z \geq N_0$, concatenate the leg of length t with a witness k_2 for M' . Since $M' \leq z$,

$$t + k_2 \leq \frac{\log(M/N_0)}{d} + \Delta_0 + C_1(\log z)^p + C'(\log z)^{3/5}.$$

If $M > N_0$ and $z < N_0$, use $k_2 = 0$; in this case $\log(N_0/z)/d \leq \Delta_0$ by $z \geq N_0^{1/\alpha^3}$, so $(**)$ gives the same bound. Choose C_1 and z_* so that, for $z \geq z_*$,

$$C'(\log z)^{3/5} \leq C_1(\alpha^p - 1)(\log z)^p.$$

Thus in both cases $M \in E_{z^\alpha}^\bullet$. The upper estimate for $T_{z^\alpha}(N)$ also supplies the time clause of $B_{z^\alpha}^\bullet$, after increasing C_1 if necessary. We have proved the actual inclusion

$$G \cap \{\text{Pass}_z(N) \in E_z^\bullet\} \subseteq B_{z^\alpha}^\bullet.$$

Apply the first-passage law comparison to the *fixed* subset $E_z^\bullet \subset O_z$, subtract $\mathbb{P}(G^c)$, and use $B_z^\bullet \subseteq \{\text{Pass}_z(U_{z^\alpha}) \in E_z^\bullet\}$. This proves the probability inequality. No independence between passage location and time was assumed. $\square$

Proposition 23 (A timed estimate on every power interval). *There is an absolute constant C_K such that, with*

$$K(N) = \frac{\log N}{d} + C_K(\log(N+2))^{4/5},$$

for every integer $N_0 \geq 2$ and every sufficiently large y ,

$$U_y\{N : \nexists k \in \mathbb{N}, k \leq K(N), \text{Syr}^k(N) \leq N_0\} \ll_c (\log N_0)^{-c}.$$

The clock K is independent of N_0 and c .

Proof. Bounded targets are absorbed into the constant as before, and $y^\alpha \leq N_0$ is trivial. Otherwise use exactly the ladder $z_0, \dots, z_J$ in Proposition 21. All scales exceed N_0^{1/α^3} , and $z_0 < N_0$. The first part of Lemma 22 starts the induction; its second part and the same geometric error sum show $\mathbb{P}(B_{z_J}^\bullet) \geq 1 - O_c((\log N_0)^{-c})$. Here $z = z_J = y^{1/\alpha}$ and the input law is U_y .

For an input in that event, choose $k = 0$ if $N \leq N_0$. Otherwise, if $z \geq N_0$, concatenate the first passage with the $E_z^\bullet$ witness and bound $(\log(\text{Pass}_z(N)/N_0))_+ \leq \log(z/N_0)$. If $z < N_0$, the passage itself is a witness, and $\log(N_0/z)/d \leq \Delta_0$. In either case an actual witness satisfies

$$k \leq \frac{\log(N/N_0)}{d} + \Delta_0 + 2C_1(\log z)^p = \frac{\log N}{d} - \frac{\alpha^{-3} \log N_0}{d} + 2C_1(\log z)^p \leq K(N).$$

We used $N \geq y = z^\alpha > z$ and may take $C_K = 2C_1$. This also handles the band $N_0^{1/\alpha} \leq y < N_0$ without silently assuming that every input is already above N_0 . $\square$

15 Exact coverage and the raw Collatz clock

A single top interval would leave a proportion $O(X^{1/\alpha-1}) = O(X^{-1/1001})$ untreated. That term is not necessary: the timed estimate holds on *every* power interval, and the intervals below the top one can be included too.

Lemma 24 (Disjoint coverage with closed-interval estimates). *Let $\alpha > 1$, $H \geq 1$, and suppose $t^\alpha - t > 2$ for every $t \geq H^{1/\alpha}$. Let A be a set of odd integers, disjoint from $[1, H]$, such that*

$$\#(A \cap [t, t^\alpha]) \leq \varepsilon \#((2\mathbb{N} + 1) \cap [t, t^\alpha]) \quad (t \geq H^{1/\alpha}).$$

Then for every real $X \geq 1$,

$$\#(A \cap [1, X]) \leq 2\varepsilon \#((2\mathbb{N} + 1) \cap [1, X]).$$

Proof. For $X \leq H$ the left side vanishes. Otherwise put $t_j = X^{\alpha^{-j}}$ and let J be least such that $t_J \leq H$. For $0 \leq j < J$, $t_{j+1} > H^{1/\alpha}$ and $t_j = t_{j+1}^\alpha$. The intervals $(t_{j+1}, t_j]$ are pairwise disjoint and cover $(t_J, X]$. Each has length greater than two, hence contains at least one odd integer. Passing from this half-open interval to $[t_{j+1}, t_j]$ adds at most its lower endpoint, hence at most one integer. Its closed-interval odd count is therefore at most twice its half-open odd count. Sum the given estimate over the closed intervals and then the disjoint half-open counts. The remaining segment $[1, t_J]$ is contained in $[1, H]$ and contains no element of A . This proves the assertion, with all real or integer endpoint coincidences included. $\square$

Corollary 25 (Timed fixed-target count, without a cutoff error). *For every $0 < c < 1/17.232$ there is C_c such that for every real $X \geq 2$ and every integer $N_0 \geq 2$,*

$$\frac{1}{X} \# \{N \leq X : N \text{ odd}, \nexists k \in \mathbb{N}, k \leq K(N), \text{Syr}^k(N) \leq N_0\} \leq C_c (\log N_0)^{-c}. \quad (***)$$

In particular $c = 1/18$ is admissible. Dropping the time restriction gives the same estimate for orbits that never attain N_0 .

Proof. For sufficiently large N_0 , apply Lemma 24 with $H = N_0$ to the failure set in Proposition 23. It is disjoint from $[1, N_0]$ because time zero is allowed. The elementary function $t^\alpha - t$ is increasing for $t \geq 1$ and tends to infinity, so the width hypothesis holds after one fixed lower bound on N_0 . All source intervals are then within the range of the proposition. Finally $2\#((2\mathbb{N}+1) \cap [1, X]) \leq X+1 \leq 3X/2$ for $X \geq 2$. For targets below that fixed lower bound, enlarge C_c so that $C_c(\log N_0)^{-c} \geq 1$. This covers every target and cutoff. $\square$

This is a consequence of Allikvere's timed iteration with the stronger kernel estimate proved above, and of the scale-covering mechanism already used by Tao in [8, §3]. Relative to Allikvere's stated Theorem `thm:quant`, it replaces the displayed exponent $1/29$ by every $c < 1/17.232$ and removes the extra $X^{-1/2000}$ term. The removal is a counting consequence, not a new mixing theorem.

Define the unaccelerated map C by $C(n) = 3n+1$ for odd n and $C(n) = n/2$ for even n . These are the individual steps counted in the following statement.

Corollary 26 (One threshold-independent clock). *There is a constant C_R for which the clock*

$$R(N) = \left(\frac{3}{d} + \frac{1}{\log 2} \right) \log N + C_R (\log(N+2))^{4/5}$$

satisfies, for every $X \geq 2$, $N_0 \geq 2$ and $0 < c < 1/17.232$,

$$\frac{1}{X} \#\{N \leq X : \nexists m \in \mathbb{N}, m \leq R(N), C^m(N) \leq N_0\} \ll_c (\log N_0)^{-c}.$$

Consequently, for each function $f : \mathbb{N}_{\geq 1} \rightarrow \mathbb{R}$ with $f(N) \rightarrow +\infty$,

$$\#\{N \leq X : \nexists m \leq R(N), C^m(N) < f(N)\} = o(X).$$

For positive odd inputs the same conclusion holds with Syr and K in place of C and R . The clocks are independent of f ; the exceptional set is permitted to depend on f .

Proof. For odd M write $M_i = \text{Syr}^i(M)$ and $s_k = \sum_{i=0}^{k-1} v_2(3M_i + 1)$. Concatenating the actual odd step and its subsequent halvings proves $C^{k+s_k}(M) = M_k$. Moreover

$$2^{s_k} M_k = \prod_{i=0}^{k-1} (3 + 1/M_i) M \leq 4^k M, \quad s_k \leq 2k + \log_2 M.$$

The empty product covers $k = 0$. For an arbitrary positive integer $N = 2^a M$ with M odd, an odd-map witness $k \leq K(M)$ therefore gives a raw-map witness at

$$m = a + k + s_k \leq 3k + a + \log_2 M = 3k + \log_2 N \leq R(N)$$

with $C_R = 3C_K$. The exact coordinate map $M \mapsto 2^a M$ identifies odd integers $M \leq X/2^a$ with inputs $N \leq X$ of valuation exactly a . Hence the raw failure count is at most the sum of the odd failure counts at cutoffs $X/2^a$. By (***) that sum is at most

$$C_c X (\log N_0)^{-c} \sum_{a \geq 0} 2^{-a} = 2C_c X (\log N_0)^{-c}.$$

Cutoffs below two contribute nothing, since their only possible positive odd input is $1 \leq N_0$. This proves the uniform fixed-target estimate without a separate tail-limit argument.

For a diverging f , put $b(X) = \inf_{n \geq \sqrt{X}} f(n)$. For large X this is finite and exceeds three, and $b(X) \rightarrow \infty$. Choose the integer $N_0(X) = \lceil b(X) \rceil - 1$, so that $2 \leq N_0(X) < b(X)$ and $N_0(X) \rightarrow \infty$. Every input $N \geq \sqrt{X}$ that reaches $N_0(X)$ reaches a value strictly less than $f(N)$. The failure proportion is therefore at most

$$X^{-1/2} + O_c((\log N_0(X))^{-c}) = o(1).$$

The identical argument uses (***) for odd inputs. In particular, discarding the time bound recovers Allikvere's natural-density conclusion. This is a written reconstruction from the cited analytic inputs, not an inference from the finite trajectory tests. $\square$

16 Mazur's terminal error: retaining the square-root scales

Mazur's version-2 manuscript [5] gives another natural-density argument, with an explicit logarithmic clock. Its Theorems 1.1–1.2 state the odd-return constant $501501/(5000 \log 2)$, the raw constant $1509503/(5000 \log 2)$, and every fixed-target exponent $0 < d < 5/143$. These are the source's claims; the terminal scalar calculation below is not a certificate for the whole source package.

The two restrictions in its Lemmas 3.4–3.5 have different origins. Phase averaging requires $d < 1/(2\kappa)$ when $\|q \log_2 3\| \geq cq^{1-\kappa}$. Terminal reconstruction is stated with the additional restriction $d < 1/20$. The latter is introduced when square-root scales are replaced by larger pure powers. Keeping the original expressions improves this part of the argument without changing its reference level, tube, or measures.

Literal parameters and source locators. Let B be a positive integer, $L = \log B$, and fix $C > 0$. Put

$$\tau = \frac{1}{100000}, \quad q = \frac{1}{10 \log 2}, \quad f = \frac{340}{100000}, \quad a = \frac{1}{200000}, \quad b = \frac{1}{50000},$$

$$n = \lfloor qL \rfloor, \quad m_0 = \lfloor \tau L \rfloor, \quad m = \lceil m_0 L^{-2/5} \rceil, \quad S = \sqrt{m \log m}, \quad W = C \sqrt{n \log n}, \quad F = fL.$$

Here n is the local passage horizon, m_0 the step-back depth, and m the smaller common reference level. They are not interchangeable. These are the unnumbered schedule following (19) in Section 3.3 and the tube definition preceding (26) of the paper, and the definitions in `A5ReferenceSchedule.lean`, `A5Physical.lean` and `A5ReferenceEta.lean`, in its `Erdos1135/ND/Band` directory. The executable definition $n = \lfloor \lfloor \log_2 B \rfloor / 10 \rfloor$ equals $\lfloor qL \rfloor$ by integer division. The exact nonnegative scalar majorant in the last file is

$$\eta_B(C) = \frac{57SW}{F} + \frac{1021S^2}{F} + \frac{8}{m^3}. \quad (29)$$

All following assertions concern large B , where $m, n \geq 1$.

Proposition 27 (a sharper terminal majorant). *For $L \geq 10^{10}$ the literal parameters satisfy*

$$\eta_B(C) \leq \frac{57C\sqrt{bq}}{f} L^{-1/5} \log L + \frac{1021b}{f} L^{-2/5} \log L + 8a^{-3} L^{-9/5}. \quad (30)$$

Moreover, with C fixed and positive,

$$\frac{\eta_B(C)}{L^{-1/5} \log L} \longrightarrow A_C := \frac{57C}{f} \sqrt{\frac{3}{5}} \tau q > 0. \quad (31)$$

In particular, for every $d < 1/5$ and $\varepsilon > 0$, $\eta_B(C) \leq \varepsilon L^{-d}$ for all sufficiently large B .

Proof. The floor and ceiling give

$$\tau L - 1 < m_0 \leq \tau L, \quad m_0 L^{-2/5} \leq m < m_0 L^{-2/5} + 1.$$

Since $L \geq 10^{10}$, these imply $aL^{3/5} \leq m \leq bL^{3/5}$: for the lower bound use $\tau L - 1 \geq \tau L/2$; for the upper bound use $1 \leq \tau L^{3/5}$. Also $1 \leq m \leq L$ and $1 \leq n \leq qL < L$. The last inequality uses $\log 2 > 1/2$, obtained by integrating $1/t > 1/2$ on $(1, 2)$. Monotonicity of the logarithm now gives

$$S^2 \leq bL^{3/5} \log L, \quad W^2 \leq C^2 qL \log L.$$

All factors are nonnegative. Taking square roots, multiplying, and dividing by the original $F = fL$ gives the first term of (30). The same estimate for S^2 gives its second term. The lower bound for m gives its third. No rescaling of either law or tube is involved.

For the asserted limit, the same floor and ceiling bounds give

$$\frac{m}{L^{3/5}} \rightarrow \tau, \quad \frac{n}{L} \rightarrow q, \quad \frac{\log m}{\log L} \rightarrow \frac{3}{5}, \quad \frac{\log n}{\log L} \rightarrow 1.$$

Consequently

$$\frac{SW}{L^{4/5} \log L} = C \sqrt{\frac{m}{L^{3/5}} \frac{n \log m \log n}{L \log L \log L}} \longrightarrow C \sqrt{\frac{3}{5}} \tau q.$$

After division by $L^{-1/5} \log L$, the second term in (29) is $O(L^{-1/5})$ and its third is $O(L^{-8/5}/\log L)$; both tend to zero. This proves (31). Finally, if $\delta = 1/5 - d > 0$, then $L^{-\delta} \log L \rightarrow 0$. For example, $\log L \leq (2/\delta)L^{\delta/2}$ for $L \geq 1$, by integrating $t^{-1} \leq t^{\delta/2-1}$. This proves the final assertion. $\square$

Retained scales	Source's pure-power bounds
$S = O(L^{3/10}\sqrt{\log L})$	$S \leq L^{7/20}$
$W = O_C(L^{1/2}\sqrt{\log L})$	$W \leq L^{3/5}$
$F = fL$	$F = fL$
$SW/F = O_C(L^{-1/5} \log L)$	$SW/F = O(L^{-1/20})$
The leading coefficient of the actual scalar majorant is $A_C = (57C/f)\sqrt{(3/5)\tau q}$, not an unspecified change of normalization. The lower two terms in (29) decay faster.	

Figure 1: Where the terminal exponent is lost, and how it is recovered. This is a proof diagram for Proposition 27, not sampled orbit data. The original parameters and coefficients are Mazur's; the retained-scale comparison is the present refinement.

Proposition 28 (the complete scalar terminal bound). *Fix $C > 0$ and $K \geq 0$. Define the two nonproportional errors and terminal scalar bounds by the source's exact expressions*

$$p_H(m, K) = 20K/m^{11} + 80/m^3, \quad p_F(m, K) = 4p_H(m, K),$$

$$\mathcal{R}_\sigma(B) = \frac{86}{25}\eta_B(C)(1 + 11B^{-9/10}) + 14Wp_\sigma(m, K), \quad \sigma \in \{H, F\}.$$

Then

$$0 \leq \mathcal{R}_H(B) \leq \mathcal{R}_F(B), \quad \frac{\mathcal{R}_\sigma(B)}{L^{-1/5} \log L} \longrightarrow \frac{86}{25}A_C.$$

For every $0 < d < 1/5$ and $\varepsilon > 0$ both bounds are at most $\varepsilon(\log B)^{-d}$ for all sufficiently large B . The threshold may depend on C, K, d, ε , but not on a source branch, a band index, an endpoint, or an output event.

Proof. The factor $86/25$ is twice the original absorption coefficient $43/25$; it has not been discarded. Positivity and $p_H \leq p_F$ give the first assertion. With the unchanged schedules,

$$W/m^{11} = O_C(L^{-61/10}\sqrt{\log L}), \quad W/m^3 = O_C(L^{-13/10}\sqrt{\log L}).$$

Indeed $W = O_C(L^{1/2}\sqrt{\log L})$ and $m \asymp L^{3/5}$, so the exponents are $1/2 - 33/5 = -61/10$ and $1/2 - 9/5 = -13/10$. Both quantities are $o(L^{-1/5} \log L)$. Since $B^{-9/10} \rightarrow 0$, Proposition 27 therefore proves the two limits, and its final elementary logarithmic comparison proves the rate assertion. All four parameters were fixed before taking B large. The displayed scalar expressions contain none of the later branch, band, endpoint or event parameters, which proves the stated uniformity. $\square$

Propagation through the source estimates. The source theorem `eventually_ndA5ReferenceTerminalAggregateRHS_le_rate` bounds exactly $\mathcal{R}_H, \mathcal{R}_F$ above, but only for $d < 1/20$. Proposition 28 proves that scalar assertion for every $d < 1/5$. Its next theorem compares each nominal terminal sum with the physical reference profile by these scalar bounds; the fixed constant K is chosen before B and before the band and event. Thus the improved estimate remains uniform when used there.

The subsequent genuine-band comparison is a four-term triangle inequality. Its coverage error is at most L^{-10} , its exact-to-nominal error at most $11B^{-9/10}$, its terminal aggregation error is bounded by Proposition 28, and its analytic full-profile error is at most $6L^{-d}$. The first two errors

are respectively at most $L^{-d}/4$ and $L^{-d}/2$ for large B and $d < 1/5$; take $\varepsilon = 1/4$ for the third. The same displayed sum in the source is therefore $(1/4 + 1/2 + 1/4 + 6)L^{-d} = 7L^{-d}$. The full-profile estimate still requires the separate phase restriction $d < 1/(2\kappa)$. Consequently the restriction introduced by this terminal calculation is $d < 1/5$, not $d < 1/20$. This deduction uses the source's finite terminal comparison and full-profile estimate at their stated scopes; those entire dependency chains have not been independently replayed here.

For the source's $\kappa = 143/10$, the phase ceiling $5/143 < 1/20$ already dominates, so this refinement does not change its printed headline exponent. For the published Rhin input $\kappa = 8.616$ used earlier in this note, the phase ceiling is $1/17.232$: the improved terminal bound no longer prevents that exponent from passing through this part of Mazur's argument. This is a precise compatibility result between the two arguments, not a new claim that a 599-file Lean package has been checked locally.

Equation (31) also shows that this particular positive scalar majorant is not $O(L^{-1/5})$. That statement is about the retained upper-bound expression. It is not a lower bound on the actual passage-law error and does not exclude a sharper argument using cancellation.

Version and formal status. The manuscript is version 2, 21 July 2026. It names source commit `f386357d453ac4dcf91242b76252d88a5a729906`; the downloaded source and its evidence records name `ca3dd0d63920411213403092aecc6946619eb082`. The latter's main file hash agrees with its advertised hash. The source evidence reports the standard axioms `propext`, `Classical.choice`, `Quot.sound` and no `sorryAx`. These are the publisher's records, not a build performed in this workbench. The archive contains the first-party Lean closure but no bundled Mathlib, lake manifest, or toolchain file. No formal verification status is transferred from that evidence to the refinement above.

17 Finite terminal records, boundary mass, and sharp absorption

The terminal scalar has a specific probabilistic meaning. It measures the comparison between a finite sum of nominal affine weights and a reference profile. Before this comparison can be used, one must retain the actual integer sources and show why summing over many terminal records does not multiply a probability mass by the number of records. We reconstruct that finite mechanism here, then sharpen its algebraic absorption step. The source locators are Mazur [5], Appendix C.2–C.3, pp. 21–23, and the Lean files named below; all refer to the downloaded commit already identified in Section 16.

The exact inverse, with compatibility retained. Write $T = \text{Syr}$ on positive odd integers. Fix an integer threshold $B \geq 1$, a positive integer h , a finite set J of integer times greater than h , and a finite set $\mathcal{E}$ of positive odd integers whose first hitting time of $[1, B]$ is exactly h . For $t \in J$ put $\nu = t - h$. A terminal record consists of (t, a, M) , where $a = (a_1, \dots, a_\nu)$ has positive integer entries and $M \in \mathcal{E}$. Restrict these records to any finite collection; Mazur uses the additional strict total-valuation tube $|\ell - 2\nu| < W$, where $\ell = \sum_i a_i$.

Put $s_i = a_1 + \dots + a_i$, $s_0 = 0$, and

$$c_a = \sum_{i=0}^{\nu-1} 3^{\nu-1-i} 2^{s_i}, \quad F_a = c_a / 2^\ell, \quad N_a(M) = \frac{2^\ell M - c_a}{3^\nu}, \quad N_a^0(M) = \frac{2^\ell M}{3^\nu}.$$

Compatibility means $2^\ell M \equiv c_a \pmod{3^\nu}$. Retain only compatible records with $M > F_a$. Then $N_a(M)$ is a positive odd integer: c_a is odd, the numerator is positive and odd, and its odd divisor is 3^ν . This is the inverse of the original affine equation $T^\nu N = (3^\nu N + c_a)/2^\ell = M$ on the indicated word cylinder, not a change of the map.

For clarity, terminal oddness proves the intermediate exponents too. The one-letter assertion is immediate from $3N + 1 = 2^{a_1} M$. For a longer word, use $c_a = 3^{\nu-1} + 2^{a_1} c_{a'}$ to obtain $2^{a_1} \mid 3N + 1$. The resulting integer $y = (3N + 1)/2^{a_1}$ is positive and odd: the suffix equation $3^{\nu-1} y + c_{a'} = 2^{\ell-a_1} M$ has odd $c_{a'}$ and even right side. Induction on the suffix proves all remaining exponents. Conversely actual iteration gives the affine equation. This gives both directions and the unique inverse. It is the arithmetic content of `A5TerminalAtomDenominatorComparison.lean`, `ndA5TerminalAtomCandidate_odd_and_affine`; the source's endpoint-room estimate supplies $M > F_a$.

Proposition 29 (one source per terminal record, across times). *Suppose $\max J - \min J \leq h$ when J is nonempty. The map from compatible positive terminal records to $N_a(M)$ is injective. In particular, restricting to any physical sampling band preserves disjointness, and boundary candidates outside that band also have multiplicity at most one.*

Proof. Suppose the same N has records at $t \leq t'$. The affine decoding gives $M = T^{t-h} N$ and $M' = T^{t'-h} N = T^{t'-t} M$. Put $d = t' - t$. Since $0 \leq d \leq h$ and the first hit from M is at h , the first hit from $T^d M$ is exactly $h - d$: its earlier iterates are the corresponding earlier iterates of M , and its iterate at $h - d$ is $T^h M \leq B$. But $M' \in \mathcal{E}$ has first-hit time h , so $d = 0$. The times and word lengths coincide. Deterministic iteration from N then gives the same ordered word and the same endpoint. No band-membership assumption entered the proof. $\square$

For the source's particular band, with $d = \log(4/3)$,

$$J = \left[\left\lfloor \frac{\log(z/B)}{d} - 3W \right\rfloor, \left\lceil \frac{\log(U/B)}{d} + 3W \right\rceil \right] \cap \mathbb{N} \implies \text{diam } J \leq \frac{\beta}{d} + 6W.$$

The floor and ceiling can only reduce this diameter. With fixed C , $W = o(\log B)$ and $h = m_0 \sim (\log B)/100000$, so the diameter is at most h eventually. Also $z \geq B^{1001/1000}$ gives $\min J \geq (\log B)/(1000d) - 3W > h$ eventually: $d < 1/3$ implies $1/(1000d) > 3/1000 > 1/100000$. Thus the time hypotheses hold for the unchanged bands and schedules, not for an arbitrary long interval of times.

The source proof is `A5TerminalAtoms.lean`, `ndA5TerminalAtomIndex_eq_of_common_affineSource`. This is a statement about terminal records, not yet identification of their union with the genuine first-passage event. For example, the actual orbit of 63 first reaches $[1, 50]$ at time 35, with value 23; its values at times 36, 37, 38 are 35, 53, 5. Thus an endpoint at time 37 whose own first hit is one step later does not imply that the source's first hit is time 38. Also the endpoints at times 34 and 37 are 61 and 53, both with first-hit time one; their corresponding terminal times differ by three and therefore violate the proposition's diameter hypothesis. These exact examples explain why the source has a separate coverage argument and a short time-window condition.

Actual and nominal weights. Take a nonempty band $\mathcal{B} = \{N > 0 : N \text{ odd}, z \leq N < U\}$, where $z > 0$, $U = ze^\beta$, and $\beta > 0$. Write $D = \#\mathcal{B}$, $H = \sum_{N \in \mathcal{B}} N^{-1}$. For a compatible record put

$$\alpha_a = F_a/M, \quad \delta_a = -\log(1 - \alpha_a), \quad u_a = \log(N_a^0(M)/z).$$

Then $0 < \alpha_a < 1$ and the exact identities are

$$N_a = (1 - \alpha_a)N_a^0, \quad \log(N_a/z) = u_a - \delta_a, \quad \frac{3^\nu}{2^\ell M} = \frac{1}{N_a^0} = \frac{1 - \alpha_a}{N_a}. \quad (*)$$

Let $P_a = \mathbf{1}_{\{z \leq N_a < U\}}$ and $Q_a = \mathbf{1}_{\{z < N_a^0 < U\}}$; their endpoint conventions differ deliberately. The actual masses of the union and the nominal sums are

$$U_H = \sum_a \frac{P_a}{HN_a}, \quad U_F = \sum_a \frac{P_a}{D}, \quad I_H = \sum_a \frac{Q_a}{HN_a^0}, \quad I_F = \sum_a \frac{Q_a}{D}.$$

Here and below the index a in sums abbreviates the entire terminal record; different times have not been identified. Proposition 29 gives $0 \leq U_H, U_F \leq 1$.

Proposition 30 (boundary comparison with no record-count loss). *If all retained records have $\delta_a \leq \varepsilon$ and $\alpha_a \leq \rho$, define the odd-integer shells*

$$S_- = [ze^{-\varepsilon}, z) \cap (2\mathbb{N} + 1), \quad S_+ = [Ue^{-\varepsilon}, U) \cap (2\mathbb{N} + 1).$$

Then

$$|U_F - I_F| \leq \frac{\#(S_- \cup S_+)}{D}, \quad |U_H - I_H| \leq \rho U_H + \frac{1}{H} \sum_{N \in S_- \cup S_+} \frac{1}{N}.$$

There are also the one-sided bounds

$$I_F \leq U_F + \frac{\#S_-}{D}, \quad I_H \leq U_H + \frac{1}{H} \sum_{N \in S_-} \frac{1}{N}.$$

Proof. If $Q_a = 1$ and $P_a = 0$, then $0 < u_a < \beta$ while $u_a - \delta_a < 0$; consequently $N_a \in S_-$. If $P_a = 1$ and $Q_a = 0$, positivity of δ_a excludes $u_a \leq 0$, so $\beta \leq u_a < \beta + \delta_a$ and $N_a \in S_+$. Injectivity charges each mismatching record to a distinct shell integer. This proves the flat bound. For the harmonic bound insert the actual-support nominal sum $I_H^{\text{phys}} = \sum_a P_a / (HN_a^0)$. Identity (*) gives

$$0 \leq U_H - I_H^{\text{phys}} = \sum_a \frac{P_a \alpha_a}{HN_a} \leq \rho U_H.$$

For support mismatch, $1/N_a^0 \leq 1/N_a$, so injectivity bounds $|I_H^{\text{phys}} - I_H|$ by the stated shell sum. The triangle inequality proves the assertion. For the one-sided bounds only nominal-only records can increase the nominal sum; these lie in S_- , while on the common support the nominal harmonic weight is no greater than the actual one. $\square$

Mazur's finite band estimates supply $\varepsilon = B^{-9/10}$, $\rho = B^{-19749/20000}$, and the respective shell ratios $< 11B^{-9/10}$ and $\leq 5B^{-9/10}$. The precise source declarations are in `A5TerminalBoundaryShell.lean` and `A5TerminalAtomNominalization.lean`. Thus both nominal sums have the common cap $1 + 11B^{-9/10}$; for the harmonic sum the triangle estimate alone already gives $1 + 6B^{-9/10}$. The one-sided statement above retains still more information if the individual lower shell is to be used. These estimates explain, rather than assume, why the cap is for the entire time-indexed sum.

The two integer rows and the flat exponential. Let $\theta = \log_2 3$, $A = \log_2(z/M)$, and

$$b_\nu = A + \nu\theta, \quad \ell_{\nu,r} = [b_\nu] + 1 + r, \quad u_{\nu,r} = (\ell_{\nu,r} - b_\nu) \log 2 = (r + 1 - \{b_\nu\}) \log 2, \quad r = 0, 1.$$

The $+1$ is retained even when b_ν is integral. The source bands have $0 < \beta < 2 \log 2$ for $\log B \geq 300000$: their logarithmic length before subdivision is at least 300, so $\beta < 301/300 < 2 \log 2$.

Proposition 31 (exact two-row reindexing). *For $\nu \geq 1$ and $0 < W \leq \nu/8$, let $q_{\nu,\ell} = \binom{\ell-1}{\nu-1} 2^{-\ell}$ for $\ell \geq \nu$ and zero otherwise. For every real-valued function f on integers,*

$$\sum_{\substack{\nu \leq \ell \leq 3\nu \\ |\ell - 2\nu| < W \\ 0 < (\ell - b_\nu) \log 2 < \beta}} q_{\nu,\ell} f(\ell) = \sum_{r=0}^1 \mathbf{1}_{\{|\ell_{\nu,r} - 2\nu| < W\}} \mathbf{1}_{\{0 < u_{\nu,r} < \beta\}} q_{\nu,\ell_{\nu,r}} f(\ell_{\nu,r}).$$

Proof. The integers strictly above b_ν begin at $\lfloor b_\nu \rfloor + 1$. Since $\beta/\log 2 < 2$, only this integer and its successor can satisfy the displayed band inequality. The map $r \mapsto \lfloor b_\nu \rfloor + 1 + r$ is injective, with inverse $\ell \mapsto \ell - \lfloor b_\nu \rfloor - 1$. The tube and $W \leq \nu/8$ imply $\nu < \ell < 3\nu$, so no additional endpoint is lost. Each summand is identical after this bijection; neither positivity of f nor an approximation is used. The expression for $q_{\nu,\ell}$ follows by counting the $\binom{\ell-1}{\nu-1}$ positive compositions, each of geometric probability $2^{-\ell}$. $\square$

This is the finite identity in `A5TerminalAtomReconstruction.lean`, `sum_ndA5TerminalTotal s_endpointMass_strict_eq_range_two`. Before any mixing estimate, let $\rho_{\nu,\ell}(M)$ be the fraction of those compositions satisfying the original compatibility congruence. Summing the harmonic nominal weights over one fixed (ν, ℓ, M) cell gives exactly

$$\frac{3^\nu}{MH} q_{\nu,\ell} \rho_{\nu,\ell}(M).$$

The flat expression is the same before the harmonic normalizer, multiplied instead by $ze^{(\ell-b_\nu)\log 2}/D$. Indeed $(3^\nu/(2^\ell M))ze^{(\ell-b_\nu)\log 2} = 1$. Thus the exponential appears exactly once; omitting it would change the flat law to a different weighting.

Replacing the conditioned residue factor by its common coarse law is the separate mixing step. Its resulting endpoint kernel is $K_m(M) = 3^m p_m(M \bmod 3^m)/M$, with p_m the source's Syracuse offset law at the fixed reference depth m . The two reference profiles retain this same kernel and multiply it by

$$\frac{1}{H} \sum_{\nu,r} Q_{\nu,r} \quad \text{and} \quad \frac{z}{D} \sum_{\nu,r} Q_{\nu,r} e^{u_{\nu,r}},$$

respectively, where $Q_{\nu,r}$ is the product of the two indicators and $q_{\nu,\ell_{\nu,r}}$ above, and $\nu = t - h$, $t \in J$. The inverse time reindex is $t = \nu + h$. These are the literal profiles in `A5ReferenceTerminalAggregate.lean`, not an identification of the two sampling measures.

Lemma 32 (sharp finite proportional absorption). *Let a finite collection have $Y_i, Z_i, E_i \geq 0$ and $0 \leq a_i \leq \lambda < 1$, with $|Y_i - Z_i| \leq E_i + a_i Z_i$ for each i . Then*

$$\sum_i |Y_i - Z_i| \leq \frac{\lambda \sum_i Y_i + \sum_i E_i}{1 - \lambda}.$$

In particular this bounds the absolute difference of the two sums. The coefficients cannot be decreased using only these hypotheses.

Proof. Put $D_* = \sum_i |Y_i - Z_i|$. Summation and $Z_i \geq 0$ give $D_* \leq \sum_i E_i + \lambda \sum_i Z_i$. Also $\sum_i Z_i \leq \sum_i Y_i + D_*$. Hence $(1 - \lambda)D_* \leq \sum_i E_i + \lambda \sum_i Y_i$; division is permitted since $1 - \lambda > 0$. For one cell, choose any $Y, E \geq 0$ and $Z = (Y + E)/(1 - \lambda)$. Then $Z \geq Y$ and $Z - Y = E + \lambda Z$, so equality holds. This proves sharpness for this finite comparison problem, not saturation by the Collatz distributions. $\square$

In the source cell inequality, $i = (t, \ell)$, $Z_i = q_{\nu, \ell} R_i$ and $E_i = q_{\nu, \ell} p_\sigma(m, K)$, where R_i is the nonnegative coarse-reference expectation. The coefficient is

$$a_i = \frac{43}{25} \left(\text{eps}_{m, \nu, \ell} + \frac{8}{m^3} \right), \quad \lambda_B = \frac{43}{25} \eta_B(C).$$

Here $\text{eps}_{m, \nu, \ell}$ denotes exactly `ndFixedTotalRatioEps` in the source, not a new choice. The common terminal packet gives $0 \leq a_i \leq \lambda_B \leq 1/2$. Since $\sum_\ell q_{\nu, \ell} \leq 1$, summing the errors first over totals and then over times costs at most $\#J p_\sigma$. The nominal mass cap, on the other hand, is applied once after both sums. With $\#J \leq 7W$, Lemma 32 therefore replaces the source's bound by the smaller explicit scalar

$$\mathcal{R}_\sigma^*(B) = \frac{\lambda_B(1 + 11B^{-9/10}) + 7W p_\sigma(m, K)}{1 - \lambda_B}.$$

This is an algebraic refinement of the local-cell estimate used in `A5ReferenceFixedTimeAbsorption.lean`, followed by the same finite aggregation. It does not purport to prove the analytic conditioned-mixing estimate used to obtain those cells.

Corollary 33 (leading constant of the refined terminal scalar). *For the unchanged schedules, fixed $C > 0$ and $K \geq 0$, eventually $0 \leq \lambda_B < 1$, and*

$$\frac{\mathcal{R}_\sigma^*(B)}{(\log B)^{-1/5} \log \log B} \longrightarrow \frac{43}{25} A_C, \quad \frac{\mathcal{R}_\sigma^*(B)}{\mathcal{R}_\sigma(B)} \longrightarrow \frac{1}{2}, \quad \sigma = H, F.$$

For $0 \leq \lambda_B \leq 1/2$, $\mathcal{R}_\sigma^* \leq \mathcal{R}_\sigma$.

Proof. Proposition 27 gives $\lambda_B \rightarrow 0$ and its scaled limit $(43/25)A_C$. Proposition 28 proves that $W p_\sigma$ is negligible on that scale. Thus the denominator tends to one, proving the first limit. The exact relation is $\mathcal{R}_\sigma^* = \mathcal{R}_\sigma / [2(1 - \lambda_B)]$; the second limit and the finite comparison follow. $\square$

The admissible terminal exponents remain every $d < 1/5$. In the four-term genuine-band estimate of the preceding section, $\mathcal{R}_\sigma$ may be replaced by $\mathcal{R}_\sigma^*$; all other terms and the separate phase restriction remain unchanged. The finite inverse, boundary comparison and two-row identities above have written proofs and exact arithmetic regression tests. The coverage and conditioned-mixing dependencies are reconstructed separately below; Section 25 now composes them with this finite mechanism. No finite test certifies those analytic proofs or supplies a formal replay of the source package.

18 Fixed-total descent with fibre cancellation

The terminal comparison needs more than equidistribution of a residue. Its endpoint weights depend on the valuation total, several endpoints may occupy the same residue, and the reference law has forgotten that total. These are three distinct operations. The following reconstruction keeps their maps separate and improves the error by using the signs of the discarded masses. The source is Mazur [5], `ND/Fourier/FixedTotalFiniteResolvedDescent.lean`, `FixedTotalSourceLaw.lean`, `FixedTotalRatioWindow.lean`, `FixedTotalRatioNormalization.lean`, and `ND/Band/A5FixedCellTerminalDescent.lean`, in the pinned source release specified above.

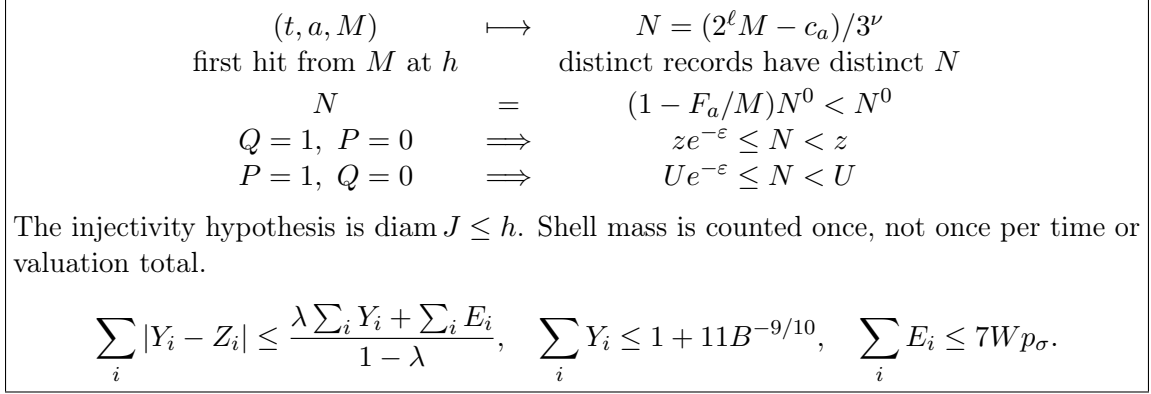


Figure 2: Exact terminal records and the two different aggregation costs. Propositions 29–31 reconstruct Mazur’s finite mechanism; Lemma 32 retains the denominator in its final comparison. The diagram shows formulas and set membership, not sampled orbit geometry.

Masses and tests have different projection maps. Write $G_j = \mathbb{Z}/3^j\mathbb{Z}$, let $0 \leq m \leq n$, and put $k = 3^{n-m}$. The reduction $\pi : G_n \rightarrow G_m$ has the fibre $\{x + 3^m j : 0 \leq j < k\}$ over the standard representative x . For real masses p on G_n , masses q on G_m , and real tests c on G_n , define

$$(\pi_* p)(x) = \sum_{\pi y = x} p(y), \quad (Uq)(y) = k^{-1} q(\pi y), \quad (Ac)(x) = k^{-1} \sum_{\pi y = x} c(y). \quad (32)$$

Here π_* sums masses; A averages tests. With the counting pairing,

$$\pi_* U = I, \quad \langle Uq, c \rangle = \langle q, Ac \rangle.$$

The map U is an ℓ^1 isometry, π_* an ℓ^1 contraction, and A a positive unital supremum-norm contraction. All statements follow by summing over the displayed k -element fibres. The pullback $\pi^* g = g \circ \pi$ satisfies $A\pi^* = I$. The kernel of A consists exactly of tests whose sum on every fibre is zero. Thus A is onto, and loses precisely these within-fibre variations, not the total weight assigned to a fibre.

For a finite endpoint index set I , arbitrary addresses $a_i \in G_n$ and coefficients d_i , put $c(y) = \sum_{a_i = y} d_i$. Finite interchange gives

$$\langle p, c \rangle = \sum_{i \in I} p(a_i) d_i, \quad (Ac)(x) = k^{-1} \sum_{\pi a_i = x} d_i.$$

Coincident addresses add their weights. No injectivity or choice of one endpoint per residue is permitted. In the terminal cell the harmonic coefficient is $d_M = 3^n/(MH)$ and the flat coefficient is $d_M = z3^n e^{u_{n,\ell}(M)}/(MD)$, with the strict endpoint set and $u_{n,\ell}(M) = (\ell - \log_2(z/M) - n \log_2 3) \log 2$ as in the preceding section. Averaging changes 3^n to 3^m ; it changes neither that endpoint set nor the flat exponential.

Proposition 34 (cancellation on each residue fibre). *For a probability mass p on G_n , put $q = \pi_* p$, $r = p - Uq$ and $\Omega = \|r\|_1$. For every real test c ,*

$$|\langle p, c \rangle - \langle q, Ac \rangle| \leq \frac{1}{2} \sum_{x \in G_m} \text{osc}_{\pi^{-1}x}(c) \sum_{\pi y = x} |r(y)|. \quad (33)$$

In particular, $0 \leq c \leq H$ gives the bound $H\Omega/2$. The coefficient $1/2$ is best possible in this general statement.

Proof. The sum of r on each fibre is zero, since $\pi_* Uq = q$. On that fibre subtract the midpoint between the largest and smallest values of c without changing its pairing with r . The absolute value of the resulting test is at most half its oscillation. The triangle inequality, summed over fibres, proves the estimate. For sharpness take $m = 0, n = 1, p = (1, 0, 0)$ and $c = (H, 0, 0)$, with $H > 0$. Then $Uq = (1/3, 1/3, 1/3)$, $\Omega = 4/3$, and the pairing difference is $2H/3$. Equality holds. $\square$

This uses the full ℓ^1 distance, as in the source, not a redefinition of that distance as total variation. The source's $H\Omega$ bound is valid; the zero fibre sums give the smaller displayed bound.

The exact law after conditioning. Let a_i be independent positive integers with $\mathbb{P}(a_i = j) = 2^{-j}$, and let $S_j = a_1 + \dots + a_j$. For $j \geq 1$ set

$$b_j(u) = \begin{cases} \binom{u-1}{j-1} 2^{-u} & u \geq j, \\ 0 & u < j, \end{cases} \quad X_j = \sum_{i=1}^j 3^{i-1} 2^{-S_i} \pmod{3^j}.$$

The inverses of 2 here exist in G_j . Write $\mu_{j,u} = \text{Law}(X_j \mid S_j = u)$ when $u \geq j$.

Proposition 35 (conditioned prefix mixture and its likelihood). *For $1 \leq m < n$ and $\ell \geq n$, define probabilities on the integers by*

$$P(u) = \frac{b_m(u) b_{n-m}(\ell - u)}{b_n(\ell)}, \quad Q(u) = b_m(u).$$

Then

$$\pi_* \mu_{n,\ell} = \sum_u P(u) \mu_{m,u}, \quad \text{Law}(X_m) = \sum_u Q(u) \mu_{m,u}. \quad (34)$$

Zero-weight terms with $u < m$ can use any probability law. On $u \geq m$, the likelihood is $R(u) = P(u)/Q(u) = b_{n-m}(\ell - u)/b_n(\ell)$. For $u + (n - m) < \ell$ its positive adjacent ratio is exactly

$$\frac{R(u+1)}{R(u)} = \frac{2(\ell - u - (n - m))}{\ell - u - 1}, \quad \frac{R(u+1)}{R(u)} - 1 = \frac{(\ell - 2n) - (u - 2m) + 1}{\ell - u - 1}. \quad (35)$$

Proof. Every positive composition of ℓ into n parts has probability $2^{-\ell}$; there are $\binom{\ell-1}{n-1}$ such compositions. Splitting after m parts is a bijection onto the disjoint union over u of a composition of u into m parts and a composition of $\ell - u$ into $n - m$ parts. Its inverse concatenates the two ordered lists. Counting proves the formula for P and $\sum_u P(u) = 1$. For fixed u , the prefix is uniform among its positive compositions. Reduction modulo 3^m discards exactly the terms with $i > m$ in X_n , so it equals X_m of that prefix. This proves the first mixture. The second is ordinary partition by S_m , with an absolutely convergent sum when paired with a bounded test. Finally $b_j(v-1)/b_j(v) = 2(v-j)/(v-1)$ for $v > j$. Use $j = n - m, v = \ell - u$ and subtract one. The strict suffix room ensures positive numerator and denominator; the $+1$ is not omitted. $\square$

The affine endpoint offset uses the reversed word: if $F_a = \sum_{i=1}^n 3^{n-i} 2^{-(a_i + \dots + a_n)}$, then $F_{a^{\text{rev}}} = X_n(a)$. Word reversal is its own inverse and preserves the geometric probability and total valuation. It therefore identifies the two fixed-total offset laws. It does not preserve arbitrary prefix events. This is exactly the separately stated reversal map in `ndAffineFixedTotalSubmass_eq_section7`, not an identification of an unreversed prefix with an affine tail.

Lemma 36 (likelihood comparison retaining the actual masses). *Let P, Q be probabilities on a countable set and G a set on which both are positive. Put $p_0 = P(G^c) < 1$, $q_0 = Q(G^c) < 1$. Suppose the oscillation of $\log(P/Q)$ on G is at most $\epsilon \geq 0$. For every $u \in G$,*

$$e^{-\epsilon} \frac{1-p_0}{1-q_0} \leq \frac{P(u)}{Q(u)} \leq e^{\epsilon} \frac{1-p_0}{1-q_0}. \quad (36)$$

If $p_0 \leq \theta_P < 1$ and $q_0 \leq \theta_Q < 1$, set

$$a = \max \left(\frac{e^{\epsilon}}{1-\theta_Q} - 1, 1 - e^{-\epsilon}(1-\theta_P) \right).$$

For every $0 \leq \phi \leq H$,

$$\left| \sum_u (P(u) - Q(u)) \phi(u) \right| \leq a \mathbb{E}_Q \phi + H \max(\theta_P, \theta_Q). \quad (37)$$

When $\theta_P = \theta_Q = \theta$, one can take $a = e^{\epsilon}/(1-\theta) - 1$.

Proof. For fixed $u \in G$, exponentiation compares $R(v) = P(v)/Q(v)$ to $R(u)$ within the factors $e^{\pm\epsilon}$ at every $v \in G$. Average over $Q(v)/Q(G)$. The exact average is $P(G)/Q(G)$. Rearranging proves the first inequality and then $|R(u) - 1| \leq a$. The central contribution is bounded by $a \mathbb{E}_Q(\phi \mathbf{1}_G) \leq a \mathbb{E}_Q \phi$. The outside signed contribution lies in $[-Hq_0, Hp_0]$. Its absolute value is at most $H \max(p_0, q_0)$, not the sum of these two bounds. All sums converge absolutely because ϕ is bounded. For equal tails put $v = e^{\epsilon}/(1-\theta) \geq 1$; $v - 1 \geq 1 - v^{-1}$ since $(v-1)^2/v \geq 0$. $\square$

The source's literal central window. Let $m \geq 150$, $8m \leq n$, $D_{\ell} = |\ell - 2n| \leq n/8$, and put

$$S_m = \sqrt{m \log m}, \quad R_m = \lceil 16S_m \rceil, \quad G = \{u \geq m : |u - 2m| \leq R_m\},$$

$$\epsilon_{m,n,\ell} = \frac{57S_m D_{\ell} + 1021S_m^2}{n}.$$

This is precisely `ndFixedTotalRatioEps`: its first term is $57\sqrt{\log m} (mD_{\ell}/n)/\sqrt{m} = 57S_m D_{\ell}/n$.

For completeness the window constants follow from $\log m \leq m/29$ and $S_m \geq 2$. The function $\log x/x$ decreases for $x \geq 150$, and $\log 150 = 7\log 2 + \log(75/64) < 49/10 + 11/64 < 51/10 < 150/29$. Here $\log 2 < 7/10$ follows from the first five terms of the exponential series at $7/10$, whose sum exceeds 2. Thus $S_m \leq m/\sqrt{29}$ and $16m/\sqrt{29} \leq 3m - 1$ for $m \geq 150$ (square the positive sides). Consequently

$$R_m \leq 3m, \quad R_m + 1 \leq 18S_m, \quad 2R_m + 1 \leq 34S_m.$$

Every $u \in G$ satisfies $u \leq 5m$ and $u + (n - m) < \ell$. The denominator in (35) is at least $15n/8 - 5m - 1 \geq 5n/4 - 1 \geq 6n/5$. Its numerator after subtracting one has absolute value at most $D_{\ell} + R_m + 1 \leq n/2 + 1 \leq 27n/50$, since $n \geq 1200$. The ratio is therefore within $9/20$ of one. Integration of $1/t$ on $[11/20, 29/20]$ gives $|\log t| \leq 2|t - 1|$ there. Adjacent log differences are at most $5(D_{\ell} + R_m + 1)/(3n)$. For $u < v$ in G , telescope over the half-open integer interval $[u, v)$, all of whose indices remain in G . There are at most $34S_m$ terms, so

$$|\log R(u) - \log R(v)| \leq \frac{170S_m(D_{\ell} + 18S_m)}{3n} \leq \epsilon_{m,n,\ell}.$$

This reconstructs the source's pairwise likelihood estimate with its literal signs, ceiling, and 57, 1021 constants.

The source tail estimates used at this point are

$$P(G^c) \leq 2/m^3, \quad Q(G^c) \leq 2/m^3 \quad \text{when } mD_\ell/n \leq S_m. \quad (38)$$

Their exact declarations are `ndFixedTotalHeadTotalPMF_outside_ratioGood_le_two_div_cube` and `ndGeom2EndpointPMF_outside_ratioGood_le_two_div_cube` in `FixedTotalRatioTails.lean`. The conditioned proof applies its concentration estimate about $m\ell/n$ at radius $15S_m$, using the displacement bound $mD_\ell/n \leq S_m$, and obtains $2e^{-225S_m^2/(63m)} \leq 2/m^3$. The unconditioned proof obtains $2e^{-\min(R_m^2/(32m), R_m/8)} \leq 2/m^3$. These are Mazur's source estimates. Complete written reconstructions, including their concentration inputs, now appear in Corollary 41 below. The finite checks do not certify the analytic inequalities or replay their Lean formalization.

Corollary 37 (refined fixed-cell descent). *Use the preceding window and displacement hypotheses. For any test $0 \leq c \leq H$ on G_n , put*

$$\Omega = \|\mu_{n,\ell} - U\pi_*\mu_{n,\ell}\|_1, \quad V = \mathbb{E}_{\text{Law}(X_m)}Ac, \quad a^\dagger = \frac{\exp(\epsilon_{m,n,\ell})}{1 - 2/m^3} - 1.$$

Using the source tail estimates (38),

$$|\mathbb{E}_{\mu_{n,\ell}}c - V| \leq H\Omega/2 + a^\dagger V + 2H/m^3. \quad (39)$$

Proof. The first replacement, from $\mu_{n,\ell}$ to $U\pi_*\mu_{n,\ell}$, costs at most $H\Omega/2$ by Proposition 34. By Proposition 35, the remaining difference is $\sum_u (P(u) - Q(u))\phi(u)$, where $\phi(u) = \mathbb{E}_{\mu_{m,u}}Ac$ lies in $[0, H]$. Its Q -expectation is V . Apply Lemma 36 with $\theta = 2/m^3$. $\square$

At a terminal cell the source bounds the assembled tests by 20 (harmonic) and 80 (flat). The harmonic bound uses the endpoint coefficient bound 5 and band harmonic mass at least $1/4$; the flat bound uses pointwise domination by four times the harmonic test. Their original guards include the source passage-window facts, positive band count, $m_0 + n \leq n_0$, and, for the flat case, $\log B \geq 300000$. We retain these guards and do not substitute a single-endpoint estimate for the assembled-test bound. With $\Omega \leq K/m^{11}$ the nonproportional errors become

$$p_H^\dagger = 10K/m^{11} + 40/m^3 = p_H/2, \quad p_F^\dagger = 40K/m^{11} + 160/m^3 = p_F/2.$$

The two halves have different reasons: zero fibre mass for the first term, and the sign of the two outside contributions for the second. This does not strengthen the Fourier theorem itself.

The mixing input on the terminal schedule is also covered by Corollary 5: the fine level n lies between $f \log B$ and $n_0 = \lfloor q \log B \rfloor$, while $m \sim \tau(\log B)^{3/5}$. Hence $m \geq n^{1/2}$ eventually, and $|\ell - 2n| \leq W = C\sqrt{n_0 \log n_0} \ll_C \sqrt{n \log n}$ uniformly over these cells. Taking $A = 11$ there gives a single finite K for this range. This connects the previously reconstructed Tao-based estimate to the present descent. Also $8m = o(F)$, $W = o(F)$, and $mW/(FS) = O_C((\log B)^{-1/5}) \rightarrow 0$; hence the central-window and displacement guards hold uniformly eventually. It makes no assertion about every smaller reference scale in the source's formal package.

Corollary 38 (retaining the exponential improves the terminal constant). *Keep the literal schedules of (29). Define*

$$E_B = \frac{57SW + 1021S^2}{F}, \quad \theta_B = \frac{2}{m^3}, \quad \gamma_B = \frac{e^{E_B}}{1 - \theta_B} - 1, \quad (40)$$

$$\mathcal{R}_\sigma^\dagger(B) = \frac{\gamma_B(1 + 11B^{-9/10}) + (7/2)Wp_\sigma(m, K)}{1 - \gamma_B}.$$

For all sufficiently large B , this is the terminal comparison bound obtained from the preceding descent and the same nominal cap. On the source range $0 \leq \lambda_B = (43/25)\eta_B \leq 1/2$,

$$0 \leq \gamma_B \leq \lambda_B, \quad \mathcal{R}_\sigma^\dagger \leq \mathcal{R}_\sigma^* \leq \mathcal{R}_\sigma.$$

For fixed $C > 0$, $K \geq 0$, and $\sigma = H, F$,

$$\frac{\mathcal{R}_\sigma^\dagger(B)}{(\log B)^{-1/5} \log \log B} \rightarrow A_C, \quad \frac{\mathcal{R}_\sigma^\dagger}{\mathcal{R}_\sigma} \rightarrow \frac{25}{86}, \quad \frac{\mathcal{R}_\sigma^\dagger}{\mathcal{R}_\sigma^*} \rightarrow \frac{25}{43}.$$

Proof. On each cell, $n \geq F$ and $D_\ell \leq W$ give $\epsilon_{m,n,\ell} \leq E_B$. Therefore $a^\dagger \leq \gamma_B$. Multiplication by the single endpoint mass $q_{n,\ell}$ gives a cell error with absolute part $q_{n,\ell} p_\sigma / 2$. Sum first over ℓ , using $\sum_\ell q_{n,\ell} \leq 1$, and then over times, using $\#J \leq 7W$. Apply Lemma 32 and the nominal mass cap $1 + 11B^{-9/10}$ once. This gives the displayed quotient.

Since $\eta_B = E_B + 4\theta_B$, the stated range implies $0 \leq \theta_B < 1/2$ and $E_B + 2\theta_B \leq 1$. Integration gives $-\log(1 - \theta_B) \leq 2\theta_B$. For $0 \leq x \leq 1$, convexity gives $e^x - 1 \leq (e - 1)x < (43/25)x$. For instance the exponential series through degree 6 has remaining tail at most $(1/7!)/(1 - 1/8)$, proving $e < 68/25$. Consequently

$$\gamma_B \leq e^{E_B + 2\theta_B} - 1 \leq (43/25)(E_B + 2\theta_B) \leq \lambda_B.$$

The function $(aM + e)/(1 - a)$ increases in $a, e \geq 0$ for $a < 1$, $M \geq 0$, proving the finite comparison with $\mathcal{R}_\sigma^*$.

Write $s_B = (\log B)^{-1/5} \log \log B$. The earlier scalar proof gives $E_B/s_B \rightarrow A_C$, $\theta_B/s_B \rightarrow 0$ and $Wp_\sigma/s_B \rightarrow 0$. Since $E_B \rightarrow 0$, $\gamma_B = (e^{E_B} - 1 + \theta_B)/(1 - \theta_B)$ has $\gamma_B/s_B \rightarrow A_C$. The denominator and nominal cap tend to one. The last two limits follow from the previously proved coefficients $(86/25)A_C$ and $(43/25)A_C$, both positive. $\square$

The four-term genuine-band estimate can therefore use $\mathcal{R}_\sigma^\dagger$ in place of $\mathcal{R}_\sigma^*$. Its boundary, exceptional-event and phase terms are unchanged. The terminal exponent range is still $d < 1/5$; the source's separate phase restriction is not removed by this constant improvement. The tail estimates and genuine first-passage identification are proved separately in Section 19; their dependencies remain distinct from the finite terminal comparison.

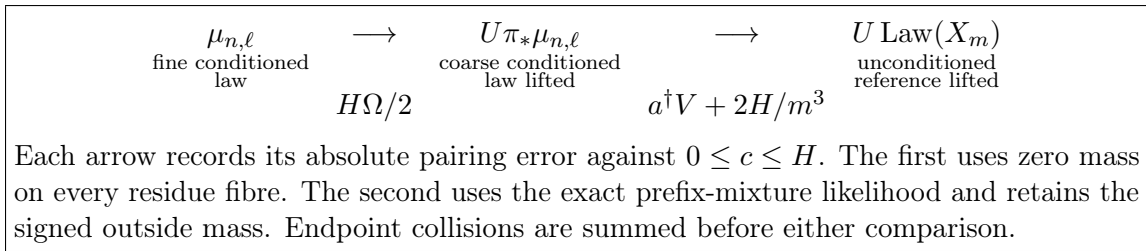


Figure 3: Two different reasons for error in the fixed-cell comparison. The arrows are comparisons of expectations, not deterministic maps between sample outcomes. The actual mass and test maps are (32); the error is proved in Corollary 37, with the tail bounds proved in Corollary 41. Mazur's source construction is retained with finer error accounting.

19 Concentration from cuts and the actual passage event

There are two remaining identifications behind the preceding comparison. First, conditioning on a total valuation changes the distribution of an initial block; its exceptional mass must be proved small. Second, a terminal affine record need not by itself describe the *first* passage of its source. This section reconstructs both mechanisms. The first uses finite counting and exponential moments. The second uses the actual ordered valuation word and a lower bound on the step-back endpoint. Neither is inferred from a numerical orbit survey.

The primary source locators are Mazur [5], ND/Fourier/FixedTotalBars.lean, FixedTotalConcentration.lean, FixedTotalPGF.lean, FixedTotalMGF.lean, FixedTotalTail.lean, FixedTotalT3.lean; and ND/Band/A5TerminalAtomCoverage.lean, A5GoodTimeLocalization.lean, Tao/Section5/ReversePrefix.lean, ReversePrefixScalar.lean, and PassLostWindow.lean. The written proofs below expose the numerical content of their guards. They are source-level reconstructions, not a claim to have replayed the Lean package or to have discovered sampling-without-replacement bounds.

Positive compositions as cuts. For $1 \leq n \leq \ell$, let

$$\mathcal{C}_{n,\ell} = \{(a_1, \dots, a_n) \in \mathbb{N}_{\geq 0}^n : a_1 + \dots + a_n = \ell\}.$$

The map to subsets of $\{1, \dots, \ell - 1\}$ is

$$(a_1, \dots, a_n) \mapsto \{a_1, a_1 + a_2, \dots, a_1 + \dots + a_{n-1}\}. \quad (41)$$

Its inverse takes consecutive differences after adjoining 0 and ℓ . The cuts are strictly increasing, so all inverse differences are positive; telescoping proves both compositions are the identity. There are $\binom{\ell-1}{n-1}$ such words. Each has the same geometric probability $2^{-\ell}$ before conditioning, hence the fixed-total law is exactly the uniform cut-set law. The source stores a physical cut s as $s - 1 \in \text{Fin}(\ell - 1)$. Thus its strict zero-based test $s - 1 < q$ is our inclusive physical test $s \leq q$. For $n = \ell = 1$ both carriers consist of one empty-cut configuration. No corresponding identification with a uniform cut set is asserted for the infeasible positive-total, zero-length carrier.

Proposition 39 (finite count and inclusive cut-count tails). *Let $\mathcal{B}$ be a uniform K -subset of $\{1, \dots, N\}$, where $N \geq 1$ and $0 \leq K \leq N$. Put $H_q = \#(\mathcal{B} \cap \{1, \dots, q\})$, $0 \leq q \leq N$, and $\rho = K/N$. For $0 \leq r \leq N$,*

$$\mathbb{E} \binom{H_q}{r} = \binom{q}{r} \frac{\binom{K}{r}}{\binom{N}{r}} \leq \binom{q}{r} \rho^r. \quad (42)$$

For $q \geq 1$ and $v \geq 0$,

$$\mathbb{P}(H_q - q\rho \geq v) \leq e^{-2v^2/q}, \quad \mathbb{P}(q\rho - H_q \geq v) \leq e^{-2v^2/q}. \quad (43)$$

Both events include equality. For $q = 0$ the count is identically zero.

Proof. Count pairs $(R, \mathcal{B})$ with $R \subseteq \{1, \dots, q\}$, $|R| = r$, and $R \subseteq \mathcal{B}$. For $r \leq K$ their number is $\binom{q}{r} \binom{N-r}{K-r}$; for $r > K$ it is zero. Dividing by $\binom{N}{K}$ proves the identity, with binomial zeros retained. For $r \leq K$ the ratio equals $\prod_{i=0}^{r-1} (K-i)/(N-i) \leq (K/N)^r$; for $r > K$ its numerator is zero. More generally a prefix extending beyond N uses $\min(q, N)$ in this count; it is not an additional population.

For $x \geq 1$, expand $x^{H_q} = (1 + (x - 1))^{H_q}$ and extend by the zero binomial terms to $r = q$. Positivity of $(x - 1)^r$ gives

$$\mathbb{E}x^{H_q} \leq (1 + \rho(x - 1))^q.$$

For a Bernoulli variable of mean ρ , set $g(t) = \log(1 - \rho + \rho e^t) - \rho t$. For $0 < \rho < 1$ its second derivative is $p_t(1 - p_t) \leq 1/4$, where $p_t = \rho e^t / (1 - \rho + \rho e^t)$. Since $g(0) = g'(0) = 0$, integration twice gives $g(t) \leq t^2/8$. For $\rho = 0, 1$, $g = 0$ and the same bound holds. At $x = e^t$, $t \geq 0$, it follows that $\mathbb{E}e^{t(H_q - q\rho)} \leq e^{qt^2/8}$. On the event $H_q - q\rho \geq v$, the exponential is at least e^{tv} . Markov's inequality with $t = 4v/q$ proves the upper-tail bound, including $v = 0$. Complementation $\mathcal{B} \mapsto \mathcal{B}^c$ is a bijection of the uniform K - and $(N - K)$ -subset carriers. It changes H_q to $q - H_q$ and ρ to $1 - \rho$, so the same argument proves the lower tail. No independence of the individual membership indicators was assumed. $\square$

Proposition 40 (the fixed-total prefix tail with its integer endpoints). *Let $n \geq 3$, $1 \leq j < n$, $n \leq \ell \leq 3n$, and let $U_j = a_1 + \dots + a_j$ under the uniform law on $\mathcal{C}_{n,\ell}$. For $8 \leq u \leq 3j$,*

$$\mathbb{P}\left(\left|U_j - \frac{j\ell}{n}\right| \geq u\right) \leq 2 \exp\left(-\frac{u^2}{63j}\right). \quad (44)$$

Proof. The variable U_j is the j th physical cut. Its mean is $c = j\ell/n$: permutation of the parts preserves the uniform composition law, so every part has mean ℓ/n . Set $\rho = (n - 1)/(\ell - 1)$. Direct calculation gives

$$\rho \geq \frac{1}{4}, \quad j - 1 \leq c\rho \leq j, \quad c \leq 3j.$$

Indeed $4(n - 1) \geq \ell - 1$ follows from $n \geq 3$, $\ell \leq 3n$; also $j - c\rho = j(\ell - n)/(n(\ell - 1)) \in [0, 1)$.

For the lower deviation put $q = \lfloor c - u \rfloor$. If $q < j$ the event is empty, since $U_j \geq j$. Otherwise $1 \leq q \leq 3j$ and $q < \ell$. The exact event is $\{U_j \leq q\} = \{H_q \geq j\}$. As $q\rho \leq c\rho - u\rho \leq j - u/4$, Proposition 39 bounds its probability by $e^{-u^2/(8q)} \leq e^{-u^2/(24j)} \leq e^{-u^2/(63j)}$.

For the upper deviation put $v = \lceil c + u \rceil$. The support bound is $U_j \leq \ell - (n - j)$; if v exceeds it, the event is empty. Otherwise put $q = v - 1$. Here $1 \leq q \leq \ell - 1$, $q < c + u \leq 6j$, and $q \geq c + u - 1$. The exact event is $\{U_j \geq v\} = \{H_{v-1} \leq j - 1\}$. In particular the ceiling is followed by a subtraction of one, also when $c + u$ is integral. Since $q\rho \geq j - 1 + (u - 1)/4$, the lower cut-count tail gives $e^{-(u-1)^2/(8q)} \leq e^{-(u-1)^2/(48j)}$. For $u = 8 + w$, $w \geq 0$,

$$63(u - 1)^2 - 48u^2 = 15w^2 + 114w + 15 \geq 0.$$

This proves the claimed upper-deviation bound. The two one-sided events cover the absolute deviation; their sum proves the assertion. $\square$

Corollary 41 (both exceptional masses in the fixed-cell descent). *Keep $m, n, \ell, S_m, R_m, G, P, Q$ from Section 18, with $m \geq 150$, $8m \leq n$, $|\ell - 2n| \leq n/8$, and $m|\ell - 2n|/n \leq S_m$. Then*

$$P(G^c) \leq 2m^{-25/7} \leq 2m^{-3}, \quad Q(G^c) \leq 2m^{-8} \leq 2m^{-3}. \quad (45)$$

Thus both tail inputs in Corollary 37 now have complete written proofs.

Proof. The law P is the law of U_m conditional on the total ℓ . Its mean differs from $2m$ by at most S_m . Outside G its support still has $u \geq m$, so $|u - 2m| > R_m$. Since $R_m \geq 16S_m$, this implies $|u - m\ell/n| \geq 15S_m$. The previously proved $S_m \geq 2$ and $R_m \leq 3m$ imply $8 \leq 15S_m \leq 3m$. Also $n \leq \ell \leq 3n$. Proposition 40 gives

$$P(G^c) \leq 2e^{-225S_m^2/(63m)} = 2m^{-25/7}.$$

For the unconditioned law Q , the centered one-step moment is exactly

$$\mathbb{E}e^{t(a_i-2)} = \frac{e^{-t}}{2-e^t}, \quad t < \log 2.$$

Its logarithm $h(t) = -t - \log(2 - e^t)$ satisfies $h(0) = h'(0) = 0$ and $h''(t) = 2e^t/(2 - e^t)^2$. For $|t| \leq 1/4$, $e^t \leq e^{1/4} < 4/3$, because $\log(4/3) = \int_1^{4/3} x^{-1} dx > 1/4$. Thus $h''(t) \leq 6$, and integration gives $h(t) \leq 3t^2 \leq 8t^2$ on this interval. Independence now gives $\mathbb{E}e^{t(\sum_{i=1}^m a_i - 2m)} \leq e^{8mt^2}$. Using $t = \min(R/(16m), 1/4)$ for the upper deviation and $-t$ for the lower one yields, for every $R > 0$,

$$Q\left(\left|\sum_{i=1}^m a_i - 2m\right| \geq R\right) \leq 2 \exp\left(-\min\left(\frac{R^2}{32m}, \frac{R}{8}\right)\right).$$

In the first case $R \leq 4m$, the exponent is $-R^2/(32m)$; in the second, $-R/4 + m/2 \leq -R/8$. At $R = R_m \leq 3m$ the quadratic term is the minimum, and $R_m^2/(32m) \geq 8 \log m$. This proves the second bound. All infinite geometric sums converge in the displayed range of t . $\square$

The sharper, unequal tails can also be inserted directly into Lemma 36. For the same test as before, let $\theta_P = 2m^{-25/7}$, $\theta_Q = 2m^{-8}$ and

$$a^\sharp = \max\left(\frac{e^{\epsilon_{m,n,\ell}}}{1 - \theta_Q} - 1, 1 - e^{-\epsilon_{m,n,\ell}}(1 - \theta_P)\right).$$

Then the error is at most $H\Omega/2 + a^\sharp V + H\theta_P$. This is no greater than the previous bound because each tail is smaller. In particular the harmonic and flat absolute cell errors may be $10K/m^{11} + 40m^{-25/7}$ and $40K/m^{11} + 160m^{-25/7}$, respectively. The coefficient $a^\sharp$ is retained as a maximum; unequal tails do not justify silently replacing it by its first entry.

What a terminal record has to prove about the orbit. We now use the positive odd-return map $T(N) = (3N + 1)/2^{v_2(3N+1)}$. Write $x_k = T^k N$, $a_k = v_2(3x_{k-1} + 1)$ and $S_k = a_1 + \dots + a_k$. For any consecutive block of length r , the exact affine identity gives

$$x_{j+r} = \frac{3^r}{2^{S_{j+r}-S_j}} x_j + \sum_{i=1}^r 3^{r-i} 2^{-(a_{j+i} + \dots + a_{j+r})}. \quad (46)$$

The offset is nonnegative and at most 3^r . For $r = 0$ it is zero. For $r \geq 1$, bounding each inverse power by one bounds the sum by $(3^r - 1)/2 < 3^r$. The *upper* offset bound, not merely its positivity, is needed to exclude an earlier passage.

Fix $C > 0$. Use the unchanged schedules

$$L = \log B, \quad n_0 = \lfloor L/(10 \log 2) \rfloor, \quad h = \lfloor L/100000 \rfloor,$$

$$W = C \sqrt{n_0 \log n_0}, \quad H_0 = L^{3/5}, \quad s_0 = L^{7/10}.$$

Put $d = \log(4/3)$, $\delta = d/\log 2$, and

$$D_- = (4/3)^h B e^{-s_0}, \quad D_+ = (4/3)^h B e^{s_0}.$$

For a target set E of positive odd integers, define the original finite endpoint set

$$E'_B(E) = \{M \text{ odd} : \lceil D_- \rceil \leq M \leq \lfloor D_+ \rfloor, \tau_B(M) = h, T^h M \in E\},$$

where τ_B is the first time at which the orbit is at most B , and is ∞ if it never reaches that threshold. These are exactly the closed integer endpoints of `taoSection5EPrime`, not a changed window.

Take a band of positive odd integers $z \leq N < U$ with

$$B^{1001/1000} \leq z < U \leq B^{(1001/1000)^3}, \quad 0 < \log(U/z) < 2 \log 2,$$

which includes the source's two branches and their exact sub-bands. Set

$$J = \left[\left\lfloor \frac{\log(z/B)}{d} - 3W \right\rfloor, \left\lfloor \frac{\log(U/B)}{d} + 3W \right\rfloor \right] \cap \mathbb{N}, \quad \mathcal{G} = \{N : |S_k - 2k| < W \ (0 \leq k \leq n_0)\}.$$

Let $\mathcal{A}_E$ be the event of a finite first passage into E from this band. Let $\mathcal{U}_E$ be its terminal-record union: a member is an odd N in the band for which some $t \in J$, $r = t - h$, positive word a of length r and total ℓ , and $M \in E'_B(E)$ satisfy

$$r \leq \ell \leq 3r, \quad |\ell - 2r| < W, \quad \text{Aff}_a(N) = M.$$

The endpoint bound makes the family finite. The list coordinates are equivalent to the source's extras multiset: the multiplicity of index i is $a_i - 1$, with inverse $a_i = 1 + \text{multiplicity}(i)$. This is the same unrestricted affine union used above; no first-passage condition on N has been inserted in its definition.

Proposition 42 (actual first passage equals the terminal union on the good event). *For all sufficiently large B , depending on C but uniformly in the band and the target E ,*

$$\mathcal{A}_E \cap \mathcal{G} = \mathcal{U}_E \cap \mathcal{G}. \quad (47)$$

The map from a good source to its terminal record is its actual first-hit time t , actual first $t - h$ valuations, their total, and $M = T^{t-h}N$. Its inverse is the integer affine inverse of Proposition 29.

Proof. We first verify the scalar guards; they are not assumptions about an unknown orbit. Uniformly over the stated bands, as $L \rightarrow \infty$,

$$W = o(L), \quad W = o(H_0), \quad H_0 = o(s_0), \quad s_0 = o(h), \quad 2 \cdot 3^{n_0} \leq B.$$

The last assertion follows from $3^{n_0} \leq B^{\log 3/(10 \log 2)} < B^{1/5}$. Also $h < \min J \leq \max J < n_0$ whenever J is nonempty, and $W \leq (t - h)/8$ for all $t \in J$ eventually. For the lower edge use $\log(z/B) \geq L/1000$ and $d < 1/3$, giving $\min J - h \geq (3/1000 - 1/100000)L - 3W$. For the upper edge use $(1001/1000)^3 < 101/100$, $d > 1/4$ and $n_0 > L/10 - 1$, giving $\max J \leq L/25 + 3W < n_0$. Floors and ceilings have the displayed favourable directions.

If $N \in \mathcal{G}$ lies in the band, (46) gives

$$T^{n_0}N \leq (3/4)^{n_0} 2^W U + 3^{n_0} \leq B$$

eventually. Indeed the first term is at most $(4/3)B^{(1001/1000)^3 - d/(10 \log 2)} 2^W$; its exponent is less than $101/100 - 1/40 < 1$ before the $o(1)$ contribution of 2^W . Thus a first hit exists with $1 \leq t \leq n_0$.

The final inequality $T^t N \leq B$, offset positivity, and $S_t < 2t + W$ imply

$$t > \frac{\log(N/B)}{d} - \frac{W}{\delta}.$$

At its predecessor $T^{t-1}N > B$, the upper offset bound and $2 \cdot 3^{n_0} \leq B$ imply $B/2 < 3^{t-1}N/2^{S_{t-1}}$. Using $S_{t-1} > 2(t-1) - W$ therefore gives

$$t < \frac{\log(N/B)}{d} + \frac{W+1}{\delta} + 1.$$

The exact integer inequality $3^{29} < 2^{46}$ implies $\delta > 12/29$. For $W \geq 6$, $W/\delta < 3W$ and $(W+1)/\delta + 1 < 3W$. The band bounds now place t in the original J .

Put $r = t - h$ and $M = T^r N$. Differences of the prefix inequalities give $2k - 2H_0 \leq S_{r+k} - S_r \leq 2k + 2H_0$ for $k \leq h$. Since $T^h M \leq B$,

$$M \leq (4/3)^h 2^{2H_0} B.$$

Since $T^{h-1}M > B$ and its offset is at most $B/2$,

$$M > (3/8)(4/3)^h 2^{-2H_0} B.$$

For large B , $2(\log 2)H_0 + \log(8/3) \leq s_0$. Consequently $D_- \leq M \leq D_+$, with exactly the stated integer rounding. First passage shifts to $\tau_B(M) = h$, and the landing point is unchanged. The actual prefix word has $|\ell - 2r| < W \leq r/8$, hence $r \leq \ell \leq 3r$. This constructs a terminal record and proves the forward inclusion.

Conversely take a record for $N \in \mathcal{U}_E \cap \mathcal{G}$. Its odd affine endpoint decodes to the actual positive valuation word and $T^r N = M$, by the affine inverse already proved. Suppose some $j < r$ had $T^j N \leq B$. The prefix deviations give $S_r - S_j \geq 2(r-j) - 2H_0$, and (46) then gives

$$M \leq (3/4)^{r-j} 2^{2H_0} B + 3^{r-j} \leq 2^{2H_0} B + 3^{n_0}.$$

But $M \geq D_-$, whereas

$$2^{2H_0} B + 3^{n_0} < D_-$$

eventually: divide by B and use $hd - s_0 - 2(\log 2)H_0 \rightarrow +\infty$ and $3^{n_0}/B \rightarrow 0$. This is a contradiction. There is no earlier hit before r ; the endpoint's first hit is exactly h steps later, so $\tau_B(N) = t$ and the landing point belongs to E . This proves the reverse inclusion.

Finally $\text{diam } J \leq \log(U/z)/d + 6W = o(h)$, so Proposition 29 gives uniqueness of the record. The construction and the affine inverse are inverse on these good events. All eventual bounds used only C and the fixed band envelopes, not E or the individual source. $\square$

The distinction matters: without the good-prefix and endpoint bounds, an orbit can hit $[1, B]$, leave it, and later supply a terminal record. The earlier exact example with starting value 63 illustrates precisely that possibility. The reverse argument excludes it here by an explicit upper bound on the later endpoint of any orbit that had already hit.

Corollary 43 (one exceptional-event charge for both sampling laws). *For every probability measure p on the positive odd integers and all sufficiently large B as above,*

$$|p(\mathcal{A}_E) - p(\mathcal{U}_E)| \leq p(\mathcal{G}^c). \quad (48)$$

In particular this applies separately to the harmonic and uniform probability laws on a nonempty band. The exceptional mass is charged once after taking the full union, not once per time or valuation total.

Proof. The two event indicators agree on $\mathcal{G}$ and differ in absolute value by at most one on its complement. Integrate this pointwise inequality. For a measure supported on the band, the same proof can use $p(\text{band} \cap \mathcal{G}^c)$, which is equal to $p(\mathcal{G}^c)$. $\square$

For the original A5 sub-bands, with the band-mass and assembled-test cap guards stated in Section 18, the genuine-band comparison is now connected by proved maps and event identities:

$$|p_\sigma(\mathcal{A}_E) - \text{common reference profile}| \leq p_\sigma(\mathcal{G}^c) + \text{the original boundary error} + \mathcal{R}_\sigma^\dagger.$$

This inequality combines the actual-event identification here, the nominal boundary comparison, and the refined terminal descent. The band-measure estimates for $p_\sigma(\mathcal{G}^c)$ are proved in Section 20 below, through an exact residue decoder and separate estimates for both sampling laws. Evaluation of the common reference profile remains a separate mathematical step.

20 Actual band laws and a maximal prefix bound

The good-prefix event is a property of an actual integer orbit. It is therefore necessary to transport the geometric valuation law to the integer sampling law, with a quantitative error. There is also a separate improvement in the geometric estimate: summing a tail over all times is unnecessary when the event asks whether a partial sum ever crosses one common threshold.

The source comparison here is with Mazur [5], `ND/Band/A5HarmonicResidue.lean`, `A5HarmonicProp19.lean`, `A5HarmonicFullGood.lean`, `A5BandCoverageFullGood.lean`, and `Tao/Probability/Geom2PrefixTypical.lean`, at the same pinned commit. Its strict-truncation mechanism is in `Tao/Syracuse/Prop19Laws.lean`, `Prop19Truncation.lean`, and `TruncatedValuationTV.lean`. The underlying valuation approximation is Tao’s Proposition 1.9 [8], source label `rach`; the proof is in his Section 4, source lines 597–647. The following argument reconstructs the finite transport, then uses the exact geometric moment to sharpen this particular prefix estimate. It makes no priority claim for exponential stopping arguments.

The two sampling laws on the same integers. Let

$$I = \{N \in 2\mathbb{N} + 1 : z \leq N < U\}, \quad D = \#I > 0, \quad H = \sum_{N \in I} \frac{1}{N}, \quad z \geq 1.$$

The upper endpoint remains excluded, even when it is an odd integer. The uniform and harmonic laws are respectively

$$p_F(N) = D^{-1} \mathbf{1}_I(N), \quad p_H(N) = \frac{\mathbf{1}_I(N)}{NH}.$$

On their common support the exact density ratio is $p_F(N)/p_H(N) = NH/D$, with inverse $D/(NH)$. These are distinct probability measures on the same carrier; the estimates below are proved separately for both.

Proposition 44 (exact band residue errors). *Let $M \geq 1$, $q = 2^M$, and let ν_M be the uniform law on the $q/2$ odd residue classes modulo q . Let $\pi_M(N) = N \bmod q$. Then, with full ℓ^1 distance,*

$$\|\pi_{M*} p_F - \nu_M\|_1 \leq \frac{q}{2D}, \quad \|\pi_{M*} p_H - \nu_M\|_1 \leq \frac{q}{2\lceil z \rceil H}. \quad (49)$$

The estimates retain the actual D, H and real band endpoints.

Proof. Enumerate I as $N_i = N_1 + 2(i-1)$, $1 \leq i \leq D$. For an odd residue b , put $a = q/2$, $d_i = \mathbf{1}_{\{N_i \bmod q=b\}} - 1/a$ and $F_j = \sum_{i=1}^j d_i$. The residue repeats once every a indices. Writing $j = ka + r$, $0 \leq r < a$, proves $|F_j| \leq 1$, including $a = 1$. In particular the flat atom error is at most $1/D$. Summing over the a classes gives the first bound.

For $w_i = 1/N_i$, finite summation by parts is the exact identity

$$\sum_{i=1}^D d_i w_i = F_D w_D + \sum_{i=1}^{D-1} F_i (w_i - w_{i+1}).$$

The weights decrease and are nonnegative. Hence the absolute value is at most $w_D + \sum_{i < D} (w_i - w_{i+1}) = w_1 \leq 1/\lceil z \rceil$. Divide by H and sum over the same a classes. No asymptotic count or endpoint replacement has been used. $\square$

Proposition 45 (strict decoding and the single ideal overflow mass). *Let $n, M \geq 1$, let p be any probability law on positive odd integers, and write P_n for the law of its first n actual valuations. Let $Q_n = \text{Geom}(2)^n$, and set $\Delta_M = \|\pi_{M*}p - \nu_M\|_1$. Then*

$$\|P_n - Q_n\|_1 \leq \Delta_M + 2\Theta(n, M), \quad \Theta(n, M) = 2^{-(M-1)} \sum_{j=0}^{n-1} \binom{M-1}{j}. \quad (50)$$

Consequently every set A of length- n positive words satisfies

$$|P_n(A) - Q_n(A)| \leq \frac{\Delta_M}{2} + \Theta(n, M). \quad (51)$$

Terms outside the binomial support are zero.

Proof. Let V_n send an odd integer to its actual ordered valuation word. Let $\mathcal{W}_{n,M}$ be the finite set of such positive words with total $s < M$, and define $T_{n,M}$ on all positive length- n words by retaining the word in $\mathcal{W}_{n,M}$ and sending every other word to one symbol $\perp$.

For an accepted word a , Proposition 2 gives its unique odd cylinder $u_a \bmod 2^{s+1}$. As $s+1 \leq M$, membership is determined by a residue modulo 2^M . Distinct accepted words have disjoint residue fibres: a common odd residue has a positive lift, which cannot have two actual valuation words. Define $d_{n,M}$ on the odd residues to output the accepted word whose cylinder contains the residue, or $\perp$ if there is none. This proves the commuting identity

$$d_{n,M} \circ \pi_M = T_{n,M} \circ V_n.$$

The accepted fibre over a has exactly 2^{M-s-1} residues, namely the lifts of u_a modulo 2^M . Its uniform probability is $2^{M-s-1}/2^{M-1} = 2^{-s}$, exactly the geometric word probability. All remaining residues form the one overflow fibre. Thus $d_{n,M*}\nu_M = T_{n,M*}Q_n$.

Pushforward contracts full ℓ^1 : group the absolutely summable differences in each fibre and apply the triangle inequality. It follows that $\|T_{n,M*}P_n - T_{n,M*}Q_n\|_1 \leq \Delta_M$. If b_P, b_Q are the two overflow masses, the full untruncated distance is at most the accepted-atom distance plus $b_P + b_Q$. The truncated distance already contains $|b_P - b_Q|$, and

$$b_P + b_Q - |b_P - b_Q| = 2 \min(b_P, b_Q) \leq 2b_Q.$$

This proves the first inequality with $b_Q = Q_n(s \geq M)$; no separate bound for the actual orbit's overflow was assumed.

To evaluate b_Q , realize each geometric variable as the waiting time to the next success in independent fair Bernoulli trials. A run of $a_i - 1$ failures followed by one success has probability 2^{-a_i} ; disjoint runs multiply to 2^{-s} . The n th success occurs at time at least M precisely when at most $n - 1$ successes occur among the first $M - 1$ trials. Counting those finite Bernoulli strings gives the displayed formula, also when $M \leq n$ and that probability is one.

Finally $P_n - Q_n$ has total mass zero. Its positive and negative masses are equal, each half its full ℓ^1 norm. The mass on any event is between the negative of this number and this number. This proves (51) without changing the source's full- ℓ^1 convention. $\square$

One maximal estimate instead of a union over times. Let A_i be independent with $\mathbb{P}(A_i = a) = 2^{-a}$, $a \geq 1$, and write $X_k = \sum_{i=1}^k (A_i - 2)$. Its exact one-step logarithmic moment is

$$h(t) = \log \mathbb{E} e^{t(A_i - 2)} = -t - \log(2 - e^t), \quad t < \log 2.$$

It has $h(0) = h'(0) = 0$, $h''(t) > 0$, and therefore $h(t) \geq 0$.

Proposition 46 (maximal geometric tail with the exact rate). *For $n \geq 2$ and real $0 < w < n$, define*

$$\begin{aligned} I(v) &= (1 + v) \log(1 + v) - (2 + v) \log(1 + v/2) \quad (-1 < v < \infty), \\ \Psi(n, w) &= e^{-nI(w/n)} + e^{-nI(-w/n)}. \end{aligned} \quad (52)$$

Then

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |X_k| \geq w\right) \leq \Psi(n, w). \quad (53)$$

There is no factor n multiplying this estimate. Also, for every $n \geq 1$,

$$\Theta(n, 3n) \leq (27/32)^n. \quad (54)$$

Proof. For fixed $t < \log 2$, $Z_k = \exp(tX_k - kh(t))$ has expectation one and satisfies $\mathbb{E}(Z_{k+1} \mid A_1, \dots, A_k) = Z_k$. This follows directly from independence and the displayed moment. For $t > 0$, stop at the first $k \leq n$ with $X_k \geq w$, or at n if no crossing occurs. Call that bounded time τ . The identity

$$Z_{(k+1) \wedge \tau} - Z_{k \wedge \tau} = \mathbf{1}_{\{\tau > k\}}(Z_{k+1} - Z_k)$$

has zero expectation: the indicator depends only on the first k variables. Summing this finite identity proves $\mathbb{E}Z_\tau = 1$. All its terms are integrable; for example $Z_\tau \leq \sum_{k=0}^n Z_k$. On a crossing, $Z_\tau \geq e^{tw - nh(t)}$, since $h(t) \geq 0$. Thus the upper maximum has probability at most $e^{nh(t) - tw}$.

For the lower maximum apply the same argument at a negative parameter t , stopping when $X_k \leq -w$. Its bound is $e^{nh(t) + tw}$. Writing $r = w/n$, choose respectively

$$t_+ = \log \frac{2(1+r)}{2+r}, \quad t_- = \log \frac{2(1-r)}{2-r}.$$

Here $0 < t_+ < \log 2$ and $t_- < 0$, because $0 < r < 1$. Since

$$h'(t) = \frac{2(e^t - 1)}{2 - e^t},$$

these parameters solve $h'(t_+) = r$ and $h'(t_-) = -r$. Substitution, with no discarded terms, gives $rt_+ - h(t_+) = I(r)$ and $-rt_- - h(t_-) = I(-r)$. The union of the two crossing events proves the result.

For the overflow only the final total is needed. Exponential Markov at $t = \log(4/3)$ gives

$$Q_n(X_n \geq n) \leq e^{-nt+nh(t)} = \left(\frac{27}{32}\right)^n, \quad h(\log(4/3)) = \log(9/8).$$

The event is exactly the strict-decoder overflow $s \geq 3n$. $\square$

For comparison, using merely $h(\pm t) \leq 8t^2$ for $0 \leq t \leq 1/4$ in the same stopped argument gives $2e^{-\min(w^2/(32n), w/8)}$, still without a union factor. The optimized rate above retains more of the exact geometric law. Its behavior near zero is explicit:

$$I(0) = I'(0) = 0, \quad I''(v) = \frac{1}{(1+v)(2+v)}, \quad I(v) = v^2 \int_0^1 \frac{1-u}{(1+uv)(2+uv)} du.$$

On $|v| \leq 1/2$ the integrand and its v -derivative are bounded, so $I(v) = v^2/4 + O(|v|^3)$, with an absolute remainder constant.

Corollary 47 (full-prefix failure under both actual band laws). *Use the unchanged schedules of Section 19, and put $n = n_0$, $W = C\sqrt{n \log n}$ and $w = \lceil W \rceil$. For every fixed $C > 0$, sufficiently large B gives $1 \leq w < n$. For every nonempty band I as above, let $\mathcal{G}$ retain the literal strict inequalities $|S_k - 2k| < W$ for $0 \leq k \leq n$. Then*

$$p_H(\mathcal{G}^c) \leq \Psi(n, w) + \frac{2^{3n-2}}{\lceil z \rceil H} + \Theta(n, 3n), \quad (55)$$

$$p_F(\mathcal{G}^c) \leq \Psi(n, w) + \frac{2^{3n-2}}{D} + \Theta(n, 3n). \quad (56)$$

On the source's exact A5 sub-bands these imply, uniformly in the branch and valid band index,

$$p_\sigma(\mathcal{G}^c) \leq \Psi(n, w) + B^{-701/1000} + (27/32)^n \leq (2 + o_C(1))n^{-C^2/4}, \quad \sigma = H, F. \quad (57)$$

The last expression is an asymptotic upper bound, not an asymptotic formula for the actual exceptional probability.

Proof. Every deviation $S_k - 2k$ is integral. Consequently $|S_k - 2k| \geq W$ is exactly $|S_k - 2k| \geq w$, including integral W . Equivalently it is strictly greater than the source radius $\lceil W \rceil - 1$. The zero prefix is never bad when $W > 0$. Thus $\mathcal{G}^c$ is the preimage under V_n of the single maximal event in Proposition 46. Apply (51) at $M = 3n$, with the two separate residue estimates (49). This proves both finite-band bounds.

Here are the band constants, with their original endpoints retained. For an A5 branch $y = B^\alpha$ or B^{α^2} , $\alpha = 1001/1000$, put

$$b = (\alpha - 1) \log y, \quad K = \lfloor b \rfloor, \quad \beta = b/K, \quad z = ye^{j\beta}, \quad U = ze^\beta, \quad 0 \leq j < K.$$

Eventually $K \geq 1$, $\beta \geq 1$, $\beta \rightarrow 1$, $z \geq B^\alpha$ and $z \geq 6$. In particular $U \geq ez$. The first odd point is less than $z + 2$, and the successor of the last odd point is at least U . Comparing the decreasing function $1/t$ on each interval of length two gives

$$H \geq \frac{1}{2} \log \frac{U}{z+2} \geq \frac{1}{2} \left(1 - \frac{2}{z}\right) \geq \frac{1}{3} \geq \frac{1}{4}.$$

Direct lattice counting gives $D > (U - z)/2 - 1 \geq (e - 1)z/2 - 1 \geq z/2$ for $z \geq 6$; for the last inequality $e > 8/3$ follows from its first five Taylor terms. Since $n = \lfloor \log B/(10 \log 2) \rfloor$,

$$2^{3n} \leq B^{3/10}, \quad \frac{2^{3n-2}}{\lceil z \rceil H} \leq B^{-701/1000}, \quad \frac{2^{3n-2}}{D} \leq \frac{1}{2} B^{-701/1000}.$$

Keep the exact denominators in the finite bounds even though the common upper bound uses the larger of these last two quantities.

Finally $w = C\sqrt{n \log n} + O(1)$. The integral expression for I gives

$$nI(\pm w/n) = \frac{C^2}{4} \log n + O_C\left(\frac{(\log n)^{3/2}}{\sqrt{n}}\right) + o(1).$$

Hence $\Psi(n, w) = (2 + o_C(1))n^{-C^2/4}$. The remaining two terms are negligible relative to every fixed negative power of n : n is comparable to $\log B$, and

$$(27/32)^n \leq (32/27)B^{-\kappa}, \quad \kappa = \frac{\log(32/27)}{10 \log 2} > 0.$$

All constants used the band envelopes, not its index or a target set. □

This replaces the source's positive-prefix union bound

$$2n \exp \left[-\min \left(\frac{R^2}{32n}, \frac{R}{8} \right) \right] + 4 \cdot 2^{-n/128}, \quad R = \lceil W \rceil - 1,$$

whose first term is $(2 + o_C(1))n^{1-C^2/32}$. The improvement has three explicit causes: a maximum is controlled without summing over times; the exact geometric moment is retained; and an event costs half the full ℓ^1 error. The uniform band is treated directly, so no factor e^β from harmonic domination is needed. The new component estimate decays for every fixed $C > 0$; this is not a change to the source's separate phase estimate or a claim of a stronger final natural-density theorem.

For the original A5 bands, Corollary 43 now bounds the genuine-passage/terminal-union discrepancy by the appropriate right-hand side of (55) or (56). Combining with the earlier boundary comparison and $\mathcal{R}_\sigma^\dagger$ is therefore justified with this quantitative exceptional term. A convex combination of band laws retains their common bound because its weights sum to one; it does not multiply the error by the number of bands. The excluded upper endpoint of the full branch interval is not in that combination. Its own singleton mass is retained in Corollary 67. The common profile is evaluated in Section 24; the resulting actual passage comparison is Theorem 66.

Corollary 48 (an explicit prefix constant in the Allikvere window). *Let $x \rightarrow \infty$, $L = \log x$, $\alpha = 1001/1000$, $y = x^\alpha$ or x^{α^2} , and $n = \lfloor L/(10 \log 2) \rfloor$. If N is uniform on the positive odd integers in the closed interval $[y, y^\alpha]$, then*

$$\mathbb{P} \left(\max_{0 \leq k \leq n} |S_k(N) - 2k| \geq L^{3/5} \right) \leq (2 + o(1)) \exp \left[-\frac{5 \log 2}{2} L^{1/5} \right]. \quad (58)$$

The error is uniform in the two branches.

Proof. Keep the closed upper endpoint by taking the equivalent half-open integer carrier $y \leq N < \lfloor y^\alpha \rfloor + 1$. Its odd count D_y is comparable to y^α . Propositions 44 and 45, with $M = 3n$, give event error at most $2^{3n-2}/D_y + \Theta(n, 3n)$. Both terms decrease as a positive power of x . Apply Proposition 46 at $w = \lceil L^{3/5} \rceil < n$. The integral rate identity gives

$$nI(\pm w/n) = \frac{w^2}{4n} + O(w^3/n^2) = \frac{5 \log 2}{2} L^{1/5} + o(1).$$

Here $w^3/n^2 = O(L^{-1/5})$ and the floor/ceiling changes also contribute $o(1)$. Summing the two exponentials proves the result; the power errors in x are negligible relative to this exponential. $\square$

This supplies an explicit constant for the earlier uniform-input prefix bound. It does not replace the separate stronger power bound for the event that no passage occurs by n_0 : those are different events, and their established estimates are both retained.

21 The common endpoint profile with exact band corrections

The common center is not obtained by identifying uniform and harmonic sampling. Their raw sums have different phase functions and different means. This section reconstructs the finite comparison before estimating the oscillatory sums. In particular, it keeps the shifted raw mass, the two band denominators and the exterior center visible.

The source locators are Mazur [5], `A5TwoProfileRawQ.lean`, `A5TwoProfilePhysicalA6.lean`, `A5TwoProfileNormalizedA6.lean`, `A5ReferencePhysicalSplit.lean`, `A5ReferenceInteriorAggregate.lean`, and `A5ReferenceFullCenterRate.lean`, at the pinned commit already specified. The kernel is Tao's common term, represented in the package by `Tao/Section5/CommonZ.lean`. The identities below are finite, and do not presume the interior equidistribution or exterior tail estimates.

Exact lattice corrections. Keep the original odd band $I = [z, U) \cap (2\mathbb{N} + 1)$, $z \geq 1$, $U > z$, $D = \#I > 0$, $H = \sum_I 1/N$, and $\beta = \log(U/z)$. Define the signed discrepancies

$$\epsilon_D = D - \frac{U - z}{2}, \quad \epsilon_H = H - \frac{\beta}{2}.$$

They are not discarded by replacing the band with a continuous interval.

Proposition 49 (band discrepancies and retained denominator errors). *For the band just defined,*

$$|\epsilon_D| < 1, \quad |\epsilon_H| \leq \frac{1}{z}. \quad (59)$$

For $d = \log(4/3) > 0$, put $a = 2/d$. For any real raw values R_H, R_F , the following are exact:

$$\frac{R_H}{H} - a = \frac{R_H - \beta/d}{H} - \frac{2\epsilon_H}{dH}, \quad (60)$$

$$\frac{z}{D} R_F - a = \frac{z}{D} \left(R_F - \frac{e^\beta - 1}{d} \right) - \frac{2\epsilon_D}{dD}. \quad (61)$$

Thus the absolute denominator charges are at most $2/(zdH)$ and strictly less than $2/(dD)$, respectively. If $z \geq 6$ and $\beta \geq 1$, then $H \geq 1/3$ and $D > z/2$, so these charges are at most $6/(zd)$ and less than $4/(zd)$.

Proof. For $z \leq t \leq U$, write

$$C(t) = \#\{N \text{ positive odd} : z \leq N < t\}, \quad \Delta(t) = C(t) - (t - z)/2.$$

For $b \geq a$, the exact count of integers in $[a, b]$ is $\lceil b \rceil - \lceil a \rceil$. Apply this at $a = (z-1)/2$, $b = (t-1)/2$. The two errors $\lceil u \rceil - u$ belong to $[0, 1)$, so $|\Delta(t)| < 1$ and the claimed bound for $\epsilon_D = \Delta(U)$ follows, including integral odd endpoints.

For each $N \in I$ the identity $1/N = 1/U + \int_N^U t^{-2} dt$ gives, after a finite sum,

$$H = \frac{D}{U} + \int_z^U \frac{C(t)}{t^2} dt.$$

The choices $N < t$ and $N \leq t$ differ only at finitely many points and give the same ordinary integral. The continuous expression

$$\frac{U - z}{2U} + \int_z^U \frac{t - z}{2t^2} dt = \frac{1}{2} \log(U/z)$$

is verified by integration. Subtracting proves the signed formula

$$\epsilon_H = \frac{\Delta(U)}{U} + \int_z^U \frac{\Delta(t)}{t^2} dt.$$

Its absolute value is at most $1/U + \int_z^U t^{-2} dt = 1/z$.

Equations (60) and (61) now follow by substitution and $U - z = z(e^\beta - 1)$. Finally $H \geq \beta/2 - 1/z \geq 1/3$ and $D > (e - 1)z/2 - 1 > z/2$ for $z \geq 6$; for the last strict inequality $e > 8/3$. All denominators are the original positive sums. $\square$

Compared with the source estimates $|H - \beta/2| \leq 3/(2z)$, $1/H \leq 4$ and $z/D < 3$, the displayed proof retains the exact signed errors and yields the bounds $|H - \beta/2| \leq 1/z$, $1/H \leq 3$ and $z/D < 2$ under the same $z \geq 6, \beta \geq 1$ conditions. In particular an absolute raw harmonic error E_H costs $E_H/H + 2/(zdH) \leq 3E_H + 6/(zd)$, and an absolute raw flat error E_F costs $(z/D)E_F + 2/(dD) \leq 2E_F + 4/(zd)$. The original coarse constants $4E_H + 12/(zd)$ and $3E_F + 6/(zd)$ are preserved in the source comparison, not silently substituted into its unreviewed consumers.

The literal two phase functions. Let $\gamma = \log_2 3$, $\delta = 2 - \gamma = d/\log 2$, and suppose $\log 2 < \beta < 2 \log 2$. These inequalities hold on the source A5 bands for sufficiently large B . For $W > 0$, $A = \log_2(z/M)$ and $\nu \geq 1$, define

$$s_\nu = \lfloor A + \nu\gamma \rfloor + 1, \quad \theta_\nu = \{A + \nu\gamma\}, \quad b = 2 - \beta/\log 2 \in (0, 1),$$

and, for every integer s ,

$$p_\nu(s) = \begin{cases} \binom{s-1}{\nu-1} 2^{-s}, & s \geq \nu, \\ 0, & s < \nu. \end{cases}$$

Retain the two distinct indicated atoms

$$q_0(\nu) = p_\nu(s_\nu) \mathbf{1}_{\{|s_\nu - 2\nu| < W\}}, \quad q_1(\nu) = p_\nu(s_\nu + 1) \mathbf{1}_{\{|s_\nu + 1 - 2\nu| < W\}}.$$

The strict integer anchor is used even at integral phase. These formulas agree with the source's raw atoms at A and $A + 1$; for $\nu > 0$, clipping a negative signed level to zero leaves its probability zero. The tube's finite upper cap removes no index: $|\lfloor A - \delta\nu \rfloor + 1| < W$ implies $\nu \leq (A + \lceil W \rceil)/\delta$. Let V be the unchanged positive physical time carrier $\{\nu > 0 : \nu + h \in J\}$.

Proposition 50 (two distinct raw profiles and their phase means). *For $\theta \in [0, 1)$, set*

$$f_H(\theta) = 1 + \mathbf{1}_{\{\theta > b\}}, \quad f_F(\theta) = 2^{1-\theta} + \mathbf{1}_{\{\theta > b\}} 2^{2-\theta}.$$

The two raw sums on V are exactly

$$R_H(M) = \sum_{\nu \in V} (q_0(\nu) + \mathbf{1}_{\{\theta_\nu > b\}} q_1(\nu)), \quad (62)$$

$$R_F(M) = \sum_{\nu \in V} (2^{1-\theta_\nu} q_0(\nu) + \mathbf{1}_{\{\theta_\nu > b\}} 2^{2-\theta_\nu} q_1(\nu)). \quad (63)$$

Equivalently,

$$R_H(M) = \sum_V q_0(\nu) f_H(\theta_\nu) + \sum_V \mathbf{1}_{\{\theta_\nu > b\}} (q_1(\nu) - q_0(\nu)), \quad (64)$$

$$R_F(M) = \sum_V q_0(\nu) f_F(\theta_\nu) + \sum_V \mathbf{1}_{\{\theta_\nu > b\}} 2^{2-\theta_\nu} (q_1(\nu) - q_0(\nu)). \quad (65)$$

Their ordinary integrals are

$$\int_0^1 f_H(t) dt = \frac{\beta}{\log 2}, \quad \int_0^1 f_F(t) dt = \frac{e^\beta - 1}{\log 2}. \quad (66)$$

The gated upper-cell weight is strictly below e^β ; at $\theta = b$ the upper cell is absent.

Proof. The base position $(1 - \theta) \log 2$ belongs to $(0, \beta)$ throughout $[0, 1)$. The successor position belongs to that interval exactly when $(2 - \theta) \log 2 < \beta$, or $\theta > b$. This proves the first two formulas directly from Proposition 31; the flat exponential is attached once to its own cell. Add and subtract the gated q_0 term to obtain the next pair. No equality of the two tube supports or their masses is asserted.

The harmonic integral is $1 + (1 - b) = \beta / \log 2$. The two flat integrals are

$$\int_0^1 2^{1-t} dt = \frac{1}{\log 2}, \quad \int_b^1 2^{2-t} dt = \frac{2^{2-b} - 2}{\log 2}.$$

Since $2^{2-b} = e^\beta$, their sum is as claimed. The strict gate also gives $2^{2-\theta} < e^\beta$ when active. Its endpoint exclusion changes neither integral but remains essential to the finite sums. $\square$

This explains the two raw centers: multiplying the respective means by the source's raw-mass center $1/\delta$ gives β/d and $(e^\beta - 1)/d$. It does not establish that an individual raw sum equals its center. Both the weighted rotation error and the signed neighbour correction in (65) must still be estimated on their actual supports.

The common kernel and the landing map. For a fixed depth m , retain

$$K_m(M) = \frac{3^m}{M} p_m(M \bmod 3^m) \quad (M > 0),$$

where p_m is the source's Syracuse offset probability law. Let $\mathcal{E}_*$ be the original positive odd endpoint window with $\tau_B(M) = h$, without imposing a landing target. Its lower and upper integer endpoints

remain $\lceil (4/3)^h B e^{-(\log B)^{7/10}} \rceil$ and $\lfloor (4/3)^h B e^{(\log B)^{7/10}} \rfloor$. Define $\lambda(M) = T^h M \in [1, B] \cap (2\mathbb{N} + 1)$. Then the original $\mathcal{E}(E)$ is $\mathcal{E}_* \cap \lambda^{-1}(E)$.

The weighted finite endpoint measure $\mu_m = \sum_{\mathcal{E}_*} K_m(M) \delta_M$ therefore has the exact landing pushforward

$$Z(E) = \frac{2}{d} \lambda_* \mu_m(E) = \frac{2}{d} \sum_{M \in \mathcal{E}(E)} K_m(M).$$

This is the source common profile. It has no band or sampling-law parameter. It is not declared a probability measure: its total mass has not been set to one. The landing map can have multiple endpoints in a fibre, and that fibre has precisely the sum of their weights. The map from endpoint measures to landing measures is linear and positive; it loses the distribution within each landing fibre.

Proposition 51 (signed common-center identity on the actual carrier). *For $S = \mathcal{E}(E)$, put $Z_0(S) = \sum_S K_m(M)$ and*

$$r_H(M) = R_H(M) - \beta/d, \quad r_F(M) = R_F(M) - (e^\beta - 1)/d.$$

The actual reference profiles

$$P_H(S) = \sum_S K_m(M) R_H(M)/H, \quad P_F(S) = \sum_S K_m(M) (z/D) R_F(M)$$

satisfy

$$P_H(S) - a Z_0(S) = \frac{1}{H} \sum_S K_m(M) r_H(M) - \frac{2\epsilon_H}{dH} Z_0(S), \quad (67)$$

$$P_F(S) - a Z_0(S) = \frac{z}{D} \sum_S K_m(M) r_F(M) - \frac{2\epsilon_D}{dD} Z_0(S). \quad (68)$$

These formulas retain signs before the absolute-value estimate. They hold for the empty target as well. In particular, the denominator charge occurs once against $Z_0(S)$, not once per endpoint or per time.

Proof. Substitute (60) or (61) at every endpoint, multiply by $K_m(M)$, and sum. The band discrepancies are independent of M . The kernel is nonnegative because $M > 0$ and p_m is a probability mass function, but positivity is not needed for the identities. It gives the corresponding absolute error bounds by applying the triangle inequality to these finite sums. $\square$

An explicit all-endpoint envelope. For the same finite positive time set V , define

$$G_V = \sum_{\nu \in V} \frac{\binom{2\nu-2}{\nu-1}}{2^{2\nu-2}}, \quad L_H = \frac{G_V}{H}, \quad L_F = \frac{ze^\beta G_V}{D}.$$

These quantities depend on the actual time carrier and band, but not on the endpoint M .

Proposition 52 (finite envelope and sharp exterior centering). *For every positive endpoint M ,*

$$0 \leq R_H(M)/H \leq L_H, \quad 0 \leq (z/D) R_F(M) \leq L_F, \quad G_V \leq \sum_{\nu \in V} \nu^{-1/2}.$$

Split the original endpoint set at its closed interior condition

$$S_{\text{in}} = \{M \in S : |\log M - \log B - hd| \leq 4W/25\}, \quad S_{\text{out}} = S \setminus S_{\text{in}}.$$

For either $\sigma = H, F$, write $P_{\sigma, \text{out}}$ for its profile restricted to S_{out} and $Z_{\text{out}} = \sum_{S_{\text{out}}} K_m(M)$. Then

$$|P_{\sigma, \text{out}} - aZ_{\text{out}}| \leq \max\{a, L_\sigma - a\} Z_{\text{out}}. \quad (69)$$

The constant is sharp given only $0 \leq P_{\sigma, \text{out}} \leq L_\sigma Z_{\text{out}}$. For the full profile one may combine (68) on S_{in} with this exterior term, retaining the exact denominators and $Z_0(S_{\text{in}})$ in the interior charge.

Proof. The adjacent endpoint masses obey, for $s \geq \nu$,

$$\frac{p_\nu(s+1)}{p_\nu(s)} = \frac{s}{2(s+1-\nu)}.$$

For $\nu \geq 2$ this ratio is at least one exactly when $s \leq 2\nu - 2$. The two maximal atoms are at $2\nu - 2$ and $2\nu - 1$, with common mass

$$b_\nu = \frac{\binom{2\nu-2}{\nu-1}}{2^{2\nu-1}}.$$

For $\nu = 1$, $p_1(s) = 2^{-s}$ has maximum $1/2 = b_1$ at $s = 1$. Thus each of the two indicated raw atoms is at most b_ν . Each active flat weight is less than e^β . Summing gives $R_H \leq 2 \sum_V b_\nu = G_V$ and $R_F \leq e^\beta G_V$, without replacing an exterior physical window by an intrinsic tube.

For completeness, let $c_k = \binom{2k}{k}/4^k$. The recurrence $c_{k+1}/c_k = (2k+1)/(2k+2)$ and

$$4(k+1)^3 - (2k+1)^2(k+2) = 3k+2 > 0$$

prove by induction from $c_0 = 1$ that $c_k \leq 1/\sqrt{k+1}$. This proves the bound for G_V .

Nonnegative kernel weighting gives $0 \leq P_{\sigma, \text{out}} \leq L_\sigma Z_{\text{out}}$. If $Z_{\text{out}} = 0$ the desired statement is immediate. Otherwise $P_{\sigma, \text{out}}/Z_{\text{out}}$ lies in $[0, L_\sigma]$. The largest distance of this interval from $a > 0$ is $\max\{a, |L_\sigma - a|\} = \max\{a, L_\sigma - a\}$. Both endpoints of the interval realize the corresponding distances, which proves sharpness at this information level. Finally S_{in} and S_{out} are a disjoint partition, including equality in the interior, so the two signed errors add exactly before applying the triangle inequality. $\square$

The source full-center argument instead uses a coarser common envelope and the sum of that envelope with a . The positive interval argument above retains the best coefficient supplied by an envelope; it does not assert sharpness for actual Collatz orbits. The finite identities explain the common center and carry the exact weights into its next analytic stage. They do not yet replace the source's interior weighted-rotation estimate or its exterior coefficient-tail argument, and do not establish a new final natural-density exponent.

22 The shifted tube and the weighted phase sum

The remaining phase calculation has two distinct sources of error. First, moving the integer level from s_ν to $s_\nu + 1$ changes both the mass and the set of permitted times. Second, the phases in the original, unequally weighted sum need not have their integral distribution. We give finite formulas for each effect. These formulas retain the original masses; no probability normalization is imposed.

The source arguments are Mazur [5], ND/Band/A5ShiftedRawComparison.lean, A5Regularity.lean, and ND/Discrepancy/A5Padding.lean, A5PaddedProfile.lean, A5TwoProfileTail.lean. Their endpoint discrepancy majorant is defined in PhaseToDbar.lean. All refer to the pinned source commit. The recurrences and padding mechanism below reconstruct those arguments, with exact finite terms retained.

The two supports. Keep $\gamma = \log_2 3$, $\delta = 2 - \gamma$, and p_ν, s_ν, q_0, q_1 from Section 21. Put

$$R = \lceil W \rceil - 1, \quad d_\nu = s_\nu - 2\nu = \lfloor A - \delta\nu \rfloor + 1.$$

Thus R is a nonnegative integer. Define finite sets of positive integers

$$I_0 = \{\nu : -R \leq d_\nu \leq R\}, \quad I_1 = \{\nu : -R \leq d_\nu + 1 \leq R\}.$$

These are the indicator supports, even where an endpoint mass vanishes.

Proposition 53 (exact support change and signed shift). *The common part, lost part and gained part are respectively*

$$I_c = \{\nu : -R \leq d_\nu \leq R - 1\}, \quad I_- = \{\nu : d_\nu = R\}, \quad I_+ = \{\nu : d_\nu = -R - 1\}.$$

For every integer r ,

$$\{\nu \geq 1 : d_\nu = r\} = \mathbb{Z}_{\geq 1} \cap \left(\frac{A - r}{\delta}, \frac{A - r + 1}{\delta} \right]. \quad (70)$$

In particular $\#I_-, \#I_+ \leq 3$. Suppose $I_0 \cup I_1$ is nonempty and its least element ℓ satisfies $\ell > R + 1$. For every real function g on this union,

$$\sum_{\nu} g_\nu(q_1(\nu) - q_0(\nu)) = - \sum_{\nu \in I_c} g_\nu q_0(\nu) \frac{d_\nu + 2}{2(\nu + d_\nu + 1)} - \sum_{\nu \in I_-} g_\nu q_0(\nu) + \sum_{\nu \in I_+} g_\nu q_1(\nu). \quad (71)$$

Define the exact nonnegative quantity

$$\mathcal{E} = \sum_{I_c} q_0(\nu) \frac{|d_\nu + 2|}{2(\nu + d_\nu + 1)} + \sum_{I_-} q_0(\nu) + \sum_{I_+} q_1(\nu).$$

It equals $\sum_{\nu} |q_1(\nu) - q_0(\nu)|$. For $R \geq 1$, put

$$c_R(\ell) = \max \left\{ \frac{(R - 2)_+}{2(\ell - R + 1)}, \frac{R + 1}{2(\ell + R)} \right\}; \quad c_0(\ell) = 0.$$

Writing $Z = \sum_{I_0} q_0(\nu)$, we have the finite bound

$$\mathcal{E} \leq c_R(\ell)Z + \sum_{I_-} q_0(\nu) + \sum_{I_+} q_1(\nu) \leq c_R(\ell)Z + \frac{3}{\sqrt{\ell}}. \quad (72)$$

Proof. For an integer d , $|d| < W$ is equivalent to $-R \leq d \leq R$. Intersection and set subtraction give the three displayed supports. The floor inequalities $r - 1 \leq A - \delta\nu < r$ give (70), with its strict left endpoint. The inequalities $1/3 < \delta < 1$ follow from $(4/3)^3 > 2$ and $1 < 4/3 < 2$. Hence the interval length $1/\delta$ is less than three and cannot contain four integers.

On I_c , $s_\nu \geq \nu$ and the adjacent binomial ratio gives

$$\frac{q_1(\nu)}{q_0(\nu)} - 1 = \frac{2\nu + d_\nu}{2(\nu + d_\nu + 1)} - 1 = -\frac{d_\nu + 2}{2(\nu + d_\nu + 1)}.$$

Our hypothesis on ℓ makes these denominators and both masses positive. On the other two parts, precisely one indicator is zero. This proves the signed identity and the exact expression for $\mathcal{E}$. If $R = 0$, the common part is empty. Otherwise the function $|d + 2|/[2(\nu + d + 1)]$, for $-R \leq d \leq R - 1$, decreases up to $d = -2$ and increases thereafter: the derivative of $(d + 2)/(\nu + d + 1)$ is $(\nu - 1)/(\nu + d + 1)^2 > 0$. Its maximum is therefore bounded by the two endpoint expressions in $c_R(\nu)$, each nonincreasing in ν . Finally Proposition 52 proves $p_\nu(s) \leq 1/(2\sqrt{\nu})$ for every integer s . There are at most six boundary indices, proving the last bound. The exact boundary sums, rather than this ceiling, remain available. $\square$

For the harmonic gate take $g_\nu = \mathbf{1}_{\{\theta_\nu > b\}}$; for the flat gate take $g_\nu = \mathbf{1}_{\{\theta_\nu > b\}} 2^{2-\theta_\nu}$. Their absolute corrections are at most $\mathcal{E}$ and $e^\beta \mathcal{E}$, respectively. The signed expression (71) retains cancellation, including the zero common-part correction at $d_\nu = -2$.

Variation along the original time coordinate. Assume I_0 is nonempty, write $l = \min I_0$, $h = \max I_0$, and suppose $l > R$. The floor d_ν decreases by zero or one at each step, since $0 < \delta < 1$. Thus $I_0 = [l, h] \cap \mathbb{Z}$. Write $q_\nu = q_0(\nu)$, $Z = \sum_l^h q_\nu$ and

$$\mathcal{V} = q_l + q_h + \sum_{\nu=l}^{h-1} |q_{\nu+1} - q_\nu|.$$

Both boundary terms occur, also when $l = h$.

Proposition 54 (raw recurrences and full variation). *For $l \leq \nu < h$, $s_{\nu+1} - s_\nu$ is one or two, and*

$$\frac{q_{\nu+1}}{q_\nu} - 1 = \begin{cases} d_\nu/(2\nu), & s_{\nu+1} = s_\nu + 1, \\ (d_\nu^2 + d_\nu - 2\nu)/(4\nu(\nu + d_\nu + 1)), & s_{\nu+1} = s_\nu + 2. \end{cases} \quad (73)$$

In particular, with

$$\eta(l, R) = \max \left\{ \frac{R}{2l}, \frac{R^2 + R + 2l}{4l(l - R + 1)} \right\},$$

one has

$$\mathcal{V} \leq q_l + q_h + \eta(l, R)Z \leq \frac{1}{2\sqrt{l}} + \frac{1}{2\sqrt{h}} + \eta(l, R)Z. \quad (74)$$

Under the explicit additional scalar guards

$$W > 1, \quad W \leq l/8, \quad Z \geq 1/32, \quad W \geq (37/20)\sqrt{l},$$

the same raw variation satisfies $\mathcal{V} \leq 20ZW/l$.

Proof. Consecutive levels rise by one or two because $1 < \gamma < 2$. Cancelling factorials in the two cases gives ratios $s_\nu/(2\nu)$ and $s_\nu(s_\nu + 1)/[4\nu(s_\nu + 1 - \nu)]$, respectively. Substitution of $s_\nu = 2\nu + d_\nu$ proves (73). The first absolute defect is at most $R/(2l)$. The second is at most

$$\frac{R^2 + R}{4\nu(\nu - R + 1)} + \frac{1}{2(\nu - R + 1)} \leq \frac{R^2 + R + 2l}{4l(l - R + 1)}.$$

Multiplication by q_ν and summation proves the first bound; the atom ceiling proves the second.

For the last statement it is enough to retain the source's coarser bound $2W/\nu$ on each relative defect. Here is a direct check. The first defect is less than $W/(2\nu)$. For the second, $|d_\nu| < W \leq \nu/8$, $\nu > 8$, and $W > 1$ give

$$|d_\nu^2 + d_\nu - 2\nu| \leq W^2 + W + 2\nu \leq (9/4)W\nu, \quad 4\nu(\nu + d_\nu + 1) \geq (7/2)\nu^2.$$

The ratio is therefore at most $(9/14)W/\nu < 2W/\nu$. The internal variation is at most $2WZ/l$. The two endpoint masses sum to at most $1/\sqrt{l}$, which is at most $32Z/\sqrt{l} \leq (640/37)ZW/l < 18ZW/l$. Adding proves the constant 20, without replacing Z by one. $\square$

An exact padding map and its error formula. Put $s = h - l + 1$ and $u_k = q_{l+k}$ for $0 \leq k < s$. For any integer $P \geq 1$, pad on the left by P zero entries:

$$\omega_j = 0 \ (0 \leq j < P), \quad \omega_{P+k} = u_k \ (0 \leq k < s), \quad \phi' = A + (l - P)\gamma \pmod{1}.$$

The map $\mathbb{R}^s \rightarrow \mathbb{R}^{P+s}$ so defined is linear, injective, and preserves the sum and the ℓ^1 norm. Its inverse on its image discards the zero head. The circle translation is invertible as well. In particular,

$$\{\phi' + (P + k)\gamma\} = \{A + (l + k)\gamma\}.$$

Negative $l - P$ causes no loss: this is an integer circle translation, not truncated subtraction of natural numbers.

Proposition 55 (weighted phase error with both endpoints retained). *For $\sigma = H, F$, let $\mu_\sigma = \int_0^1 f_\sigma$ and define*

$$S_m^\sigma = \sum_{j=0}^{m-1} (f_\sigma(\{\phi' + j\gamma\}) - \mu_\sigma).$$

The exact weighted error is

$$\sum_{\nu=l}^h q_\nu f_\sigma(\theta_\nu) - \mu_\sigma Z = u_{s-1} S_{P+s}^\sigma - u_0 S_P^\sigma + \sum_{k=0}^{s-2} (u_k - u_{k+1}) S_{P+k+1}^\sigma. \quad (75)$$

Define the actual weak endpoint discrepancies

$$D_m = \sup_{0 \leq t \leq 1} \left| \frac{1}{m} \# \{0 \leq j < m : \{\phi' + j\gamma\} \leq t\} - t \right|, \quad D_* = \max_{P \leq m \leq P+s} D_m,$$

and the exact Abel coefficient

$$\mathcal{B}_P = P|u_0| + (P + s)|u_{s-1}| + \sum_{k=0}^{s-2} (P + k + 1)|u_k - u_{k+1}|.$$

Then

$$\left| \sum_{\nu=l}^h q_\nu f_\sigma(\theta_\nu) - \mu_\sigma Z \right| \leq C_\sigma D_* \mathcal{B}_P, \quad C_H = 1, \quad C_F = 2e^\beta - 1, \quad \mathcal{B}_P \leq (P + s)\mathcal{V}. \quad (76)$$

All quantities are finite and defined from the original phase orbit.

Proof. Write each centered summand as $S_{j+1}^\sigma - S_j^\sigma$, multiply by ω_j , and telescope. Only the jump at P , the internal differences and the terminal term survive, giving (75). This also proves the identity for $s = 1$.

Let $E_m(t) = \#\{j < m : \{\phi' + j\gamma\} \leq t\} - mt$. For f_H , the strict upper gate gives $S_m^H = -E_m(b)$. For f_F , the jump immediately to the right of b has size e^β . On the two open pieces its derivative is $-\log 2 f_F(t)$. Integrating the identity $f_F(x) = f_F(1) - \int_x^1 f_F'(t) dt - e^\beta \mathbf{1}_{\{x \leq b\}}$ for $0 \leq x < 1$ and then summing it at the finite sample points yields

$$S_m^F = - \int_0^1 E_m(t) f_F'(t) dt - e^\beta E_m(b).$$

Here $f_F(1)$ denotes its left limit, namely 3. The total integral of $|f_F'|$ is $(2 - e^\beta/2) + (3e^\beta/2 - 3) = e^\beta - 1$. Thus $|S_m^\sigma| \leq C_\sigma m D_m$, with the weak endpoint convention accounting exactly for a sample at b . Inserting these bounds into the exact identity proves (76). Each length coefficient in $\mathcal{B}_P$ is at most $P + s$, which proves its bound by the full variation. $\square$

This calculation explains a constant already available inside the source. For its equal padding $P = s$, a nonincreasing uniform endpoint majorant $D(m)$ gives $D_* \leq D(s)$. Its proved constant-20 variation estimate therefore gives $40 ZsWD(s)/l$ for the harmonic error, and $(2e^\beta - 1)$ times this for the flat error. The later source adapter instead relaxes 20 to 32 and consequently uses 64. This is a direct consequence of its stronger lemma, not a new equidistribution theorem. The source uses $D(m) = \min\{1, (6 + 2/(\pi c))m^{-1/\kappa}\}$ under its specified Diophantine input. The finite proof above uses the actual D_m and does not certify that analytic input by computation.

For the full intrinsic two-cell sums $\mathcal{R}_H, \mathcal{R}_F$ over $I_0 \cup I_1$, the exact decomposition now reads

$$\mathcal{R}_\sigma - \mu_\sigma/\delta = \left(\sum_{I_0} q_0 f_\sigma - \mu_\sigma Z \right) + \sum_{I_0 \cup I_1} g_\nu^\sigma(q_1 - q_0) + \mu_\sigma(Z - 1/\delta).$$

Consequently the three respective absolute charges are $C_\sigma D_* \mathcal{B}_P$, $\mathcal{E}$ or $e^\beta \mathcal{E}$, and $\mu_\sigma |Z - 1/\delta|$. The raw-mass error is still present. These intrinsic sums equal the original physical sums only when their two supports lie in the physical time window. For an arbitrary physical window, insert its indicator into every displayed finite sum; neither missing times nor their weights can be dropped. The source's interior containment, raw-mass asymptotic, and exterior coefficient tail remain separate parts of the analytic reconstruction.

23 Raw mass, Gaussian coordinates, and physical interior profiles

The total weight in the tube is not one. Its limit is $1/\delta$, and this factor is essential to both raw phase centers. We now prove the mass estimate with explicit finite errors and then evaluate the original physical sums. This connects Mazur's A6 calculation to the Gaussian envelope in Lemma 14; the envelope here is exactly half of that earlier function. The truncation scales differ, so their tail estimates must not be identified.

The source comparison is with Mazur [5], `ND/Band/A6PhysicalRate.lean`, `A6PhysicalInteriorRate.lean`, `A5Physical.lean`, and `ND/Discrepancy/A5TwoProfilePhysicalA6.lean`, at the pinned commit. The factorial estimate used below is Robbins's original formula [4, (1)–(2), p. 26], not an asymptotic formula without its remainder.

Proposition 56 (an exact atom formula and a finite logarithmic error). *For integers $\nu \geq 1$, k with $|k| \leq \nu/2$, put $t = k/\nu$ and*

$$I(t) = (1+t) \log(1+t) - (2+t) \log(1+t/2).$$

Let r_n be defined by $n! = \sqrt{2\pi n}(n/e)^n e^{r_n}$. Then

$$p_\nu(2\nu + k) = \frac{e^{-\nu I(t) + r_{2\nu+k} - r_\nu - r_{\nu+k}}}{2\sqrt{\pi\nu} \sqrt{(1+t/2)(1+t)}}. \quad (77)$$

For $G_\nu(k) = (2\sqrt{\pi\nu})^{-1} e^{-k^2/(4\nu)}$,

$$\left| \log \frac{p_\nu(2\nu + k)}{G_\nu(k)} \right| \leq \frac{4|k|}{3\nu} + \frac{16|k|^3}{27\nu^2} + \frac{1}{4\nu}. \quad (78)$$

Now retain the strict tube I_0 , $R = \lceil W \rceil - 1$, $d_\nu = \lfloor A - \delta\nu \rfloor + 1$, and $l = \min I_0$. Suppose I_0 is nonempty and $W \leq l/2$. Put $\xi = A/\delta$ and

$$G_\xi(u) = \frac{1}{2\sqrt{\pi u}} \exp\left(-\frac{\delta^2(u - \xi)^2}{4u}\right) \quad (u > 0), \quad \varepsilon = \frac{11W}{6l} + \frac{16W^3}{27l^2} + \frac{1}{2l}.$$

For every $\nu \in I_0$, $q_0(\nu) > 0$ and

$$e^{-\varepsilon} G_\xi(\nu) \leq q_0(\nu) \leq e^\varepsilon G_\xi(\nu). \quad (79)$$

The exact rounding term in its logarithm is $-(2d_\nu e_\nu - e_\nu^2)/(4\nu)$, where $e_\nu = 1 - \{A - \delta\nu\} \in (0, 1]$.

Proof. Set $s = 2\nu + k$ and $m = \nu + k \geq \nu/2 > 0$. The exact identity $p_\nu(s) = (\nu/s) \binom{s}{\nu} 2^{-s}$, followed by Robbins's factorial formula at s, ν, m , gives (77). In particular the factor ν/s is not removed. Robbins proves $1/(12n+1) < r_n < 1/(12n)$. Since $s \geq \nu + 1$, $r_s < 1/(12s) < 1/(12\nu + 1) < r_\nu$. Thus

$$-\frac{1}{4\nu} < -(r_\nu + r_m) < r_s - r_\nu - r_m < 0.$$

On $[-1/2, 1/2]$ the derivative of $-\frac{1}{2} \log(1+t/2) - \frac{1}{2} \log(1+t)$ has absolute value at most $1/3+1 = 4/3$. Also $I(0) = I'(0) = 0$, $I''(0) = 1/2$, and

$$I''(t) = \frac{1}{(1+t)(2+t)}, \quad |I'''(t)| = \frac{1}{(1+t)^2} - \frac{1}{(2+t)^2} \leq \frac{32}{9}.$$

The last expression decreases with t . Taylor's integral remainder therefore gives $|I(t) - t^2/4| \leq 16|t|^3/27$. Taking logarithms in the exact identity proves the first bound. This is the same rate function I used in Proposition 46, now in a local atom formula.

For the tube, $|d_\nu| < W \leq \nu/2$ and $d_\nu = A - \delta\nu + e_\nu$. Hence $d_\nu^2 - (A - \delta\nu)^2 = 2d_\nu e_\nu - e_\nu^2$, whose absolute value is at most $2W + 1$. Adding this divided by 4ν to (78), and replacing ν by its lower bound l , gives precisely ε . At integral phase $e_\nu = 1$, not zero; the same identity and estimate apply. $\square$

Proposition 57 (finite tube mass and its uniform asymptotic). *Assume $R \geq 1$, $\xi > R/\delta$, and $W \leq l/2$. Set*

$$a = \xi - \frac{R}{\delta}, \quad b = \xi + \frac{R+1}{\delta}, \quad t_* = \sqrt{\xi^2 + \delta^{-4}} - \delta^{-2}, \quad M_\xi = G_\xi(t_*),$$

$$u_- = \frac{R}{\delta\sqrt{a}}, \quad u_+ = \frac{R+1}{\delta\sqrt{b}}, \quad T = \frac{2}{\delta^2\sqrt{\pi}} \left(\frac{e^{-\delta^2 u_-^2/4}}{u_-} + \frac{e^{-\delta^2 u_+^2/4}}{u_+} \right), \quad Q = 2M_\xi + T.$$

The strict tube is exactly $(a, b] \cap \mathbb{Z}$, and its original mass $Z = \sum_{I_0} q_0(\nu)$ satisfies

$$\left| Z - \frac{1}{\delta} \right| \leq (e^\varepsilon - 1) \left(\frac{1}{\delta} + Q \right) + Q. \quad (80)$$

In particular, fix $C \geq 1/2$. As $N \rightarrow \infty$, for

$$W = C\sqrt{N \log N}, \quad \frac{N}{43} \leq \xi \leq \frac{N}{13},$$

all the preceding guards hold eventually and, uniformly over every real center in this interval,

$$Z = \frac{1}{\delta} + O_C \left(\frac{(\log N)^{3/2}}{\sqrt{N}} \right). \quad (81)$$

Here C is fixed before N tends to infinity.

Proof. The floor inequalities in (70) give the exact interval $(a, b]$. Its length $(2R+1)/\delta > 1$ ensures nonemptiness; $a > 0$ ensures positive indices. Lemma 14, with $z = \xi$, gives $\int_0^\infty G_\xi = 1/\delta$ and the unique maximum at t_* . The zero-extended function $G_\xi \mathbf{1}_{(a,b]}$ is unimodal, with total variation at most $2M_\xi$, including its two endpoint jumps. For each integer cell, the difference between its left-endpoint value and its integral is bounded by that cell's variation. Summing over cells gives

$$\left| \sum_{\nu \in I_0} G_\xi(\nu) - \int_a^b G_\xi(u) du \right| \leq 2M_\xi.$$

This estimate respects an integral a or b with the displayed strict/weak conventions.

For the continuous tails use the explicit increasing bijection

$$v = \sqrt{u} - \xi/\sqrt{u}, \quad u = \left(\frac{v + \sqrt{v^2 + 4\xi}}{2} \right)^2.$$

It sends $(0, \infty)$ onto $\mathbb{R}$ and a, b to $-u_-, u_+$. Its exact density transformation is

$$G_\xi(u) du = \frac{e^{-\delta^2 v^2/4}}{2\sqrt{\pi}} \left(1 + \frac{v}{\sqrt{v^2 + 4\xi}} \right) dv.$$

The parenthesis lies strictly between zero and two. For $u_0 > 0$, integration of $(v/u_0)e^{-\delta^2 v^2/4}$ on $[u_0, \infty)$ proves

$$\int_{u_0}^\infty e^{-\delta^2 v^2/4} dv \leq \frac{2e^{-\delta^2 u_0^2/4}}{\delta^2 u_0}.$$

Thus the two discarded continuous tails sum to at most T . Writing $Z_G = \sum_{I_0} G_\xi(\nu)$, we have $|Z_G - 1/\delta| \leq Q$. By (79), $|Z - Z_G| \leq (e^\varepsilon - 1)Z_G$. These two bounds prove (80), including its product error.

For the asymptotic, $W = o_C(N)$, $R = W + O(1)$ and $l = \xi + O_C(W+1) \asymp N$, uniformly in the stated center interval. In particular $a > 0$, $l > 2W$ and $R \geq 1$ eventually. The explicit expression for ε is $O_C((\log N)^{3/2}/\sqrt{N}) = o_C(1)$. Also $t_* \geq \xi/2$ eventually, so $M_\xi \leq (2\pi\xi)^{-1/2} = O(N^{-1/2})$. Both tail exponents satisfy, uniformly,

$$\frac{\delta^2 u_\pm^2}{4} \geq \left(\frac{13C^2}{4} + o_C(1) \right) \log N \geq \frac{2}{3} \log N$$

for sufficiently large N , because $13C^2/4 \geq 13/16 > 2/3$. Moreover $u_\pm \gg_C \sqrt{\log N}$. Consequently $T = O_C(N^{-2/3}/\sqrt{\log N})$ and $Q = O_C(N^{-1/2})$. Substitution in the finite bound proves the asserted rate. These tails are polynomially small; the different cutoff in Lemma 14 had smaller tails and was not used to justify this truncation. $\square$

Corollary 58 (the two intrinsic centers from the actual raw weights). *Under the asymptotic schedule of Proposition 57, let $\log 2 < \beta < 2\log 2$, $\mu = 8.616$ and $d = \log(4/3)$. The literal two-cell sums over $I_0 \cup I_1$ satisfy*

$$\mathcal{R}_H = \frac{\beta}{d} + O_C\left(N^{-1/(2\mu)}(\log N)^{1/2-1/(2\mu)}\right), \quad (82)$$

$$\mathcal{R}_F = \frac{e^\beta - 1}{d} + O_C\left(N^{-1/(2\mu)}(\log N)^{1/2-1/(2\mu)}\right). \quad (83)$$

Both errors are uniform in the real center and in β . In particular each is $O_{C,c}(N^{-c})$ for every $0 < c < 1/17.232$.

Proof. The base support has length $s = O_C(W)$ and endpoints comparable to N . Proposition 57 bounds Z uniformly. Put $w_\nu = G_\xi(\nu)\mathbf{1}_{I_0}(\nu)$. Its full variation is at most $2M_\xi = O(N^{-1/2})$. Write the exact difference

$$\sum_{I_0} q_0(f_\sigma - \mu_\sigma) = \sum_{I_0} w(f_\sigma - \mu_\sigma) + \sum_{I_0} (q_0 - w)(f_\sigma - \mu_\sigma).$$

The second sum is bounded in absolute value by $\|f_\sigma - \mu_\sigma\|_\infty(e^\varepsilon - 1)(1/\delta + Q)$. Thus every original atom is retained with an explicit error; no equality between it and the Gaussian is assumed. Use Lemma 13, whose Diophantine input is Rhin [7, p. 160, (8)] and whose discrepancy inequalities are those of Kuipers–Niederreiter [6, Ch. 2]. As periodic functions, f_H, f_F have variations 2 and $2e^\beta \leq 8$. These include the wrap-around jump at zero; the interval coefficient $2e^\beta - 1$ in the exact weak-discrepancy formula is a different quantity. The finite bound on the first sum is $C_\gamma \text{Var}(f_\sigma) s^{1-1/\mu} 2M_\xi$. The total centered weighted error is consequently at most

$$O_C\left(s^{1-1/\mu} N^{-1/2} + (\log N)^{3/2} N^{-1/2}\right) = O_C\left(N^{-1/(2\mu)} (\log N)^{1/2-1/(2\mu)}\right).$$

Both centered functions have uniformly bounded sup norm. No division by Z occurs. Using only the raw-variation ceiling would instead give the larger logarithmic power $1 - 1/(2\mu)$; the explicit Gaussian comparison saves a factor $\sqrt{\log N}$.

For the shift correction, $\ell = \min(I_0 \cup I_1) \asymp N$ and $R = O_C(W)$, so (72) gives $\mathcal{E} = O_C(\sqrt{\log N/N})$. Its flat multiplier e^β is bounded by four. Finally both phase means are bounded, and (81) bounds the raw-mass correction by $O_C((\log N)^{3/2}/\sqrt{N})$. Insert these three estimates in the exact decomposition at the end of Section 22. The last two errors are smaller than the first. The means in (66), together with $\delta = d/\log 2$, give precisely the two displayed centers. The stated weaker powers follow since a fixed power of $\log N$ is $o(N^\eta)$ for every $\eta > 0$. $\square$

Proposition 59 (transport to the original physical window). *Let $B, z, M > 0$, $m_0 \in \mathbb{Z}_{\geq 0}$, $\tau = \log M - \log B - m_0 d$, $A = \log(z/M)/\log 2$, $t_0 = \log(z/B)/d$, and $\log 2 < \beta < 2 \log 2$. For $W > 0$, suppose*

$$\frac{|\tau|}{d} + \frac{W+1}{\delta} \leq 3W. \quad (84)$$

The map $\nu \mapsto n = \nu + m_0$ embeds $I_0 \cup I_1$ in

$$J = \mathbb{Z}_{\geq 0} \cap [t_0 - 3W, t_0 + \beta/d + 3W].$$

It is bijective onto its image with inverse $n \mapsto n - m_0$; all image points have $n > m_0$. Zero extension gives exact equality between each intrinsic two-cell sum and its original physical sum on $\{\nu > 0 : \nu + m_0 \in J\}$.

For the original source schedule

$\alpha = 1001/1000$, $L = \log B$, $N = \lfloor L/(10 \log 2) \rfloor$, $m_0 = \lfloor L/100000 \rfloor$, $W = C\sqrt{N \log N}$, fix $C \geq 1/2$, let $y = B^\alpha$ or B^{α^2} , and put

$$K = \lfloor (\alpha - 1) \log y \rfloor, \quad \beta = (\alpha - 1) \log y / K, \quad z = ye^{j\beta} \quad (0 \leq j < K).$$

Uniformly in the branch, j , and every $M > 0$ with $|\tau| \leq 4W/25$, for sufficiently large B ,

$$R_H(M) = \beta/d + O_C\left(N^{-1/17.232} (\log N)^{1/2-1/17.232}\right), \quad (85)$$

$$R_F(M) = (e^\beta - 1)/d + O_C\left(N^{-1/17.232} (\log N)^{1/2-1/17.232}\right). \quad (86)$$

These are interior estimates on the original physical carrier.

Proof. The exact affine identity is

$$\xi = A/\delta = t_0 - m_0 - \tau/d.$$

The union of the two indicator supports consists exactly of the positive integers in $((A - R)/\delta, (A + R + 2)/\delta]$. Thus every image point obeys

$$n > t_0 - \tau/d - R/\delta, \quad n \leq t_0 - \tau/d + (R + 2)/\delta.$$

The first inequality and the margin put n above $t_0 - 3W$. For the upper endpoint, $\beta/d > 1/\delta$ and $R < W$ give

$$|\tau|/d + (R + 2)/\delta < \beta/d + |\tau|/d + (W + 1)/\delta \leq \beta/d + 3W.$$

This proves containment of both supports without asserting that they are equal. Translation is injective, its displayed inverse is exact on its image, and all summands outside the union vanish. This proves the equality of the finite sums, not just an estimate of discarded endpoints. In particular the one-unit shift is absorbed by the upper band width; a symmetric enlargement by that unit is unnecessary for this containment argument.

For the source schedule, $K \rightarrow \infty$ and $\beta \rightarrow 1$ uniformly in the two branches; thus the required interval for β holds eventually. Since $z \in [y, y^\alpha]$, the exact identity for ξ gives uniformly

$$\frac{\alpha - 1}{d} - \frac{1}{100000} + o_C(1) \leq \frac{\xi}{L} \leq \frac{\alpha^3 - 1}{d} + o_C(1).$$

Here the floor error is at most $1/L$, and $|\tau|/L \leq 4W/(25L) = o_C(1)$. Since $N/L \rightarrow 1/(10 \log 2)$, these bounds put ξ in $[N/43, N/13]$ eventually. For an explicit check of the strict room, use

$$0.6931 < \log 2 < 0.6932, \quad 0.2876 < d < 0.2877,$$

which give

$$10(0.6931) \left(\frac{0.001}{0.2877} - 0.00001 \right) > 1/43, \quad 10(0.6932) \frac{(1001/1000)^3 - 1}{0.2876} < 1/13.$$

The logarithm intervals follow from $\log x = 2 \sum_{k=0}^{J-1} u^{2k+1}/(2k+1) + E_J$, $u = (x-1)/(x+1)$, with $0 < E_J < 2u^{2J+1}/((2J+1)(1-u^2))$: take $J = 12$ for $x = 2$ and $J = 6$ for $x = 4/3$. This also gives

$$\frac{4}{25d} + \frac{1}{\delta} = \frac{4/25 + \log 2}{d} < \frac{0.16 + 0.6932}{0.2876} < 3.$$

Consequently the margin holds for all sufficiently large W , uniformly under $|\tau| \leq 4W/25$. All required guards of the mass and intrinsic-profile results therefore hold with C fixed before $B \rightarrow \infty$. Exact transport proves the physical assertions. $\square$

The same band denominators from Proposition 49 now give $R_H/H = 2/d + O_{C,c}(N^{-c})$ and $(z/D)R_F = 2/d + O_{C,c}(N^{-c})$ throughout the interior, for $0 < c < 1/17.232$. The signed denominator errors and the exact kernel residual in Proposition 51 are unchanged. The exterior coefficient estimate and full kernel summation are proved in Section 24 below. Mazur's source uses its stated $\kappa = 14.3$ phase input; the rate here uses the independently cited Rhin bound, already used in the Allikvere reconstruction. No improved final Mazur natural-density theorem or local Lean replay is asserted.

24 Exterior endpoint weights and the full reference profile

The interior estimate must be summed against the endpoint kernel, not against a probability measure obtained by dividing that kernel by its mass. The remaining issue is therefore quantitative: how much of the original weighted endpoint set lies outside the interior? The native reciprocal-fibre estimate contains a linear factor in the last valuation. Summing that factor exactly, rather than bounding it by a multiple of $\log B$, gives a substantially smaller exterior error.

The source route is Mazur [5], `Tao/Section5/EndpointValuationFiber.lean`, `EndpointRatio.lean`, `EndpointLowRoom.lean`, `EndpointFiberEnvelope.lean`, `EndpointCoefficientBound.lean`, and `Geom2HighWeightScalar.lean`; then `ND/Band/A5ReferenceFilteredCoefficient.lean`, `A5ReferenceExteriorTupleTail.lean`, `A5ReferenceTerminalCap.lean`, and `ND/Probability/Geom2FiniteTwoEventTail.lean`. The final comparison is with `ND/Discrepancy/A5ReferenceFullCenterRate.lean`. These are the pinned v2 sources, not a replay of their formal build. The geometric law and first-passage mechanism originate in Tao [8], the definition of “Geometric random variable” (v7 source label `geom`) and the Section 5 coefficient argument (source label `loam`). We reconstruct the needed finite estimates before taking limits.

Original endpoints and their exact fibres. Write $h = \lfloor L/100000 \rfloor$, $L = \log B$, $N = \lfloor L/(10 \log 2) \rfloor$, and $d = \log(4/3)$. For a finite set S of positive odd integers with first passage at time h under the odd Syracuse map T , define

$$c_q(S, x) = 3^q \sum_{\substack{M \in S \\ M \equiv x \pmod{3^q}}} \frac{1}{M}.$$

In the application S is the original $\mathcal{E}(E)$ or its interior/exterior subset; its rounded lost-window endpoints remain unchanged. For the chronological valuation word $b = (b_1, \dots, b_h)$, put

$$s = \sum_{i=1}^h b_i, \quad s_- = \sum_{i=1}^{h-1} b_i, \quad a = b_h, \quad w(b) = 2^{-s}, \quad b_0(b) = (1 + a/2)2^{-s}.$$

The fibre $F_{b,x}$ consists of precisely the $M \in S$ with word b and residue x . Only nonempty fibres enter the finite word carrier. Thus $c_q(S, x) = \sum_b 3^q \sum_{F_{b,x}} 1/M$ exactly. The map $M \mapsto (b(M), M)$ is a bijection from the residue-restricted endpoint set to the disjoint union of these fibres, with inverse $(b, M) \mapsto M$. Projection to the word alone is generally many-to-one; its pushforward retains the reciprocal sum in each fibre.

Proposition 60 (native endpoint coefficient with its decaying remainder). *Suppose $B > 1$, $h \geq 1$, $2 \cdot 3^{h-1} \leq B$, and $h + q \leq N$. For every such finite S and every $x \in \mathbb{Z}/3^q\mathbb{Z}$,*

$$c_q(S, x) \leq \sum_{b: F_{b,x} \neq \emptyset} b_0(b) + \epsilon_B, \quad \epsilon_B = 3B^{-4997/600000}. \quad (87)$$

In particular $0 \leq c_q(S, x) \leq 2 + \epsilon_B$. The assertion holds for filtered original endpoint sets without requiring the enlarged same-word fibres to obey the filter.

Proof. The exact affine iteration has positive offset

$$T^j M = 3^j 2^{-s_j} M + F_j, \quad F_j = \sum_{i=1}^j \frac{3^{j-i}}{2^{b_i + \dots + b_j}} \leq (3/2)^j - 1 < 3^j.$$

This includes $j = 0$ with $F_0 = 0$. Since $T^{h-1}M > B$ and $B \geq 2 \cdot 3^{h-1}$, the affine main term at time $h - 1$ exceeds $B/2$. At time h , positivity and $T^h M \leq B$ give

$$B2^{s-} < 2 \cdot 3^{h-1} M, \quad 3^h M \leq B2^s.$$

Moreover $T^{h-1}M < 2 \cdot 3^{h-1} 2^{-s-} M$ and $2^a \leq 3T^{h-1}M + 1 \leq 4T^{h-1}M$, so $2^s < 8 \cdot 3^{h-1} M$. Consequently each fibre lies in the inclusive integer interval $[A_0, U]$, and also in $[A_1, U]$, where

$$A_0 = \left\lfloor \frac{B2^{s-}}{2 \cdot 3^{h-1}} \right\rfloor + 1, \quad U = \left\lfloor \frac{B2^s}{3^h} \right\rfloor, \quad A_1 = \max \left\{ A_0, \left\lfloor \frac{2^s}{8 \cdot 3^{h-1}} \right\rfloor + 1 \right\}.$$

For a nonempty fibre these lower endpoints are at most U . The strict quotient inequalities give $U < 2^a A_0$, hence $\log(U/A_i) \leq a \log 2 \leq a$.

The exact valuation cylinder is one odd class modulo 2^{s+1} . Indeed the one-step condition is $3M + 1 \equiv 2^{b_1} \pmod{2^{b_1+1}}$. Inductively the next prescribed odd residue lifts its unique previous class to a unique class modulo the next power of two: solving $3M + 1 = 2^{b_1} M_1$ modulo 2^{s+1} is possible and unique since 3 is invertible modulo that power. This also proves the converse by dividing each congruence by its prescribed power of two. Combining this class with $M \equiv x \pmod{3^q}$ gives one class modulo $Q = 2^{s+1} 3^q$, since the two moduli are coprime. No endpoints or word multiplicities are identified in this step.

For any nonempty portion of a Q -progression in $[A, U]$, $A \geq 1$, decreasingness of $1/t$, integrated between successive points, gives

$$\sum \frac{1}{M} \leq \frac{1}{A} + \frac{1}{Q} \log(U/A) \leq \frac{1}{A} + \frac{1}{Q} \{1/A + \log(U/A)\}.$$

To verify the first inequality, retain the first point separately; for each later point M its contribution is at most $Q^{-1} \int_{M-Q}^M dt/t$. The intervals concatenate, and their union lies in $[A, U]$. This also bounds every subset of the progression.

Call a word high when $2^s > B^{1/2}$, strictly. On the low branch, $4^s \leq B$. Also $2^{4(h+q)} \leq B$, because $h + q \leq N$. Thus $2^s 2^{2(h+q)} \leq B$ by multiplying the two square bounds. As $a \leq s$ and $h \geq 1$, this implies $2^{a+2} 3^{h+q-1} \leq B$. Multiplication by 2^{s-} shows $Q \leq \lfloor B2^{s-}/(2 \cdot 3^{h-1}) \rfloor < A_0$. The retained progression estimate is therefore at most

$$3^q \{Q^{-1} + Q^{-1}(1+a)\} = (1+a/2)2^{-s}.$$

On the high branch, use A_1 . Its outer term satisfies $3^q/A_1 < (8/3)3^{h+q}2^{-s} \leq 3 \cdot 3^{h+q}2^{-s}$. The remaining term is at most $(1+a)2^{-s-1} \leq b_0(b)$. This proves the pointwise envelope $b_0(b) + 3 \cdot 3^{h+q}2^{-s} \mathbf{1}_{\{2^s > B^{1/2}\}}$.

Here and below enlarging a finite word carrier to all positive words of length h is legitimate by nonnegativity, without duplicating words. For independent X_i with $\Pr(X_i = r) = 2^{-r}$, $r \geq 1$, the centered one-coordinate moment generating function is

$$g(t) = \mathbb{E}e^{t(X_i-2)} = \frac{e^{-t}}{2 - e^t}, \quad t < \log 2.$$

Since $h \leq L/100000$ and $\log 2 < 7/10$, the high event implies $\sum_i (X_i - 2) > (7/10)L$. For $0 \leq t \leq 1/4$, $e^t < 4/3$ and $(\log g)'' = 2e^t/(2 - e^t)^2 \leq 6$. Here $\log(4/3) > 1/4$ follows directly by integrating $1/x$. As $(\log g)(0) = (\log g)'(0) = 0$, $\log g(1/4) \leq 3/16 < 1/2$. Markov's inequality now bounds the high mass by $e^{-7L/40+h/2} \leq B^{-34999/200000}$. Finally $3^{6(h+q)} \leq 2^{10(h+q)} \leq B$, so the total high charge is at most ϵ_B . This proves the first assertion.

The elementary series $\sum_{a \geq 1} 2^{-a} = 1$ and $\sum_{a \geq 1} a 2^{-a} = 2$ show $\sum_{b \in \mathbb{N}_{>0}^h} b_0(b) = 2$. This proves the last bound, retaining the smaller $2 + \epsilon_B$ rather than replacing the decaying remainder by 3. The proof never enlarged the original endpoint filter before retaining its nonempty word carrier. $\square$

Proposition 61 (exact weighted two-event estimate). *Let $h \geq 2$, $u > 0$, $0 < v < h - 1$, and let $\mathcal{C}$ be any finite set of positive words of length h such that every $b \in \mathcal{C}$ satisfies at least one of*

$$s - 2h > u, \quad s_- - 2(h - 1) < -v.$$

With the rate function from Section 20,

$$I(t) = (1 + t) \log(1 + t) - (2 + t) \log(1 + t/2) \quad (t > -1),$$

one has

$$\sum_{b \in \mathcal{C}} (1 + b_h/2) 2^{-|b|} \leq \left(2 + \frac{u}{2h}\right) e^{-hI(u/h)} + 2e^{-(h-1)I(-v/(h-1))}. \quad (88)$$

Proof. For $t < \log 2$, summing the terminal factor exactly gives

$$\mathbb{E}[(1 + X_h/2) e^{t \sum_{i=1}^h (X_i - 2)}] = g(t)^h \left(1 + \frac{1}{2 - e^t}\right).$$

This follows by differentiating the absolutely convergent geometric series: $\mathbb{E}[X_h e^{t(X_h - 2)}] = 2e^{-t}/(2 - e^t)^2$. The upper event is bounded, for $0 < t < \log 2$, by $e^{-tu} g(t)^h (1 + (2 - e^t)^{-1})$. Take

$$e^t = \frac{2(1 + u/h)}{2 + u/h}.$$

This lies strictly between 1 and 2; direct substitution gives $e^{-tu} g(t)^h = e^{-hI(u/h)}$ and $1 + (2 - e^t)^{-1} = 2 + u/(2h)$.

For the lower event, the terminal coordinate is independent of the first $h - 1$ coordinates and its weighted expectation equals 2. Taking $e^t = 2(1 - v/(h - 1))/(2 - v/(h - 1)) \in (0, 1)$ in the ordinary lower-tail Markov estimate gives its displayed contribution. Apply the union inequality to these two nonnegative weighted events. It remains valid when they overlap; no independence of the two events is asserted. Absolute convergence follows from $e^t < 2$ and the linear terminal factor. No terminal cutoff or factor $\log B + 1$ has been introduced. $\square$

Corollary 62 (exterior coefficient on the original carrier). *Retain Proposition 60's schedule. For $W > 0$ put*

$$u = \frac{4W}{25 \log 2}, \quad \kappa_8 = \frac{\log(8/3)}{\log 2}, \quad v = u - \kappa_8.$$

Suppose $h \geq 2$ and $0 < v < h - 1$. Let $S_{\text{out}} = \{M \in S : |\log M - L - hd| > 4W/25\}$. Then, uniformly in x ,

$$c_q(S_{\text{out}}, x) \leq \left(2 + \frac{u}{2h}\right) e^{-hI(u/h)} + 2e^{-(h-1)I(-v/(h-1))} + \epsilon_B =: \mathcal{T}_B(W). \quad (89)$$

For fixed $C > 0$ and $W = C\sqrt{N \log N}$, all the guards hold eventually, and the displayed majorant satisfies

$$\mathcal{T}_B(W) = (4 + o_C(1)) N^{-A_C}, \quad A_C = \frac{64C^2}{(\log 2)^3}. \quad (90)$$

This is an asymptotic for the explicit majorant, not an equality for the actual exterior coefficient.

Proof. The cleared first-passage inequalities in the preceding proof give the exact corridor

$$(s_- - 2(h-1)) \log 2 - \log(8/3) < \log M - L - hd \leq (s - 2h) \log 2.$$

An exterior endpoint thus supplies either the strict upper event $s - 2h > u$ or the strict predecessor event $s_- - 2(h-1) < -v$. Each retained word has such an endpoint witness, so the word satisfies the event even after its reciprocal fibre is enlarged. Apply (87) and (88). The exact buffer κ_8 has not been relaxed to the source's 3.

For the limit, $h = L/100000 + O(1)$ and $N = L/(10 \log 2) + O(1)$, with C fixed before B grows. In particular $u/h \rightarrow 0$, $v/(h-1) \rightarrow 0$, and $v > 0$ eventually. The proved derivative identity $I''(t) = 1/((1+t)(2+t))$ gives $I(t) = t^2/4 + O(t^3)$ near zero. Therefore

$$hI(u/h) = \frac{u^2}{4h} + O(u^3/h^2) = A_C \log N + o_C(1),$$

since $u^3/h^2 = O_C((\log L)^{3/2}/\sqrt{L}) \rightarrow 0$ and the floor errors in N/h contribute $O_C(\log L/L)$. Replacing u by $u - \kappa_8$ and h by $h - 1$ changes the exponent by $O_C(\sqrt{\log L/L} + \log L/L) = o_C(1)$. Both prefactors tend to 2. Also $\epsilon_B = o_C(N^{-A_C})$. This proves the stated asymptotic. The room condition follows from $(h-1) \log 3 + \log 2 = L \log 3/100000 + O(1) < L$. $\square$

Proposition 63 (kernel assembly without changing its mass). *For the original schedule of Proposition 59, fix $C \geq 1/2$, put $q = \lceil hL^{-2/5} \rceil$, and retain the original offset probability p_q and $K_q(M) = 3^q p_q(M \bmod 3^q)/M$. For the original endpoint set $S = \mathcal{E}(E)$, write $Z_q(S) = \sum_S K_q(M)$ and $a_* = 2/d$. Then*

$$0 \leq Z_q(S) \leq 2 + \epsilon_B, \quad 0 \leq Z_q(S_{\text{out}}) \leq \mathcal{T}_B(W).$$

If e_σ is any valid uniform bound for the interior error $|P_\sigma(M) - a_*|$, where $P_H(M) = R_H(M)/H$ and $P_F(M) = (z/D)R_F(M)$, the exact finite estimate is

$$\left| \sum_S K_q(M) P_\sigma(M) - a_* Z_q(S) \right| \leq e_\sigma Z_q(S_{\text{in}}) + \max\{a_*, L_\sigma - a_*\} Z_q(S_{\text{out}}). \quad (91)$$

Here L_H, L_F are the proved original-carrier envelopes of Proposition 52. For the two source branches, all bands, and all landing targets E , uniformly as $B \rightarrow \infty$,

$$\sum_S K_q(M) P_\sigma(M) = \frac{2}{d} Z_q(S) + O_C\left(N^{-1/17.232} (\log N)^{1/2-1/17.232}\right), \quad \sigma = H, F. \quad (92)$$

Proof. Grouping endpoints by residue gives the exact identity

$$Z_q(S) = \sum_{x \in \mathbb{Z}/3^q \mathbb{Z}} p_q(x) c_q(S, x).$$

The summation weights are nonnegative with sum one. Thus the residue-uniform coefficient bounds pass to Z_q without a factor 3^q and without declaring Z_q to have mass one. The reference level obeys $q = O(L^{3/5})$, hence $h + q \leq N$ eventually. The preceding propositions apply.

The closed interior and strict exterior partition S exactly. On the former, the triangle inequality gives $e_\sigma Z_q(S_{\text{in}})$. On the latter, the proved positive interval bound gives the second term in (91). This proves that estimate before any upper bound on the two masses is substituted.

The actual positive time carrier is $V = \{\nu > 0 : \nu + h \in J\}$, of cardinality at most $6W + \beta/d + 1 = O_C(W)$. Uniformly over the two branches and bands, its minimum is at least $N/43$ for large B : use $\nu \geq t_0 - h - 3W$ and the same strict lower constant calculation in Proposition 59, with $3W/L = o_C(1)$. This assertion is independent of M . Hence $G_V \leq \#V/\sqrt{\min V} = O_C(\sqrt{\log N})$. The retained denominator estimates give $H \geq 1/3$, $z/D \leq 2$, and $e^\beta < 4$, so $L_H, L_F = O_C(\sqrt{\log N})$. For $C \geq 1/2$, $A_C \geq 16/(\log 2)^3 > 48$; the last strict bound follows already from $\log 2 < 0.6932$. The entire exterior charge is therefore $O_C(N^{-A_C} \sqrt{\log N})$.

On the interior, Proposition 59 and the exact signed denominator identities give

$$e_\sigma = O_C\left(N^{-1/17.232}(\log N)^{1/2-1/17.232}\right).$$

The denominator contribution $O(1/z)$ is smaller uniformly. Since $Z_q(S_{\text{in}}) \leq 2 + \epsilon_B$, combining these estimates proves (92). $\square$

The common center here is the same landing measure $(2/d)\lambda_*\mu_q$ already constructed, not a new probability law. This completes the reference-profile input using Rhin's cited Diophantine estimate. The source's terminal-coordinate cap is unnecessary for the reconstructed exterior estimate, and its coarse total coefficient 5 can be retained more accurately as $2 + \epsilon_B$. Section 25 now combines this estimate with actual first passage and the whole-source mixtures. The original clock and threshold-iteration consumers remain to be reconstructed. Neither a full-package formal replay nor a new final Mazur natural-density exponent is asserted here.

25 From reference profiles to genuine passage laws

The reference profile is useful because it approximates an actual question: where does a Collatz orbit first enter the integers at most B ? This section completes that passage from the profile to the orbit. Both ways of sampling the starting integer, harmonic and uniform, are kept. The finite boundary charges and the possible upper endpoint of the source interval are kept too.

The source comparison is Mazur [5], `ND/Band/A5ExactAffineCorrection.lean`, `A5TerminalBoundaryShell.lean`, `A5FixedCellTerminalDescent.lean`, `A5SourceMixture.lean`, and `ND/Discrepancy/A5ReferenceGenuineBandRate.lean`, `A5ReferenceGenuineSourceRate.lean`, at the pinned downloaded commit. The last two files use $C = 40$, a factor-seven band estimate and a factor-eight whole-source estimate, with exponent restrictions $c < 1/20$ and $c < 1/(2\kappa)$. We reconstruct their four comparisons using the sharper estimates proved above. The analytic inputs remain Tao's cited mixing theorem and Rhin's cited Diophantine estimate; there is no new Fourier decay theorem here.

Throughout this section put

$$L = \log B, \quad N = \lfloor L/(10 \log 2) \rfloor, \quad h = \lfloor L/100000 \rfloor, \quad m = \lceil hL^{-2/5} \rceil, \\ W = C\sqrt{N \log N}, \quad d = \log(4/3).$$

Fix $C \geq 1/2$. Write $\alpha = 1001/1000$, and for either $y = B^\alpha$ or $y = B^{\alpha^2}$ set

$$b_y = (\alpha - 1) \log y, \quad k_y = \lfloor b_y \rfloor, \quad \beta = b_y/k_y, \quad z_j = ye^{j\beta}, \quad U_j = z_{j+1}, \quad 0 \leq j < k_y.$$

All statements below concern sufficiently large integer B ; the threshold may depend on C but not on the branch, band index or target set. Write D_j, H_j for the exact odd-band count and reciprocal sum. The original target set is $E'_B(E)$ from Section 19. In particular its endpoints have actual first-passage time h . Its lower bound $(4/3)^h B e^{-L^{7/10}}$ exceeds B eventually.

Proposition 64 (explicit boundary charges and their constants). *For a retained band suppress j and put*

$$\varepsilon = B^{-9/10}, \quad \rho = B^{-19749/20000}.$$

Every retained compatible terminal record satisfies $0 < F_a/M \leq \rho$ and $0 < -\log(1 - F_a/M) \leq \varepsilon$. Define the following positive majorants, using the original z, U, D, H :

$$\begin{aligned} b_{H,-} &= \frac{\varepsilon/2 + e^\varepsilon/z}{H}, & b_H &= \frac{\varepsilon + e^\varepsilon(1/z + 1/U)}{H}, \\ b_{F,-} &= \frac{z(1 - e^{-\varepsilon})/2 + 1}{D}, & b_F &= \frac{(z + U)(1 - e^{-\varepsilon})/2 + 2}{D}. \end{aligned}$$

For the terminal union masses U_σ and nominal sums I_σ ,

$$|U_H - I_H| \leq \rho U_H + b_H, \quad |U_F - I_F| \leq b_F, \quad 0 \leq I_\sigma \leq 1 + b_{\sigma,-}. \quad (93)$$

Uniformly in the branches and bands,

$$\frac{b_{H,-}}{\varepsilon} \rightarrow 1, \quad \frac{b_H}{\varepsilon} \rightarrow 2, \quad \frac{b_{F,-}}{\varepsilon} \rightarrow \frac{1}{e-1}, \quad \frac{b_F}{\varepsilon} \rightarrow \frac{e+1}{e-1}.$$

These are limits of the displayed bounds, not of the actual errors.

Proof. The original padded time interval gives

$$\frac{L}{1000d} - 3W - h \leq \nu = t - h \leq \frac{(\alpha^3 - 1)L}{d} + 3W.$$

The directed bounds $0.2876 < d < 0.2877$ established above imply

$$\frac{340}{100000} < \frac{1}{1000d} - \frac{1}{100000}, \quad \frac{\alpha^3 - 1}{d} < \frac{105}{10000}.$$

Since $W = o(L)$, eventually $F = (340/100000)L \leq \nu \leq (105/10000)L$, uniformly over all retained times. The elementary bound $F_a \leq 3^\nu$ and $\log 3 < 11/10$ give $F_a \leq B^{231/20000}$. The latter logarithmic inequality follows, for example, from the exponential series through degree five at $11/10$, whose sum exceeds 3. Since $M > B \geq B^{999/1000}$, division gives $F_a/M \leq \rho$. Here $\rho/\varepsilon = B^{-1749/20000} \rightarrow 0$. Eventually $\rho \leq 1/2$ and $2\rho \leq \varepsilon$, so

$$0 < -\log(1 - F_a/M) \leq \frac{F_a/M}{1 - F_a/M} \leq 2\rho \leq \varepsilon.$$

Strict positivity uses $\nu \geq 1$.

For either endpoint $x = z, U$, the odd shell $[xe^{-\varepsilon}, x)$ has fewer than $x(1 - e^{-\varepsilon})/2 + 1$ points. Proposition 49, applied with lower endpoint $xe^{-\varepsilon} \geq 1$, bounds its reciprocal sum by $\varepsilon/2 + e^\varepsilon/x$; the empty case satisfies the same inequality directly. Apply Proposition 30. Using the sum of both shell bounds overcovers their union; using just the lower shell gives the one-sided cap. Injectivity of the compatible record-to-source map is essential here and was proved in Proposition 29. It holds on the original time interval because $\beta/d + 6W \leq h$ eventually.

Finally $\beta \rightarrow 1$ uniformly, $z \geq B^\alpha$, and the exact discrepancies give $H = \beta/2 + O(1/z)$ and $D = z(e^\beta - 1)/2 + O(1)$. Thus $1/(z\varepsilon) \rightarrow 0$ uniformly. Also $(1 - e^{-\varepsilon})/\varepsilon \rightarrow 1$ and $e^\varepsilon \rightarrow 1$. Substitution proves all four limits. $\square$

Proposition 65 (assembled cell tests with the endpoint mass retained). Put $\epsilon_B = 3B^{-4997/600000}$ as in Proposition 60. At any retained (ν, ℓ) , let

$$S_{\nu, \ell} = \{M \in E'_B(E) : z < 2^\ell M/3^\nu < U\}.$$

On the fine residue group $G_\nu = \mathbb{Z}/3^\nu \mathbb{Z}$ define

$$c_H(x) = \sum_{\substack{M \in S_{\nu, \ell} \\ M \equiv x \pmod{3^\nu}}} \frac{3^\nu}{MH},$$

$$c_F(x) = \sum_{\substack{M \in S_{\nu, \ell} \\ M \equiv x \pmod{3^\nu}}} \frac{3^\nu}{M} \frac{2^\ell M/3^\nu}{D}.$$

Then

$$0 \leq c_\sigma \leq A_\sigma, \quad A_H = \frac{2 + \epsilon_B}{H}, \quad A_F = (2 + \epsilon_B) \frac{U}{D}. \quad (94)$$

The exact projection $\pi : G_\nu \rightarrow G_m$, $m \leq \nu$, and its fibre average $Ac(x) = 3^{m-\nu} \sum_{\pi v=x} c(v)$ give respectively

$$Ac_H(x) = \frac{3^m}{H} \sum_{\substack{M \in S_{\nu, \ell} \\ M \equiv x \pmod{3^m}}} \frac{1}{M}, \quad Ac_F(x) = \frac{3^m}{D} \sum_{\substack{M \in S_{\nu, \ell} \\ M \equiv x \pmod{3^m}}} \frac{2^\ell M/3^\nu}{M}.$$

All endpoint collisions are retained. Uniformly, $A_H \rightarrow 4$ and $A_F \rightarrow 4e/(e-1)$.

Let K_C be the single fixed-total mixing constant furnished by Corollary 5 with exponent 11 on this schedule. Keep E_B, γ_B from (40), and the original finite time set J . Then the nominal sums and full physical reference profiles $\mathcal{P}_\sigma$ obey

$$|I_\sigma - \mathcal{P}_\sigma| \leq \widehat{\mathcal{R}}_\sigma := \frac{\gamma_B(1 + b_{\sigma,-}) + \#J A_\sigma \{K_C/(2m^{11}) + 2/m^3\}}{1 - \gamma_B}. \quad (95)$$

In particular $\widehat{\mathcal{R}}_\sigma = O_C(L^{-1/5} \log L)$, and its ratio to $L^{-1/5} \log L$ tends to the positive terminal constant A_C of (31).

Proof. The retained times have $h + \nu \leq N$ eventually, by the bounds in the preceding proof. The room inequality $23^{h-1} \leq B$ also holds. Every subset $S_{\nu, \ell}$ is therefore covered by Proposition 60 at depth ν . Division by H proves the harmonic cap. On strict support $2^\ell M/3^\nu < U$; keeping this factor inside the original sum and then bounding it by U proves the flat cap. Summing first over residue fibres proves the two projection identities: every labelled M appears once, and $3^{m-\nu} 3^\nu = 3^m$. This is an averaging map on tests, not replacement of the endpoint measure by a probability measure. The denominator limits in the preceding proof prove the limits of A_σ .

To obtain the terminal estimate, condition a positive geometric word of length ν on its total ℓ . Its total probability is $q_{\nu, \ell} = \binom{\ell-1}{\nu-1} 2^{-\ell}$. The fine conditional offset law paired with c_σ gives exactly the nominal cell mass divided by $q_{\nu, \ell}$, by the compatibility calculation in Section 17. Its projected unconditioned pairing gives the corresponding coarse reference cell. Apply Corollary 37 with test cap A_σ , $\Omega \leq K_C/m^{11}$ and the uniform likelihood coefficient γ_B . Multiplication by $q_{\nu, \ell}$ gives

$$|Y_{\nu, \ell} - Z_{\nu, \ell}| \leq q_{\nu, \ell} A_\sigma \{K_C/(2m^{11}) + 2/m^3\} + \gamma_B Z_{\nu, \ell}.$$

The hypotheses are supplied by the original scales: $F \leq \nu \leq N$, $m \asymp L^{3/5} \geq \sqrt{\nu}$ eventually, $|\ell - 2\nu| < W = O_C(\sqrt{\nu \log \nu})$, $8m = o(F)$, $W = o(F)$ and $mW/(F\sqrt{m \log m}) \rightarrow 0$. Thus the

mixing constant is chosen before B , uniformly in all cells; the two tail estimates used in descent were proved in Corollary 41.

Now sum over totals using $\sum_{\ell} q_{\nu,\ell} \leq 1$, then over times. Lemma 32 and $I_{\sigma} \leq 1 + b_{\sigma,-}$ give (95). The exact two-row reindexing and time translation $t = \nu + h$ identify the reference sum with $\mathcal{P}_{\sigma}$, without changing either band weight. Since $\#J = O_C(W)$ and $A_{\sigma} = O(1)$, the nonproportional numerator is $O_C(L^{-13/10} \sqrt{\log L}) = o(L^{-1/5} \log L)$. The limit of $\gamma_B/(L^{-1/5} \log L)$ was proved in Corollary 38; both remaining factors tend to one. This proves the limit. Eventually $A_H \leq 20$, $A_F \leq 80$, $b_{\sigma,-} \leq 11\varepsilon$ and $\#J \leq 7W$, so this bound also refines $\mathcal{R}_{\sigma}^{\dagger}$. $\square$

Theorem 66 (the unchanged genuine passage measures). *For each band let $p_{\sigma,j}$ be its actual harmonic or uniform probability law, and define the finite-passage subprobability*

$$G_{\sigma,j}(E) = p_{\sigma,j}\{n : \tau_B(n) < \infty, T^{\tau_B(n)}n \in E\}.$$

No assertion that every orbit reaches B is implicit in this definition. Keep the original, unnormalized common measure

$$\mathcal{Z}_B(E) = \frac{2}{d} \sum_{M \in E'_B(E)} \frac{3^m p_m(M \bmod 3^m)}{M}.$$

Let $\chi_{\sigma,j}$ denote the corresponding exact full-prefix majorant in (55) or (56), and let $Q_{\sigma,j}(E)$ denote the right side of (91). The explicit finite conclusion is

$$\begin{aligned} |G_{H,j}(E) - \mathcal{Z}_B(E)| &\leq \chi_{H,j} + \rho U_H + b_H + \widehat{\mathcal{R}}_H + Q_{H,j}(E), \\ |G_{F,j}(E) - \mathcal{Z}_B(E)| &\leq \chi_{F,j} + b_F + \widehat{\mathcal{R}}_F + Q_{F,j}(E). \end{aligned} \quad (96)$$

In particular, with $b_ = 1/17.232$,*

$$\sup_{\text{branches}, j, E, \sigma} |G_{\sigma,j}(E) - \mathcal{Z}_B(E)| = O_C\left(L^{-b_*}(\log L)^{1/2-b_*}\right). \quad (97)$$

This holds for every fixed $C \geq 1/2$, including $C = 1/2$.

Proof. Insert, in order, the actual terminal union mass U_{σ} , nominal sum I_{σ} , and physical reference profile $\mathcal{P}_{\sigma}$. The resulting four differences are bounded by Corollary 43 with Corollary 47, the preceding two propositions, and Proposition 63. This proves the finite inequalities and explains each term; it is not a substitution of the reference event for actual first passage.

The estimates already proved are uniform in E , including targets depending on B . Coverage contributes $O_C(N^{-C^2/4})$; the boundary and affine terms are $O(B^{-9/10})$; the terminal term is $O_C(L^{-1/5} \log L)$. The full reference error is $O_C(N^{-b_*}(\log N)^{1/2-b_*})$. Since $C^2/4 \geq 1/16 > b_*$ and $N \sim L/(10 \log 2)$, the first three terms are of smaller order than the last. For example their quotient for the terminal term is $O_C(L^{-(1/5-b_*)}(\log L)^{1/2+b_*}) \rightarrow 0$. This proves (97) and the stated order of quantifiers: one threshold and constant for all branches, bands and landing targets, once C is fixed. $\square$

Corollary 67 (whole-source laws and exact completion at nonpassage). *On each original closed source block*

$$\mathcal{I}_y = [y, y^{\alpha}] \cap (2\mathbb{N} + 1), \quad y = B^{\alpha} \text{ or } B^{\alpha^2},$$

let $G_{\sigma,y}$ be its genuine finite-passage subprobability under the harmonic or uniform starting law. Then

$$\sup_{\sigma, y, E} |G_{\sigma,y}(E) - \mathcal{Z}_B(E)| = O\left(L^{-b_*}(\log L)^{1/2-b_*}\right).$$

For every $0 < c < b_*$, the source's factor-seven and factor-eight conclusions therefore hold eventually with L^{-c} in place of its rate, without the additional restriction $c < 1/20$. The implicit constants above may be fixed by taking $C = 1/2$.

Define the actual total landing map

$$\Lambda_B : 2\mathbb{N} + 1 \longrightarrow \mathcal{Y}_B := \{r > 0 : r \text{ odd}, r \leq B\} \sqcup \{\partial\}, \quad \Lambda_B(n) = \begin{cases} T^{\tau_B(n)}n, & \tau_B(n) < \infty, \\ \partial, & \tau_B(n) = \infty. \end{cases}$$

Here ∂ is one extra point, not a numerical landing value. For any two of the four whole-source starting laws, their pushforwards $\widehat{G}_1, \widehat{G}_2$ satisfy

$$\|\widehat{G}_1 - \widehat{G}_2\|_{\text{TV}} := \sup_{A \subseteq \mathcal{Y}_B} |\widehat{G}_1(A) - \widehat{G}_2(A)| = \sup_{E \subseteq \mathcal{Y}_B \setminus \{\partial\}} |G_1(E) - G_2(E)| = O\left(L^{-b_*}(\log L)^{1/2-b_*}\right). \quad (98)$$

In particular it is at most $16L^{-c}$ eventually for every $0 < c < b_*$. For every real function f on $\mathcal{Y}_B$, the difference of its expectations is bounded by $(\max f - \min f)\|\widehat{G}_1 - \widehat{G}_2\|_{\text{TV}}$.

The original common mass itself satisfies

$$\mathcal{Z}_B(\mathcal{Y}_B \setminus \{\partial\}) = 1 + O\left(L^{-b_*}(\log L)^{1/2-b_*}\right),$$

or, equivalently, the unscaled original kernel mass on $E'_B([1, B])$ is $d/2 + O(L^{-b_*}(\log L)^{1/2-b_*})$. No division by that mass is used.

Proof. The half-open bands partition $[y, y^\alpha)$, with exactly one additional point if y^α is an odd integer. Write H_y, D_y for the reciprocal sum and count of the closed block. The respective band weights are H_j/H_y and D_j/D_y ; the possible top weights are $p_H = (1/y^\alpha)/H_y$ and $p_F = 1/D_y$ when that point exists, and zero otherwise. In either case $\sum_j w_{\sigma,j} + p_\sigma = 1$ exactly. For any event its whole-source mass is $\sum_j w_{\sigma,j}G_{\sigma,j}(E) + t_\sigma(E)$, where $0 \leq t_\sigma(E) \leq p_\sigma$.

For any nonnegative center Z and valid band errors r_j this gives the finite identity and bound

$$G_{\sigma,y}(E) - Z = \sum_j w_{\sigma,j}(G_{\sigma,j}(E) - Z) + (t_\sigma(E) - p_\sigma Z),$$

$$|G_{\sigma,y}(E) - Z| \leq \sum_j w_{\sigma,j}r_j + p_\sigma \max\{Z, |1 - Z|\}.$$

The last factor is exact for a variable $t_\sigma/p_\sigma \in [0, 1]$; when $p_\sigma = 0$ its term is zero. The first band gives $H_y \geq 1/3$ and $D_y > y/2$, hence $p_H \leq 3/B$ and $p_F < 2/B$. The endpoint coefficient bound gives $0 \leq \mathcal{Z}_B(E) \leq (2/d)(2 + \epsilon_B)$. Thus the top costs $O(B^{-1})$ uniformly, not the number of bands. This proves the whole-source rate. Because $L^{-(b_*-c)}(\log L)^{1/2-b_*} \rightarrow 0$, each band and source error is $o(L^{-c})$; the particular source constants 7 and 8 follow.

For the completion, restriction to the finite landing points gives G_i exactly, and the mass at ∂ is $1 - G_i([1, B])$. If a set A contains ∂ , its complementary set does not, and equality of the total probability masses gives $|\widehat{G}_1(A) - \widehat{G}_2(A)| = |G_1(\mathcal{Y}_B \setminus A) - G_2(\mathcal{Y}_B \setminus A)|$. This proves the equality of the two suprema. The triangle inequality through the *unchanged* common measure proves the rate and constant 16; there is no separate factor for the nonpassage atom. For the expectation bound, subtract $\min f$ and write $f - \min f = \int_0^{\max f - \min f} \mathbf{1}_{\{f - \min f > s\}} ds$. The domain is finite, so summation and integration interchange directly.

Finally the good-prefix event ensures passage by time N , as proved in Proposition 42. Band failure mass is at most $\chi_{\sigma,j}$; whole-source failure mass is at most its mixture plus p_σ . At $C = 1/2$

this is $O(N^{-1/16}) + O(B^{-1})$. Therefore $G_{\sigma,y}([1, B]) = 1 + O(N^{-1/16})$. Its comparison with $\mathcal{Z}_B([1, B])$ proves the asserted mass limit. $\square$

These results strengthen the genuine passage-law stage of Mazur’s argument and connect it to the previously reconstructed Allikvere passage comparison. Section 26 carries them through real thresholds and the original scheduled iteration to a timed natural-density theorem. No Lean replay is inferred from the written proofs.

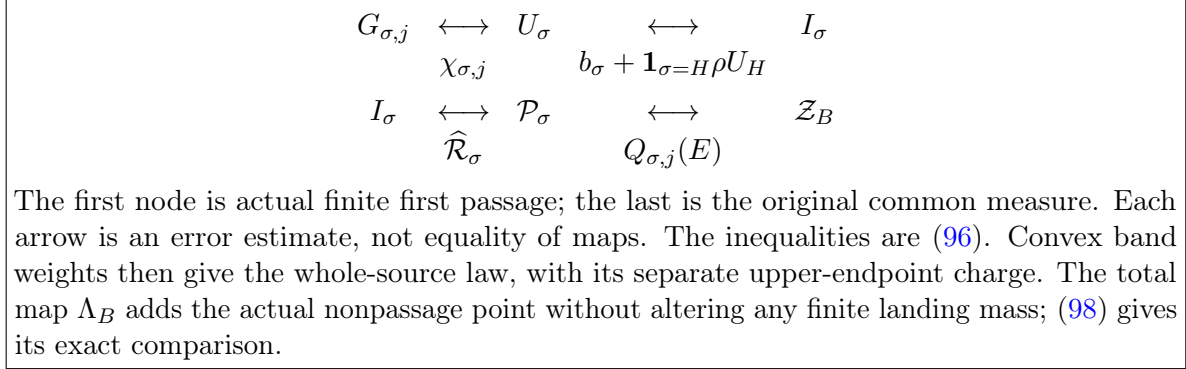


Figure 4: How the four estimates reach actual Collatz passage laws. This retains Mazur’s original comparisons with the quantitative refinements proved here from the cited Tao and Rhin inputs. The endpoints of the diagram and all intermediate quantities are defined in this section; the arrows do not denote deterministic orbit maps.

26 Real thresholds and the timed density conclusion

The preceding rate concerns genuine landing probabilities. To use it in the scale iteration, two changes must be made explicitly: the source formalization sends nonpassage to the number 1, and its iteration uses real thresholds and real source windows. Neither change requires a loss in the logarithmic exponent. Keeping the exact geometric sum of the original scheduled times also improves the source’s clock constants.

The source route is Mazur [5], ND/Probability/PassTotalizer.lean, ND/Discrepancy/A5ReferenceTotalizedSourceRate.lean, A5ReferenceRealFloorPass.lean, and A5ReferenceRealLogBlock.lean. Its scale definitions are in Tao/Section3/AmbientPartialTrace.lean; its clock comparison is in Tao/Section3/TimeBudget.lean and ND/LogTime/Clock.lean. All refer here to the pinned source commit specified in the bibliography. The underlying passage method is Tao’s [8]; the quantitative input remains the written Tao–Rhin comparison above. The following is a written reconstruction and refinement, not a claim to have replayed the whole Lean package.

Put

$$\alpha = \frac{1001}{1000}, \quad b = \frac{1}{17.232}, \quad a = \frac{1}{2} - b, \quad R(L) = L^{-b}(1 + \log L)^a \quad (L \geq \log 2).$$

This positive rate is asymptotic to $L^{-b}(\log L)^a$. It only extends the displayed scalar rate to small positive logarithms; it does not alter a measure, source window, or passage map. For large L , R is decreasing, since its logarithmic derivative with respect to L is $-b + a/(1 + \log L) < 0$.

Proposition 68 (the exact effect of sending nonpassage to one). *For integer $B \geq 1$, let $\mathcal{Y}_B$ and Λ_B be as in Corollary 67. The map*

$$r_B : \mathcal{Y}_B \longrightarrow \{0, 1, \dots, B\}, \quad r_B(s) = s \ (s \neq \partial), \quad r_B(\partial) = 1$$

has image precisely the positive odd points of its codomain. Every nonempty fibre is a singleton except $r_B^{-1}(1) = \{1, \partial\}$. The source's totalized map is exactly $F_B = r_B \circ \Lambda_B$.

For probability measures μ, ν on $\mathcal{Y}_B$, write $\Delta = \mu - \nu$, $u = \Delta(\{1\})$, $v = \Delta(\{\partial\})$. With $\|\Delta\|_1 = \sum_s |\Delta(\{s\})|$,

$$\|\Delta\|_1 - \|(r_B)_* \Delta\|_1 = |u| + |v| - |u + v| = \begin{cases} 2 \min(|u|, |v|), & uv < 0, \\ 0, & uv \geq 0. \end{cases} \quad (99)$$

Moreover $\|\Delta\|_1 = 2 \sup_A |\Delta(A)|$. Consequently each natural-source pair in Corollary 67 has totalized full ℓ^1 distance $O(R(\log B))$, and at most $32(\log B)^{-c}$ eventually for every $0 < c < b$.

Proof. First passage, if it exists, has one uniquely determined time and landing point. This proves the map formula and fibres, including the zero masses at even points and at zero. Pushforward adds only the two coordinates u, v ; all other coordinates are unchanged, proving the first equality in (99). Splitting according to the signs of u, v proves the second.

Since $\sum_s \Delta(\{s\}) = 0$, the positive and negative parts each have mass $\|\Delta\|_1/2$. The event consisting of the positive coordinates attains that value, and no event exceeds it in absolute value. This proves the convention comparison and the rate. The linear kernel of $(r_B)_*$ consists exactly of $t(\delta_1 - \delta_\partial)$, $t \in \mathbb{R}$. Thus no inverse on arbitrary laws exists: δ_1 and δ_∂ have the same image. If the nonpassage mass is provided separately, reconstruction is exact: retain every other coordinate, set the ∂ -coordinate to that mass, and subtract it from the image mass at 1. An admissible mass lies between zero and that image mass. This describes precisely what was lost. $\square$

Proposition 69 (exact overlap for unchanged finite source laws). *Let S, T be nonempty finite subsets of an ambient set and let $w > 0$ on $S \cup T$. Write $W_A = \sum_{n \in A} w(n)$ and $p_A(n) = w(n)\mathbf{1}_A(n)/W_A$. Then*

$$\sup_E |p_S(E) - p_T(E)| = 1 - \frac{W_{S \cap T}}{\max(W_S, W_T)} = \frac{\max(W_{S \setminus T}, W_{T \setminus S})}{\max(W_S, W_T)}. \quad (100)$$

Their full ℓ^1 distance is twice this number. For every common map $f : S \cup T \rightarrow Z$, the output event distance is at most (100); its exact value is

$$\frac{1}{2} \sum_{z \in f(S \cup T)} \left| \sum_{n: f(n)=z} (p_S(n) - p_T(n)) \right|.$$

Proof. For any two probability vectors p, q , $\frac{1}{2} \sum |p - q| = 1 - \sum \min(p, q)$, by $|p - q| = p + q - 2 \min(p, q)$. On the intersection here, $\min(p_S(n), p_T(n)) = w(n)/\max(W_S, W_T)$; elsewhere it is zero. This proves the first equality. Writing $W_S = W_{S \cap T} + W_{S \setminus T}$ and similarly for T proves the second. Summing differences on each fibre gives the displayed output formula. The triangle inequality on each fibre proves contraction. This applies to the existing probabilities without replacing either denominator. For an injective f equality holds; cancellation of opposite signs within fibres describes every strict loss. $\square$

Proposition 70 (real/floor transport with both endpoints retained). *For $x \geq 1$, set $B = \lfloor x \rfloor$. For either $\beta = \alpha$ or $\beta = \alpha^2$, put $q = \alpha\beta$ and*

$$S = [B^\beta, B^q] \cap (2\mathbb{N} + 1), \quad T_x = [x^\beta, x^q] \cap (2\mathbb{N} + 1).$$

Assume $S, T_x \neq \emptyset$. Define $\epsilon_F(x, \beta)$ and $\epsilon_H(x, \beta)$ by the exact right side of (100), using respectively $w(n) = 1$ and $w(n) = 1/n$. For all sufficiently large B , uniformly in $B \leq x < B + 1$ and the two branches,

$$\epsilon_F(x, \beta) \leq \frac{(x^q - B^q)/2 + 1}{(B^q - B^\beta)/2 - 1}, \quad (101)$$

$$\epsilon_H(x, \beta) \leq \frac{(q/2) \log(x/B) + B^{-\beta}}{((q - \beta)/2) \log B - B^{-\beta}}. \quad (102)$$

The displayed denominators are positive in this range. In particular

$$\limsup_{B \rightarrow \infty} \sup_{B \leq x < B+1} B \epsilon_F(x, \beta) \leq q, \quad \limsup_{B \rightarrow \infty} \sup_{B \leq x < B+1} B \log B \epsilon_H(x, \beta) \leq \frac{q}{q - \beta} = 1001.$$

The exact errors vanish at $x = B$, even though these upper envelopes need not vanish there.

Proof. On an interval of length $v - u$, the number of odd integers differs from $(v - u)/2$ by at most 1, for any choice of endpoint inclusion. For completeness, if $A(t)$ counts those odd points between u and t , then $|A(t) - (t - u)/2| \leq 1$. Summation by parts gives

$$\sum_{n \in [u, v] \text{ odd}} \frac{1}{n} = \frac{A(v)}{v} + \int_u^v \frac{A(t)}{t^2} dt.$$

The same formula uses the appropriate A for a deleted endpoint. Replacing $A(t)$ by $(t - u)/2$ gives $\frac{1}{2} \log(v/u)$; the total absolute error is at most $1/v + \int_u^v t^{-2} dt = 1/u$. This also handles an atom at u .

Eventually $x^\beta \leq B^q$, uniformly in $x < B + 1$, because $(B + 1)^\beta / B^q \rightarrow 0$. The two exclusive sets are then exactly $[B^\beta, x^\beta] \cap (2\mathbb{N} + 1)$ and $(B^q, x^q] \cap (2\mathbb{N} + 1)$. The larger of their cardinalities is at most $(x^q - B^q)/2 + 1$: indeed $t \mapsto t^q - t^\beta$ is increasing for $t \geq 1$, since $q > \beta > 1$. The full mass of S is at least $(B^q - B^\beta)/2 - 1$. Equation (100) proves (101).

The exclusive reciprocal masses are at most $\frac{\beta}{2} \log(x/B) + B^{-\beta}$ and $\frac{q}{2} \log(x/B) + B^{-q}$. Their maximum is bounded by the latter logarithmic term plus $B^{-\beta}$. The reciprocal mass of S is at least the denominator of (102), proving that bound.

Finally $x^q - B^q \leq q(B + 1)^{q-1}(x - B)$ and $\log(x/B) \leq (x - B)/B$. Divide numerator and denominator by the indicated powers. The additive cardinal error contributes $O(B^{1-q})$ after multiplication by B , and the harmonic error contributes $O(B^{1-\beta})$ after multiplication by $B \log B$. Both vanish. These estimates are uniform in $x - B \in [0, 1)$. $\square$

Theorem 71 (real-source passage and its actual scheduled failure). *Whenever the unchanged closed odd block $[y, y^\alpha]$ is nonempty, let H_y, F_y be its harmonic and uniform probability laws. For real x let*

$$n_x = \left\lfloor \frac{\log \lfloor x \rfloor}{10 \log 2} \right\rfloor, \quad F_x(n) = \begin{cases} T^{\tau_x(n)} n, & \tau_x(n) < \infty, \\ 1, & \tau_x(n) = \infty. \end{cases}$$

For either source branch $y = x^\alpha, x^{\alpha^2}$ and either law, the probability of $\tau_x > n_x$, including nonpassage, is $O((\log x)^{-1/16})$. Every pair of the four totalized landing laws has event distance $O(R(\log x))$ and full ℓ^1 distance $O(R(\log x))$, uniformly as $x \rightarrow \infty$. No hypothesis about the existence of passage on every orbit is made.

Proof. Since every iterate is an integer,

$$T^j n \leq x \iff T^j n \leq \lfloor x \rfloor \quad \text{for every } j.$$

Thus first-hit times, nonpassage events, and landing values agree exactly at x and $B = \lfloor x \rfloor$. The identity map on $\{0, \dots, B\}$ is the carrier equivalence; there is no rounding error in the dynamics.

Let $\mu_{\sigma, \beta}^x$ denote a real source law and $\mu_{\sigma, \beta}^B$ its floor source law. Propositions 69 and 70 give event distance $\epsilon_\sigma(x, \beta)$ before the common map F_B , hence at most that distance after it. If $D_B(i, j)$ is the exact event distance between the natural-source outputs, the finite pairwise bound is

$$D_x(i, j) \leq \epsilon_i + D_B(i, j) + \epsilon_j. \quad (103)$$

Full ℓ^1 is twice this entire expression, not twice just its middle term. In particular the natural bound $32(\log B)^{-c}$ from the totalizer proposition remains the middle full- ℓ^1 term. The source's safe outer estimates were $96x^{-99/100}$ for uniform events and $96000/(B \log B)$ for harmonic events. Equations (101)–(102) replace those outer terms using the original finite sets.

The good-prefix argument in Proposition 42 proves passage by $n_B = \lfloor \log B / (10 \log 2) \rfloor$, not just eventual passage. Corollary 47, with $C = 1/2$, bounds its complement by $O((\log B)^{-1/16})$ on every band. The exact closed-block mixture in Corollary 67 adds at most the top endpoint mass $O(B^{-1})$. Applying the source overlap bound to the single event $\{\tau_B > n_B\}$ costs only ϵ_σ , with no full- ℓ^1 factor. This proves the scheduled failure estimate for real windows. Finally $\log B / \log x \rightarrow 1$; $\epsilon_F = O(B^{-1})$, $\epsilon_H = O((B \log B)^{-1})$, and $1/16 > b$. Thus both the source perturbation and scheduled failure are of smaller order than $R(\log x)$. Equation (103) proves the asserted rate. $\square$

Proposition 72 (the actual time-carrying scale iteration). *There are constants C_0, B_0 , independent of B and y , such that for every integer $B \geq B_0$ and every $y \geq 2$ with a nonempty odd source block, the harmonic law H_y satisfies*

$$H_y \left\{ n : \tau_B(n) > \frac{\log y}{10\alpha(\alpha - 1) \log 2} \right\} \leq C_0 R(\log B). \quad (104)$$

In fact its integer time budget is the exact finite sum below.

For a seed $y_0 > 1$, define $y_j = y_0^{\alpha^j}$, $q_j = y_j^{1/\alpha}$, and $K_j = \sum_{i < j} n_{q_i}$. Let d_j be the event distance between $(F_{q_j})_ H_{y_j}$ and $(F_{q_j})_* H_{y_{j+1}}$, and put $s_j = H_{y_{j+1}} \{\tau_{q_j} > n_{q_j}\}$. Whenever $y_0^\alpha \leq B$,*

$$H_{y_J} \{\tau_B > K_J\} \leq \sum_{j < J} (d_j + s_j), \quad K_J \leq \frac{\log q_0 (\alpha^J - 1)}{10(\alpha - 1) \log 2}. \quad (105)$$

The finite sum of the critical comparison rates is bounded, uniformly in J , by

$$\sum_{j < J} R(\alpha^j L) \leq R(L) \left(\frac{1}{1-r} + \frac{a \log \alpha}{1 + \log L} \frac{r}{(1-r)^2} \right), \quad r = \alpha^{-b}, \quad L \geq 1. \quad (106)$$

For the infinite sum, its ratio to $R(L)$ tends to $1/(1-r)$ as $L \rightarrow \infty$.

Proof. For $K \in \mathbb{N}$, let $A_{B,K} = \{n \text{ odd} : \tau_B(n) > K\}$. For every $q \geq 1$ and positive odd n ,

$$\begin{aligned} \mathbf{1}_{A_{B,K}}(F_q(n)) &\leq \mathbf{1}_{A_{B,K}}(n), \\ \mathbf{1}_{A_{B,K+N}}(n) &\leq \mathbf{1}_{A_{B,K}}(F_q(n)) + \mathbf{1}_{\{\tau_q > N\}}(n) \quad (N \in \mathbb{N}). \end{aligned} \quad (107)$$

For the first inequality, suppose $\tau_B(n) \leq K$. At nonpassage $F_q(n) = 1$, which already lies below B . If first passage to q occurs before the hit at B , the residual time to B is at most K . If it occurs later, then the earlier hit at B had value above q ; hence $q < B$, and the landing at q is itself below B . This proves the first inequality without an assumed ordering between q and B . For the second, when $\tau_q \leq N$ and the landing hits B within K , concatenation supplies a hit within $N + K$. Otherwise its right side is at least one whenever necessary.

Integrating the second inequality under $H_{y_{j+1}}$, comparing its landing event with the one under H_{y_j} , and applying the first inequality gives

$$H_{y_{j+1}}(A_{B, K_j + n_{q_j}}) \leq H_{y_j}(A_{B, K_j}) + d_j + s_j.$$

These are the literal two source branches at threshold q_j : $q_j^\alpha = y_j$ and $q_j^{\alpha^2} = y_{j+1}$. All points in the seed block are at most B , so its bad probability at $K_0 = 0$ is zero. Induction proves the first part of (105). Since $n_{q_j} \leq \log q_j / (10 \log 2)$ and $\log q_j = \alpha^j \log q_0$, summing the finite geometric series proves its clock bound, with no extra or unused step.

For (106), write $h = \log \alpha$. Then exactly

$$\frac{R(\alpha^j L)}{R(L)} = r^j \left(1 + \frac{jh}{1 + \log L} \right)^a.$$

Here $0 < a < 1$. Concavity, or integration of $a(1+t)^{a-1} \leq a$, bounds the last factor by $1 + ajh/(1 + \log L)$. Summing $\sum r^j = (1-r)^{-1}$ and $\sum jr^j = r/(1-r)^2$ proves the upper bound. The lower bound for the infinite sum is $(1-r)^{-1}$, because the parenthesized factor is at least one; squeezing proves the limit. All terms are nonnegative, so no interchange of conditionally convergent sums occurs.

Now fix y with $y^\alpha > B$. Choose the least integer $J \geq 1$ such that $y_0 = y^{\alpha^{-J}}$ satisfies $y_0^\alpha \leq B$. Existence follows from $y^{\alpha^{-J}} \rightarrow 1$. Minimality gives $y_0^{\alpha^2} > B$, whence

$$B^{1/\alpha^2} < y_0 \leq B^{1/\alpha}, \quad B^{1/\alpha^3} < q_0 \leq B^{1/\alpha^2}.$$

Thus one sufficiently large B_0 , chosen before y, J , puts all q_j in the range of Theorem 71. That theorem gives $d_j + s_j \leq CR(\log q_j)$ with one C . The geometric bound is $O(R(\log q_0))$. Since $\log q_0 \in (\alpha^{-3} \log B, \alpha^{-2} \log B]$, direct substitution gives $R(\log q_0) \leq \alpha^{3b} R(\log B)$ for large B . The denominator $1-r$ is fixed and strictly positive. This proves the probability estimate, uniformly in the trace length. Finally $\alpha^J \log q_0 = (\log y)/\alpha$, proving (104). If $y^\alpha \leq B$, all starts already hit at time zero and the same statement holds. Both this case and an empty trace have zero cost. $\square$

Theorem 73 (timed natural density from the original Mazur schedules). *There is an absolute C such that for every integer $B \geq 2$ and real $X \geq 2$,*

$$\frac{1}{X} \# \left\{ 1 \leq n \leq X : n \text{ odd}, \tau_B(n) > \frac{100}{\log 2} \log n \right\} \leq CR(\log B), \quad (108)$$

$$\frac{1}{X} \# \left\{ 1 \leq n \leq X : \tau_B^{\text{raw}}(n) > \frac{301}{\log 2} \log n \right\} \leq 2CR(\log B). \quad (109)$$

Here the ordinary Collatz map is $n/2$ for even n and $3n+1$ for odd n ; τ_B^{raw} is its first hitting time of $[1, B]$. These are not accelerated raw steps.

In particular every function $f : \mathbb{N}_{>0} \rightarrow \mathbb{R}$ tending to infinity is exceeded downwards by an iterate within $(301/\log 2) \log n$ ordinary steps for a set of natural density one: the iterate is strictly

below $f(n)$. The corresponding odd-relative-density statement holds for functions tending to infinity on the odd integers with the Syracuse clock $(100/\log 2) \log n$. For every $0 < c < b$, the right sides of the fixed-target estimates are also $O_c((\log B)^{-c})$.

These constants refine the source's $501501/(5000 \log 2)$ and $1509503/(5000 \log 2)$ within its own scheduled construction. They are not claimed to be the best clocks in this workbench: the earlier Allikvere-based reconstruction has a much smaller leading Syracuse coefficient.

Proof. First take B large enough for the preceding results. When $X \leq B$ the counted sets are empty. For $X > B$, put $y = X^{1/\alpha}$ and $q = X^{1/\alpha^2}$. The top odd block is exactly $[y, X] \cap (2\mathbb{N} + 1)$, including both integer endpoints. If $y^\alpha \leq B$ there is nothing to prove; otherwise choose the harmonic trace ending at H_y as in the preceding proposition, with its integer budget K . Integrating (107) under F_y , comparing its landing law at q with that of H_y , then using the first inequality there, gives

$$F_y\{\tau_B > K + n_q\} \leq H_y\{\tau_B > K\} + D((F_q)_*F_y, (F_q)_*H_y) + F_y\{\tau_q > n_q\}.$$

Here D denotes event-supremum distance. All three terms are $O(R(\log B))$, uniformly in $X > B$: the first by the trace, and the last two by the real passage theorem and $\log q = \alpha^{-2} \log X$. For the comparison to $R(\log B)$, use eventual monotonicity at $\alpha^{-2} \log B$, followed by direct substitution.

Write $k = 10 \log 2$. The exact trace bound and the final schedule yield

$$K + n_q \leq \frac{\log y}{k\alpha(\alpha - 1)} + \frac{\log q}{k} = \frac{\log X}{k\alpha(\alpha - 1)}.$$

There is no use of $\log q \leq \log X$ in this equality. Every point in the top block satisfies $\log X \leq \alpha \log n$. It follows that $K + n_q \leq \log n / [k(\alpha - 1)] = (100/\log 2) \log n$. Thus the top-block bad fraction is $O(R(\log B))$. There are at most $y + 1$ positive integers below its lower boundary, so the contribution of all omitted odd points divided by X is at most $X^{-(1-1/\alpha)} + X^{-1}$. Uniformly for $X > B$ this is $O(R(\log B))$, since every negative power of B is $o(R(\log B))$. The number of odd points in the top block is at most X for $X \geq 2$, so changing its probability to count divided by X does not increase its bound. This proves (108) for large B . For the finitely many integers $2 \leq B < B_0$, the counted ratio is at most one and $\min_{2 \leq B < B_0} R(\log B) > 0$; increasing one C handles all of them. The same odd count bound also holds for $0 < X < 2$, since the only possible positive odd input, 1, is good at time zero.

For the raw clock, write uniquely $n = 2^v u$, u positive odd. Suppose $u = u_0, \dots, u_t$ is a Syracuse trajectory with $u_t \leq B$ and $t \leq (100/\log 2) \log u$. If $a_i = v_2(3u_i + 1)$, then

$$2^{\sum_{i < t} a_i} u_t = \prod_{i < t} (3 + 1/u_i) u \leq 4^t u.$$

Taking base-two logarithms and using $u_t \geq 1$ gives $\sum a_i \leq 2t + \log_2 u$. The ordinary trajectory reaches u_t in exactly $v + t + \sum a_i$ steps. Therefore its first hit is no later than

$$v + 3t + \log_2 u \leq \frac{300}{\log 2} \log u + \frac{\log n}{\log 2} \leq \frac{301}{\log 2} \log n.$$

This is an exact map from a Syracuse path to the ordinary path: each odd edge expands into one $3u + 1$ edge followed by its a_i halving edges, after the v initial halvings. There are no deleted labels or valuations.

Consequently every raw-bad n has odd part u in the odd-bad set. The unique factorization gives, with no residual cutoff,

$$\#\{\text{raw-bad } n \leq X\} \leq \sum_{v \geq 0} \#\{\text{odd-bad } u \leq X/2^v\} \leq CR(\log B) \sum_{v \geq 0} X/2^v = 2CX R(\log B).$$

The actual count sum is finite. The bound also covers its terms with $X/2^v < 2$, whose counts vanish. This proves the raw estimate.

For the strict growing-threshold conclusion, fix any integer $B \geq 2$. Outside a finite set, $f(n) > B$. Thus failure to have an iterate strictly below $f(n)$ within the stated time is contained in the fixed- B bad set plus that finite set. Take the upper density in X first, then let $B \rightarrow \infty$; $R(\log B) \rightarrow 0$. For odd relative density the denominator is $\#\{n \leq X : n \text{ odd}\} \sim X/2$, changing only a factor two. Finally $R(L) = o(L^{-c})$ for every $0 < c < b$, which proves the last rate assertion. No pointwise assertion that every orbit is bounded follows from these density estimates. $\square$

The probability recurrence, rather than a sequence of independent random orbits, is what permits the iteration. Its test event changes with the accumulated integer budget K_j ; uniformity over all landing events in Theorem 71 is therefore essential. The construction does not presume stationarity or independence at successive scales. The two identities $q_j^\alpha = y_j$, $q_j^{\alpha^2} = y_{j+1}$ identify every source law used, and the pointwise inequalities identify exactly how an actual path passes between the tests.

The measures and the two clocks. The exact source overlap controls real/floor transport. Actual landing with ∂ maps to the source's landing-with-1; only its $\{1, \partial\}$ fibre loses information. At each threshold q_j , compare the two harmonic landing laws, charge actual failure to land by n_{q_j} , and add precisely n_{q_j} to the clock. Their finite sum, plus the one uniform top-block passage, is at most $\log X/[10\alpha(\alpha-1)\log 2]$. The inequality $\log X \leq \alpha \log n$ gives $100 \log_2 n$ Syracuse steps. Expanding the valuation-labelled odd edges gives $301 \log_2 n$ ordinary steps. Proofs: Propositions 68–72 and Theorem 73.

27 Shaik's reverse loss and exact first-passage transport

Shaik's first-passage argument counts starting values whose later landing belongs to a specified bad set. This is the deterministic step that permits successive descents without assuming that the landing values are uniformly distributed. We reconstruct this step from [3], Lemmas 4.1–4.2, Proposition 4.3 and Lemmas 5.1–5.2 (pp. 14–17), and the pinned formal source. Two finite improvements follow: the odd-step loss has a smaller denominator bound, and intersection with the original half-open shell improves the fibre coefficient. Neither improvement changes the entropy exponent in the paper's headline theorem by itself.

Here T denotes Shaik's *shortcut* map: $T(x) = x/2$ for even x , and $T(x) = (3x+1)/2$ for odd x . It is not the odd-return map Syr used above. Write $I_M = [2^M, 2^{M+1}) \cap \mathbb{N}$. For $1 < Y < n$, a finite first passage consists of $x_j = T^j(n)$, $0 \leq j \leq h$, with $x_j > Y$ for $j < h$ and $y = x_h \leq Y$. We do not assign a finite passage to a non-hitting orbit. Let p_j be 1 when x_j is odd and 0 otherwise, and $s = \sum_{j < h} p_j$. Set

$$u_j = \frac{p_j}{2x_{j+1}}, \quad L = \sum_{j < h} u_j, \quad P = \prod_{j < h} (1 - u_j), \quad E_Y = YL, \quad G_Y = Y(1 - P).$$

Thus E_Y is Shaik's additive loss budget; G_Y measures the actual product defect. The two are different quantities.

Proposition 74 (The exact defect and the odd-step denominator). *For the first passage just defined,*

$$Y/2 < y \leq Y, \quad n = AP, \quad A = \frac{2^h y}{3^s}, \quad 0 \leq G_Y \leq E_Y. \quad (110)$$

If $o(Y)$ is the smallest positive odd integer strictly greater than Y , then

$$E_Y \leq \frac{Ys}{3o(Y)+1} \leq \frac{Y(h-1)}{3o(Y)+1} < \frac{h}{3}. \quad (111)$$

The last inequality holds also when $s = 0$. For the forward affine formula $y = 3^s n / 2^h + b$, the exact relation is

$$1 - P = \frac{b}{y}, \quad G_Y = \frac{Yb}{y}. \quad (112)$$

For two consecutive finite orbit segments, with local products P_1, P_2 , losses L_1, L_2 , and positive reference thresholds Y_1, Y_2 ,

$$\begin{aligned} L_{12} &= L_1 + L_2, & P_{12} &= P_1 P_2, \\ E_{Y_2,12} &= \frac{Y_2}{Y_1} E_{Y_1,1} + E_{Y_2,2}, \\ G_{Y_2,12} &= \frac{Y_2}{Y_1} G_{Y_1,1} + G_{Y_2,2} - \frac{G_{Y_1,1} G_{Y_2,2}}{Y_1}. \end{aligned} \quad (113)$$

Proof. An odd shortcut step strictly increases its positive input. The final crossing is therefore even, so $x_{h-1} = 2y > Y$, and $s \leq h-1$. Reversing an even step multiplies by 2. Reversing an odd step gives

$$x_j = \frac{2}{3} x_{j+1} \left(1 - \frac{1}{2x_{j+1}} \right).$$

Multiplication proves $n = AP$. All factors lie in $(0, 1]$, and induction on the factors proves $\prod(1 - u_j) \geq 1 - \sum u_j$. For an odd step before passage, x_j is an odd integer greater than Y , and

$$u_j = \frac{1}{3x_j + 1} \leq \frac{1}{3o(Y) + 1}.$$

Summing only the s odd contributions proves the first two bounds. Since $o(Y) > Y$, their common right side is less than $h/3$ for $h \geq 1$; the zero-odd case is immediate.

The forward formula follows by induction through the affine steps: with $s_i = \sum_{j < i} p_j$,

$$b = \sum_{i=0}^{h-1} \frac{p_i 3^{s-s_{i+1}}}{2^{h-i}}.$$

The identity $n = AP$ gives $P = 3^s n / (2^h y) = 1 - b/y$. Concatenation partitions the sum and product into the two consecutive segments. Substitution of $G_{Y_i,i} = Y_i(1 - P_i)$ then gives the last equality, including its negative cross term. $\square$

Equation (112) identifies an exact coordinate change, not an analogy between two errors. For fixed $Y > 0$, on the actual finite affine-path records, the map $(n, h, s, y, b) \mapsto (n, h, s, y, G_Y)$ has inverse $b = yG_Y/Y$. Both records retain the original path when path reconstruction is needed. Discarding the path and retaining only (h, y) is instead a genuine many-to-one map; its fibres are counted next.

Proposition 75 (A shell-clipped bound for a tagged fibre). *Fix $M \geq 1$, $1 < Y < 2^M$, $h \geq 1$, $y \in \mathbb{N}$, and $0 \leq \varepsilon \leq 1/3$. Among sources $n \in I_M$ with $\tau_Y(n) = h$, $T^h(n) = y$, and $1 - P(n) \leq \varepsilon$, all odd counts are equal. The number of such sources is at most*

$$1 + \lfloor \varepsilon(2^{M+1} - 1) \rfloor. \quad (114)$$

More precisely, if the fibre is nonempty and its common odd count is s , put $A = 2^h y / 3^s$. Its cardinality is at most

$$\max\{0, \min(2^{M+1} - 1, \lfloor A \rfloor) - \max(2^M, \lceil (1 - \varepsilon)A \rceil) + 1\}. \quad (115)$$

These conclusions apply in particular to Shaik's restriction $E_Y(n) \leq D$, with $\varepsilon = D/Y \leq 1/3$. For this restriction (114) is no larger than $1 + 2D 2^M / Y$, improving the coefficient 3 in his (4.8) to 2.

Proof. If two sources had $s_1 < s_2$, their scales satisfy $A_1 \geq 3A_2$. Consequently $n_1 \geq (1 - \varepsilon)A_1 \geq 2A_2 \geq 2n_2$. But $n_1 < 2^{M+1}$ and $n_2 \geq 2^M$, a contradiction. Thus s and A are constant on the fibre. Its integer sources lie in $[(1 - \varepsilon)A, A] \cap [2^M, 2^{M+1})$, proving (115) without any assertion of full occupancy.

For a nonempty fibre let a and z be its least and greatest sources. Since $z \leq A$ and $1 - \varepsilon \geq 0$,

$$a \geq (1 - \varepsilon)A \geq (1 - \varepsilon)z, \quad z - a \leq \varepsilon z \leq \varepsilon(2^{M+1} - 1).$$

An integer set between a and z has at most $z - a + 1$ elements. This proves (114). An empty fibre satisfies the bound as well. Finally $1 - P \leq E_Y / Y$ by the preceding proposition. $\square$

For example, the actual shortcut path 3, 5, 8, 4, 2, 1 first reaches $Y = 3/2$ at its last point. Its product defect is $5/32 = 50/320$, whereas $L = 1/10 + 1/16 = 13/80 = 52/320$. Thus the exact-defect filter with $\varepsilon = 51/320$ retains this path while the additive-loss filter does not. This is a strict enlargement of the admissible finite-path set, not an assertion about its asymptotic density.

The integer bound is optimal for the interval-containment argument alone: take $A = 2^{M+1} - 1$. The interval's lower endpoint remains at least 2^M when $M \geq 1$ and $\varepsilon \leq 1/3$, and its integer occupancy is exactly (114). This does not claim that a Collatz fibre fills that interval. First-passage admissibility and congruences can reduce its occupancy.

Proposition 76 (Rank-chain loss and support-sensitive transport). *Consider a finite, nonempty chain of positive-length first-passage blocks through distinct decreasing integer threshold ranks $q_0 > \dots > q_i = q \geq 1$. Block j has duration $1 \leq h_j \leq m_j$, odd count s_j , and $m_j < (q_j + 1)/r_*$, where $r_* > 0$. Its cumulative landing is the original orbit's direct first passage to 2^q . At that landing its additive scaled loss satisfies*

$$E_{2^q} \leq \sum_{j=0}^i \frac{2^q s_j}{3 \cdot 2^{q_j} + 4} < \frac{1}{3} \sum_{j=0}^i m_j 2^{q - q_j} < \frac{2(q + 2)}{3r_*}. \quad (116)$$

Let S be any finite set of allowed positive cumulative times and $B \subset (Y/2, Y] \cap \mathbb{N}$, where $1 < Y < 2^M$. The number of sources in I_M whose first-passage tag belongs to $S \times B$ and whose $E_Y \leq D$, with $0 \leq D/Y \leq 1/3$, is at most

$$|S| |B| \left(1 + \left\lfloor \frac{D}{Y} (2^{M+1} - 1) \right\rfloor \right). \quad (117)$$

In particular, for the chains above set $Y = 2^q$ and $D_q = 2(q + 2)/(3r_*)$. Whenever $D_q/2^q \leq 1/3$, their source proportion is at most

$$|S| \frac{|B|}{2^q} \left(2^{q-M} + \frac{4(q + 2)}{3r_*} \right). \quad (118)$$

All estimates persist under arbitrary restriction of the source set.

Proof. For two successive first passages to $Y_1 > Y_2$, every state before the first passage to Y_1 exceeds Y_1 and therefore Y_2 . The next segment stays above Y_2 until its endpoint. Concatenation is therefore exactly the first passage to Y_2 . Induction proves the assertion for the chain.

Since $q_j \geq 1$, the smallest odd integer exceeding 2^{q_j} is $2^{q_j} + 1$. Apply (111) in each block and the exact rescaling identity (113). This proves the first bound in (116). Each summand is smaller than $m_j 2^{q-q_j}/3$, including $s_j = 0$ because $m_j > 0$. Put $d = q_j - q$. Distinct ranks use each $d \geq 0$ at most once. Hence

$$\frac{1}{3} \sum_{j=0}^i m_j 2^{q-q_j} < \frac{1}{3r_*} \sum_{d \geq 0} (q + d + 1) 2^{-d} = \frac{2(q+2)}{3r_*}.$$

The two sums are $\sum 2^{-d} = 2$ and $\sum d 2^{-d} = 2$, obtained by summing finite geometric series and taking their limits.

The tagged fibres over distinct (h, y) are disjoint: first passage has a unique time and value. Apply Proposition 75 to each allowed tag and sum. Source restriction only removes elements. Divide by 2^M , use $1 + \lfloor D_q(2^{M+1} - 1)/2^q \rfloor \leq 1 + 2D_q 2^{M-q}$, and substitute D_q . This proves (118). $\square$

The coefficient comparison is now explicit. Shaik's (4.9) and (5.10), with the same time support, give $|S|(|B|/2^q)(2^{q-M} + 3(q+2)/r_*)$. The new rank-dependent coefficient is $4/9$ of that coefficient. The support size and the exceptional-set decay have not changed. In particular, these finite improvements alone do not lower the paper's critical polylogarithmic exponent. Proving a change there requires following the small-rank tail and time-support estimates.

Proposition 77 (The exact shortcut–Syracuse passage correspondence). *Fix an odd $n > Y > 1$, and suppose its shortcut first passage occurs at h , with landing y . Put $d = v_2(y)$ and let s be the number of odd shortcut inputs before time h . Then its first Syracuse passage below Y occurs at return s , with landing $u = y/2^d$. If $a_1, \dots, a_s$ are its actual return valuations,*

$$h + d = \sum_{\nu=1}^s a_\nu, \quad \ell_{\text{raw}} = h + s, \quad \ell_{\text{raw,odd}} = \sum_{\nu=1}^s a_\nu + s. \quad (119)$$

Here the first raw index is the raw first-passage index, and the second is the raw time of the subsequent odd landing. Conversely, given the actual first Syracuse landing u and these valuations, let d be the largest integer with $2^d u \leq Y$. Then

$$y = 2^d u, \quad h = \sum_{\nu=1}^s a_\nu - d$$

recovers the shortcut first passage. The reverse losses and product do not change during the terminal even tail. On a complete odd-return path $u_0 = n, \dots, u_s = u$,

$$L = \sum_{\nu=0}^{s-1} \frac{1}{3u_\nu + 1}, \quad P = \prod_{\nu=0}^{s-1} \frac{3u_\nu}{3u_\nu + 1}, \quad n = \frac{2^{\sum a_\nu}}{3^s} u P.$$

Proof. An odd return of valuation a expands into one odd shortcut step and $a - 1$ even shortcut steps. This expansion is bijective onto the corresponding finite paths between consecutive odd states; its inverse cuts the shortcut path at its odd states. Before shortcut passage every state is

greater than Y , so every previous odd state is too. The final even tail $y, y/2, \dots, y/2^d = u$ adds no odd input and ends at the first odd state below Y . This gives the return number s , the valuation sum and the unchanged loss/product.

Each odd shortcut step expands into two raw steps; each even shortcut step into one. Thus its raw index is $h + s$. The inserted raw state is $3x + 1 > Y$ whenever its pre-step state $x > Y$, so no earlier raw passage is introduced. Completing the even tail gives the second clock.

Conversely the last return descends along consecutive powers of two times u . Its first shortcut step is increasing, hence still exceeds Y . The largest power at most Y is therefore attained inside that final descending tail, and all earlier returns are above Y . This proves the inverse formula, including its first-passage property. The product identity follows either by expansion or by multiplying $3u_\nu + 1 = 2^{a_{\nu+1}}u_{\nu+1}$. $\square$

This is the precise connection to the actual valuations and clocks in Section 26. A Syracuse landing need not equal the shortcut landing: the integer d is the exact discrepancy, and the displayed map recovers it from the threshold and the odd landing. Likewise the raw maximum is not generally the shortcut maximum, because odd raw steps insert $3x + 1$. No height conclusion is transferred by merely identifying the endpoint clocks.

For $n = 7$, $Y = 4$, the actual shortcut path is

$$7, 11, 17, 26, 13, 20, 10, 5, 8, \underbrace{4}_{h=9}, 2, \underbrace{1}_{h+d=11}.$$

The odd-return path is $7, 11, 17, 13, 5, 1$, with valuations $1, 1, 2, 3, 4$. Thus $s = 5$, $d = 2$, the raw passage time is $9 + 5 = 14$, and the raw odd-landing time is $11 + 5 = 16$.

For the interval geometry alone, take $I_3 = [8, 16) \cap \mathbb{N}$, $A = 15$ and $\varepsilon = 1/3$. The candidate interval is $[10, 15]$, containing six integers. The source bound is 9; the clipped bound is $1 + \lfloor (16 - 1)/3 \rfloor = 6$. This interval is not asserted to be a populated Collatz fibre.

Figure 5: The clock correspondence on an actual path and the clipped integer-fibre mechanism. Complete proofs are Propositions 75 and 77. The underlying first-passage transport is Shaik’s [3], pp. 14–17. A reproducible companion image is `shaik_reverse_transport.png`.

Formal-source comparison and reading boundary. At commit `ef3410843bf58d69f771f5ba2c0571d54b54da59`, `FirstPassage.lean` contains the exact reverse identity, first-passage band and time-budget fibre estimate. `LossTransport.lean` proves the additive-loss-filtered coefficient 3; `NestedRecertification.lean` proves the direct-passage and rescaled-additivity identities; and `RankScaledLoss.lean` proves the rank budget $(q + 2)/r$. These are the source comparisons used above, not a claim to have replayed the package.

The moving endpoint in `Main.lean` explicitly retains a common time/landing/height witness and a shell exceptional estimate. The fixed-exponent wrapper read there quantifies an *existential* positive exceptional exponent. The manuscript’s Corollary 1.2(1) states every exponent in a specified interval. The next section reconstructs that manuscript range by explicit scalar and shell summation; the corresponding formal dependency-closure review remains separate. The four propositions here are complete written deterministic results. The finite rational checks test their implementations, not the manuscript’s full natural-density theorem.

28 Shaik's timeout tail, critical rate and drift clock

The preceding transport estimate counts preimages of a bad landing set. To use it one needs both the size of that set and the number of possible arrival times. Shaik's construction supplies these in different ways: a shrinking parity-prefix tolerance at large ranks, and a terminal odd-count test at small ranks. This section reconstructs [3], Section 6, pp. 19–26, preserving that distinction. It also retains the deterministic rank sequence and the summed passage-time inequalities. These give a smaller high-failure charge and an explicit error about the drift clock. The critical target exponent itself is unchanged.

Throughout this section put

$$\lambda = \log_2 3, \quad a_0 = \lambda/2, \quad g = 1 - a_0, \quad \rho = 2^{-g} = \sqrt{3}/2,$$

$$p_* = \log_3 2, \quad \kappa = 1 - H_2(p_*), \quad A_{\text{FP}} = \frac{1}{2\kappa}.$$

Here $H_2(u) = -u \log_2 u - (1 - u) \log_2 (1 - u)$; $\log$ without a subscript is natural. In particular $g \log 2 = \frac{1}{2} \log(4/3) > 0$. All paths below are paths of the shortcut map T .

Proposition 78 (Uniform prefix certificate). *For $0 < t \leq 1$, define W_t by*

$$x \in W_t \iff \rho^j x^{1-t} \leq T^j(x) \leq \rho^j x^{1+t} \quad (0 \leq j \leq \lfloor \log_2 x \rfloor).$$

With $c_0 = 1/(2\lambda^2)$, $C_0 = 2e^{4c_0}$, every integer $m \geq 0$ satisfies

$$2^{-m} |I_m \setminus W_t| \leq C_0 e^{-c_0 t^2 m}. \quad (120)$$

Proof. The parity-prefix map from $\mathbb{Z}/2^m\mathbb{Z}$ to $\{0,1\}^m$ is bijective. Here is the precise induction. Inputs congruent modulo 2^m have the same first m parities; along a shared prefix of length $j \leq m$, their iterate difference is $3^{s_j}(x' - x)/2^j$. Each residue modulo 2^m has two lifts modulo 2^{m+1} . Their time- m difference is the odd integer 3^{s_m} , so the next parities are opposite. Starting with the empty prefix proves bijectivity, and also gives its inverse by successively choosing the unique lift with the prescribed next parity. Reduction from I_m to residues is itself a bijection. Thus its uniform counting law makes the first m parity bits independent fair bits. This is a finite counting statement, not an independence assertion about an infinite Collatz orbit.

Let s_j count the first j odd inputs and $d_j = s_j - j/2$. For $\theta > 0$, $\exp(\theta d_j)/\cosh(\theta/2)^j$ is a nonnegative martingale. Stopping it at the first $d_j > h$, or at m , gives

$$\Pr\{\max_{j \leq m} d_j > h\} \leq \exp(-\theta h + m\theta^2/8).$$

Indeed $\log \cosh u \leq u^2/2$, by integrating $\tanh u \leq u$; finite stopping follows by conditioning one step at a time. Take $\theta = 4h/m$ and apply the same argument to $-d_j$. For $m > 0$,

$$\Pr\{\max_{j \leq m} |d_j| > h\} \leq 2e^{-2h^2/m}.$$

There is also a purely finite version of this argument in Shaik's `Barrier.lean`, `barrierHitCount_mul_cosh_1e`. For $a, \theta \geq 0$, let $B_a(r, y)$ count the r -bit continuations whose walk, started at $y \in \mathbb{Z}$ with increments $2\epsilon - 1$, has $|Y_i| > a$ at some $0 \leq i \leq r$. The boundary is open and the current position is included. If $|y| > a$, then $B_a(r, y) = 2^r$. Otherwise $B_a(0, y) = 0$ and

$$B_a(r+1, y) = B_a(r, y-1) + B_a(r, y+1).$$

These identities follow by splitting at the first bit. They give, by induction on r ,

$$B_a(r, y) \cosh(\theta a) \leq 2^r \cosh(\theta y) \cosh(\theta)^r.$$

At an already bad vertex, use $\cosh(\theta a) \leq \cosh(\theta y)$ and $\cosh \theta \geq 1$. At a safe vertex, add the two induction inequalities and use the exact identity $\cosh(\theta(y-1)) + \cosh(\theta(y+1)) = 2 \cosh(\theta y) \cosh \theta$. With $y = 0$, $a = 2h$, divide by $2^m \cosh(2\theta h)$, use $\cosh \theta \leq e^{\theta^2/2}$ and $\cosh(2\theta h) \geq e^{2\theta h}/2$, and choose $\theta = 2h/m$. This gives the same $2e^{-2h^2/m}$, including $h = 0$. The finite parity bijection transports this exact word count to I_m , as proved explicitly in Corollary 85 below. This comparison identifies the same boundary, length, constants and integer-to-Boolean map in the written argument and the source; it does not claim an independent build of its real-analysis imports.

The affine expansion along the actual path is

$$T^j(x) = \rho^j 3^{d_j} x + \frac{1}{2} \sum_{i < j} p_i \rho^{j-i-1} 3^{d_j-d_{i+1}}, \quad p_i = T^i(x) \bmod 2.$$

If every $|d_j| \leq h$, its nonnegative additive term is at most $3^{2h}/(2(1-\rho)) = (2+\sqrt{3})3^{2h}$. For $m \geq 4$, $tm \geq 2$, set $h = tm/(2\lambda)$. Then $3^h = 2^{tm/2}$. The multiplicative term is at least $\rho^j x^{1-t}$ and at most $2^{-tm/2} \rho^j x^{1+t}$. The unused upper-envelope allowance is at least

$$(1 - 2^{-tm/2}) \rho^m 2^{(1+t)m} \geq \frac{1}{2} 2^{(a_0+t)m}.$$

It dominates the additive term $(2+\sqrt{3})2^{tm}$, since $2^{a_0 m}/2 \geq 9/2 > 2+\sqrt{3}$. Thus failure of W_t implies a prefix deviation exceeding h . The displayed maximal inequality gives $2e^{-c_0 t^2 m}$. Outside this startup range, $t^2 m \leq 4$, and the trivial bound 1 is absorbed by $C_0 e^{-c_0 t^2 m}$. This includes $m = 0$. $\square$

Proposition 79 (An integer timeout bound and its critical tail). *For $m \geq 1$, $x \in I_m$, let $s = s_m(x)$. Then*

$$T^m(x) \leq U_m(s) := \left\lfloor \frac{3^s(2^{m+1} - 1 + 2^{m-s})}{2^m} \right\rfloor - 1. \quad (121)$$

The function U_m is strictly increasing on $0, \dots, m$. For $0 \leq q < m$, put

$$k^\sharp(m, q) = \min\{s \in \{0, \dots, m\} : U_m(s) > 2^q\}, \quad k(q) = \lfloor qp_* \rfloor.$$

This minimum exists, $k^\sharp(m, q) \geq k(q)$, and

$$|\{x \in I_m : T^j(x) > 2^q \ (0 \leq j \leq m)\}| \leq \sum_{s=k^\sharp(m, q)}^m \binom{m}{s} \leq \sum_{s=k(q)}^m \binom{m}{s}. \quad (122)$$

For fixed $K_0 > 0$, $C > 1$, let $r_L = 1 - K_0/(2L)$, $q = \lfloor r_L m \rfloor$. Uniformly for integers $L \leq m \leq CL$, once L is sufficiently large depending only on K_0, C , the timeout proportion is

$$O_{C, K_0}(m^{-1/2} 2^{-\kappa m}). \quad (123)$$

Proof. If the odd positions are $i_1 < \dots < i_s$, their contribution to the affine offset is

$$b = \sum_{\nu=1}^s \frac{3^{s-\nu}}{2^{m-i_\nu}} \leq \sum_{\nu=1}^s \frac{3^{s-\nu}}{2^{s-\nu+1}} = (3/2)^s - 1.$$

Here $i_\nu \leq m - s + \nu - 1$, and the empty sum has value zero. Combining this with the integer bound $x \leq 2^{m+1} - 1$ and taking the floor proves (121). Before taking the floor, the increment as s increases by one is

$$2 \cdot 3^s(2 - 2^{-m}) + \frac{1}{2}(3/2)^s > 1,$$

so U_m is strictly increasing. Its last value is $U_m(m) = 2 \cdot 3^m - 1 > 2^q$, proving existence of the minimum. Moreover $U_m(s) < 3^{s+1}$: it bounds the integer part of $2 \cdot 3^s - 3^s 2^{-m} + (3/2)^s - 1 < 3^{s+1}$. A timeout implies $U_m(s) > 2^q$, hence $s > qp_* - 1$, equivalently $s \geq \lfloor qp_* \rfloor$. The parity-prefix bijection in the preceding proof gives exactly $\binom{m}{s}$ sources with odd count s , proving the finite bounds. No full occupancy of the affine upper envelope is asserted.

For the asymptotic bound write $\theta = r_L m - q \in [0, 1)$, $k = k(q)$, $u = k/m$. Exact rounding gives

$$qp_* - 1 < k \leq qp_*, \quad 0 < p_* - u < \frac{K_0 p_*}{2L} + \frac{p_* + 1}{m}.$$

Consequently, eventually u belongs to the fixed interval $[u_0, p_*]$, $u_0 = (p_* + 1/2)/2 > 1/2$, and

$$m(p_* - u) < B_{C, K_0} := CK_0 p_*/2 + p_* + 1.$$

For $k \leq j < m$, the ratio of consecutive binomial coefficients is $(m - j)/(j + 1) \leq (1 - u_0)/u_0 < 1$. The upper tail is therefore at most $(1 - (1 - u_0)/u_0)^{-1} \binom{m}{k}$. Robbins' factorial bounds, already used above and cited in [4], give

$$2^{-m} \binom{m}{k} \leq \frac{e^{1/(12m)}}{\sqrt{2\pi mu(1-u)}} e^{-mD(u||1/2)}, \quad D(u||1/2) = u \log(2u) + (1-u) \log(2(1-u)).$$

Its prefactor is uniformly $O(m^{-1/2})$. Since $D'(u) = \log(u/(1-u))$, on this interval

$$0 \leq m\{D(p_*||1/2) - D(u||1/2)\} \leq \log \frac{p_*}{1-p_*} B_{C, K_0}.$$

The loss is a constant, not a loss in the coefficient of m . Finally $D(p_*||1/2) = \kappa \log 2$. This proves (123). $\square$

Proposition 80 (Deterministic ranks and one passage-time interval). *Fix $a_0 < r < 1$, $0 < \tau < r - a_0$, $D > 0$, and $S = \lceil C_{\text{sw}} \log(M + 2) \rceil$, with $D/\sqrt{C_{\text{sw}}} \leq \tau$. At a high parent rank $m \geq S$, use $t_m = D\sqrt{\log(M + 2)/m}$ and threshold rank $q = \lfloor rm \rfloor$. Execute only certified W_{t_m} blocks, stop at an upper-endpoint landing, and otherwise continue at rank $q - 1$. For all sufficiently large M , every such block has a positive first-passage duration at most m .*

Every possible high run is an initial part of the single rank sequence $m_0 = M$, $q_i = \lfloor rm_i \rfloor$, $m_{i+1} = q_i - 1$, until $m_i < S$. For blocks $0, \dots, j$, put $H = \sum h_i$ and $C_j = M - q_j - j$. Then

$$C_j - \sum_{i=0}^j t_{m_i} m_i \leq gH < C_j + \sum_{i=0}^j (t_{m_i} m_i + t_{m_i} + 2). \quad (124)$$

Thus all possible H for this fixed target lie in one interval containing at most

$$1 + \left\lceil \frac{\sum_{i=0}^j (2t_{m_i} m_i + t_{m_i} + 2)}{g} \right\rceil = O_{r, \tau, D}(\sqrt{M \log(M + 2)})$$

integers. There is no multiplicity from different rank histories.

If a nonendpoint high run enters the low phase, its entry rank is fixed by that same sequence. For integer $1 \leq L < S$, the duration of any sequence of successful low blocks before its first timeout, or before success below rank L , is at most

$$K_{S,L} = \sum_{m=L}^{S-1} m = \frac{S(S-1) - L(L-1)}{2}. \quad (125)$$

For any fixed later timeout band, the cumulative arrival times still lie in a single interval of length $O(\sqrt{M \log(M+2)}) + K_{S,L}$.

Proof. The tolerance cap is inactive because $m \geq C_{\text{sw}} \log(M+2)$. For $x \in I_m \cap W_{t_m}$,

$$T^m(x) \leq 2^{-gm} x^{1+t_m} < 2^{(a_0+\tau)m+1+\tau}.$$

Since $r - a_0 - \tau > 0$, eventually this is at most $2^{\lfloor rm \rfloor}$, uniformly for $m \geq S$. The source exceeds that threshold, so its first passage has $1 \leq h \leq m$. Its landing y lies in $(2^{q-1}, 2^q]$. Applying the two certificate inequalities at this actual h gives

$$(1 - t_m)m - q \leq gh < (1 + t_m)(m+1) - q + 1.$$

Every earlier continuing landing is strictly below its upper endpoint, and hence has rank exactly $q_i - 1$. Induction proves determinism; summing $m_i - q_i$ gives $M - q_j - j$, proving the one-sided interval.

The rank sequence satisfies $m_i \leq r^i M$. Thus $\sum \sqrt{m_i} \leq \sqrt{M}/(1 - \sqrt{r})$. While $m_i \geq 1$, its number of terms is $O_r(1 + \log(M+2))$. These estimates prove the interval-width bound. Counting integers in an interval of length w gives at most $1 + \lfloor w \rfloor$, also for a half-open interval.

If the high run does not stop at an endpoint, the first $m_i < S$ and its preceding threshold are fixed, independently of the orbit. At low ranks every successful step decreases the integer parent rank strictly, and has duration at most that parent. Each integer from L to $S-1$ is therefore charged at most once, proving (125). Add the interval $[0, K_{S,L}]$ to the high-entry interval. Restricting to paths which time out in a specified band cannot enlarge it. Since $S = O(\log(M+2))$, $K_{S,L} = O(\log^2(M+2)) = o(\sqrt{M \log(M+2)})$. $\square$

Theorem 81 (Critical shell bound with an explicit drift-clock error). *Let $L = L_M$ be an integer sequence with $L_M \asymp \log(M+2)$. For every $\beta > 0$ and every $e > 0$, there are constants $B, C, M_0 > 0$ such that, outside at most*

$$C 2^M \left\{ \sqrt{M \log(M+2)} \sqrt{L_M} 2^{-\kappa L_M} + M^{-e} \right\} \quad (126)$$

sources in I_M , there is one actual shortcut path ending at $y < 2^{L_M}$ with

$$h \leq \frac{M}{g} + B \sqrt{M \log(M+2)}, \quad \max_{j \leq h} T^j(n) \leq n^{1+\beta}. \quad (127)$$

The constants may depend on the sequence's comparison bounds, β, e ; they are independent of M . No eventual hitting assumption is used for the excluded sources.

Proof. Choose $a_0 < r < 1$, $0 < \tau < \min(r - a_0, \beta, 1)$, and $K_0 > 0$. Choose D , then C_{sw} , sufficiently large that

$$D/\sqrt{C_{\text{sw}}} \leq \tau, \quad L < S < M, \quad \phi := \min(c_0 D^2, C_{\text{sw}} \log 2) > e + 2$$

eventually. This order of choices is possible because $L = O(\log M)$. Increase the fixed startup whenever needed in the estimates below.

Start in I_M and reject an initial failed certificate. At each high certified checkpoint execute the first passage in the preceding proposition. If its landing is 2^q with $q > S$, declare a high endpoint failure; if $q \leq S$, halve until the rank is less than L and declare success. Otherwise test the next certificate when its rank is at least S , and stop at its first failure. A nonendpoint landing of rank below S enters the low phase without another certificate.

At a low checkpoint $x \in I_m$, stop successfully if $m < L$. Otherwise use $q = \lfloor (1 - K_0/(2L))m \rfloor$. If all iterates through time m exceed 2^q , declare the first timeout. If not, take the least hitting time and its physical landing. Continue or stop below rank L . Eventually $1 \leq q < m$ for every low parent $m \geq L$. A power of two always reaches the next threshold by halving within m steps, so cannot produce a timeout. Strict rank decrease makes this a terminating finite procedure, regardless of subsequent orbit behaviour.

The initial failure proportion and each high bad landing band, measured relative to 2^q , are bounded by

$$d(M) := C_0(M+2)^{-c_0 D^2} + (M+2)^{-C_{\text{sw}} \log 2} = O(M^{-\phi}).$$

For a band $q > S$, this follows from $\{2^q\} \cup (I_{q-1} \setminus W_{t_{q-1}})$: its proportion is at most $2^{-q} + (C_0/2)(M+2)^{-c_0 D^2}$. The singleton is essential here.

All high failure ranks belong to the deterministic sequence above. Let $H_M = O(\sqrt{M \log(M+2)})$ be a common bound on its time supports. Proposition 76 bounds their total preimage proportion by

$$d(M) + H_M d(M) \sum_{i: q_i > S} \left(2^{q_i - M} + \frac{4(q_i + 2)}{3r} \right) = O((1 + MH_M)d(M)). \quad (128)$$

Indeed $\sum_i q_i \leq rM/(1-r)$ and the number of terms is $O(\log M)$. The small-loss condition holds eventually since $q_i > S \rightarrow \infty$. This replaces the source's $O(H_M M^2 d(M))$ charge by $O(H_M M d(M))$, using its actual rank sequence. As $\phi > e + 2$, (128) is $O(M^{-e})$.

For a first timeout, let p be the preceding block's threshold rank. Concatenation makes its arrival the original source's direct first passage to 2^p . Its landing lies strictly between 2^{p-1} and 2^p , since the upper endpoint cannot time out; moreover $L+1 \leq p \leq S$. Because S/L is bounded, the timeout proposition with $m = p-1$ gives a band-target proportion $O(p^{-1/2} 2^{-\kappa p})$. Choose fixed $0 < r_* < r$. Eventually $r_* < 1 - K_0/(2L)$, so every successful high or low block has $m < (q+1)/r_*$. Proposition 76 gives loss $D_p = 2(p+2)/(3r_*)$, with $D_p/2^p \leq 1/3$ eventually uniformly for $p \geq L+1$. The time-corridor proposition bounds the arrival support by $O(\sqrt{M \log(M+2)})$. Hence the total low failure proportion is at most a fixed multiple of

$$\sqrt{M \log(M+2)} \sum_{p=L+1}^S (p+1) p^{-1/2} 2^{-\kappa p} \ll \sqrt{M \log(M+2)} \sqrt{L} 2^{-\kappa L}.$$

For completeness, write $p = L + d$. Since $(p+1)p^{-1/2} \leq 2\sqrt{p} \leq 2\sqrt{L}\sqrt{1+d}$, the constant in the last estimate is at most $2 \sum_{d \geq 1} \sqrt{1+d} 2^{-\kappa d} < \infty$; comparison with $\sum (1+d) 2^{-\kappa d}$ proves convergence. Combining high and low failures proves (126).

It remains to keep the witness. The upper side of (124) gives, at every high exit,

$$gH \leq M + O(\sqrt{M \log(M+2)}).$$

The low cost is bounded by $K_{S,L} = O(\log^2 M)$; an endpoint halving tail costs at most S . Both are absorbed into the same square-root error. This proves the clock in (127), without replacing it by a geometric sum of parent-rank time budgets. Each high block is bounded by its initial value to the power $1 + \tau$; checkpoint values decrease, so this is at most $n^{1+\tau}$. Each low block starts below 2^S , lasts fewer than S steps, and satisfies $T(z) \leq 2z$ for positive z . Thus its height is below 2^{2S} , a fixed power of $M + 2$. For large M that is at most $n^{1+\beta}$. The deterministic halving tail adds no height. These are the same paths already counted, not separately chosen witnesses. $\square$

Corollary 82 (Fixed target, retained endpoint rate, and three clocks). *For fixed $A > A_{\text{FP}}$, put $\delta = \kappa A - \frac{1}{2} > 0$. For every $\beta > 0$ there are constants such that, with at most*

$$O_{A,\beta} \left(X(\log X)^{-\delta} \log \log X \right) \quad (129)$$

exceptions up to X , the shortcut orbit reaches $C(\log n)^A$, staying below $n^{1+\beta}$, by time

$$\frac{2}{\log(4/3)} \log n + O_{A,\beta}(\sqrt{\log n \log \log n}).$$

The ordinary Collatz orbit has a witness with the same target and height, and time

$$\frac{3}{\log(4/3)} \log n + O_{A,\beta}(\sqrt{\log n \log \log n}).$$

For odd sources the first Syracuse landing below 2^{L_M} has at most

$$\frac{1}{\log(4/3)} \log n + O_{A,\beta}(\sqrt{\log n \log \log n})$$

returns; completing its terminal even tail does not increase height. The exception count for these odd sources is bounded by (129) as a count of integers, not asserted to have the same normalized density as the all-integer set.

Proof. Set $L_M = \lceil A \log_2(M + 2) \rceil$ and choose $e > \delta$. The shell estimate is $O(M^{-\delta} \log(M + 2))$. Also $2^{L_M} \leq 2(M + 2)^A \ll_A (\log n)^A$. For $J = \lfloor \log_2 X \rfloor$, shells below $J/2$ contain $O(2^{J/2})$ integers. In shells $J/2 \leq M \leq J$, including the possibly incomplete last shell by its full bad count,

$$\sum 2^M O(M^{-\delta} \log(M + 2)) \ll 2^J J^{-\delta} \log(J + 2).$$

Finite startup shells are absorbed. This proves (129) with the exponent δ retained, rather than replacing it by any smaller γ . Since $M \leq \log_2 n$, the shortcut clock follows from the theorem.

Here is the exact map behind the other clocks. Expand each odd shortcut edge $x \mapsto (3x + 1)/2$ into the two ordinary edges through $3x + 1$, leaving each even edge unchanged. Cutting those inserted edges is its inverse on these actual finite-path records. A path with h shortcut steps and s odd inputs has ordinary length $h + s$; each inserted height is twice the following shortcut height. The reverse identity of Proposition 74, valid by the same multiplication for any finite positive path, gives

$$n = \frac{2^h y}{3^s} P, \quad 0 < P \leq 1, \quad s = \frac{h + \log_2(y/n) + \log_2 P}{\lambda} \leq \frac{h - M + L_M}{\lambda}.$$

Consequently

$$h + s \leq \frac{3M}{2g} + O(\sqrt{M \log(M + 2)}), \quad s \leq \frac{M}{2g} + O(\sqrt{M \log(M + 2)}).$$

The coefficients use $1 - g = \lambda/2$, not an assumed parity frequency. Apply the theorem with $\beta/2$; for large n , the factor 2 in the ordinary height is absorbed by $2n^{1+\beta/2} \leq n^{1+\beta}$. For odd n , complete the even tail at the first shortcut crossing of 2^{L_M} and cut at odd states as in Proposition 77. Its return count is no larger than the total s just bounded; its landing is no larger than the shortcut landing, and its additional halving height only decreases. Finally $1/(2g \log 2) = 1/\log(4/3)$. $\square$

For a bounded moving sequence A_M , the same theorem recovers Shaik’s Δ_M -criterion exactly:

$$L_M = \lceil A_M \log_2(M+2) \rceil, \quad \Delta_M = \kappa L_M - \frac{1}{2} \log_2(M+2) - \log_2 \log(M+3).$$

If $\Delta_M \rightarrow \infty$, then $L_M \asymp \log(M+2)$; the shell error is $O(2^{-\Delta_M} + M^{-e})$. Boundedness of A_M , and its eventual positivity from this same inequality, make the conversion $2^{L_M} \ll (\log n)^{A_M}$ uniform. To pass to natural density without assuming any monotonicity of A_M , fix a shell cutoff after which every shell error is below ε , sum those full-shell bounds through the top shell, and then send $X \rightarrow \infty$, followed by $\varepsilon \rightarrow 0$. This recovers the manuscript’s moving endpoint with the more precise clock above. It does not claim the pure target $(\log n)^{A_{FP}}$.

In $I_4 = \{16, \dots, 31\}$, a timeout through four steps above 8 requires $T^4(n) > 8$. The integer envelope gives $U_4(0) = 1$, $U_4(1) = 6$, $U_4(2) = 18$. Thus the cut improves from $s \geq 1$ to $s \geq 2$: the two binomial counts are 15 and 11, while direct enumeration gives 9 actual timeouts.

For the rank algebra with $M = 64$, $r = 7/8$, $S = 16$:

i	0	1	2	3	4	5	6	7
m_i	64	55	47	40	34	28	23	19
q_i	56	48	41	35	29	24	20	16

The next parent is $15 < S$, and $\sum_{i=0}^7 (m_i - q_i) = 64 - 16 - 7 = 41$. This is the exact center of gH , before the one-sided errors are added. The table does not assert that every displayed block is realized by an orbit satisfying the required certificates.

Figure 6: The integer timeout refinement and the telescoping time center. Propositions 79 and 80 supply the complete proofs. The timeout mechanism and original corridor are due to Shaik [3], Section 6. The finite enumeration and companion image `shaik_timeout_mechanism.png` are reproducible.

Relation to the earlier work and to the formal source. The sharper finite transport coefficient from the preceding section enters both high and low charges. Deterministic ranks further improve only the high charge; the critical factor remains $\sqrt{M \log M} \sqrt{L} 2^{-\kappa L}$, so A_{FP} is unchanged. The odd-return leading coefficient agrees with the Allikvere-based clock reconstructed earlier. The present ordinary clock has coefficient $3/\log(4/3)$, whereas the earlier threshold-independent raw clock has the larger coefficient $3/\log(4/3) + 1/\log 2$. These accompany different conclusions: the present result has a polylogarithmic target and a same-witness height bound; the earlier one permits arbitrary diverging targets and a uniform fixed-target exception estimate. Neither statement is silently substituted for the other.

In the pinned commit, `TimeoutDensity.lean` proves the exact parity/binomial count, the integer cutoff $k(q)$, and an entropy bound with rate $D(p_* \| 1/2) - O(1/L)$. Its unrestricted $m \geq L$ bound retains an exponential multiplier $e^{O(m/L)}$. The restriction $m \leq CL$ above is what makes

that multiplier fixed. `TimeoutEndpointProfile.lean` sums the source's loss coefficient and all possible rank indices, giving its $(M + 1)^2$ high charge. The reconstruction here proves the smaller charge for the actual deterministic run, not for arbitrary elements of that larger envelope. The written endpoint reconstruction does not certify the full Lean dependency closure or compilation, nor does it claim priority for these refinements. Those are distinct checks.

A kernel-checked exact subargument. The counting argument depends on actual integer endpoints, not just on their real-valued majorants. The following recurrence gives a second proof of the integer envelope and specifies the deterministic part now checked in Lean. It uses the original shortcut map T , the raw map $R(x) = x/2$ for even x and $R(x) = 3x + 1$ for odd x ; both maps take 0 to 0.

Proposition 83 (Exact endpoint, completion and clock certificate). *Put $x_j = T^j(n)$, $\epsilon_j = x_j \bmod 2$, and $s_j = \sum_{i < j} \epsilon_i$, for $n, j \in \mathbb{N}$. Define integer numerators by*

$$A_0 = 0, \quad A_{j+1} = \begin{cases} A_j, & \epsilon_j = 0, \\ 3A_j + 2^j, & \epsilon_j = 1. \end{cases}$$

Then, for every j ,

$$2^j x_j = 3^{s_j} n + A_j, \quad A_j + 2^j \leq 3^{s_j} 2^{j-s_j}. \quad (130)$$

In particular $n < 2^{m+1}$ alone suffices for

$$T^m(n) \leq U_m(s_m), \quad [T^j(n) > 2^q \ (0 \leq j \leq m)] \implies U_m(s_m) > 2^q.$$

The lower shell bound $n \geq 2^m$ is needed for the complete-shell count, but not for this endpoint inequality.

Write $\text{FP}_f(n, Y, h)$ for $f^h(n) \leq Y$ and $f^i(n) > Y$ for every $i < h$. For an arbitrary map $f : \mathbb{N} \rightarrow \mathbb{N}$, $Z \leq Y$, and actual passages $\text{FP}_f(n, Y, a)$ and $\text{FP}_f(f^a(n), Z, b)$, one has $\text{FP}_f(n, Z, a+b)$. Consequently, if $L \leq q$, $\text{FP}_T(n, 2^q, h)$, and $T^h(n) = 2^q$, then

$$\text{FP}_T(n, 2^L, h + q - L).$$

For $c_n(h) = h + s_h$ the exact raw-clock correspondence is

$$R^{c_n(h)}(n) = T^h(n), \quad \text{FP}_T(n, Y, h) \iff \text{FP}_R(n, Y, c_n(h)).$$

For every integer $B \geq 0$, a shortcut prefix bounded by B expands to a raw prefix bounded by $2B$; conversely, a raw prefix bounded by B has all its shortcut points bounded by B .

Proof. The identity in (130) follows by induction: an even step gives $2x_{j+1} = x_j$, and an odd step gives $2x_{j+1} = 3x_j + 1$. At an odd step the two sides of the numerator inequality evolve by the same factor 3, since

$$3A_j + 2^j + 2^{j+1} = 3(A_j + 2^j).$$

At an even step,

$$A_j + 2^{j+1} \leq 2(A_j + 2^j).$$

These prove the inequality from equality at $j = 0$; $s_j \leq j$ follows directly from $\epsilon_i \in \{0, 1\}$. Adding 2^m to the affine identity now gives

$$2^m(T^m(n) + 1) \leq 3^{s_m}(n + 2^{m-s_m}) \leq 3^{s_m}(2^{m+1} - 1 + 2^{m-s_m}).$$

Division by the positive integer 2^m , followed by taking the integer part, gives precisely $U_m(s_m)$, including its final minus one. The timeout implication uses the inequality at $j = m$.

For passage composition, times $i < a$ lie above $Y \geq Z$. Every time $a \leq i < a + b$ has the form $a + t$, $t < b$, so its value is $f^t(f^a(n)) > Z$. The terminal value is at most Z . For $i \leq q$, induction gives $T^i(2^q) = 2^{q-i}$. The first index with $2^{q-i} \leq 2^L$ is $q - L$; the composition result proves the asserted completion on the same path.

Each even shortcut step is one raw step; each odd step is two, with inserted value $3x + 1 = 2T(x)$. Induction gives the clock identity and strict increase of c_n . More precisely, every raw time strictly before $c_n(h)$ is either a shortcut point $T^i(n)$, $i < h$, or the inserted image $3T^i(n) + 1$ of an odd such point. This proves the forward first-passage implication: the inserted image exceeds its source, which already exceeds Y . The converse follows by restricting the raw prefix to the strictly increasing times $c_n(i)$. The same classification proves the height bound, since each inserted image equals twice the following shortcut point; that following point still belongs to the prefix. Restriction proves the reverse height statement. $\square$

The independent file `formal_shaik_core/ClockAndRanks.lean` checks these statements for all natural inputs and indices, using Lean 4.15.0 and only its `Init` library. It also checks the exact rank telescoping used above: if $q_i \leq m_i$ for $i \leq j$ and $m_{i+1} + 1 = q_i$ for $i < j$, then

$$\sum_{i=0}^j (m_i - q_i) + q_j + j = m_0.$$

There is no division or discarded endpoint correction in this identity. Its proof is induction on j : at the next index substitute $m_{j+1} = q_j - 1$ and cancel q_j . The natural subtractions are legitimate because of the stated inequalities.

The formal definitions agree with Shaik's `Basic.lean` and `RawDynamics.lean`: the maps are identical formulas. The identity map on each starting value extends to the identity on every finite orbit record, by the zero and successor recurrences. Its inverse is the same identity; all intermediate values and parity bits are retained. Induction on the parity sum also identifies the two odd counts. The certificate proves uniqueness of recursive iteration; this comparison with the source's Mathlib iteration is a written comparison using its `orbit_zero` and `orbit_succ`, not an imported Mathlib build. Shaik's `TimeoutRun.lean`, `TimeoutExecution.lean` and `TimeoutEndpointWitness.lean` explicitly retain nested first passage and the dyadic completion. The certificate independently checks that exact deterministic mechanism; it does not repair an omitted completion in those source files.

Seventeen exported theorem axiom reports were checked. Sixteen use only propositional extensionality and quotient soundness; recursive iteration uniqueness uses no axioms. No admitted proof appears in their dependency reports. The compiler run, source, output, 232 core-module source/object pairs and bundled binary hashes are pinned in the accompanying receipts. The binomial law, entropy estimates, analytic time corridor and full natural-density dependency closure are outside this certificate's scope.

Where the additive clock already occurs in the source. For a fixed admissible parameter package, Shaik's `TimeoutOrbitCeiling.lean`, in the proof of `eventually_timeoutRun_elapsed_add_switch_lt_shellClock`, already obtains the intermediate inequality named `htime`:

$$\text{elapsed} \leq \frac{M}{g} + \frac{1}{g} + C_{\text{time}} \sqrt{(M+2) \log(M+2)}.$$

Here C_{time} is its explicitly defined `timeoutTimeSupportConstant`, fixed independently of M . The proof subsequently absorbs the error and the switch cost into an arbitrary strictly larger leading clock. Thus the additive square-root shortcut error exposed above is already present in Shaik’s formal argument. It is not a newly discovered clock estimate. The deterministic-rank charge and integer envelope refinements must be distinguished from that existing bound. The comparison above keeps its error visible and transports the same witness to the other clocks; the new certificate verifies only the exact integer and path components just specified.

Constructive parity coordinates and their exact fibres. The timeout count uses a deterministic fact before it uses probability: each binary itinerary of length m occurs at exactly one point of I_m . The same coordinates identify the finite modular Collatz graph with a binary De Bruijn graph. Laarhoven–de Weger state this graph isomorphism in [2, Section 2]; Shaik proves the parity bijection in `Parity.lean`, then transports filtered cardinalities to the positive shell in `Barrier.lean`, `card_dyadicShell_filter_mod_mem` and `card_shellBarrierHit`. The construction below gives an executable inverse and an independent core certificate for the integer identities. It reconstructs this existing mechanism, rather than asserting a new parity or graph theorem.

Proposition 84 (Decoder, integer lifts and complete parity fibres). *For the shortcut map T , put*

$$\epsilon_i(n) = T^i(n) \bmod 2, \quad s_i(n) = \sum_{a < i} \epsilon_a(n), \quad Q_m(n) = \sum_{i < m} 2^i \epsilon_i(n).$$

The earliest orbit bit is the least significant binary digit. For $n, d, m \in \mathbb{N}$ and $0 \leq i \leq m$, one has the exact integer identities

$$\begin{aligned} s_i(n + 2^m d) &= s_i(n), \\ T^i(n + 2^m d) &= T^i(n) + 2^{m-i} 3^{s_i(n)} d. \end{aligned} \tag{131}$$

In particular $Q_m(n) = Q_m(n \bmod 2^m) < 2^m$. Define $D_0(0) = 0$. For $0 \leq w < 2^{m+1}$, set

$$v = w \bmod 2^m, \quad r = D_m(v), \quad D_{m+1}(w) = \begin{cases} r, & Q_{m+1}(r) = w, \\ r + 2^m, & Q_{m+1}(r) \neq w. \end{cases}$$

Then D_m is the inverse of Q_m on $\{0, \dots, 2^m - 1\}$, and

$$\{n \in \mathbb{N} : Q_m(n) = w\} = \{D_m(w) + 2^m d : d \in \mathbb{N}\}. \tag{132}$$

The positive-shell inverse is $w \mapsto 2^m + D_m(w)$, from $\{0, \dots, 2^m - 1\}$ onto I_m . Both inverse identities hold, including $m = 0$. If $\text{wt}_m(w)$ denotes the sum of the m binary digits of w , then

$$\text{wt}_m(Q_m(n)) = s_m(n).$$

Proof. For $x, e \in \mathbb{N}$, the two branches of T give $T(x + 2e) = T(x) + 3^{x \bmod 2} e$. At stage $i < m$, the offset in (131) is even; hence both iterates have the same parity. Applying this one-step identity proves the next iterate formula and adds the same bit to both odd counts. The initial case is immediate, so induction proves (131).

For $d = 1$ and $i = m$, the terminal difference is $3^{s_m(n)}$, which is odd. Thus n and $n + 2^m$ have the same first m bits and opposite bits at index m . Their $(m + 1)$ -codes are exactly $Q_m(n)$ and $Q_m(n) + 2^m$, in one order. Their codes of length m agree for every d , proving residue invariance. The code bound follows by summing $\sum_{i < m} 2^i = 2^m - 1$.

Induct on m to construct the inverse. Once v is fixed, the induction hypothesis gives exactly one $r < 2^m$ with $Q_m(r) = v$. Every possible inverse of w modulo 2^{m+1} is either r or $r + 2^m$: reduce its code modulo 2^m , then use the inductive injectivity. The opposite next bits show that exactly one candidate has code w . The displayed recursion chooses it. This proves both existence and uniqueness without a cardinality argument.

For arbitrary n , Euclidean division writes $n = (n \bmod 2^m) + 2^m \lfloor n/2^m \rfloor$. Residue invariance and the inverse just proved identify its residue with $D_m(w)$ exactly when its code equals w , proving (132). Translation by 2^m then gives the shell inverse. The weight identity follows directly from the binary expansion of Q_m ; alternatively, split off its last bit and induct. $\square$

Corollary 85 (Counting measure and conditioned integer images). *Under uniform counting measure on I_m , the full parity word is uniform on $\{0, 1\}^m$. Consequently*

$$\#\{n \in I_m : s_m(n) = s\} = \binom{m}{s} \quad (0 \leq s \leq m). \quad (133)$$

More precisely, fix $0 \leq j \leq m$, a prefix code $0 \leq u < 2^j$, and put $r = D_j(u)$, $a = \text{wt}_j(u)$. The prefix fibre inside I_m and its image after j actual iterations are

$$\{n \in I_m : Q_j(n) = u\} = \{r + 2^j d : 2^{m-j} \leq d < 2^{m+1-j}\}, \quad (134)$$

$$T^j \{n \in I_m : Q_j(n) = u\} = \{T^j(r) + 3^a d : 2^{m-j} \leq d < 2^{m+1-j}\}. \quad (135)$$

The map between these displayed sets is bijective. Each suffix word of length $m - j$ occurs once, and, for $0 \leq b \leq m - j$,

$$\#\{n \in I_m : Q_j(n) = u, s_m(n) - s_j(n) = b\} = \binom{m-j}{b}.$$

Proof. The shell decoder is a bijection and retains every parity digit. Restriction to any subset of words is again a bijection. In particular, words of weight s correspond bijectively to the s -element subsets of the m positions; this proves (133), not merely an asymptotic estimate.

Equation (132) gives $n = r + 2^j d$. Because $0 \leq r < 2^j$, the two shell inequalities are equivalent to the bounds on d in (134), including the closed lower and open upper endpoints. Equation (131) at $i = j$ gives (135). Its positive slope 3^a makes this image map injective; its inverse recovers $d = (y - T^j(r))/3^a$, an integer precisely on the displayed image.

A full code with the specified prefix is exactly $u + 2^j v$, $0 \leq v < 2^{m-j}$. Applying the shell decoder gives each such word once, and its last $m - j$ bits are the parity word of $T^j(n)$. Their weight gives the last count. This also proves the claim for $j = 0$ and $j = m$, with the corresponding empty word. $\square$

This distinction matters in the density argument. Conditioning on an initial parity word leaves uniform remaining parity bits for the unused part of that same complete block. Its actual integer image, however, is the explicit arithmetic progression (135); the bijection does not replace it by uniform measure on an arbitrary new dyadic shell. The landing-fibre and transport estimates above account for those actual image coordinates. On the original complete shell, (133) supplies exactly the binomial tail used in Proposition 79; the integer envelope specifies which weights can contain timeouts. No exponent or endpoint constant is changed by the decoder itself.

The directed graph correspondence in the same coordinates. For $m \geq 1$, let the modular Collatz relation on residues $0 \leq r < 2^m$ have successors

$$\mathcal{T}_m(r) = \{T(r) \bmod 2^m, T(r + 2^m) \bmod 2^m\}.$$

This is exactly the graph defined using all integer lifts in [2, Section 2]. To compare its signed lifts, extend T to $\mathbb{Z}$ by the same two parity formulas. On either parity branch, for every integer d ,

$$T(r + 2^m d) = T(r) + 2^{m-1} 3^{r \bmod 2} d.$$

Modulo 2^m , only $d \bmod 2$ remains; the two values are distinct. Thus nonnegative lifts already witness every edge, including those originally defined by negative integer lifts.

On binary codes define

$$\mathcal{S}_m(w) = \{\lfloor w/2 \rfloor, \lfloor w/2 \rfloor + 2^{m-1}\}.$$

The identity

$$Q_m(T(n)) = \lfloor Q_m(n)/2 \rfloor + 2^{m-1} \epsilon_m(n) \quad (136)$$

follows by deleting the initial bit and retaining the next bit at index m . The two lifts $r, r + 2^m$ have opposite such bits. Residue invariance therefore gives the equality of two-element sets

$$Q_m[\mathcal{T}_m(r)] = \mathcal{S}_m(Q_m(r)).$$

Together with the explicit inverse D_m , this proves the directed graph isomorphism, including both directions of the edge relation. This Q_m is Laarhoven–de Weger’s Φ_m , with no reversal of the forward arrow. It is a map of finite relations; choosing the canonical residue does not make T a single-valued self-map modulo 2^m .

code w	0	1	2	3	4	5	6	7
residue $D_3(w)$	0	5	2	3	4	1	6	7
shell point $8 + D_3(w)$	8	13	10	11	12	9	14	15
odd count s_3	0	1	1	2	1	2	2	3

$r = 1 \xrightarrow{Q_3} w = 5, \quad \mathcal{T}_3(1) = \{2, 6\} \xrightarrow{Q_3} \mathcal{S}_3(5) = \{2, 6\}.$

Figure 7: An exact finite view of the decoder and graph map, at $m = 3$. Codes use the earliest parity as the low bit: code 5 is the chronological word 1, 0, 1. Each row retains an actual shell starting integer, not only a residue. The weight multiplicities are 1, 3, 3, 1. Proposition 84 and (136) prove the displayed correspondences; the graph mechanism is from [2, Section 2]. Reproducible diagram: `draw_shaik_parity.py`.

Proposition 86 (Exact reindexing and Pascal counts). *For every $m \in \mathbb{N}$ and $f : \mathbb{N} \rightarrow \mathbb{N}$,*

$$\sum_{n=0}^{2^m-1} f(Q_m(n)) = \sum_{w=0}^{2^m-1} f(w) = \sum_{r=0}^{2^m-1} f(Q_m(2^m + r)). \quad (137)$$

Define $B(0, s) = \mathbf{1}_{\{s=0\}}$, $B(m+1, 0) = 1$, and $B(m+1, s+1) = B(m, s) + B(m, s+1)$. Then, for every $m, s \in \mathbb{N}$, including $s > m$,

$$\sum_{r=0}^{2^m-1} \mathbf{1}_{\{s_m(2^m+r)=s\}} = B(m, s) = \binom{m}{s}.$$

These are general finite-sum identities, not estimates obtained from samples.

Proof. Write $N = 2^m$. The two-lift identity says that the unordered pair $\{Q_{m+1}(n), Q_{m+1}(n+N)\}$ is $\{Q_m(n), Q_m(n) + N\}$, with its two distinct entries. Split the sum over $0 \leq n < 2N$ into the two intervals of length N . Its summands pair as $f(Q_m(n)) + f(Q_m(n) + N)$. The induction hypothesis, applied to $w \mapsto f(w) + f(w + N)$, gives $\sum_{w < N} (f(w) + f(w + N)) = \sum_{w < 2N} f(w)$. At $m = 0$ both sides are $f(0)$. Translation by 2^m preserves the first m bits, giving the positive-shell identity.

Let $C(m, s)$ count codes $0 \leq w < 2^m$ of weight s . The lower half of the codes at length $m + 1$ has the same weight as the length- m code; adding 2^m adds precisely one bit. Consequently $C(0, s) = \mathbf{1}_{\{s=0\}}$, $C(m+1, 0) = C(m, 0) = 1$, and $C(m+1, s+1) = C(m, s+1) + C(m, s)$. Induction gives $C = B$. The subsets of an m -element set have the identical recursion, obtained by separating those containing the last element from those not containing it. Thus $B(m, s) = \binom{m}{s}$, with zero outside the support. Finally apply (137) to $f(w) = \mathbf{1}_{\{\text{wt}_m(w)=s\}}$ and use weight preservation. $\square$

The formal definitions use a recursive sum over the literal integers $0, \dots, N - 1$, and the displayed Pascal recursion for B . The four exported theorems are `parity_weighted_sum`, `shell_parity_weighted_sum`, `weightCount_binomial`, and `shellOddCountCard_binomial`. The identification of this Pascal recursion with the standard binomial notation is proved above; no import of Mathlib's `Nat.choose` is being claimed. The parity counting mechanism is classical and is already present in Shaik's `Parity.lean` and `Barrier.lean`, as cited above. This is an independent formal reconstruction, not a priority claim.

Proposition 87 (Finite startup preserves every incoming logarithmic rate). *Let $S, W \subseteq \mathbb{N}_{>0}$, let $N \in \mathbb{N}$, and suppose that $n \in S$ and $n \geq N$ imply $n \in W$. Put*

$$E = \max(N - 1, 0), \quad B_S(X) = \#\{1 \leq n \leq X : n \notin S\}.$$

For every natural X ,

$$B_W(X) \leq B_S(X) + \min(X, E). \quad (138)$$

For fixed $C_0 \geq 0$, $\eta > 0$, and any $\varepsilon > 0$, an eventual bound $B_S(X) \leq C_0 X (\log X)^{-\eta}$ therefore gives

$$B_W(X) \leq (C_0 + \varepsilon) X (\log X)^{-\eta} \quad \text{for all sufficiently large } X. \quad (139)$$

In particular no replacement of η by $\min(\eta, 1)$ is needed.

Proof. If $n \leq X$ is outside W , it is outside S or it is one of the positive integers less than N . This gives (138), including the empty startup intervals when $N = 0$ or 1 . Here is an explicit scalar absorption. Choose an integer $k > \eta$ and put

$$y_0 = \max \left\{ 1, \left(\frac{\max(1, E) k!}{\varepsilon} \right)^{1/(k-\eta)} \right\}.$$

For $y = \log X \geq y_0$, the nonnegative exponential series gives $e^y \geq y^k/k!$. Hence

$$\varepsilon X (\log X)^{-\eta} = \varepsilon e^y y^{-\eta} \geq \varepsilon y^{k-\eta}/k! \geq \max(1, E) \geq E.$$

Adding this inequality to the assumed estimate, past its own starting cutoff as well, proves (139). All parameters here are fixed before X tends to infinity. $\square$

The fifth new formal statement, `finite_startup_count_bound`, proves, for $f, g : \mathbb{N} \rightarrow \mathbb{N}$, $f(i) \leq 1$, and $f(i) \leq g(i)$ for $i \geq E$,

$$\sum_{i < X} f(i) \leq \sum_{i < X} g(i) + \min(X, E).$$

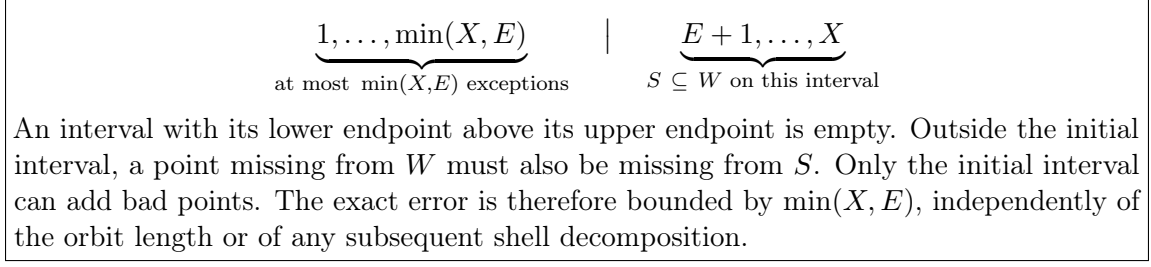


Figure 8: Finite-startup transfer on the actual positive integers. Proposition 87 proves both the count comparison and absorption without exponent loss. The source comparison is Shaik’s `FiniteStartup.lean`, lines 25–129, at the pinned commit of [3]; the sharper retained rate is the comparison here.

Its induction separates $X < E$, where the extra term costs at most one, from $X \geq E$, where it is paid by $g(X)$. Apply it to the bad-set indicators at the positive integer $i + 1$. The real exponential-series argument above is a written proof, not part of this Init-only certificate.

Effect on the published formal endpoints. At the pinned source commit, `FiniteStartup.lean`, lines 80–129, exports exponent $\min(\eta, 1)$. The fixed-polylogarithmic wrapper in `Main.lean`, lines 168–268, invokes that result after forming its literal clock/landing/height witness set. The incoming exponent from `shrinkingFixedPolylogNaturalDensityDescent` can instead be retained by Proposition 87. Increasing the common constant to dominate both the landing coefficient and $C_0 + \varepsilon$ only enlarges that witness set: the same time and intermediate-height witness still works. This changes no orbit map, time cutoff, or height condition. It removes a loss in this wrapper; it does not assert that its incoming existential exponent ranges over every exponent in the manuscript’s Corollary 1.2(1). Proposition 88 below supplies that parameter-level construction separately.

For comparison, the moving theorem in `Main.lean`, lines 63–117, retains one simultaneous witness and the shell error

$$C (2^{-\Delta_M} + (M + 2)^{-\epsilon}), \quad \Delta_M = \kappa \lceil A_M \log_2(M + 2) \rceil - \tfrac{1}{2} \log_2(M + 2) - \log_2 \log(M + 3).$$

Its assumptions are a fixed upper bound $A_M \leq A_{\max}$, $A_{\max} > 0$, $c > 2/\log(4/3)$, $\beta > 0$, and $\Delta_M \rightarrow +\infty$. The same $C > 0$ controls its target and error. `TimeoutEndpointNaturalDensity.lean` transports the shell family into this literal witness set; it does not erase the second error term. The scalar comparison in `MovingEndpointAsymptotics.lean` bounds the low contribution by

$$\sqrt{(M + 2) \log(M + 2)} \sqrt{L} e^{-(b_* - D/L)(L-1)} \leq \sqrt{A_{\max}/\log 2 + 1} e^{b_* + D} 2^{-\Delta_M},$$

where $L = \lceil A_M \log_2(M + 2) \rceil \geq 1$, $b_* = \kappa \log 2$, $D > 0$, and $0 \leq A_M \leq A_{\max}$. The $L - 1$, ceiling and logarithmic factors are retained. These statements were compared at source level; the analytic package has not been independently replayed.

The core Lean certificate now has 36 exported theorem axiom reports: the preceding 31, the four reindexing/counting statements, and the finite-startup count comparison. Of these reports, 34 use only `propext` and `Quot.sound`, the head-bit identity uses only `propext`, and iteration uniqueness uses none. The complete-shell binomial identity is now formally proved. The prefix-conditioned cardinality and packaged graph isomorphism retain their complete written proofs; those two packages are not separately exported Lean theorems. Shaik’s own finite cardinality transports are credited above. No full analytic package replay is claimed.

Prescribing the exceptional exponent in the source construction. The generic estimates in Shaik's source retain the field $P.kappa$. Its existing constructor chooses that field only after choosing a low barrier. Reversing this order makes the quantifier in Corollary 1.2(1) explicit, without changing the original run or failure envelope. We use γ for this prescribed field, to distinguish it from the entropy constant $\kappa = 1 - H_2(p_*)$.

Proposition 88 (Explicit parameters for every strict fixed-target rate). *Fix*

$$A > A_{FP}, \quad c > \frac{2}{\log(4/3)}, \quad \beta > 0, \quad 0 < \gamma < \delta := \kappa A - \frac{1}{2}.$$

There is a *ShrinkingPolylogParameterPackage*(A, c, β) with its exceptional-exponent field equal to γ . The following construction specifies every scalar choice and an integer startup for both of its stage packages.

Put $d_* = p_* - \frac{1}{2} = g/\lambda$, $b_* = \kappa \log 2$, and

$$L_* = \log \frac{p_*}{1 - p_*}, \quad \theta = \frac{(\gamma + 1/2) \log 2}{A}, \quad e = b_* - \theta > 0.$$

Choose

$$j = \left\lceil \frac{2L_* d_*}{e} \right\rceil, \quad \alpha = 1 - \frac{1}{j+2}, \quad t = g\alpha, \quad b_j = \log 2 - h\left(\frac{1}{2} + d_* \alpha^2\right), \quad b = \frac{\theta + b_j}{2}, \quad (140)$$

where $h(p) = -p \log p - (1-p) \log(1-p)$. Choose rational numbers

$$a_0 < r_{hi} < 1 - \frac{1}{c \log 2}, \quad a_0 + t < r_{lo} < 1, \quad r_* = \min(r_{hi}, r_{lo}).$$

One specified choice in either interval (u, v) is $(\lfloor 2^k u \rfloor + 1)/2^k$, with the least integer $k \geq 0$ such that $2^k(v - u) > 1$. Finally set

$$\begin{aligned} \tau &= \frac{1}{2} \min(r_{hi} - a_0, \beta, a_0), & K &= \gamma + 4, \\ D &= K/c_0 + 1, & C_{sw} &= (D/\tau)^2 + K/\log 2 + A/\log 2 + 3. \end{aligned} \quad (141)$$

The package fields are $\lambda_{lo} = \alpha$, $\eta_{lo} = t$, $b_{lo} = c_{dyadic} = b$, and $P.kappa = \gamma$. The run uses the stated $r_{hi}, r_{lo}, r_*, \tau, t, D$.

For either stage pair $(r, \eta) = (r_{hi}, \tau)$ or (r_{lo}, t) , write $v = r - a_0 - \eta > 0$. A valid *StageSetup* startup is

$$M_0 = 1 + \left\lceil \max \left\{ 1, \frac{2+\eta}{v}, \frac{2}{r}, \frac{6}{(r \log 2)^2} \right\} \right\rceil. \quad (142)$$

These choices are fixed before the outer shell rank tends to infinity. They do not give constants uniform as $\gamma \uparrow \delta$, $c \downarrow 2/\log(4/3)$, or $\beta \downarrow 0$.

Proof. Since $1 < \lambda < 2$, one has $1/2 < p_* < 1$, $0 < g < a_0 < 1$, and $0 < d_* < 1/2$. The function $F(p) = \log 2 - h(p)$ has

$$F'(p) = \log \frac{p}{1-p}, \quad 0 \leq F'(p) \leq L_* \quad (1/2 \leq p \leq p_*).$$

It is strictly increasing on $(1/2, p_*]$. As $F(1/2) = 0$ and $h(p_*) > 0$, $0 < b_* < \log 2$. The hypothesis on γ gives $0 < \theta < b_*$. For $\alpha = 1 - 1/(j+2)$ the mean-value estimate gives

$$0 < b_* - b_j \leq L_* d_* (1 - \alpha^2) < \frac{2L_* d_*}{j+2} < e. \quad (143)$$

The last strict inequality uses $j + 2 > 2L_*d_*/e$; no equality at the entropy endpoint is assumed. Consequently

$$0 < \theta < b < b_j < b_* < \log 2, \quad \gamma < \frac{Ab}{\log 2} - \frac{1}{2}.$$

Furthermore $0 < t < g < a_0$, $0 < \alpha < 1$, and the source's `adjustableEntropyRate`(α, t) is exactly b_j : its displacement is $\alpha t/\lambda = d_*\alpha^2$. This proves all low-barrier and low-rate conditions.

The high interval is nonempty because $1/(c \log 2) < g = 1 - a_0$. The low interval is nonempty because $t < g$. For the stated dyadic choice,

$$u < \frac{\lfloor 2^k u \rfloor + 1}{2^k} \leq u + 2^{-k} < v.$$

Existence of a suitable k follows from $2^k \rightarrow \infty$. Thus both rational ranks lie in $(a_0, 1)$; $\tau > 0$, $\tau < r_{\text{hi}} - a_0$, $\tau < \beta$, and $\tau < a_0$. The clock inequality is precisely

$$\frac{1}{1 - r_{\text{hi}}} < c \log 2.$$

Also $D > 1$, so

$$c_0 D^2 > c_0 D = K + c_0 > K, \quad C_{\text{sw}} \log 2 > K, \quad C_{\text{sw}} > A/\log 2.$$

As $C_{\text{sw}} > (D/\tau)^2$, $D/\sqrt{C_{\text{sw}}} < \tau$. It follows that

$$\gamma < \min(C_{\text{sw}} \log 2, c_0 D^2) - \frac{5}{2}.$$

Together with the low inequality and $r_* > 0$, these are every scalar field of the source parameter structure. In particular we have retained its original high charge $5/2$, not substituted the smaller deterministic-rank charge proved earlier.

It remains to construct the two stage structures, rather than leave their startup as an assumption. For either pair, $a_0 < r < 1$ and $0 < \eta < r - a_0 < 1$. If $M \geq M_0$, put $Y = 2^{\lfloor rM \rfloor}$. Then $rM > 0$ and $\lfloor rM \rfloor < M$, so $1 < Y < 2^M$. Since $\lfloor rM \rfloor > rM - 1$,

$$\frac{M}{2Y} \leq M e^{-(r \log 2)M} \leq \frac{2}{(r \log 2)^2 M} \leq \frac{1}{3}.$$

The middle inequality follows from $e^z \geq z^2/2$ for $z > 0$; the last one is exactly the fourth term in (142). Finally $vM > 2 + \eta$, whence

$$(a_0 + \eta)M + 1 + \eta < rM - 1 < \lfloor rM \rfloor.$$

Thus the required terminal budget is

$$\rho^M (2^{M+1})^{1+\eta} = 2^{(a_0+\eta)M+1+\eta} \leq Y.$$

This proves every `StageSetup` field, including the original floor and the rational horizon $M/(2Y)$. Assign these two stages to the run. All fields are now constructed. $\square$

For fixed A, c, β , this is a typed map from $(0, \kappa A - 1/2)$ to the source's admissible parameter packages. Projection to the `kappa` field composes with it to the identity. It is therefore injective, with singleton fibres on its image; the projection is its inverse there. No surjectivity onto all packages is claimed.

From the prescribed package to the literal orbit statement. Here is the complete rate passage through the source interfaces. Write $x = M + 2$, $L = \lceil A \log_2 x \rceil$, $S = \lceil C_{\text{sw}} \log x \rceil$, and

$$p_{\text{hi}} = \min(C_{\text{sw}} \log 2, c_0 D^2), \quad p_{\text{lo}} = Ab / \log 2.$$

The literal separated failure envelope used in `ShrinkingPolylogProfile.lean` retains an initial failed certificate, the high first-failure targets, and the low first-failure targets. Its scalar bound, with constants depending on the fixed package, is

$$d_{\text{hi}}(M) + H_M(1 + 6/r_*)((M + 1)^2 d_{\text{hi}}(M) + Q_L), \quad (144)$$

where

$$\begin{aligned} d_{\text{hi}}(M) &= 2^{-S} + C_0 x^{-c_0 D^2} \leq (1 + C_0) x^{-p_{\text{hi}}}, \\ H_M &= (C_{\text{time}} + 1) \sqrt{x \log x}, \\ Q_L &= W(\log 2, b) e^{-bL} + e^{b_j} W(b_j, b) e^{-bL}, \\ W(u, b) &= \frac{1 + 1/(u - b)}{1 - e^{-b}}. \end{aligned}$$

Here $C_0 = 2e^{4c_0}$; C_{time} is the source's fixed `shrinkingTimeSupportConstant`. All denominators in W are positive, since $0 < b < b_j < \log 2$. The two terms of Q_L are the dyadic endpoint contribution and the entropy contribution, respectively. Thus $Q_L \leq C_Q x^{-p_{\text{lo}}}$, with $C_Q = W(\log 2, b) + e^{b_j} W(b_j, b)$.

For clarity, the strict-power absorption can be made explicit. If $d > 0$, $y = \log x \geq \max(1, 2/d^2)$ gives $e^{dy} \geq (dy)^2/2 \geq y \geq \sqrt{y}$. Hence $\sqrt{\log x} \leq x^d$. Apply this with the two strictly positive gaps $p_{\text{hi}} - 5/2 - \gamma$ and $p_{\text{lo}} - 1/2 - \gamma$. Since $(M + 1)^2 \leq x^2$, each term of (144) is at most its fixed coefficient times $x^{-\gamma}$, eventually. The complete envelope therefore has shell proportion $O_{A,c,\beta,\gamma}(x^{-\gamma})$. There is no reduction of γ in this calculation.

The endpoint and height assertions concern the same successful run. In the source's `ShrinkingOrbitCeiling.lean`, its terminal rank is below L , its elapsed time is below $cM \log 2$, and every prefix value is at most $n^{1+\beta}$. The high blocks use tolerances at most $\tau < \beta$; their starting checkpoints are below the original start. The low blocks start below 2^S , so their local ceilings are at most $2^{S(1+t)}$. Since $S = O(\log M)$, eventually $S(1+t) \leq \beta M$, which gives the claimed outer height. The geometric-clock proof uses exactly the strict pressure $1/(1 - r_{\text{hi}}) < c \log 2$; its low part has $O(\log M)$ cost for these fixed rank ratios. Neither argument changes the rate or chooses a second witness. For $n \in I_M$ and $\log n \geq 1$,

$$k < cM \log 2 \leq c \log n, \quad T^k(n) < 2(M + 2)^A \leq C_{\text{tar}}(\log n)^A, \quad C_{\text{tar}} = 2(1/\log 2 + 2)^A.$$

To pass to all positive integers up to X , put $J = \lfloor \log_2 X \rfloor$. The shells below $J/2$ contribute at most $O(2^{J/2})$. For $J/2 \leq M \leq J$, the full bad shell count bounds even the possibly incomplete last shell, and

$$\sum_{J/2 \leq M \leq J} O(2^M (M + 2)^{-\gamma}) \leq O(J^{-\gamma}) \sum_{M \leq J} 2^M = O(X(\log X)^{-\gamma}).$$

The finite startup and the $O(\sqrt{X})$ term are absorbed by the same exponential-series argument as Proposition 87. This proves the literal simultaneous-witness conclusion with that prescribed γ .

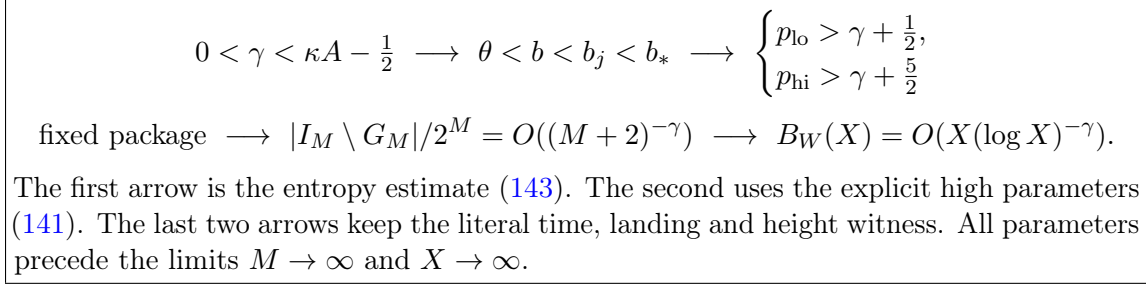


Figure 9: Where the prescribed exponent is retained. The package and failure-envelope construction are due to Idris Ali Shaik [3]; Proposition 88 gives the explicit rate-first selection and startup. This is a mathematical dependency diagram, not numerical evidence of orbit behaviour.

Enlarging a common constant to cover both C_{tar} and the exceptional-count coefficient preserves the very same witnesses.

The source locators are `FixedPolylogParameters.lean`, the definitions and convergence of `endpointLambda` and `endpointTolerance`; `ShrinkingParameters.lean`, lines 139–329; `Parameters.lean`, `exists_stageSetup`; `ShrinkingTailAsymptotics.lean`; `ShrinkingPolylogProfile.lean`; `ShrinkingOrbitCeiling.lean`; and `ShrinkingNaturalDensityDescent.lean`, at commit `ef3410843bf58d69f771f5ba2c0571d54b54da59`. The manuscript’s Corollary 1.2(1), printed p. 4, was checked visually. The existing constructor chooses an existential rate and the existing finite-startup helper caps it by 1; neither source statement is false. The construction here exposes the full prescribed range and the preceding startup proposition avoids that cap. The range is already asserted by Shaik, and is weaker than the endpoint-with-logarithm estimate in Corollary 82; no new density theorem or priority claim is being made. The explicit real-parameter construction and its analytic consumers have written proofs and source comparisons. They are not additional kernel-checked theorems in the 36-statement Init-only certificate, and the source package has not been replayed.

28.1 Critical targets with the boundary exceptional powers retained

The fixed-power estimate above leaves a logarithmic factor in its exceptional count. At the critical target power, that same factor determines which iterated logarithm is needed in the target. Keeping it exactly also identifies the exceptional exponent without decreasing it. The schedules below are Shaik’s schedules in [3], Corollary 1.2(2–3) and its proof, pp. 26–27, including the distinct shifts $M+2, M+3, M+4$. The following argument retains the boundary powers already furnished by the intermediate shell estimate.

Lemma 89 (A shell estimate with an arbitrary decreasing profile). *Let $G_M \subseteq I_M$, and suppose that*

$$|I_M \setminus G_M| \leq C2^M q(M) \quad (M \geq M_0),$$

where $C \geq 0$ and $q : \{M_0, M_0+1, \dots\} \rightarrow [0, \infty)$ is nonincreasing. Put $G = \bigcup_{M \geq 0} G_M$. For integer $X \geq 1$, write $J = \lfloor \log_2 X \rfloor$, $K = \lfloor J/2 \rfloor$. Whenever $K \geq M_0$,

$$\begin{aligned} |\{1 \leq n \leq X : n \notin G\}| &\leq 2^K - 1 + Cq(K)(2^{J+1} - 2^K) \\ &\leq 2^K + 2CXq(K). \end{aligned} \tag{145}$$

The same bounds hold for the complement of any larger good set.

Proof. There are exactly $2^K - 1$ positive integers below 2^K . In every remaining shell $K \leq M \leq J$, use $|I_M \setminus G_M| \leq C2^M q(K)$. The incomplete top shell contains no more bad integers than its full shell. Summing the geometric series gives the first line; $2^J \leq X$ gives the second. The map from the disjoint union of the labelled sets G_M to G is $(M, n) \mapsto n$, with inverse $n \mapsto (\lfloor \log_2 n \rfloor, n)$. Its fibres are singletons. Thus this is an exact counting-measure assembly, not a change to logarithmic measure. Enlargement of a good set only removes exceptions. $\square$

Corollary 90 (Boundary rates at the two critical logarithmic scales). *Put $a = A_{\text{FP}} = 1/(2\kappa)$, and fix $c > 2/\log(4/3)$, $\beta > 0$. For every fixed $B > 1/\kappa$, there is a constant $C_{\text{tar}} > 0$ such that the number of positive integers $n \leq X$ with no simultaneous witness*

$$h < c \log n, \quad T^h(n) \leq C_{\text{tar}}(\log n)^a (\log \log n)^B, \quad \max_{j \leq h} T^j(n) \leq n^{1+\beta}$$

is, for sufficiently large X ,

$$O_{B,c,\beta} \left(\frac{X}{(\log \log X)^{\kappa B - 1}} \right). \quad (146)$$

For every fixed $D > 0$, the target

$$C_{\text{tar}}(\log n)^a (\log \log n)^{1/\kappa} (\log \log \log n)^D$$

has the analogous exceptional bound

$$O_{D,c,\beta} \left(\frac{X}{(\log \log \log X)^{\kappa D}} \right). \quad (147)$$

All iterated logarithms here are restricted to a fixed sufficiently large initial value; the finitely many smaller integers can be counted as exceptions. Each assertion has its own constants and good set. The witnesses can in fact be chosen with shortcut time

$$h \leq \frac{2}{\log(4/3)} \log n + O(\sqrt{\log n \log \log n}).$$

The corresponding ordinary Collatz and odd-source Syracuse conclusions retain the same target and exceptional count, with leading clock coefficients $3/\log(4/3)$ and $1/\log(4/3)$, respectively, and the same order of additive error and the stated height bound. The latter exception estimate counts odd integers, without changing its denominator to an unproved conditional measure.

Proof. For large M , set $x = M + 2$, $v = \log(M + 3)$, $w = \log \log(M + 4)$. The two unrounded ranks are

$$H_B = a \log_2 x + B \log_2 v, \quad H_D = a \log_2 x + \kappa^{-1} \log_2 v + D \log_2 w.$$

In each case put $L_M = \lceil H \rceil$ and $r_M = L_M - H \in [0, 1)$. They satisfy $L_M \sim a \log_2(M + 2)$. For the actual critical buffer

$$\Delta_M = \kappa L_M - \frac{1}{2} \log_2(M + 2) - \log_2 \log(M + 3)$$

there are exact identities

$$\begin{aligned} \Delta_B &= (\kappa B - 1) \log_2 v + \kappa r_M, & 2^{-\Delta_B} &= 2^{-\kappa r_M} v^{-(\kappa B - 1)}, \\ \Delta_D &= \kappa D \log_2 w + \kappa r_M, & 2^{-\Delta_D} &= 2^{-\kappa r_M} w^{-\kappa D}. \end{aligned} \quad (148)$$

The ceiling multiplier lies in $(2^{-\kappa}, 1]$. In particular the buffers diverge. The moving exponent $A_M = H/\log_2(M+2)$ tends to a , so it is bounded and positive eventually. Extend its finitely many earlier values by a , if needed; this also meets the pointwise upper bound in the formal public API.

Apply Theorem 81, for instance with $e = 1$. Its main error is $O(2^{-\Delta_M})$, since $L_M = O(\log(M+2))$. Thus the literal common-witness exception is bounded by $C2^M q(M)$, where we may take

$$q_B(M) = v^{-(\kappa B-1)} + (M+2)^{-1}, \quad q_D(M) = w^{-\kappa D} + (M+2)^{-1}.$$

Here the change from M^{-1} to $(M+2)^{-1}$ costs at most a fixed factor for $M \geq 1$. Both profiles are eventually nonincreasing. Any fixed positive high-error exponent would suffice: for $p, e > 0$, $(M+2)^{-e}(\log(M+3))^p \rightarrow 0$, and likewise with $\log \log(M+4)$ in place of $\log(M+3)$. To see this directly, put $t = \log(M+2)$, bound the logarithmic factor by a fixed multiple of t^p , and choose an integer $j > p$. The inequality $e^{et} \geq (et)^j/j!$ proves the limit.

The exact target conversion also keeps the shifts. Put $u = \log n$ for $n \in I_M$, and define

$$K_1 = 1/\log 2 + 2, \quad K_2 = 1 + \log(1/\log 2 + 3), \quad K_3 = 1 + \log(1 + \log(1/\log 2 + 4)).$$

For large n , so that $\log u, \log \log u \geq 1$, the inequality $M \leq u/\log 2$ gives

$$x \leq K_1 u, \quad v \leq K_2 \log u, \quad w \leq K_3 \log \log u.$$

For example, $M+4 \leq (1/\log 2 + 4)u$; take a logarithm, factor out $\log u$, and take another logarithm to obtain the last bound. Consequently

$$2^{L_B} < 2x^a v^B \leq 2K_1^a K_2^B u^a (\log u)^B,$$

and

$$2^{L_D} < 2x^a v^{1/\kappa} w^D \leq 2K_1^a K_2^{1/\kappa} K_3^D u^a (\log u)^{1/\kappa} (\log \log u)^D.$$

These provide admissible fixed landing constants for the reconstructed witness. Its square-root clock error is $o(\log n)$, so it also satisfies the specified strict $c \log n$ bound eventually.

Apply Lemma 89. Since $K = \log X/(2 \log 2) + O(1)$, direct logarithms give

$$\log(K+3) = \log \log X - \log(2 \log 2) + o(1), \quad \log \log(K+4) = \log \log \log X + o(1).$$

Their ratios to the corresponding iterated logarithms tend to one. Also $2^K \leq \sqrt{X}$, which is absorbed by either bound in the statement: with $t = \log X$, the same exponential-series estimate dominates every fixed power of $\log t$ or $\log \log t$. This proves the two displayed rates with the boundary exponents intact.

For the other clocks use precisely the finite-path expansion in the proof of Corollary 82: an h -step shortcut path with s odd inputs becomes an ordinary path of length $h + s$. The reverse-product identity gives $s \leq (h - M + L_M)/\lambda$, and $L_M = O(\log M)$. It follows that $h + s \leq 3M/(2g) + O(\sqrt{M \log M})$ and $s \leq M/(2g) + O(\sqrt{M \log M})$. Apply the shell theorem with $\beta/2$ before inserting the ordinary odd-edge intermediate states; their factor-two height is absorbed eventually. Cutting at odd states and completing the final even tail can only decrease the target and adds no odd return. All clocks refer to the same actual finite orbit, not intersected independently chosen density-one sets. $\square$

The precise comparison with the printed corollary. Shaik states every $\gamma < \kappa B - 1$, respectively $\gamma < \kappa D$, on p. 4. Equations (6.20)–(6.23) already supply the profiles used above. No further logarithm is introduced by the half-shell sum, so the boundary powers follow from those estimates. This strengthens the displayed corollary, not its underlying shell theorem; the weaker statements remain correct. No optimality or priority is asserted.

Proposition 91 (An arbitrary diverging final factor, with its actual profile). *For every function $\Omega : [1, \infty) \rightarrow (0, \infty)$ with $\Omega(x) \rightarrow \infty$, there is a nondecreasing $\phi : [1, \infty) \rightarrow [1, \infty)$ such that*

$$\phi(x) \rightarrow \infty, \quad \phi(x) \leq \Omega(x) \text{ eventually}, \quad \phi(x) \leq \sqrt{\log x} \text{ eventually.}$$

Fix $c > 2/\log(4/3)$, $\beta > 0$. For a suitable $C_{\text{tar}} > 0$, the target

$$C_{\text{tar}}(\log n)^a(\log \log n)^{1/\kappa}\phi(\log \log n) \quad (149)$$

has simultaneous shortcut time $< c \log n$ and height $\leq n^{1+\beta}$ outside a set E_ϕ of density zero. For some C, M_0 , its prefix exception count satisfies, with J, K as in Lemma 89 and sufficiently large X ,

$$|E_\phi \cap [1, X]| \leq 2^K + 2CX \left\{ \phi(\log(K \log 2))^{-\kappa} + (K+2)^{-1} \right\}. \quad (150)$$

The target with $\Omega(\log \log n)$ in place of $\phi(\log \log n)$ therefore has no more exceptions eventually. The constructed smaller target is $n^{o(1)}$. No monotonicity, growth upper bound or effective modulus of divergence is assumed for Ω .

Proof. For each integer $j \geq 1$, choose a real x_j such that

$$x_j \geq e^{j^2}, \quad x_{j+1} \geq x_j + 1, \quad \Omega(x) \geq j \quad \text{for all } x \geq x_j.$$

The definition of divergence supplies such cutoffs recursively. Set $\phi(x) = 1$ before x_1 , and $\phi(x) = j$ on $[x_j, x_{j+1})$. This defines a nondecreasing function on the entire domain. On that interval, $1 \leq j \leq \Omega(x)$ and $j \leq \sqrt{\log x}$. Its divergence follows because every x_j is finite; moreover $\log \phi(x)/\log x \leq \frac{1}{2} \log \log x/\log x \rightarrow 0$. The construction depends on tail cutoffs for Ω ; it does not provide a universal computable modulus from arbitrary point evaluations.

For sufficiently large M put $z_M = \log(M \log 2)$ and set

$$H_\phi = a \log_2(M+2) + \kappa^{-1} \log_2 \log(M+3) + \log_2 \phi(z_M), \quad L_M = \lceil H_\phi \rceil.$$

Then $L_M \sim a \log_2(M+2)$. With $r_M = L_M - H_\phi \in [0, 1)$, the exact buffer is

$$\Delta_M = \kappa \log_2 \phi(z_M) + \kappa r_M, \quad 2^{-\Delta_M} = 2^{-\kappa r_M} \phi(z_M)^{-\kappa}. \quad (151)$$

It diverges; all the hypotheses of the shell reconstruction apply. For $n \in I_M$ we have $z_M \leq \log \log n$. Monotonicity, not an unproved regularity estimate, gives $\phi(z_M) \leq \phi(\log \log n)$. The first two target conversions in the preceding proof give

$$2^{L_M} < 2(M+2)^a (\log(M+3))^{1/\kappa} \phi(z_M) \leq 2K_1^a K_2^{1/\kappa} (\log n)^a (\log \log n)^{1/\kappa} \phi(\log \log n).$$

Use Theorem 81 with $e = 1$, absorbing its fixed shifts. Its profile is majorized by the eventually nonincreasing function $q(M) = \phi(z_M)^{-\kappa} + (M+2)^{-1}$. Lemma 89 proves (150), and its right-hand side divided by X tends to zero because $K, z_K, \phi(z_K) \rightarrow \infty$. Finally, for $u = \log n$, the logarithm of the target is at most

$$O(1) + a \log u + \kappa^{-1} \log \log u + \frac{1}{2} \log \log \log u = o(u).$$

Thus it is $n^{o(1)}$. The larger Ω -target is reached by the same witness, even when Ω itself grows too fast for that larger target to be a descent threshold. The ordinary and odd-return clocks can be transported as in the preceding proof. $\square$

Why the arguments of the slow factor are retained. The target uses $\phi(\log \log n)$; the prefix bound uses $\phi(\log(K \log 2))$, with the half-shell rank K . A nondecreasing function can have jumps, so monotonicity supplies the needed direction of comparison but not a bounded ratio between these two values. The proof above does not replace them by one another. This reconstructs the functional clause of Shaik’s Corollary 1.2, pp. 4–5 and p. 27, with a quantitative profile in terms of its chosen minorant. It does not assert a rate uniform over arbitrary Ω .

Every row uses the exact buffer $\Delta = \kappa L - \frac{1}{2} \log_2(M + 2) - \log_2 \log(M + 3)$. Write $x = M + 2$, $v = \log(M + 3)$, $w = \log \log(M + 4)$.	
$L = \lceil H \rceil$	$2^{-\Delta} / 2^{-\kappa(\lceil H \rceil - H)}$
$H = a \log_2 x + B \log_2 v$	$v^{-(\kappa B - 1)}$
$H = a \log_2 x + \kappa^{-1} \log_2 v + D \log_2 w$	$w^{-\kappa D}$
$H = a \log_2 x + \kappa^{-1} \log_2 v + \log_2 \phi(z_M)$	$\phi(z_M)^{-\kappa}$
The shell error is $C2^M q(M)$; the disjoint-shell map $(M, n) \mapsto n$ turns it into at most $2^K + 2CXq(K)$. The same finite orbit supplies the target, time and height.	

Figure 10: Exact cancellation of the critical buffer and its counting transport. Complete proofs: Lemma 89, Corollary 90, and Proposition 91. The schedules and moving shell estimate are Shaik’s [3], Corollary 1.2 and (6.20)–(6.23). The companion diagram `shaik_critical_scales.png` is reproducible.

Formal-source correspondence. In the pinned source, `MovingEndpointScalars.lean` defines exactly this integer ceiling and buffer and proves `two_rpow_neg_movingRankBuffer_eq`. `TimeoutEndpointAsymptotics.lean` supplies the shell estimate, retaining the source’s factor-two profile comparison and positive high-error exponent. `TimeoutEndpointNaturalDensity.lean`, `timeoutEndpointLiteralNaturalDensityDescent`, passes to the literal simultaneous-witness set by inclusion, and `Main.lean` exports it. The polynomial special case of the shell sum is `badCount_assembleDyadic_le_polylog_profile` in `PolylogExceptionalCount.lean`; the lemma here proves the arbitrary decreasing-profile version directly. These are source-statement and proof comparisons, not a replay of their full analytic dependency closure. The new real-variable corollaries above have complete written proofs; the existing independent 36-theorem finite core does not formalize these corollaries.

28.2 Fixed barriers: quantitative clocks for the same orbit

The fixed-barrier argument has a different use from the shrinking certificate above: its terminal rank can be a positive fraction of the initial rank. This exposes a quantitative time–descent tradeoff. We reconstruct Shaik’s Theorem 5.3 and Section 7 [3], pp. 16–18 and 27–28, before comparing his graded-clock companion (Appendix A, pp. 30–31). Write $\ell = \log 2$, retaining $g = 1 - \lambda/2$. All parameters in this subsection are fixed before the initial integer tends to infinity.

Lemma 92 (Fixed-barrier failure count and a rank-sensitive clock). *Fix $a_0 < r < 1$ and $0 < \eta < \min(r - a_0, 1)$. Put $d_\eta = c_0 \eta^2$, $c_0 = 1/(2\lambda^2)$. There are $M_0 \geq 1$ and $C > 0$, depending only on r, η , such that for all integers $M \geq L \geq M_0$ a set $G_{M,L} \subset I_M$ satisfies*

$$\frac{|I_M \setminus G_{M,L}|}{2^M} \leq C \{e^{-d_\eta M} + M(L + 1)(e^{-d_\eta L} + 2^{-L})\}. \quad (152)$$

For each $n \in G_{M,L}$, one actual shortcut path has length h , landing $y < 2^L$, and

$$h < \frac{M}{1-r}, \quad \max_{0 \leq t \leq h} T^t(n) \leq n^{1+\eta}. \quad (153)$$

If its last executed block has index j , starting the indexing at zero, then

$$gh < M - rL + \frac{\eta M}{1-r} + (1+\eta)j + 3 + \eta, \quad j \leq \frac{\log(M/L)}{\log(1/r)}. \quad (154)$$

Proof. Use W_η from Proposition 78. For $x \in I_m \cap W_\eta$, let $q = \lfloor rm \rfloor$. Once $m \geq (2+\eta)/(r-a_0-\eta)$, the upper envelope gives

$$T^m(x) < 2^{(a_0+\eta)m+1+\eta} \leq 2^{rm-1} \leq 2^q.$$

Thus the first passage to 2^q has positive duration at most m . Starting at $n_0 = n$, stop successfully when the checkpoint rank $m_i = \lfloor \log_2 n_i \rfloor$ is below L , and stop with failure when $m_i \geq L$ but $n_i \notin W_\eta$. Otherwise execute that first passage. Its landing satisfies $2^{q_i-1} < n_{i+1} \leq 2^{q_i}$, where $q_i = \lfloor rm_i \rfloor$. Consequently $m_{i+1} \leq q_i \leq rm_i < m_i$. There can be only finitely many blocks; a chain without failure reaches rank below L . The duration is less than $M \sum_{i \geq 0} r^i = M/(1-r)$. Each block's envelope bounds its height by $n_i^{1+\eta}$, and the checkpoint values strictly decrease. This proves (153) for the successful set.

To count failures, the initial failure has proportion at most $C_0 e^{-d_\eta M}$. For a later failure define $B_q = (2^{q-1}, 2^q] \cap W_\eta^c$. The half-open shell estimate, with the extra upper endpoint retained, gives

$$|B_q|/2^q \leq (C_0 e^{d_\eta}/2) e^{-d_\eta q} + 2^{-q}.$$

A failed checkpoint of rank at least L lies in a band with $q \geq L$. Every preceding block, including the block producing that checkpoint, was certified. Its concatenation is the original orbit's direct first passage to 2^q , by Proposition 76. The possible times are positive integers smaller than $M/(1-r)$; there are at most $M/(1-r) + 1$ of them. Its scaled reverse loss is less than $D_q = 2(q+2)/(3r)$. Increase M_0 until $D_q/2^q \leq 1/3$ for $q \geq M_0$. The same proposition then bounds the proportion of sources producing a failed tag at q by

$$CM(q+1)(e^{-d_\eta q} + 2^{-q}).$$

No assertion that the landing is uniformly distributed is used. For $0 < z < 1$, the exact positive series is

$$\sum_{q=L}^{\infty} (q+1)z^q = z^L \left(\frac{L+1}{1-z} + \frac{z}{(1-z)^2} \right).$$

Summing the last bound, first with $z = e^{-d_\eta}$ and then with $z = 1/2$, proves (152).

It remains to retain the rank removed in the clock. For an executed block of duration h_i , the time immediately before its first crossing satisfies

$$2^{q_i} < T^{h_i-1}(n_i) < 2^{-g(h_i-1)+(1+\eta)(m_i+1)}.$$

Since $g < 1$, this implies

$$gh_i < (1+\eta)(m_i+1) - q_i + 1.$$

Every intermediate checkpoint from which another certified block is executed has rank exactly $q_{i-1} - 1$. Indeed, the only other possibility is the value 2^{q_i-1} ; this value is not in W_η , because

at time q_{i-1} its iterate is 1, while the lower envelope equals $2^{(a_0-\eta)q_{i-1}} > 1$. Here $q_{i-1} \geq 1$ and $\eta < g < a_0$. Thus for blocks $0, \dots, j$,

$$\sum_{i=0}^j (m_i - q_i) = M - q_j - j, \quad \sum_{i=0}^j m_i < \frac{M}{1-r}.$$

Summing the block clocks gives

$$gh < M - q_j + \frac{\eta M}{1-r} + (1+\eta)j + 2 + \eta.$$

The last parent has $m_j \geq L$, so $q_j \geq rL - 1$. Also $L \leq m_j \leq r^j M$. These two inequalities give (154), with the endpoint and floor payments shown. $\square$

The rate d_η here is the explicit quadratic rate proved above. Proposition 105 below retains the full fixed-tolerance entropy rate; Corollary 107 proves its replacement in this same failure count and all its consumers. The construction, failure decomposition and common witness in the lemma are Shaik's; the retained clock telescope will also quantify the graded companion below.

Lemma 93 (Summing near the final dyadic shell). *Let $0 < \sigma < 1$, $p \geq 0$, $d > 0$, and $f(t) = t^p e^{-dt^\sigma}$ for $t \geq 1$. Suppose the exceptional subset of I_M has size at most $C2^M f(M)$ for every $M \geq M_1$. Fix $0 < \theta < \ell$ and an integer $N_0 \geq \max(M_1, 1)$ with $d\sigma N_0^{\sigma-1} \leq \theta$. For $J = \lfloor \log_2 X \rfloor \geq N_0$, the number $B(X)$ of exceptional positive integers at most X obeys*

$$B(X) \leq 2^{N_0} - 1 + \frac{C2^J J^p e^{-dJ^\sigma}}{1 - e^{\theta/2}}. \quad (155)$$

In particular

$$B(X) = O\left(X(\log X)^p \exp\left[-\frac{d}{\ell^\sigma}(\log X)^\sigma\right]\right). \quad (156)$$

The coefficient d is not decreased in the dyadic-rank estimate.

Proof. For $N_0 \leq M \leq J$, integrate the decreasing derivative $\sigma t^{\sigma-1}$:

$$d(J^\sigma - M^\sigma) \leq \theta(J - M), \quad f(M)/f(J) \leq e^{\theta(J-M)}.$$

The second inequality also uses $(M/J)^p \leq 1$. The contribution of the entire shell I_M , including the possibly incomplete final shell as an upper bound, is therefore at most $C2^J f(J)(e^{\theta/2})^{J-M}$. Sum this geometric progression and count all positive integers below 2^{N_0} by $2^{N_0} - 1$. This proves (155). For $t = (\log X)/\ell$, one has $0 \leq t - J < 1$, and $0 \leq t^\sigma - J^\sigma \leq \sigma J^{\sigma-1} \leq \sigma$. Also $2^J \leq X$ and $J^p \leq t^p$. These inequalities prove (156); its main term tends to infinity, so it absorbs the fixed startup. $\square$

Proposition 94 (The stretched target with its full exceptional power). *Fix $0 < \sigma < 1$, $\beta > 0$, and clock budgets*

$$c_T > \frac{2}{\log(4/3)}, \quad c_C > \frac{3}{\log(4/3)}, \quad c_S > \frac{1}{\log(4/3)}.$$

There are fixed r, η , $a_0 < r < 1$, $0 < \eta < \min(r - a_0, \beta/2, 1/2)$, and a retained set G such that each sufficiently large $n \in G$ reaches $\exp((\log n)^\sigma)$ in less than $c_T \log n$ shortcut steps and less

than $c_C \log n$ raw steps. For odd $n \in G$, the first odd-return landing below that target occurs in less than $c_S \log n$ returns. Each of these actual initial paths has height at most $n^{1+\beta}$. With $b = \min(c_0 \eta^2, \ell) > 0$, the complement has count

$$|[1, X] \cap \mathbb{N} \setminus G| = O\left(X(\log X)^{1+\sigma} e^{-(b/\ell)(\log X)^\sigma}\right). \quad (157)$$

Consequently the usual bound $O(Xe^{-\gamma(\log X)^\sigma})$ holds for every fixed $0 < \gamma < b/\ell$, with the same witnesses.

Proof. As $r \downarrow a_0$, the three continuous coefficients

$$\frac{1}{(1-r)\ell}, \quad \frac{1+1/\lambda}{(1-r)\ell} - \frac{1}{\lambda\ell}, \quad \frac{1}{\lambda(1-r)\ell} - \frac{1}{\lambda\ell}$$

tend respectively to the three lower clock limits in the statement. Choose r so all three are strictly below their budgets, then choose η as stated. Put

$$\nu = \ell^{\sigma-1}, \quad L_M = \lfloor \nu M^\sigma \rfloor.$$

Eventually $M \geq L_M \geq M_0$. Apply Lemma 92 with this exact integer rank. For $n \in I_M$,

$$2^{L_M} \leq \exp((M\ell)^\sigma) \leq \exp((\log n)^\sigma).$$

Moreover $L_M \geq \nu M^\sigma - 1$ and $L_M + 1 = O(M^\sigma)$. Equation (152) is at most $C' M^{1+\sigma} e^{-b\nu M^\sigma}$. Lemma 93, with $p = 1 + \sigma$ and $d = b\nu$, now proves (157), since $d/\ell^\sigma = b/\ell$. For example any integer $N_0 \geq (2d\sigma/\ell)^{1/(1-\sigma)}$, beyond the source startup, permits $\theta = \ell/2$. Absorbing the polynomial into the difference $b/\ell - \gamma > 0$ gives the last assertion.

For completeness the clock conversion uses the actual path, not a second good set. If s is the number of odd shortcut inputs, Proposition 74 gives $n = 2^h y P / 3^s$, $0 < P \leq 1$. As $n \geq 2^M$ and $y < 2^{L_M}$,

$$s \leq \frac{h - M + L_M}{\lambda}. \quad (158)$$

The raw length is $h + s$. Since $L_M = o(M)$, the three displayed coefficients and their strict margins prove the desired clocks. For an odd initial integer, complete only the decreasing even tail from y to its odd part; the number of odd returns is s . If the stated, possibly larger target is hit earlier, truncate at its first hit, which cannot increase any clock or height. Inserted raw states equal twice their following shortcut states, so the raw height is at most $2n^{1+\eta} \leq n^{1+\beta}$ eventually. The appended even tail only decreases. On odd inputs the exceptional set is its intersection with the same complement; no second exceptional-set estimate or independence premise is needed. $\square$

This proves the full exceptional power $1 - \delta = \sigma$ in Shaik's Theorem 1.3, together with its common time/landing/height witness and the raw conversion of Corollary 1.4. The theorem is already stated and proved in the manuscript. Retaining the polynomial in (157) makes the coefficient payment visible; it is not a correction or a claim of an optimal rate.

Proposition 95 (Quantitative graded descent in three clocks). *Fix $0 < \alpha < 1$, $\varepsilon > 0$, and $\beta > 0$. There is a set $G \subset \mathbb{N}_{>0}$ such that, for all sufficiently large $n \in G$, the target n^α is attained with the bounds*

clock	number of steps or returns	
shortcut	$< (2(1 - \alpha)/\log(4/3) + \varepsilon) \log n$	(159)
raw Collatz	$< (3(1 - \alpha)/\log(4/3) + \varepsilon) \log n$	
odd returns, for odd n	$< ((1 - \alpha)/\log(4/3) + \varepsilon) \log n$	

All paths through these selected landings have height at most $n^{1+\beta}$. The same retained set has the explicit exceptional bound

$$|[1, X] \cap \mathbb{N} \setminus G| = O_{\alpha, \varepsilon, \beta}(X^{1-\xi}(\log X)^2), \quad \xi = \frac{\alpha \min(c_0 \eta^2, \ell)}{\ell} \in (0, 1), \quad (160)$$

where a permissible parameter choice is

$$\begin{aligned} D &= \frac{\varepsilon g \ell}{2(1 + 1/\lambda)}, & r &= 1 - \min\{g/2, D/(4\alpha)\}, \\ \eta &= \min\{(r - a_0)/2, \beta/2, (1 - r)D/4, 1/2\}. \end{aligned} \quad (161)$$

Proof. These choices give $a_0 < r < 1$, $0 < \eta < r - a_0$, $\eta \leq \beta/2$, and

$$\mu := \alpha(1 - r) + \frac{\eta}{1 - r} \leq D/2.$$

Use the successful set of Lemma 92 with $L = \lfloor \alpha M \rfloor$, beyond $M \geq 2/\alpha$ and its startup. Then $L \geq \alpha M/2$, so the block-index remainder is bounded by $\log(2/\alpha)/\log(1/r)$, independently of M . Put

$$E = r + 3 + \eta + (1 + \eta) \frac{\log(2/\alpha)}{\log(1/r)}.$$

The floor in L and (154) give the explicit uniform clock

$$h < \frac{(1 - \alpha) + \mu}{g} M + \frac{E}{g}.$$

The target is unchanged: $y < 2^L \leq n^\alpha$. Equation (158), now with this L , gives

$$\begin{aligned} s &< \frac{1 - \alpha}{2g} M + \frac{\mu}{\lambda g} M + \frac{E}{\lambda g}, \\ h + s &< \frac{3(1 - \alpha)}{2g} M + \frac{(1 + 1/\lambda)\mu}{g} M + \frac{(1 + 1/\lambda)E}{g}. \end{aligned}$$

Here $(1/g - 1)/\lambda = 1/(2g)$, exactly. As $M\ell \leq \log n$, the largest extra leading coefficient is at most $(1 + 1/\lambda)D/(2g\ell) = \varepsilon/4$. All the constant terms are absorbed, for instance, once $M > 4(1 + 1/\lambda)E/(\varepsilon g \ell)$. Since $2g\ell = \log(4/3)$, these are precisely (159). The path expansion, odd-tail completion, possible earlier truncation, and height comparison are the same explicit maps used in the preceding proposition.

Set $b = \min(d_\eta, \ell)$. Equation (152) is bounded by $CM^2 e^{-b\alpha M}$, including the floor factor e^b . The initial term $e^{-d_\eta M}$ is absorbed because $d_\eta \geq b\alpha$. Write $a = \ell - b\alpha > 0$. For $J = \lfloor \log_2 X \rfloor$, the shell sum is at most

$$C \sum_{M=M'_0}^J M^2 e^{aM} \leq \frac{C J^2 e^{aJ}}{1 - e^{-a}},$$

plus a fixed finite startup. Since $e^{aJ} \leq X^{a/\ell} = X^{1-\xi}$ and $J^2 \leq (\log X)^2/\ell^2$, this proves (160). In particular its ratio to X tends to zero. Every constant here depends on the displayed fixed parameters; this is not uniform as α , ε , or β approaches an excluded endpoint. $\square$

Comparison with the actual formal statements. Shaik’s pinned `Main.lean` exports the quantitative stretched count for every strict power $\sigma < 1 - \delta$, without time or height in that declaration. Separate declarations retain the explicit $6.953 \log n$ shortcut witness and height, and the $10.44 \log n$ raw witness. The manuscript itself explains this distinction on pp. 29–30: the exact endpoint and combined forms are written derivations, not additional public wrapper declarations. Proposition 94 reconstructs that written derivation; a weaker exported type is not a defect in its proof.

For the graded companion, `GradedPowerDescent.lean`, `firstPassageLinearTransportGradedPower`, exports density one with the shortcut coefficient $2(1 - \alpha)/\log(4/3) + \varepsilon$. There is already a quantitative antecedent: `GradedClock.lean`, `fixedBootstrap_naturalDensityOne`, uses `bootstrapSet_powerDense` with $\chi = r/2$ and $D_c = \text{quadraticWindowDensityRate } \eta$. The count before taking the density-one projection is $C_R X^{1-D_c \chi^R}$, with $D_c \chi^R > 0$; `Bootstrap.lean` gives both the exponent and the prefactor recurrence. Thus the existence of a power-saving exceptional count for a graded clock is already present inside the source proof. Proposition 95 gives a direct fixed-barrier reconstruction with explicit ξ , height control and all three clocks. It does not assert priority or that its ξ dominates every permitted fixed-depth choice.

The shell-to-prefix map remains the counting-measure bijection $(M, n) \mapsto n$, with inverse $n \mapsto (\lfloor \log_2 n \rfloor, n)$. The successful-chain map retains the entire actual finite orbit. Raw expansion inserts $3x + 1 = 2T(x)$ at each odd shortcut input; its inverse deletes precisely those inserted states, retaining the step labels. On odd starts, cutting at odd states after completing the final even tail gives the return path. With the threshold and return valuations retained, Proposition 77 recovers the first shortcut passage. Forgetting the labels or threshold is not claimed to be invertible.

These quantitative statements do not replace the almost-bounded targets of Tao, Allikvere or Mazur by a fixed constant. They compare the same integer orbits through the path maps already proved here; their different target functions and measures remain in their original statements. The present analytic deductions are written proofs. No independent Lean certificate of these analytic closures, or of the whole source package, is claimed.

$I_M \xrightarrow[\text{certified first passages}]{W_\eta} y < 2^L, \quad gh < M - rL + \eta M/(1 - r) + (1 + \eta)j + 3 + \eta$									
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th style="padding: 5px;">terminal rank</th> <th style="padding: 5px;">landing target</th> <th style="padding: 5px;">exception count through X</th> </tr> <tr> <td style="text-align: center; padding: 5px;">$\lfloor \ell^{\sigma-1} M^\sigma \rfloor$</td> <td style="text-align: center; padding: 5px;">$e^{(\log n)^\sigma}$</td> <td style="text-align: center; padding: 5px;">$O(X(\log X)^{1+\sigma} e^{-(b/\ell)(\log X)^\sigma})$</td> </tr> <tr> <td style="text-align: center; padding: 5px;">$\lfloor \alpha M \rfloor$</td> <td style="text-align: center; padding: 5px;">n^α</td> <td style="text-align: center; padding: 5px;">$O(X^{1-\alpha b/\ell} (\log X)^2)$</td> </tr> </table>	terminal rank	landing target	exception count through X	$\lfloor \ell^{\sigma-1} M^\sigma \rfloor$	$e^{(\log n)^\sigma}$	$O(X(\log X)^{1+\sigma} e^{-(b/\ell)(\log X)^\sigma})$	$\lfloor \alpha M \rfloor$	n^α	$O(X^{1-\alpha b/\ell} (\log X)^2)$
terminal rank	landing target	exception count through X							
$\lfloor \ell^{\sigma-1} M^\sigma \rfloor$	$e^{(\log n)^\sigma}$	$O(X(\log X)^{1+\sigma} e^{-(b/\ell)(\log X)^\sigma})$							
$\lfloor \alpha M \rfloor$	n^α	$O(X^{1-\alpha b/\ell} (\log X)^2)$							
$\text{one shortcut path} \longrightarrow \begin{cases} \text{raw length } h + s, \\ \text{odd-return length } s \quad (n \text{ odd}), \end{cases} \quad s \leq (h - M + L)/\lambda.$									

Figure 11: One successful-chain construction, two rank schedules. Here $0 < \sigma, \alpha < 1$, $b = \min(c_0 \eta^2, \log 2) > 0$; the graded parameter choice is (161). The arrows are actual path expansion and odd-tail completion, not statistical resampling. Full proofs are Lemmas 92–93 and Propositions 94–95. The construction and original companion claims are due to Shaik [3], Sections 5, 7 and Appendix A.

28.3 Keeping the endpoint exponent on the original iterated sets

The preceding fixed-barrier argument stops at a prescribed rank. There is a second quantitative question: what estimate holds for the *original fixed-depth sets* in Shaik's bootstrap? These are not newly sampled shells. They are repeated inverse images under one deterministic stopped Collatz map. In `Pullback.lean`, `transportedBad_le` retains the exponent rD , with a factor M^2 ; the subsequent `transportedBad_le_paid` spends the strict gap $\chi < r$ to remove that factor. Keeping its logarithmic counterpart instead gives the following endpoint estimate. The transport mechanism is Shaik's [3], Section 5, at the pinned source revision.

Fix a source `StageSetup` with $0 < r < 1$, $0 < \eta \leq 1$ and startup rank M_0 . Explicitly, writing $Y_M = 2^{\lfloor rM \rfloor}$, its scalar conditions for $M \geq M_0$ are

$$1 < Y_M < 2^M, \quad \frac{M}{2Y_M} \leq \frac{1}{3}, \quad \rho^M (2^{M+1})^{1+\eta} \leq Y_M. \quad (162)$$

Here $W_\eta, \rho, c_0, C_0, \lambda, \ell$ are exactly the quantities already defined above; in particular $\ell = \log 2$ and $C_0 = 2e^{4c_0}$. Set $K = 2^{M_0}$, $W^* = W_\eta \cup [1, K)$, and

$$D_\eta = \frac{c_0 \eta^2}{\ell} \in (0, 1), \quad A_\eta = \frac{2C_0}{2e^{-c_0 \eta^2} - 1}.$$

For a positive integer n , put $m = \lfloor \log_2 n \rfloor$. If $m < M_0$, set $l(n) = 0$. Otherwise let $l(n)$ be the least $k \leq m$ with $T^k(n) \leq Y_m$, setting $l(n) = 0$ if no such k exists. Define

$$G(n) = T^{l(n)}(n), \quad P(S) = W^* \cap G^{-1}(S), \quad S_0 = W^*, \quad S_{R+1} = P(S_R).$$

These are the restrictions to positive integers of the source's `stageLength`, `stageMap`, `firstPassagePullback` and `bootstrapSet`. The zero value for failed bounded passage is part of the definition, not a successful landing. For $n \in W^*$ with $m \geq M_0$, the envelope and (162) prove passage by time m ; hence that failure branch is never used.

Proposition 96 (Endpoint exponent with the logarithmic cost retained). *For $S \subset \mathbb{N}_{>0}$ write $B_S(X) = |[1, X] \cap \mathbb{N} \setminus S|$. Suppose $C > 0$, $0 < D \leq D_\eta$, $p \geq 0$, and*

$$B_S(X) \leq CX^{1-D}(1 + \log X)^p \quad \text{for every integer } X \geq 1.$$

Then, for every such X ,

$$B_{P(S)}(X) \leq \Pi(C, D, p) X^{1-rD} (1 + \log X)^{p+2}, \quad \Pi(C, D, p) = \frac{\ell^{-(p+2)} (2^{rDM_0} + C_0 + \frac{5}{2} 2^D C)}{1 - 2^{rD-1}}. \quad (163)$$

Consequently, with

$$A_0 = A_\eta, \quad A_{R+1} = \Pi(A_R, D_\eta r^R, 2R), \quad (164)$$

the original sets satisfy, for every $R \geq 0$ and integer $X \geq 1$,

$$B_{S_R}(X) \leq A_R X^{1-D_\eta r^R} (1 + \log X)^{2R}. \quad (165)$$

In particular S_R has natural density one for each fixed R .

Proof. On I_M , $M \geq M_0$, a point outside $P(S)$ either fails W^* , or its actual first passage lands in $[1, Y_M] \setminus S$. The former set has at most $C_0 2^M e^{-D_\eta \ell M}$ points. Shaik's arbitrary-target transport estimate, with horizon $H = M$, bounds the latter set by

$$\frac{5}{2} M^2 \frac{2^M}{Y_M} B_S(Y_M) \leq \frac{5}{2} C M^2 2^M Y_M^{-D} (1 + \log Y_M)^p.$$

The hypotheses of that estimate are precisely the first two conditions of (162). Its target is the actual integer Y_M , not 2^{rM} . The floor gives

$$Y_M^{-D} \leq 2^D e^{-rD\ell M}, \quad 1 + \log Y_M \leq 1 + r\ell M \leq 1 + M.$$

As $D_\eta \geq rD$, the shell complement is therefore at most

$$(C_0 + \frac{5}{2} 2^D C) (1 + M)^{p+2} e^{(1-rD)\ell M}.$$

For $M < M_0$, the elementary bound 2^M is at most $2^{rDM_0} (1 + M)^{p+2} e^{(1-rD)\ell M}$. Thus the sum of these three constants covers every shell, including startup, for an arbitrary target set S .

Put $J = \lfloor \log_2 X \rfloor$ and $a = (1 - rD)\ell > 0$. Include the whole top shell to bound its possibly incomplete prefix:

$$\sum_{M=0}^J (1 + M)^{p+2} e^{aM} \leq \frac{(1 + J)^{p+2} e^{aJ}}{1 - e^{-a}}.$$

Here $e^{aJ} \leq X^{1-rD}$ and $1 + J \leq (1 + \log X)/\ell$, since $0 < \ell < 1$. This proves (163). The initial bound $B_{W^*}(X) \leq A_\eta X^{1-D_\eta}$ is Shaik's `extendedWindow_powerDense`, obtained from the same maximal barrier count by dyadic summation. The denominator defining A_η is positive because $c_0 \eta^2 < \ell$. Induction now proves (164)–(165). For fixed R , $D_\eta r^R > 0$; division by X and exponential domination of a fixed logarithmic power give density one. $\square$

There is also an explicit bound on the dependence on R . Define

$$E = 2^{rD_\eta M_0} + C_0, \quad F = \frac{5}{2} 2^{D_\eta}, \quad Q = \frac{1}{1 - 2^{rD_\eta - 1}}, \quad L = 1 + Q(E + F).$$

All are fixed by the original stage setup. Since $D_\eta r^j \leq D_\eta$,

$$A_{j+1} + 1 \leq L \ell^{-(2j+2)} (A_j + 1).$$

Indeed, $\Pi(A_j, D_\eta r^j, 2j) \leq Q \ell^{-(2j+2)} (E + F A_j)$, and $\ell^{-(2j+2)} \geq 1$. Multiplication of these R inequalities gives

$$A_R + 1 \leq (A_\eta + 1) L^R \ell^{-R(R+1)}. \tag{166}$$

Thus the estimate is available with an explicit prefactor for every finite R ; density one for a depth depending on X does not follow by treating that prefactor as fixed.

For fixed $R \geq 1$ and fixed $0 < \chi < r$, the ratio

$$\frac{X^{1-D_\eta r^R} (1 + \log X)^{2R}}{X^{1-D_\eta \chi^R}} = X^{-D_\eta (r^R - \chi^R)} (1 + \log X)^{2R} \longrightarrow 0.$$

This compares the two majorants for the *same* S_R ; it is not a lower bound for the actual exceptional count. At $R = 0$ the two exponents agree. The original pure-power theorem remains valid. The endpoint estimate keeps the polynomial that its proof deliberately paid away; neither optimality of that polynomial nor priority is asserted.

Proposition 97 (Startup is an absorbing branch, not a multiplicative loss). *For $n \in S_R$, let $z_i = G^i(n)$, $0 \leq i \leq R$, and $h_R = \sum_{i=0}^{R-1} l(z_i)$. Put $g_R = \sum_{i=0}^{R-1} r^i = (1 - r^R)/(1 - r)$, with $g_0 = 0$. Then*

$$z_R = T^{h_R}(n) \leq \max\{K, n^{r^R}\}, \quad h_R \leq g_R \log_2 n, \quad \max_{0 \leq j \leq h_R} T^j(n) \leq n^{1+\eta}. \quad (167)$$

If s_R counts odd shortcut inputs through this path, its raw expansion has the same endpoint, length $h_R + s_R$, and height at most $2n^{1+\eta}$. More explicitly,

$$s_R \leq \frac{h_R + \log_2 z_R - \log_2 n}{\lambda} \leq \frac{g_R \log_2 n + \max\{M_0, r^R \log_2 n\} - \log_2 n}{\lambda}. \quad (168)$$

For odd n , complete the final decreasing even tail and cut at odd states. This gives s_R odd returns to the odd part of z_R , with no increase of height. All these witnesses occur on the same S_R , with the count (165).

Proof. Induction in the set recurrence proves

$$n \in S_R \iff G^i(n) \in W^* \quad (0 \leq i \leq R).$$

For $z \in W^*$ with $z \geq K$, passage exists, and

$$G(z) \leq 2^{\lceil r \lceil \log_2 z \rceil \rceil} \leq z^r < z, \quad l(z) \leq \log_2 z.$$

For $z < K$, the defining branch instead gives $G(z) = z$ and $l(z) = 0$. Once that branch is entered, every later block has zero length. Before it is entered, induction gives $z_i \leq n^{r^i}$; therefore every positive block length is at most $r^i \log_2 n$. Summing proves the time bound, with no startup additive term. If startup has been entered the final state is less than K ; otherwise repeated power contraction gives $z_R \leq n^{r^R}$. This includes $R = 0$. Concatenation of the actual blocks proves $z_R = T^{h_R}(n)$.

All stage starts satisfy $z_i \leq n$. On each positive block, membership in W_η bounds its intermediate state by $\rho^j z_i^{1+\eta} \leq n^{1+\eta}$; a zero block changes nothing. This proves the height claim. The exact path identity $n = 2^{h_R} z_R P / 3^{s_R}$, $0 < P \leq 1$, gives (168). Each inserted raw state is twice its following shortcut state. Even-tail completion decreases the endpoint and adds no odd input, proving the remaining assertions. $\square$

Shaik's `Bootstrap.lean` proves the exact concatenation identity, and uses the coarser bounds $K^R n^{r^R}$ and $g_R \log_2 n + R^2 \log_2 K$. Its comment explicitly describes the former as weaker than the manuscript's bounded geometric factor. Equation (167) follows directly from the original zero-length startup branch. It is a sharpening of these displayed intermediate bounds, not a change of the stopped map or a claim to have corrected the manuscript's main theorem.

The map from $n \in S_R$ to its marked sequence of blocks is injective: its inverse reads the initial vertex n . Concatenation retains the block boundaries, and its inverse cuts at their cumulative lengths. Zero-length blocks require their multiplicities to be retained. Raw expansion is inverted by deleting the labeled inserted states. Forgetting the starting vertex, block marks or raw labels is not asserted to preserve this inverse. The landing map G itself can be many-to-one; the arbitrary-target transport estimate, rather than an injectivity assumption, controls that information loss.

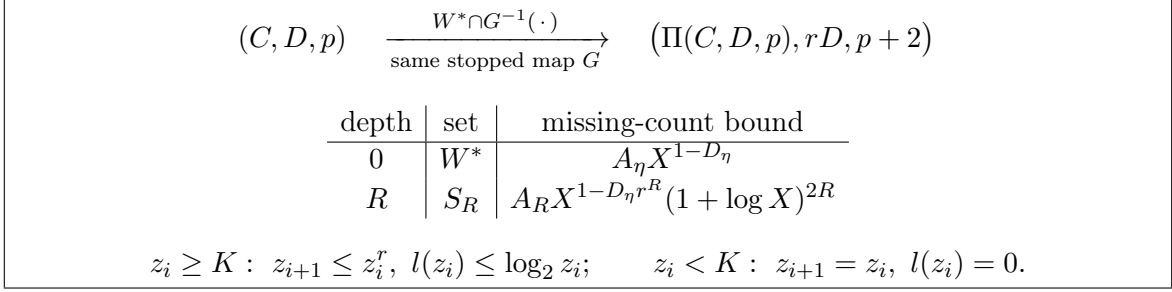


Figure 12: Endpoint exponent and absorbing startup for the original Shaik bootstrap. The upper arrow transports a missing-count estimate, not a probability law or an independent sample. The lower alternatives show why finite startup costs neither a multiplicative factor per block nor an additive time per block. The exact constants, hypotheses and proofs are Propositions 96 and 97; source: Shaik [3], `Pullback.lean` and `Bootstrap.lean`, pinned revision ef341084.

28.4 A formal joint witness, and the logarithmic schedule without a gap

The counting estimates above concern a *single* retained set of starting integers. Clock conversion must therefore transport the witness on that set; proving three unrelated density-one assertions would not express the same result. The following finite, universally quantified statement is now checked separately in Lean.

Proposition 98 (Constructive witness and exceptional-count transfer). *Let $H, S, Y, B : \mathbb{N} \rightarrow \mathbb{N}$. Write $\mathcal{T}(n)$ for the assertion that some $h \leq H(n)$ satisfies*

$$s(n, h) \leq S(n), \quad T^h(n) \leq Y(n), \quad T^j(n) \leq B(n) \quad (0 \leq j \leq h),$$

where $s(n, h)$ counts odd inputs among the first h shortcut steps. Write $\mathcal{R}(n)$ for the assertion that an actual raw Collatz path of length at most $H(n) + S(n)$ reaches $Y(n)$ and stays at most $2B(n)$, including both endpoints. Both predicates are decidable by bounded integer computations. Then $\mathcal{T}(n)$ implies $\mathcal{R}(n)$, and for every integer $X \geq 0$,

$$|\{1 \leq n \leq X : \neg \mathcal{R}(n)\}| \leq |\{1 \leq n \leq X : \neg \mathcal{T}(n)\}|. \quad (169)$$

For any decidable retained predicate P , if its actual $\mathcal{T}$ -witness is established for $n \geq N$, then

$$|\{1 \leq n \leq X : \neg \mathcal{R}(n)\}| \leq |\{1 \leq n \leq X : \neg P(n)\}| + \min\{X, (N - 1)_+\}. \quad (170)$$

For a first-passage shortcut witness of length h , the expanded raw path is also first passage, at the exact time $h + s(n, h)$.

In addition, for total maps $G, l : \mathbb{N} \rightarrow \mathbb{N}$ with $G(x) = T^{l(x)}(x)$, put $b_R(n) = \sum_{i < R} l(G^i(n))$. Then $G^R(n) = T^{b_R(n)}(n)$. If $G(n) = n$ and $l(n) = 0$, then $G^R(n) = n$ and $b_R(n) = 0$ for every R . This is the zero-length absorption used in Proposition 97.

Proof. Expand each odd shortcut step into its two raw steps. The length is exactly $h + s(n, h)$; every inserted state is twice its following shortcut state. The endpoint and height bounds follow, with the same h and starting integer. Before a shortcut first passage, the shortcut states exceed $Y(n)$; their inserted odd images are still larger, so raw first passage is retained as well. Thus failure of $\mathcal{R}$ implies failure of $\mathcal{T}$ pointwise. Summing those 0/1 indicators at the literal positive integers $1, \dots, X$ proves (169). For $n \geq N$, failure of $\mathcal{R}$ implies failure of P . The remaining positive integers number at most $\min\{X, (N - 1)_+\}$, proving (170). Finally, $T^{a+b} = T^a \circ T^b$ gives concatenation by induction on R ; the two equalities $G(n) = n, l(n) = 0$ give absorption by the same induction. $\square$

The independent module `formal_shaik_joint/JointWitness.lean` proves these statements using the identical integer maps in the existing `ClockAndRanks.lean`. Its nine reported theorems are `tagged_to_raw`, `tagged_first_passage_to_raw`, `finiteSum_mono`, `missingCount_mono`, `missingCount_startup`, `raw_exception_count`, `raw_exception_count_of_retained`, `actual_block_concat` and `zero_block_absorbs`. Bounded decidability uses no classical-choice declaration. The axiom reports contain only Lean's `propext` and `Quot.sound`; the absorption theorem uses neither. The old 36-statement source was rebuilt without modification as the sole local dependency. The compiler, 232 Init modules and nine runtime binaries are pinned; the two builds ran serially, with a 1 GiB internal cap, 2 GiB watched-tree ceiling and peak below 214 MiB. The real logarithmic bounds and exceptional-rate estimates in Propositions 94–97 are still written proofs. This new certificate verifies their integer witness/count mechanism, not their real-variable hypotheses, rates or the entire source package.

Proposition 99 (Matched landing and exceptional powers on a specified schedule). *Keep the exact stage setup and constants of Proposition 96. Fix $\omega > 0$ with*

$$a = \omega \log(1/r) \in (0, 1), \quad \sigma = 1 - a,$$

and set

$$R_M = \lfloor \omega \log(M + 4) \rfloor + 1, \quad \mathcal{G} = \{n \geq 1 : n \in S_{R_{\lfloor \log_2 n \rfloor}}\}. \quad (171)$$

The added 1 is an additional actual stopped block; it is not a change in the definition of G . For all sufficiently large $n \in \mathcal{G}$, the actual marked shortcut path reaches a value at most $\exp((\log n)^\sigma)$, has height at most $n^{1+\eta}$, and has length

$$h \leq \frac{1 - r^{R_{\lfloor \log_2 n \rfloor}}}{1 - r} \log_2 n \leq \frac{\log n}{(1 - r)\ell}. \quad (172)$$

For every fixed $0 < c < rD_\eta \ell^a$,

$$|[1, X] \cap \mathbb{N} \setminus \mathcal{G}| = O_{r, \eta, M_0, \omega, c} \left(X e^{-c(\log X)^\sigma} \right). \quad (173)$$

The raw expansion and odd-return path use the same retained set. For any fixed $\varepsilon > 0$ their respective eventual clocks are

$$\left(H + \frac{H - \ell^{-1}}{\lambda} + \varepsilon \right) \log n, \quad \left(\frac{H - \ell^{-1}}{\lambda} + \varepsilon \right) \log n, \quad H = \frac{1}{(1 - r)\ell},$$

where the odd-return statement is restricted to odd starts. The raw height is at most $2n^{1+\eta}$; consequently all three heights are at most $n^{1+\beta}$ eventually whenever $\beta > \eta$.

Proof. Put $x = M + 4$. The strict and weak sides of the floor give

$$rx^{-a} \leq r^{R_M} \leq x^{-a}, \quad R_M \leq \omega \log x + 1. \quad (174)$$

For $n \in I_M \cap S_{R_M}$, Proposition 97 gives

$$\log G^{R_M}(n) \leq \max\{M_0 \ell, r^{R_M} \log n\} \leq \max\{M_0 \ell, \ell(M + 4)^\sigma\}.$$

Both terms are at most $(\ell M)^\sigma \leq (\log n)^\sigma$ once

$$M \geq \max \left\{ 1, \frac{4}{\ell^{-a/\sigma} - 1}, \frac{(M_0 \ell)^{1/\sigma}}{\ell} \right\}. \quad (175)$$

The denominator is positive because $0 < \ell < 1$. The clock and height statements are the exact same-path bounds of that proposition.

For the exceptional count, use the all-prefix bound (165) at 2^{M+1} and divide by 2^M . With L as in (166), the shell ratio is at most

$$\begin{aligned} & 2(A_\eta + 1)L^{R_M}\ell^{-R_M(R_M+1)}(1 + (M+1)\ell)^{2R_M}e^{-D_\eta r^{R_M}\ell M} \\ & \leq \exp\{A(\log(M+4))^2 - d_0M(M+4)^{-a}\}, \end{aligned} \quad (176)$$

where

$$d_0 = rD_\eta\ell, \quad A = \log(2(A_\eta + 1)) + (\omega + 1)\log L + (\omega + 1)(\omega + 2)\log(1/\ell) + 2(\omega + 1).$$

Indeed $u = \log(M+4) > 1$, $R_M \leq (\omega + 1)u$, $R_M + 1 \leq (\omega + 2)u$, and $\log(1 + (M+1)\ell) \leq u$. Every positive term in the first line is therefore covered by Au^2 . The negative term uses the lower bound in (174); no exponential gap $\chi < r$ has been paid.

For clarity an explicit absorption cutoff is available. Fix $0 < d < d_0$, put $\Delta = d_0 - d$, and take

$$M \geq \max\left\{4, \frac{8ad_0}{\Delta}, \exp(\max\{1, 48A/(\Delta\sigma^3)\})\right\}. \quad (177)$$

Convexity gives $(1+t)^{-a} \geq 1 - at$, $t \geq 0$. Hence $d_0M(M+4)^{-a} \geq (d + \Delta/2)M^\sigma$. For $v = \log M$, $\log(M+4) \leq 2v$, and $e^{\sigma v} \geq (\sigma v)^3/6$. The last condition in (177) gives $4Av^2 \leq (\Delta/2)e^{\sigma v}$. Thus the complete shell ratio is at most e^{-dM^σ} .

The shell selection in (171) is literal: on I_M , membership in $\mathcal{G}$ is membership in S_{R_M} . Lemma 93, with polynomial power zero, therefore sums these counts to $O(Xe^{-(d/\ell^\sigma)(\log X)^\sigma})$, including the finite startup. For the stated c , choose $c\ell^\sigma < d < d_0$. Such a d exists precisely because $c < rD_\eta\ell^a$. This proves (173).

Finally $\log_2 z \leq (\log n)^\sigma/\ell$ at the selected endpoint. The exact product used in (168) yields

$$s \leq \frac{H - \ell^{-1}}{\lambda} \log n + \frac{(\log n)^\sigma}{\lambda\ell}.$$

Since $\sigma < 1$, the last term is smaller than $\varepsilon \log n$ eventually. Raw length is $h + s$. For an odd start, completing the decreasing even tail gives s odd returns. The height factor 2 is absorbed once $\log n \geq \ell/(\beta - \eta)$. These operations do not introduce a second bad set. $\square$

The source route is exact. `Scalar.lean` defines `stageCount` as $\lfloor \omega \log(M+4) \rfloor$; `GlobalAssembly.lean` uses the same shellwise set selection. `BootstrapSchedule.lean` retains the scalar negative power $1 - \omega \log(1/\chi)$ and the full quadratic-logarithmic prefactor before exporting a strict smaller power. `StretchedLogLanding.lean` uses the strict landing condition $\delta < \omega \log(1/r)$. Here the logarithmic factor in the original transport bound is kept, and one extra block removes the remaining floor factor in landing. This proves a matched endpoint on the explicitly stated schedule, not on a silently changed one. The full manuscript endpoint was already reconstructed in Proposition 94. Its direct fixed-barrier method gives a stronger available coefficient than the range $c < rD_\eta\ell^a$ just obtained. The purpose of the present calculation is to carry the original iterated-set construction, its prefactor, clocks and formal interfaces through the endpoint comparison; it is not a stronger headline theorem or a priority claim.

$R_M = \lfloor \omega \log(M+4) \rfloor + 1, \quad r(M+4)^{-a} \leq r^{R_M} \leq (M+4)^{-a}.$	
same set on shell I_M	controlled quantity
$I_M \cap S_{R_M}$	$\log z \leq \max\{M_0 \ell, \ell(M+4)^{1-a}\}$
$I_M \setminus S_{R_M}$	$ I_M \setminus S_{R_M} /2^M \leq e^{A \log^2(M+4) - r D_\eta \ell M (M+4)^{-a}}$
actual shortcut witness $\xrightarrow[\text{height} \leq 2B]{\text{length } h+s}$ actual raw witness, $B_{\mathcal{R}}(X) \leq B_{\mathcal{T}}(X).$	

Figure 13: The floor has two distinct roles: an upper bound controls the landing, a lower bound controls the surviving density exponent. One added block retains both on the declared schedule. The upper two rows are written analytic results (Proposition 99); the last arrow and positive-integer count inequality are kernel checked (Proposition 98). The original construction is Shaik’s [3], with exact module locators in the text.

28.5 Sharp shell assembly and the integer boundary

The preceding schedule calculation uses a dyadic summation step that is independent of Collatz dynamics. Shaik’s `StretchedExceptionalCount.lean` gives a useful uniform bound with coefficient $\min(c, \ell)/8$ in natural-log coordinates and prefactor $1 + 2A$. Its proof cuts the shells at half the terminal rank. Keeping the final shells individually retains more of the incoming estimate. At the linear boundary, however, the fact that one is counting integers must also be retained.

Proposition 100 (Full stretched coefficient and sharp universal prefactor). *Let $E \subset \mathbb{N}_{>0}$, $I_M = [2^M, 2^{M+1}) \cap \mathbb{N}$, $A > 0$, $c > 0$, $0 < \alpha < 1$, and $M_0 \in \mathbb{N}$. Suppose*

$$q_M := |E \cap I_M| \leq A 2^M e^{-c(M+4)^\alpha} \quad (M \geq M_0). \quad (178)$$

For $0 < \theta < \ell$, choose an integer $N_0 \geq M_0$ with $c\alpha(N_0 + 4)^{\alpha-1} \leq \theta$. Set $E_0 = |E \cap [1, 2^{N_0})|$. For every integer $X \geq 2^{N_0}$, with $J = \lfloor \log_2 X \rfloor$,

$$\begin{aligned} |E \cap [1, X]| &\leq E_0 + \frac{A 2^J e^{-c(J+4)^\alpha}}{1 - e^{\theta/2}} \\ &\leq E_0 + \frac{A}{1 - e^{\theta/2}} X \exp \left[-\frac{c}{\ell^\alpha} (\log X)^\alpha \right]. \end{aligned} \quad (179)$$

Consequently, for every $\epsilon > 0$, all sufficiently large integers X satisfy

$$|E \cap [1, X]| \leq (2A + \epsilon) X \exp \left[-\frac{c}{\ell^\alpha} (\log X)^\alpha \right]. \quad (180)$$

The constant $2A$ is the best universal asymptotic prefactor obtainable from (178) alone: there is such a set E for which the ratio to $X \exp[-c(\log_2 X)^\alpha]$ has $\limsup 2A$. This sharpness statement concerns arbitrary integer sets, not actual Collatz exceptional sets.

Proof. For $N_0 \leq M \leq J$, the derivative of ct^α is decreasing, so

$$c((J+4)^\alpha - (M+4)^\alpha) \leq \theta(J-M).$$

Thus the M -th shell contributes at most $A 2^J e^{-c(J+4)^\alpha} (e^{\theta/2})^{J-M}$. A geometric sum proves the first bound. Since $\log_2 X < J+1 < J+4$ and $2^J \leq X$, the second bound retains exactly c/ℓ^α ; the shift need not be discarded.

Here are explicit choices for the eventual assertion:

$$\theta = \log \left(1 + \frac{\epsilon}{4A + \epsilon} \right), \quad N_0 \geq \max \left\{ M_0, 0, \left(\frac{c\alpha}{\theta} \right)^{1/(1-\alpha)} - 4 \right\}, \quad N_0 \in \mathbb{N}.$$

Then $0 < \theta < \ell$ and $A/(1 - e^\theta/2) = 2A + \epsilon/2$. For $u = \log X$, if

$$X \geq 2^{N_0}, \quad u \geq (2c/\ell^\alpha)^{1/(1-\alpha)}, \quad X \geq (2E_0/\epsilon)^2,$$

then $Xe^{-c(u/\ell)^\alpha} \geq \sqrt{X}$ and $E_0 \leq (\epsilon/2)\sqrt{X}$. These give (180), including $E_0 = 0$.

For sharpness choose M_1 so $v_M = A2^M e^{-c(M+4)^\alpha} < 2^M$ for $M \geq M_1$. Let $q_M = \lfloor v_M \rfloor$ and place precisely the first q_M integers of I_M in E ; take no earlier points. Since $\ell M - c(M+4)^\alpha \rightarrow \infty$, $q_M/v_M \rightarrow 1$. For fixed $k \geq 0$, $v_{J-k}/v_J \rightarrow 2^{-k}$. For all sufficiently large $J - k$, the derivative bound above dominates this ratio by $(e^\theta/2)^k$, with $\theta < \ell$. Splitting off the finitely many earlier indices, whose total divided by v_J tends to zero, and then summing the geometric tail proves $\sum_{M=M_1}^J v_M/v_J \rightarrow 2$. Replacing v_M by its floor costs at most $J + 1$, which is $o(v_J)$. At $X_J = 2^J + q_J - 1$ the entire J -th exceptional block has been counted, and $X_J/2^J \rightarrow 1$. Moreover $\log_2 X_J = J + o(1)$, whence $(J+4)^\alpha - (\log_2 X_J)^\alpha \rightarrow 0$. Therefore

$$\frac{|E \cap [1, X_J]|}{X_J e^{-c(\log_2 X_J)^\alpha}} \rightarrow 2A.$$

The upper bound already proved gives the matching limsup. $\square$

Proposition 101 (The linear boundary has three integer regimes). *Keep (178), now with $\alpha = 1$, and put $\gamma = Ae^{-4c}$ and $E_0 = |E \cap [1, 2^{M_0}]|$. For $X \geq 2^{M_0}$, $J = \lfloor \log_2 X \rfloor$, the following hold:*

1. *If $0 < c < \ell$, then*

$$|E \cap [1, X]| \leq E_0 + \gamma \frac{e^{(\ell-c)(J+1)} - e^{(\ell-c)M_0}}{e^{\ell-c} - 1} \leq E_0 + \frac{\gamma}{1 - e^c/2} X^{1-c/\ell}.$$

2. *If $c = \ell$, then*

$$|E \cap [1, X]| \leq E_0 + \lfloor A/16 \rfloor (J - M_0 + 1).$$

In particular, if $A < 16$, there are no exceptions in any shell $M \geq M_0$.

3. *If $c > \ell$, put*

$$M_* = \max \left\{ M_0, 0, \left\lfloor \frac{\log \gamma}{c - \ell} \right\rfloor + 1 \right\}.$$

There are no exceptions in shells $M \geq M_$, and*

$$|E| \leq E_0 + \sum_{M=M_0}^{M_*-1} \left\lfloor \gamma e^{-(c-\ell)M} \right\rfloor.$$

An empty sum is zero.

Proof. The hypothesis is the integer inequality $q_M \leq \lfloor \gamma e^{(\ell-c)M} \rfloor$. For $c < \ell$, discard the floor and sum the growing geometric progression; $e^{(\ell-c)J} \leq X^{1-c/\ell}$ gives the stated prefix bound. For $c = \ell$, each shell has at most $\lfloor \gamma \rfloor = \lfloor A/16 \rfloor$ points. For $c > \ell$, every $M \geq M_*$ satisfies $\gamma e^{-(c-\ell)M} < 1$, forcing $q_M = 0$.

These distinctions are necessary. At $c = \ell$, $A = 16$, the set $\{2^M : M \geq M_0\}$ has one exception per shell and unbounded logarithmic prefix count. A nonempty finite exceptional set may also occur below 2^{M_0} when $c > \ell$. It cannot satisfy a global bound $O(X^{1-c/\ell})$, whose right side tends to zero. The correct conclusion there is finiteness, with the startup set retained. $\square$

Corollary 102 (Unclipped count on Shaik’s unshifted stage schedule). *Use the original `StageSet` `up` $p = (r, \eta, M_0)$, $0 < \chi < r < 1$, $0 < D_c \leq \min\{1, D_\eta\}$, $\chi D_c < 1$, and $\omega > 0$. Fix $0 < \delta < 1$ and $0 < \sigma < 1$ satisfying*

$$\delta < \omega \log(1/r), \quad \sigma < 1 - \omega \log(1/\chi).$$

Here, without the extra block of Proposition 99, put

$$R_M = \lfloor \omega \log(M + 4) \rfloor, \quad G = \{n \geq 1 : n \in S_{R_{\lfloor \log_2 n \rfloor}}\}, \quad \kappa = \ell D_c / 20.$$

For every $\epsilon > 0$, all sufficiently large X satisfy

$$|[1, X] \cap \mathbb{N} \setminus G| \leq (4 + \epsilon)X \exp \left[-\frac{D_c \ell^{1-\sigma}}{20} (\log X)^\sigma \right]. \quad (181)$$

Every sufficiently large $n \in G$ has one actual shortcut path to $\exp((\log n)^{1-\delta})$, of length at most $H \log n$ and height at most $n^{1+\eta}$, where $H = 1/((1-r)\ell)$. On the same starts the raw clock has leading coefficient $H + (H - \ell^{-1})/\lambda$, with arbitrary fixed positive slack. For odd starts the odd-return coefficient is $(H - \ell^{-1})/\lambda$, again with arbitrary fixed positive slack. The raw height is at most $2n^{1+\eta}$. The shortcut and raw witness predicates, and their simultaneous common-witness predicate, have positive-prefix missing counts bounded by (181) eventually. Odd-return failures are counted only in $[1, X] \cap (2\mathbb{Z} + 1)$; their absolute count obeys the same bound. In a predicate on all starts, the odd-return clause is imposed only when the start is odd.

Proof. The proof of `eventuallyShellBootstrapRatioLeStretched` in `BootstrapSchedule.lean` makes the explicit choices $c_0 = \ell D_c / 5$ and $c = c_0 / 4 = \kappa$. With the exact hypotheses above it establishes

$$|I_M \setminus S_{R_M}| / 2^M \leq 2e^{-\kappa(M+4)^\sigma}$$

for every sufficiently large M . This is a bound on the original bootstrap set, not on independent samples. Proposition 100, with $A = 2, c = \kappa, \alpha = \sigma$, proves the count for G .

The original `eventuallyShellLanding` gives the displayed target under the strict δ -inequality. Proposition 97 gives the actual clock and height for that very same S_{R_M} . Its exact odd-input identity gives

$$s \leq \frac{H - \ell^{-1}}{\lambda} \log n + \frac{(\log n)^{1-\delta}}{\lambda \ell}.$$

As $\delta > 0$, the second term is $o(\log n)$. Raw expansion and, for odd starts, decreasing even-tail completion therefore give all the asserted witnesses together. If $Q(n)$ denotes their desired conjunction, then $G \cap [N, \infty) \subseteq Q$ for a fixed integer N . Use the exceptional estimate for G first with $\epsilon/2$; the at most $N - 1$ additional startup points are absorbed by the remaining $\epsilon/2$ margin, exactly as in the explicit startup argument above. This proves the statement for the actual witness predicate without conflating separate existential witnesses. $\square$

Exact source comparison. Shaik’s `targetCertifiedSet` is literally `assembledBootstrap` $\cap \{n : \text{Good}(n)\}$. The identity on starting integers identifies these sets; on a shell where all retained starts satisfy `Good`, their exceptional sets coincide. Thus the source’s certification step loses no exceptional mass there. The numerical loss occurs later, in its generic half-rank summation:

$$\kappa \mapsto \frac{\min(\kappa, \ell)}{8} = \frac{\ell D_c}{160}.$$

The revised coefficient is $D_c \ell^{1-\sigma}/20$, a factor $8\ell^{-\sigma}$ larger for the same fixed parameters, same schedule and same incoming shell estimate. The asymptotic prefactor is $4+\epsilon$ instead of the source's 5, for any fixed $0 < \epsilon < 1$; the finite thresholds change and are retained. These are refinements of displayed majorants, not lower bounds on actual Collatz failures or a correction of the original valid estimate.

The exact $6.953 \log n$ source clock follows whenever $r < \text{clockThreshold}$, since then $H < 6.953$. The full target/exponent endpoint and stronger fixed-barrier bounds already proved in Proposition 94 remain the stronger available headline statements. The new calculation exposes what the original formal interfaces already supply quantitatively; neither a new natural-density theorem nor priority is claimed. All three proofs in this subsection are written analytic/counting arguments, not additions to the independent integer-only Lean certificate.

$\frac{ E \cap I_M }{2^M} \leq A e^{-c(M+4)^\alpha} \xrightarrow[\text{literal positive prefixes}]{0 < \alpha < 1} E \cap [1, X] \leq (2A + \epsilon) X e^{-c(\log_2 X)^\alpha}.$		
$\alpha = 1$	integer shell cap	prefix behaviour
$c < \ell$	$\lfloor A e^{-4c} e^{(\ell-c)M} \rfloor$	$O(X^{1-c/\ell})$
$c = \ell$	$\lfloor A/16 \rfloor$	$O(\log X)$
$c > \ell$	0 after M_*	finite exceptional set

Figure 14: The shell-to-prefix map retains the full incoming coefficient below the linear boundary. At the boundary the integer shell cap, not a formal continuation of the exponent, determines the conclusion. Startup is retained in every case. Proofs: Propositions 100 and 101; application: Corollary 102. Source comparison: Shaik [3], `Density.lean`, `StretchedExceptionalCount.lean` and `QuantitativeNaturalDensityDescent.lean`, revision ef341084. Sharpness concerns arbitrary integer exceptional sets.

28.6 Clipped horizons and the closed fixed-power endpoint

The fixed-depth construction can retain its original stopping times while using a smaller bound for them. This matters because a time bound is not an instruction to replace the orbit: the endpoint and the exceptional set must still belong to the same starting integer. We use Shaik's actual `StageSetup`, W^* , G , S_R from (162), and his all-prefix envelope [3]. Put $g = 1 - \frac{1}{2} \log_2 3 > 0$, as above.

Proposition 103 (The integer horizon clipped to the available envelope). *For $z \in W^*$, $z \geq K$, put*

$$M = \lfloor \log_2 z \rfloor, \quad q = \lfloor rM \rfloor, \quad H(z) = \left\lceil \frac{(1+\eta) \log_2 z - q}{g} \right\rceil, \quad L(z) = \min\{M, H(z)\}.$$

Then the original first-passage time satisfies

$$\begin{aligned} 0 < l(z) &\leq L(z), & T^{L(z)}(z) &\leq 2^q, \\ l(z) &\leq \min\{\log_2 z, a \log_2 z + c_r\}, \\ a &= \frac{1+\eta-r}{g}, & c_r &= \frac{1+r}{g} + 1. \end{aligned} \tag{182}$$

Below K , define $L(z) = 0$; both the original block and its clock remain the identity and zero. No extra startup rank is needed.

For $n \in S_R$, let ν be the first index $i < R$ with $z_i = G^i(n) < K$, or R if there is no such index. Writing $g_\nu = \sum_{i < \nu} r^i$, the same actual path obeys

$$h_R \leq \sum_{i < \nu} L(z_i) \leq \min\{g_\nu \log_2 n, ag_\nu \log_2 n + \nu c_r\}, \quad (183)$$

$$s_R \leq \min \left\{ \frac{h_R}{2} + \frac{\eta g_\nu \log n}{\log 3}, \frac{h_R + \log_2 z_R - \log_2 n}{\lambda} \right\}. \quad (184)$$

The raw clock is exactly $h_R + s_R$, not a new independent witness. The endpoint, heights and odd-return interpretation are those of Proposition 97.

Proof. The source stage hypotheses give $2^q < 2^M \leq z$; hence the numerator defining $H(z)$ is positive. In fact the terminal stage condition gives $2^{-gM} z^{1+\eta} \leq 2^q$, so $H(z) \leq M$ for this exact horizon. The same clipped-witness argument also works with the source's coarser horizon $\lceil ((1+\eta-r)M + 2 + \eta)/g \rceil$, which need not be at most M : when that horizon exceeds M , use the terminal stage witness. For the exact horizon, the existing envelope is valid at that integer time and gives

$$T^{H(z)}(z) \leq 2^{-gH(z)} z^{1+\eta} \leq 2^q.$$

Thus $L(z) = H(z) \leq M$ is a successful witness. Minimality gives $l(z) \leq L(z)$, and $l(z) > 0$ because the starting state exceeds the target. This argument never evaluates the envelope past its proved horizon. Nor does it say that passage remains successful at every later time.

To obtain the real bound, write $u = \log_2 z - M \in [0, 1)$ and $v = rM - \lfloor rM \rfloor \in [0, 1)$. Exactly,

$$(1 + \eta) \log_2 z - q = (1 + \eta - r) \log_2 z + ru + v.$$

Since $ru + v < r + 1$ and $\lceil x \rceil < x + 1$, $H(z) < a \log_2 z + c_r$. Together with $l(z) \leq M \leq \log_2 z$ this proves (182).

Before the first startup block, the original contraction gives $z_i \leq n^{r^i}$; afterwards every block has zero length. Summing the two separate upper bounds in (182), and using $a > 0$, proves (183). Notice that we sum lengths of the original blocks. Concatenating the longer $L(z_i)$ -prefixes instead would generally change the next starting state and is not used.

For a positive block of length $l_i = l(z_i)$ and odd-input count s_i , the exact affine iterate formula has a nonnegative additive term, so

$$\frac{3^{s_i} z_i}{2^{l_i}} \leq T^{l_i}(z_i) \leq 2^{-gl_i} z_i^{1+\eta}.$$

Taking logarithms yields $s_i \log 3 \leq l_i \log 3/2 + \eta \log z_i$. Sum over $i < \nu$, use $\sum_{i < \nu} \log z_i \leq g_\nu \log n$, and use exact additivity of odd-input counts to obtain the first bound in (184). The second is the whole-path affine bound (168). Both bounds concern the same integer s_R , so their minimum is valid. The raw and odd-return maps are exactly the previously proved path maps. $\square$

Corollary 104 (Closed exponent, three clocks and one exceptional count). *Fix the original stage setup with $\eta < r - a_0$, and fix $R \geq 1$. Put $\alpha = r^R$, $g_R = (1 - \alpha)/(1 - r)$, and*

$$C_T = \frac{ag_R}{\ell}, \quad C_S = \frac{C_T}{2} + \frac{\eta g_R}{\log 3}, \quad C_U = \frac{3C_T}{2} + \frac{\eta g_R}{\log 3}. \quad (185)$$

For every $n \in S_R$ with $n^\alpha \geq K$, the original path has

$$T^{h_R}(n) = z_R \leq n^\alpha, \quad h_R \leq C_T \log n + Rc_r, \quad (186)$$

$$s_R \leq C_S \log n + \frac{1}{2}Rc_r, \quad h_R + s_R \leq C_U \log n + \frac{3}{2}Rc_r. \quad (187)$$

Its shortcut and raw heights are at most $n^{1+\eta}$ and $2n^{1+\eta}$, respectively. For odd n , there is a path of s_R odd returns to the odd part of z_R , hence to an integer at most n^α . Completing its last even tail does not change the number s_R ; the raw bound above refers to the endpoint z_R before that completion.

Here is an explicit simultaneous asymptotic version. For fixed $\varepsilon_T, \varepsilon_S, \varepsilon_U > 0$, take any integer $N \geq 2$ such that

$$\log N \geq \frac{\log K}{\alpha}, \quad \log N > \max \left\{ \frac{Rc_r}{\varepsilon_T}, \frac{Rc_r}{2\varepsilon_S}, \frac{3Rc_r}{2\varepsilon_U} \right\}. \quad (188)$$

Every $n \in S_R$, $n \geq N$, satisfies the corresponding strict budgets $(C_T + \varepsilon_T) \log n$, $(C_S + \varepsilon_S) \log n$ and $(C_U + \varepsilon_U) \log n$, together with the endpoint and heights. If Q is the set of positive integers possessing these simultaneous witnesses, with the odd-return clause imposed only on odd starts, then for every integer $X \geq 1$,

$$|[1, X] \setminus Q| \leq A_R X^{1-D_\eta \alpha} (1 + \log X)^{2R} + \min\{X, N - 1\}. \quad (189)$$

The same absolute bound holds for failures among odd starts; it is not a count divided by the number of odd inputs.

Proof. The absorbing-startup bound gives $z_R \leq \max(K, n^\alpha) \leq n^\alpha$. Use $\nu \leq R$, $g_\nu \leq g_R$ and (183) to obtain the shortcut bound. The first odd-count bound in (184) gives the second line, and addition gives the raw budget. All constants are positive; $\eta < r - a_0$ also makes $a < 1$, identifying the shortened graded leading coefficient. Heights and the odd path are preserved by the already proved maps. The cutoff pays precisely the displayed additive constants. Finally, $S_R \cap [N, \infty) \subset Q$, so its complement in $[1, X]$ has at most $B_{S_R}(X) + \min(X, N - 1)$ elements. Apply (165) with $r^R = \alpha$. Restriction to odd starts can only decrease that absolute count. $\square$

For comparison with the original graded result, exact algebra gives

$$C_T = \frac{2(1-\alpha)}{\log(4/3)} + \frac{2\eta g_R}{\log(4/3)}, \quad C_S = \frac{1-\alpha}{\log(4/3)} + \eta g_R \left(\frac{1}{\log(4/3)} + \frac{1}{\log 3} \right),$$

$$C_U = \frac{3(1-\alpha)}{\log(4/3)} + \eta g_R \left(\frac{3}{\log(4/3)} + \frac{1}{\log 3} \right).$$

These are identities, not a license to set $\eta = 0$ in the density estimate. The quantitative exponent $D_\eta \alpha$ tends to zero as η tends to zero. Parameters remain fixed before X tends to infinity.

The source locators are Shaik's `HeightSensitiveClock.lean`, `heightSensitiveHorizon` and `stageLength_le_heightSensitiveHorizon`; `GradedClock.lean`, `stageClock_le_heightClock` and `fixedDepthGradedPowerDescent`; and `RawClockBudget.lean`, `oddCount_stageClock_le`, at revision `ef341084`. The source uses the larger one-block payment

$$E_{\text{src}} = \max \left\{ M_0, \left\lceil \frac{2+\eta}{r-a_0-\eta} \right\rceil \right\} + \frac{2+\eta}{g} + 1$$

and the accumulated payment $RE_{\text{src}} + aR^2M_0$. In fact $E_{\text{src}} - c_r = \max\{M_0, \lceil (2+\eta)/(r-a_0-\eta) \rceil\} + a > 0$. The new payment is just Rc_r . The original formal interface also requires $r^R < \alpha_{\text{target}}$

to absorb K^R . Corollary 104 admits equality $\alpha_{\text{target}} = r^R$, without changing the retained set. The first term in (184) similarly removes the $R^2 \log K$ charge in the source odd-count bound.

The full reading of `PowerDescent.lean` confirms that its $6.953 \log n$ fixed-power wrapper is a qualitative consequence of the stretched target. `RawNaturalDensityDescent.lean` chooses a smaller positive η , keeps the same constructed orbit within that setup, and obtains the $10.44 \log n$ raw budget. It does not assert that its existentially chosen setup is identical to the shortcut wrapper’s chosen setup. Our simultaneous estimates fix one setup from the outset. The results above refine finite bounds and a fixed-depth interface; they do not correct the source’s headline density theorem or replace Proposition 95. The real-variable arguments here are written proofs, not additional Lean exports.

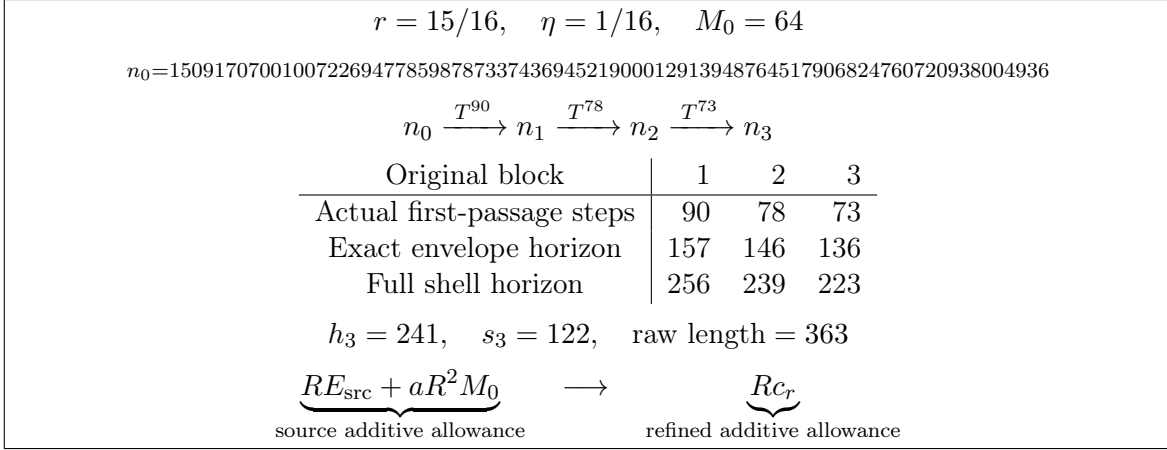


Figure 15: Clipping the envelope horizon bounds the existing first-passage clock; it does not redefine the stopped map. The table shows exact integer calculations on one retained block chain at $r = 15/16, \eta = 1/16, M_0 = 64$: actual block lengths, clipped envelope horizons and full shell horizons. The final line compares cumulative additive payments, not observed orbit times. The uniform bounds are Proposition 103 and Corollary 104. Source mechanism: Shaik [3], the three modules cited above. Reproducible data and a plotted version accompany this text.

28.7 One bit of slack retains the full entropy rate

Shaik’s entropy estimate already keeps the exact binary relative entropy. A loss remains when its adjustable barrier is held at a fixed fraction of the available tolerance. The finite-rank statement permits that fraction to vary with the rank. We make this choice explicitly, without changing the orbit-envelope set W_η or any later stopped map.

For $0 \leq u < 1/2$, set

$$\mathcal{B}(u) = \log 2 + \left(\frac{1}{2} + u\right) \log\left(\frac{1}{2} + u\right) + \left(\frac{1}{2} - u\right) \log\left(\frac{1}{2} - u\right).$$

Keep $\lambda = \log_2 3$, $a_0 = \lambda/2$ and $\ell = \log 2$. For $0 < \eta < a_0$, define

$$u_\eta = \eta/\lambda, \quad b_\eta = \mathcal{B}(u_\eta), \quad J_\eta = \frac{1}{\lambda} \log \frac{1/2 + u_\eta}{1/2 - u_\eta}, \quad (190)$$

$$m_\eta = \left\lceil \max \left\{ 1, \frac{1}{\eta}, \frac{\log_2(2(2 + \sqrt{3})) - 2}{a_0 - \eta} \right\} \right\rceil, \quad C_\eta = \max\{2e^{J_\eta}, e^{b_\eta(m_\eta-1)}\}, \quad \mathcal{D}_\eta = b_\eta/\ell.$$

Proposition 105 (Full fixed-tolerance entropy rate with explicit startup). *For every integer $M \geq 0$,*

$$|I_M \setminus W_\eta| \leq C_\eta 2^M e^{-b_\eta M}, \quad 0 < b_\eta < \ell. \quad (191)$$

For $M \geq m_\eta$ the smaller constant $2e^{J_\eta}$ suffices. Moreover

$$b_\eta > 2(\eta/\lambda)^2 = 4c_0\eta^2. \quad (192)$$

Thus the exponent improves the earlier quadratic majorant by a factor strictly greater than four at each fixed positive η . This compares exponents in upper bounds, not actual failure frequencies.

Proof. For $M \geq m_\eta$, choose the source barrier fraction and height as

$$\alpha_M = 1 - \frac{1}{\eta M} \in [0, 1), \quad h_M = \frac{\alpha_M \eta M}{\lambda} = \frac{\eta M - 1}{\lambda}.$$

These are finite-rank choices; no eventual theorem with a changing parameter is invoked. The two hypotheses of Shaik's `orbit_envelope_of_adjustableBarrier` hold because

$$(1 - \alpha_M)\eta M = 1, \quad (a_0 + \eta - 2\alpha_M\eta)M = (a_0 - \eta)M + 2 \geq \log_2(2(2 + \sqrt{3})).$$

Here the first equality pays precisely the factor two in the upper main term. To recall the deterministic reason for the second one, the affine correction under $|s_k - k/2| \leq h_M$, $k \leq M$, is at most $(2 + \sqrt{3})3^{2h_M}$. The main affine term is at most $\rho^k n 3^{h_M}$. The displayed startup inequalities respectively place these terms below one half of $\rho^k n^{1+\eta}$. The lower affine term is at least $\rho^k n 3^{-h_M} \geq \rho^k n^{1-\eta}$. These estimates follow by summing the geometric correction with ratio $\rho = \sqrt{3}/2$; see the full reconstruction above and Shaik's `Envelope.lean, normalizedCorrection_scaled_le_maximal`. Consequently failure of W_η in I_M is contained in failure of this maximal parity barrier.

For clarity, its entropy count comes from the finite Boolean tree, not an independence assumption along unbounded integer orbits. The parity bijection identifies I_M with the 2^M binary words of length M . For the walk $Y_k = 2s_k - k$, the pruned-tree cosh inequality gives

$$\#\{\max_{k \leq M} |Y_k| > 2vM\} \leq 2^{M+1} \exp\{M \log \cosh \theta - 2vM\theta\} \quad (\theta \geq 0).$$

Use $\theta = \frac{1}{2} \log((1/2+v)/(1/2-v))$ for $0 \leq v < 1/2$. Direct substitution gives $2v\theta - \log \cosh \theta = \mathcal{B}(v)$; the value at $v = 0$ is also included. This is exactly `EntropyBarrier.lean, barrierHitCount_le_binaryEntropy`. Here $v = v_M = u_\eta - 1/(\lambda M) \in [0, u_\eta]$. Since

$$\mathcal{B}'(u) = \log \frac{1/2 + u}{1/2 - u}, \quad 0 \leq \mathcal{B}'(u) \leq \mathcal{B}'(u_\eta) \quad (0 \leq u \leq u_\eta),$$

integration yields

$$M\mathcal{B}(v_M) \geq M\mathcal{B}(u_\eta) - J_\eta.$$

The exact entropy count is therefore at most $2e^{J_\eta}2^Me^{-b_\eta M}$. For $0 \leq M < m_\eta$, the trivial count 2^M is covered by the other term in $\mathcal{C}_\eta$. This proves (191) at every rank.

Finally $\mathcal{B}(0) = \mathcal{B}'(0) = 0$ and $\mathcal{B}''(u) = 4/(1-4u^2) > 4$ for $0 < u < 1/2$. Two integrations prove $\mathcal{B}(u) > 2u^2$ for $u > 0$. Positivity of the binary entropy at $1/2 + u_\eta \in (1/2, 1)$ gives $b_\eta < \ell$. $\square$

All ingredients preceding the rank-dependent choice are Shaik's: `EntropyBarrier.lean`, `AdjustableEnvelope.lean` and `AdjustableBarrierDensity.lean`, revision ef341084. The latter's finite-rank theorem allows $\alpha_M \leq 1$; its eventual wrapper instead fixes $\alpha < 1$. We have used the finite-rank theorem with verified hypotheses at every rank. The auxiliary barrier changes with M , but the set W_η , the fixed tolerance η , and the integer orbit do not. The startup rank is not uniform as η approaches zero or a_0 .

Lemma 106 (The terminal profile without spending a rate margin). *For $0 < z < 1$, $r > 0$, and integer $L \geq 1$, put*

$$F_r(z, L) = z^L \left\{ \frac{1 + 3(L+2)/r}{1-z} + \frac{3z/r}{(1-z)^2} \right\}, \quad F_r(z; L, U) = F_r(z, L) - F_r(z, U+1).$$

Fix $0 < \eta < a_0$. Let r be positive rational, $1 \leq L \leq U < M$, $H \geq 0$ integer, and suppose $(q+2)/(r2^q) \leq 1/3$ for $L \leq q \leq U$. Write $\mathcal{E}$ for the source's

$$\text{generatedFirstBadSources}(M, L, U, H, r, \eta) :$$

these are starts in I_M with a certified rank chain of elapsed time $1 \leq h \leq H$, target rank $q \in [L, U]$, and endpoint in $(2^{q-1}, 2^q] \setminus W_\eta$. Then

$$\frac{|\mathcal{E}|}{2^M} \leq H \left\{ F_r(\tfrac{1}{2}; L, U) + \frac{\mathcal{C}_\eta e^{b_\eta}}{2} F_r(e^{-b_\eta}; L, U) \right\}. \quad (193)$$

Dropping the two subtracted tail terms gives a positive infinite-tail bound with $F_r(z, L)$. Neither entropy nor dyadic rate is reduced.

Proof. For $q \geq 1$, the landing band differs from the rank- $(q-1)$ half-open shell by its endpoints. Retaining its upper endpoint gives

$$\frac{|(2^{q-1}, 2^q] \setminus W_\eta|}{2^q} \leq 2^{-q} + \frac{\mathcal{C}_\eta e^{b_\eta}}{2} e^{-b_\eta q}.$$

A certified rank chain supplies a direct first-passage witness and scaled reverse loss at most $(q+2)/r$, by the original chain transport. Shaik's `FirstBadEnvelope.lean`, `generatedFirstBadSources_subset_envelope` and `generatedFirstBadSources_card_le`, therefore give

$$\frac{|\mathcal{E}|}{2^M} \leq H \sum_{q=L}^U \left(1 + \frac{3(q+2)}{r} \right) \left(2^{-q} + \frac{\mathcal{C}_\eta e^{b_\eta}}{2} e^{-b_\eta q} \right).$$

The assumptions above are exactly the positive-target, target-below-source and small-loss conditions for that bound. Different tags may correspond to the same start; this is a union upper bound, not an injective count of tags. Finally, write $q = L + j$ and sum z^j and jz^j to obtain

$$\sum_{q=L}^{\infty} \left(1 + \frac{3(q+2)}{r} \right) z^q = F_r(z, L).$$

Subtract the tail starting at $U + 1$. This proves (193), including both integer cutoffs.

The source's `TerminalProfile.lean` first replaces the multiplier by $(1 + 6/r)(q + 1)$, then `TerminalTail.lean` pays strict margins $b' < b$ and $c' < \log 2$. Those bounds remain valid. The finite expression above keeps the original multiplier and sums it exactly; it does not require or introduce either strict rate margin. $\square$

Corollary 107 (Entropy-rate propagation on the existing orbit sets). *For a source `StageSetup`, keep K, M_0, G, P, S_R unchanged. The following replacements are valid.*

First, for a positive-integer set S with $B_S(X) \leq CX^{1-D}(1 + \log X)^p$, every integer $X \geq 1$, $C > 0$, $p \geq 0$, and $0 < D \leq \mathcal{D}_\eta$, one has

$$B_{P(S)}(X) \leq \frac{\ell^{-(p+2)}(2^{rDM_0} + \mathcal{C}_\eta + \frac{5}{2}2^D C)}{1 - 2^{rD-1}} X^{1-rD}(1 + \log X)^{p+2}. \quad (194)$$

Define $\mathcal{A}_0 = \mathcal{C}_\eta / (1 - e^{b_\eta}/2)$ and define $\mathcal{A}_{R+1}$ by the coefficient in (194), using $C = \mathcal{A}_R, D = \mathcal{D}_\eta r^R, p = 2R$. Then

$$B_{S_R}(X) \leq \mathcal{A}_R X^{1-\mathcal{D}_\eta r^R} (1 + \log X)^{2R}. \quad (195)$$

In the closed-exponent conclusion (189), replace A_R, D_η by $\mathcal{A}_R, \mathcal{D}_\eta$. The cutoff, endpoint, clocks, heights and witnesses are unchanged.

Second, for the successful stopped runs $G_{M,L}$ of Lemma 92, at sufficiently large fixed startup,

$$\frac{|I_M \setminus G_{M,L}|}{2^M} \leq C\{e^{-b_\eta M} + M(L+1)(e^{-b_\eta L} + 2^{-L})\}, \quad M \geq L. \quad (196)$$

The source sets and all deterministic clock estimates in that lemma are unchanged. Therefore the same G and same parameter choices in Proposition 94 have

$$|[1, X] \setminus G| = O\left(X(\log X)^{1+\sigma} e^{-\mathcal{D}_\eta(\log X)^\sigma}\right). \quad (197)$$

The graded conclusion (160) instead has $\xi = \alpha \mathcal{D}_\eta$, with the same chosen $\alpha, \varepsilon, \beta, r, \eta$ and the same three clocks. All constants depend on the fixed parameters.

Proof. The assumption $0 < \eta < a_0$ is automatic for a source stage. Its rank M_0 is positive because $1 < Y_{M_0} < 2^{M_0}$. Taking logarithms in its terminal budget gives

$$(a_0 + \eta - r)M_0 + 1 + \eta \leq 0.$$

Thus $\eta < r - a_0 < 1 - a_0 < a_0$, using $\lambda > 1$. The proposition applies without changing any stage data.

Summing (191) over complete shells through $J = \lfloor \log_2 X \rfloor$ gives

$$B_{W^*}(X) \leq \frac{\mathcal{C}_\eta}{1 - e^{b_\eta}/2} X^{1-\mathcal{D}_\eta},$$

since $b_\eta < \ell$ and adding the startup interval only reduces the complement. On a large shell I_M , a failed pullback either fails W^* , at cost $\mathcal{C}_\eta 2^M e^{-b_\eta M}$, or lands in the prescribed bad target. The unchanged arbitrary-target transport costs at most

$$\frac{5}{2} C M^2 2^M 2^D e^{-rD\ell M} (1 + M)^p.$$

Because $rD \leq \mathcal{D}_\eta$, the initial failure is bounded at this same exponential rate. Small shells are covered by 2^{rDM_0} . The exact weighted geometric summation used in (163) proves (194); induction proves (195). The common-witness inclusion $S_R \cap [N, \infty) \subset Q$ from the closed-exponent proof is unchanged, proving its stated replacement.

For the second part, the initial failure now costs $\mathcal{C}_\eta e^{-b_\eta M}$. The later first-bad band has

$$|(2^{q-1}, 2^q] \setminus W_\eta|/2^q \leq \frac{1}{2} \mathcal{C}_\eta e^{b_\eta} e^{-b_\eta q} + 2^{-q}.$$

The endpoint term 2^{-q} has not disappeared. The original direct-first-passage transport and finite time support then give $CM(q+1)(e^{-b_\eta q} + 2^{-q})$. Use the exact positive weighted series

$$\sum_{q=L}^{\infty} (q+1)z^q = z^L \left(\frac{L+1}{1-z} + \frac{z}{(1-z)^2} \right) \quad (0 < z < 1)$$

for $z = e^{-b_\eta}$ and $z = 1/2$. This proves (196). For $L = \lfloor \ell^{\sigma-1} M^\sigma \rfloor$, its bound is $O(M^{1+\sigma} e^{-b_\eta \ell^{\sigma-1} M^\sigma})$; the rate-preserving final-shell lemma gives (197). For $L = \lfloor \alpha M \rfloor$, the bound is $O(M^2 e^{-b_\eta \alpha M})$. The geometric sum has positive exponent $\ell - \alpha b_\eta$; it gives $O(X^{1-\alpha \mathcal{D}_\eta} (\log X)^2)$. These are precisely the previous summations with the stronger seed estimate, not new dynamics or independently selected good sets. $\square$

The depth-dependent prefactor remains explicit. With

$$\mathcal{E} = 2^{r\mathcal{D}_\eta M_0} + \mathcal{C}_\eta, \quad \mathcal{F} = \frac{5}{2} 2^{\mathcal{D}_\eta}, \quad \mathcal{Q} = (1 - 2^{r\mathcal{D}_\eta - 1})^{-1}, \quad \mathcal{L} = 1 + \mathcal{Q}(\mathcal{E} + \mathcal{F}),$$

the same induction gives

$$\mathcal{A}_R + 1 \leq (\mathcal{A}_0 + 1) \mathcal{L}^R \ell^{-R(R+1)}.$$

Consequently the explicitly shifted schedule in Proposition 99, $R_M = \lfloor \omega \log(M+4) \rfloor + 1$, still has its declared extra block, and its exceptional coefficient range becomes $0 < c < r\mathcal{D}_\eta \ell^{\omega \log(1/r)}$. Indeed its shell calculation uses the initial rate only through $D_\eta r^{R_M} \ell M$, and the displayed prefactor has exactly the same $O((\log M)^2)$ logarithm. Substitute $\mathcal{D}_\eta$ there and follow the two floor bounds and final-shell summation unchanged. This does not extend the source's existing Lean wrapper, whose literal type still names its quadratic rate.

The exact entropy seed is greater than four times the earlier quadratic seed, so these displayed decay exponents improve by the same factor at fixed r, η . Prefactors and startup bounds have also been stated, and need not improve uniformly. The fixed-tolerance entropy endpoint is not the critical target endpoint A_{FP} : none of these deductions lowers the latter or proves convergence to 1. The finite-stage and analytic deductions are complete written arguments; they are not new Lean exports.

28.8 A square-root prefactor, uniformly down to a shrinking tolerance

The entropy exponent can be retained together with a square-root prefactor. The distinction between an endpoint count and a maximal walk count matters here: the orbit envelope requires every prefix of the parity word to satisfy its barrier. We first give the exact map between the two counts, then retain its parameter dependence.

Lemma 108 (Reflection with both endpoint conventions). *Let $\Omega_M = \{-1, 1\}^M$, $Y_j(w) = \sum_{i=1}^j w_i$, and let $a \geq 1$ be an integer. There is a bijection*

$$\{w : \max_{j \leq M} Y_j(w) \geq a, Y_M(w) < a\} \longleftrightarrow \{w : Y_M(w) > a\}.$$

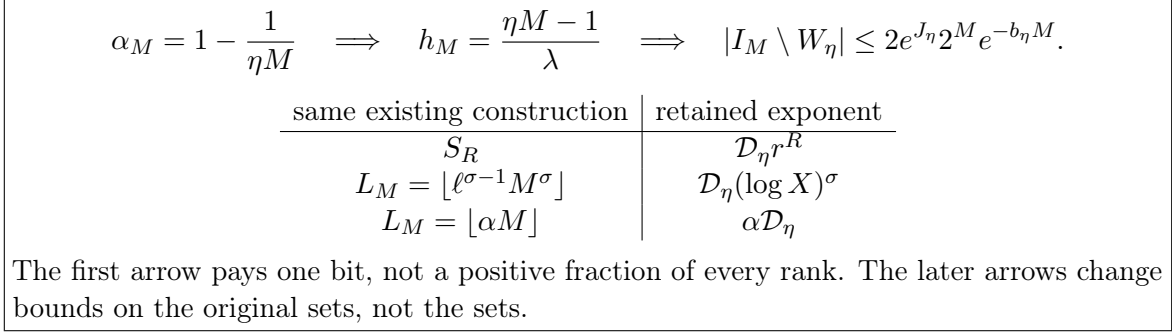


Figure 16: Where the full entropy rate goes. Proposition 105 supplies the finite startup and endpoint rate; Corollary 107 proves the three displayed transfers and preserves their common witnesses. The entropy potential and adjustable-envelope mechanism are due to Shaik [3], in the source modules identified above. A companion plot gives exact finite Boolean-tree counts; those counts test the finite mechanism, not the infinite analytic conclusions.

Writing $T(M, k) = \sum_{j=k}^M \binom{M}{j}$, with empty tails zero, $k = \lceil (M+a)/2 \rceil$ and $k^+ = \lfloor (M+a)/2 \rfloor + 1$, gives

$$\begin{aligned} \#\{\max_{j \leq M} Y_j \geq a\} &= T(M, k) + T(M, k^+), \\ \#\{\max_{j \leq M} |Y_j| \geq a\} &\leq 2\{T(M, k) + T(M, k^+)\} \leq 4T(M, k). \end{aligned} \quad (198)$$

Proof. For a word in the left-hand domain, let $t = \min\{j : Y_j = a\}$. Keep its first t increments and negate every later increment. The image has

$$Y'_j = \begin{cases} Y_j & j \leq t, \\ 2a - Y_j & j > t. \end{cases}$$

Its endpoint exceeds a , its first visit to a is still t , and repeating this operation restores every coordinate. Conversely every word ending above a visits a , because its increments are ± 1 , and reflection there ends below a . This proves both the domain and the inverse, including $a > M$, when both sets are empty. Words ending at or above a already hit a ; partition the hitting set according to its endpoint. Counting the number of positive increments gives the two tails. Negating all increments treats the lower barrier. Their union need not be disjoint, which is why the two-sided statement is an inequality. When $M+a$ is odd, $k^+ = k$; when it is even, $k^+ = k+1$. $\square$

Here is an explicit version of the factorial constant needed below. Put $c(n) = \sqrt{2n}(n/e)^n$ for $n \geq 1$. Then

$$\frac{e^{3/4}}{\sqrt{2}} c(n) \leq n! \leq \frac{e}{\sqrt{2}} c(n). \quad (199)$$

Indeed, for $s_n = n!/c(n)$, direct logarithmic subtraction and the convergent series for $\log((1+x)/(1-x))$, at $x = (2n+1)^{-1}$, give

$$\log s_n - \log s_{n+1} = \sum_{j=1}^{\infty} \frac{(2n+1)^{-2j}}{2j+1} \in \left[0, \frac{1}{4n(n+1)}\right].$$

The upper bound sums a geometric series; its denominator is $(2n+1)^2 - 1 = 4n(n+1)$. Telescoping from 1 to $n-1$ gives $0 \leq \log s_1 - \log s_n \leq (1 - 1/n)/4 \leq 1/4$, and $s_1 = e/\sqrt{2}$ proves (199). This

follows the logarithmic proof in Mathlib's `Stirling.lean`, by Moritz Firsching, Fabian Kruse and Nikolas Kuhn, specifically `log_stirlingSeq_diff_hasSum` and `log_stirlingSeq_diff_le_geo_sum`, at Shaik's pinned Mathlib revision 9837ca9d65d9de6fad1ef4381750ca688774e608. The telescoping bound above makes the constants explicit; no limit formula for π is needed.

Proposition 109 (Uniform sharp-prefactor envelope bound). *Fix $0 < \tau < a_0$, and put*

$$u_\tau = \tau/\lambda, \quad \delta_\tau = (1/2 - u_\tau)/2, \quad C_* = \sqrt{2/e},$$

$$N_\tau = \left\lceil \max \left\{ 1, \frac{\log_2(2(2 + \sqrt{3})) - 2}{a_0 - \tau}, \frac{2}{1/2 - u_\tau} \right\} \right\rceil, \quad K_\tau = \frac{4\lambda C_* e^{J_\tau}}{\sqrt{\delta_\tau}}.$$

For every $0 < \eta \leq \tau$ and every integer $M \geq N_\tau$ with $\eta M \geq 2$,

$$\frac{|I_M \setminus W_\eta|}{2^M} \leq \frac{K_\tau}{\eta \sqrt{M}} e^{-M\mathcal{B}(\eta/\lambda)}. \quad (200)$$

In particular, for fixed η , take $\tau = \eta$, and define

$$N_\eta^* = \max\{N_\eta, \lceil 2/\eta \rceil\}, \quad C_\eta^* = \max \left\{ \frac{\sqrt{2}K_\eta}{\eta}, \sqrt{N_\eta^*} e^{b_\eta(N_\eta^* - 1)} \right\}.$$

At every integer $M \geq 0$,

$$|I_M \setminus W_\eta| \leq C_\eta^* 2^M (M + 1)^{-1/2} e^{-b_\eta M}. \quad (201)$$

Proof. Use the one-bit barrier from Proposition 105. Its deterministic startup inequalities hold uniformly because $\eta \leq \tau$, $M \geq N_\tau$, and $\eta M \geq 2$. Write $u = \eta/\lambda$, $v = u - 1/(\lambda M)$. Then $v \geq u/2 > 0$, and failure of the orbit envelope implies $\max |Y_j| > 2vM$. The first excluded integer is $a = \lfloor 2vM \rfloor + 1$. With $k = \lceil (M + a)/2 \rceil$ and $p = k/M$, integer rounding gives

$$v < p - \frac{1}{2} \leq v + \frac{1}{M} \leq u + \frac{1}{M}.$$

Consequently $p \geq 1/2$ and $1 - p \geq 1/2 - u_\tau - 1/M \geq \delta_\tau > 0$. In particular $0 < k < M$; the binomial denominators below are nonzero.

For completeness, the finite upper-tail estimate used by Shaik is

$$(2k - M)T(M, k) \leq k \binom{M}{k} \quad (M/2 < k \leq M).$$

To prove it, sum $(2j - M)\binom{M}{j} = j\binom{M}{j} - (j+1)\binom{M}{j+1}$ from $j = k$ to M . The right-hand side telescopes to $k\binom{M}{k}$, and every coefficient on the left is at least $2k - M$. Thus $T(M, k) \leq (2v)^{-1}\binom{M}{k} \leq u^{-1}\binom{M}{k}$. The factorial bounds (199), applied at $M, k, M - k$, give

$$\frac{\binom{M}{k}}{2^M} \leq \frac{C_*}{\sqrt{2Mp(1-p)}} e^{-M\mathcal{B}(p-1/2)} \leq \frac{C_*}{\sqrt{M\delta_\tau}} e^{-M\mathcal{B}(v)}.$$

The exponential identity follows by cancelling the three e^{-n} factors and taking logarithms of $M^M/(k^k(M-k)^{M-k})$; it does not approximate k by a real threshold. Monotonicity of $\mathcal{B}'$ on $[0, u_\tau]$ gives $M\mathcal{B}(v) \geq M\mathcal{B}(u) - J_\tau$. Combine this with (198) and $1/u = \lambda/\eta$. This proves (200), with constants uniform over the stated two-parameter region.

For $M \geq N_\eta^*$, use $M^{-1/2} \leq \sqrt{2}(M+1)^{-1/2}$. For $M < N_\eta^*$, the trivial proportion 1 is bounded because $\sqrt{M+1}e^{b_\eta M} \leq \sqrt{N_\eta^*}e^{b_\eta(N_\eta^*-1)}$. This proves (201). $\square$

Shaik's `SharpEntropyBarrier.lean`, revision `ef341084`, proves the weighted tail estimate and the factorial-to-entropy calculation used here. Its final declaration concerns a binomial tail with fixed positive displacement and $k, M - k \geq M/4$. We used its unrestricted factorial ratio, proved the needed maximal-event comparison explicitly, and kept $1/\eta$ outside the constant. This permits a varying tolerance; substituting a varying η into a fixed- η eventual statement would not justify that conclusion.

The supplementary source `Extras/Unreachable/SharpEntropyBarrier.lean`, at the same revision, already supplies the four-times-tail maximal comparison: `integerBarrierCount_le_four_endpointUpperCount`, followed by `exists_barrierHitCount_real_normalized_le_binaryBarrierRate`. Its header identifies it as outside the canonical referee-facing roots; that is a dependency-location statement, not a mathematical dismissal. Its recursive proof uses complementary endpoint counts at y and $2a - 1 - y$, monotonicity in the initial coordinate, and induction on the remaining length. This proves the same four-times bound. Lemma 108 additionally writes the coordinate bijection and the exact two-tail identity. The maximal comparison itself is therefore already in Shaik's source; the present propagation must not be described as discovering an absent source argument.

Corollary 110 (Prefactor propagation on the original constructions). *Keep the source `Stage Setup` and all sets and witnesses from Corollary 107. Write $b = b_\eta$, $D = \mathcal{D}_\eta = b/\ell$, $C^* = C_\eta^*$, $v_r = \ell r/2$, and $Z_r(p) = v_r^{\min(p,0)}$. For $p \geq -1/2$, $0 < d \leq D$, and $B_S(X) \leq CX^{1-d}(1 + \log X)^p$, the pullback bound is*

$$B_{P(S)}(X) \leq \frac{\ell^{-(p+2)}(2^{rdM_0} + C^* + \frac{5}{2} 2^d Z_r(p)C)}{1 - 2^{rd-1}} X^{1-rd}(1 + \log X)^{p+2}. \quad (202)$$

Set

$$A_0^* = \frac{C^* \sqrt{1 + \ell}}{(1 - e^b/2)^2}, \quad p_R = 2R - \frac{1}{2},$$

and define A_{R+1}^* by the coefficient in (202), using $C = A_R^*$, $d = Dr^R$, $p = p_R$. Then

$$B_{S_R}(X) \leq A_R^* X^{1-Dr^R} (1 + \log X)^{2R-1/2}. \quad (203)$$

The closed-target bound (189) has this same right-hand side plus $\min(X, N - 1)$, with its original cutoff N .

Under the finite-chain assumptions of Lemma 106, its sharper finite profile is

$$\frac{|\mathcal{E}|}{2^M} \leq H \left\{ F_r(\tfrac{1}{2}; L, U) + \frac{C^* e^b}{2} \sum_{q=L}^U \left(1 + \frac{3(q+2)}{r} \right) q^{-1/2} e^{-bq} \right\}. \quad (204)$$

For the unchanged stopped runs $G_{M,L}$, the resulting bound is

$$\frac{|I_M \setminus G_{M,L}|}{2^M} \leq C' \{(M+1)^{-1/2} e^{-bM} + M\sqrt{L+1} e^{-bL} + M(L+1)2^{-L}\}. \quad (205)$$

Thus the same stretched and graded constructions, parameters and simultaneous witnesses satisfy, respectively,

$$|[1, X] \setminus G| = O\left(X(\log X)^{1+\sigma/2} e^{-D(\log X)^\sigma}\right), \quad (206)$$

$$|[1, X] \setminus G_\alpha| = O\left(X^{1-\alpha D} (\log X)^{3/2}\right). \quad (207)$$

Proof. Put $J = \lfloor \log_2 X \rfloor$, $z = e^b/2 < 1$. Writing $j = J - M$, $\sqrt{(J+1)/(M+1)} \leq \sqrt{j+1} \leq j+1$, so

$$\sum_{M=0}^J \frac{2^M e^{-bM}}{\sqrt{M+1}} \leq \frac{2^J e^{-bJ}}{\sqrt{J+1}(1-z)^2}.$$

Since $2^{J(1-D)} \leq X^{1-D}$ and $1 + \log X \leq (1+\ell)(J+1)$, this proves the claimed initial bound for W^* , hence S_0 .

The negative initial logarithmic power requires an explicit comparison. For a large source shell the target is $Y = 2^{\lfloor rM \rfloor} > 1$. Thus $rM \geq 1$, $\lfloor rM \rfloor \geq rM/2$, and

$$v_r(1+M) \leq 1 + \log Y \leq 1 + M, \quad (1 + \log Y)^p \leq Z_r(p)(1+M)^p.$$

The original arbitrary-target transport is therefore bounded by $\frac{5}{2} 2^d C Z_r(p) M^2 2^M e^{-rd\ell M} (1+M)^p$. Initial failure costs at most $C^* 2^M e^{-rd\ell M} (1+M)^{p+2}$: here $rd \leq D$ and $p+2 \geq 3/2 > -1/2$. Small shells cost at most $2^{rdM_0} 2^M e^{-rd\ell M} (1+M)^{p+2}$. For $a = 1 - rd > 0$, summing complete shells and using $1+J \leq \ell^{-1}(1+\log X)$ gives

$$\sum_{M=0}^J 2^{aM} (1+M)^{p+2} \leq \frac{\ell^{-(p+2)}}{1-2^{-a}} X^a (1+\log X)^{p+2}.$$

This proves (202). Induction proves (203). The established inclusion $S_R \cap [N, \infty) \subset Q$ uses the same paths as before, so its cutoff, heights and three clocks do not change.

For a landing rank $q \geq 1$, the half-open-shell estimate is now

$$|(2^{q-1}, 2^q] \setminus W_\eta|/2^q \leq 2^{-q} + \frac{1}{2} C^* e^b q^{-1/2} e^{-bq}.$$

Apply the same finite tagged union as in Lemma 106. This proves (204), with both cutoffs and the upper power-of-two endpoint retained. For $q \geq 1$ its entropy multiplier is $O_r(\sqrt{q+1})$. The elementary estimate

$$\sum_{q=L}^{\infty} \sqrt{q+1} z^q \leq \frac{\sqrt{L+1} z^L}{(1-z)^2} \quad (0 < z < 1)$$

follows from $\sqrt{1+j/(L+1)} \leq j+1$. Use it at $z = e^{-b}$, and use the exact linear-weight sum at $z = 1/2$. The original $O(M)$ time support proves (205). For $L = \lfloor \ell^{\sigma-1} M^\sigma \rfloor$, this is $O(M^{1+\sigma/2} e^{-b\ell^{\sigma-1} M^\sigma})$; the dyadic term is smaller since $b < \ell$. The existing rate-preserving final-shell summation proves (206). For $L = \lfloor \alpha M \rfloor$, sum the bound $O(M^{3/2} e^{-\alpha b M})$; the positive shell exponent $\ell - \alpha b$ proves (207). $\square$

The depth constants also remain controlled. Define

$$E = 2^{rDM_0} + C^*, \quad F = \frac{5}{2} 2^D v_r^{-1/2}, \quad Q = (1 - 2^{rD-1})^{-1}, \quad L_* = 1 + Q(E + F).$$

Since $Z_r(p_R) \leq v_r^{-1/2}$, induction in the displayed recurrence gives

$$A_R^* + 1 \leq (A_0^* + 1) L_*^R \ell^{-(R^2+R/2)}.$$

The exponent is $\sum_{j=0}^{R-1} (2j+3/2) = R^2 + R/2$. Consequently Proposition 99 retains its range $0 < c < rD\ell^{\omega \log(1/r)}$ and its actual extra block: the logarithm of this prefactor is still $O((\log M)^2)$. The new polynomial factor alone does not enlarge that coefficient range.

Corollary 111 (Uniform high-phase estimate for the shrinking schedule). *Use the high phase of Theorem 81, with amplitude $D_h > 0$ (the parameter denoted D there), cap $0 < \tau < a_0$, and $S = \lceil C_{\text{sw}} \log(M+2) \rceil$, where $D_h/\sqrt{C_{\text{sw}}} \leq \tau$. Put $V = \log(M+2)$. If*

$$V \geq \max \left\{ \frac{N_\tau}{C_{\text{sw}}}, \frac{2}{D_h \sqrt{C_{\text{sw}}}} \right\},$$

then, uniformly for every integer $m \geq S$, the original tolerance $t_{M,m} = D_h \sqrt{V/m}$ satisfies

$$\frac{|I_m \setminus W_{t_{M,m}}|}{2^m} \leq \Delta_M := \frac{K_\tau}{D_h \sqrt{V}} (M+2)^{-2D_h^2/\lambda^2}. \quad (208)$$

For a high landing band $q > S$, its bad proportion relative to 2^q is at most $2^{-q} + \Delta_M/2$. With the same actual high rank sequence and time supports as in (128), the total high failure proportion is

$$O(\Delta_M + H_M \{M\Delta_M + (S+2)2^{-S}\}) = O\left(M^{3/2}(M+2)^{-4c_0D_h^2} + M^{1/2}(\log(M+2))^{3/2}(M+2)^{-C_{\text{sw}}\ell}\right). \quad (209)$$

In particular its $O(M^{-e})$ requirement is met by

$$4c_0D_h^2 > e + \frac{3}{2}, \quad C_{\text{sw}}\ell > e + \frac{1}{2},$$

alongside the original cap, switch, startup and stage assumptions. The low timeout estimate, target threshold A_{FP} , final clock and height statements of that theorem are unchanged.

Proof. The cap is inactive by Shaik's `shrinkingHighTolerance_eq_formula`. We have $m \geq N_\tau$ and $t_{M,m}m = D_h \sqrt{V}m \geq D_h \sqrt{C_{\text{sw}}}V \geq 2$. Apply Proposition 109, using $t_{M,m}\sqrt{m} = D_h \sqrt{V}$ and $m\mathcal{B}(t_{M,m}/\lambda) \geq 2D_h^2V/\lambda^2$. This proves the uniform bound without exchanging a fixed-parameter limit with a moving parameter. Apply it at $m = q - 1$ and retain the single upper endpoint to get the landing-band bound.

The high ranks are $q_i = \lfloor rm_i \rfloor$, $m_0 = M$, $m_{i+1} = q_i - 1$, until the switch. The unchanged clipped transport multiplier is $w_i = 2^{q_i-M} + 4(q_i+2)/(3r)$. Thus the finite charge, before extending any sum, is

$$\Delta_M + H_M \sum_{i:q_i>S} w_i(2^{-q_i} + \Delta_M/2).$$

The actual ranks are distinct and $\sum_i q_i \leq rM/(1-r)$; hence $\sum_i w_i = O_r(M)$. Also $w_i \leq C_r(q_i+2)$, so summing over all integers $q > S$ gives $\sum_i w_i 2^{-q_i} = O_r((S+2)2^{-S})$. This proves the first expression in (209). Insert $H_M = O(\sqrt{MV})$, $S = O(V)$, and $2D_h^2/\lambda^2 = 4c_0D_h^2$ to obtain the second. Strict margins in the two displayed parameter inequalities absorb the remaining logarithmic factor. No change to the orbit procedure or to its low-rank timeout calculation was made. $\square$

For the source's less selective union over every high rank, the same uniform seed gives the still-valid charge $O(\Delta_M + H_M \{M^2\Delta_M + (S+2)2^{-S}\})$. Thus the earlier source-run comparison also receives the stronger seed; its larger rank union must not be silently replaced by the deterministic sequence unless that sequence is part of the construction. These estimates refine high-phase constants and logarithmic prefactors, not the low-phase rate κ which controls the critical target.

j	0	1	2	3	4	5	6	7	8	
Y_j	0	1	2	3	2	1	2	1	0	$a = 3, \quad t = 3.$
Y'_j	0	1	2	3	4	5	4	5	6	

After the first hit at $t = 3$, reflection is exactly $Y'_j = 6 - Y_j$. The endpoint $0 < 3$ becomes $6 > 3$, and the same operation reverses it.

$$\frac{e^{-M\mathcal{B}(\eta/\lambda)}}{\eta\sqrt{M}} \quad \begin{cases} \eta \text{ fixed :} & M^{-1/2} \text{ retained,} \\ \eta = D_h\sqrt{V/M} : & V^{-1/2}e^{-2D_h^2V/\lambda^2} \text{ retained.} \end{cases}$$

Figure 17: Exact reflection and the two parameter regimes. Lemma 108 proves the map and inverse; Proposition 109 supplies the omitted fixed constants and finite domain restrictions. The binomial and entropy inputs are reconstructed from Shaik [3] and the attributed Mathlib Stirling proof.

28.9 A literal finite certificate and a low-barrier estimate without an auxiliary reserve

Proposition 112 (Finite reflection certificate). *The maps and counts in Lemma 108 have a separate Lean certificate, valid for every finite word length, rather than only the enumerated test lengths. In addition to the two inverse maps, it proves that the recursive hitting and endpoint counts are the counts of their stated predicates on all Boolean words of that length.*

Proof. Here is the recurrence-level argument, including its connection with the objects being counted. Enumerate words by

$$\Omega_0 = [[]], \quad \Omega_{r+1} = [0w : w \in \Omega_r] ++ [1w : w \in \Omega_r],$$

where the two branches mean concatenation of lists, not identification of words. Induction proves that every word of length r occurs exactly once: membership is equivalent to that length, each branch is injective, and their first coordinates differ. For a walk starting at y , write $E_a(r, y)$ for its endpoint count at or above a , and $U_a(r, y)$ for its count of visits at or above a , including time zero. With $m_r = 2^r$,

$$E_a(0, y) = U_a(0, y) = \mathbf{1}_{a \leq y}, \quad E_a(r+1, y) = E_a(r, y-1) + E_a(r, y+1).$$

If $y \geq a$, $U_a(r, y) = m_r$; otherwise it has the same two-branch recursion. These are exactly predicate counts on the preceding duplicate-free enumeration, by separating the first bit. Endpoint reflection gives, for every integer y ,

$$E_a(r, y) + E_{a+1}(r, 2a - y) = m_r.$$

Induct on r : at $r = 0$ the inequalities partition the integer line; at $r + 1$ the two reflected branches exchange places. At $y = a$ this proves $U_a = E_a + E_{a+1}$. At $y < a$, both $y - 1$ and $y + 1$ are at most a , so induction in the two branches proves the same identity. The two-sided hitting count $V_a(r, y)$ is at most $U_a(r, y) + U_a(r, -y)$, by a union count.

The endpoint of a Boolean word w , with $j(w)$ positive bits, is $y + 2j(w) - r$. Thus the endpoint count is the literal tail

$$E_a(r, y) = T\left(r, \left\lfloor \frac{r + a - y + 1}{2} \right\rfloor\right), \quad T(r, k) = \#\{w \in \Omega_r : j(w) \geq k\}.$$

Here $T(r, k) = 2^r$ for $k \leq 0$, and it is zero for $k > r$. Its recursion is $T(r+1, k) = T(r, k) + T(r, k-1)$; the literal-count equality is proved, not assumed as an interpretation of that recursion. Setting $y = 0$ gives exactly the two tails and the factor-four bound in (198). The counting identity also holds at $a = 0$, with the initial visit included. The strict-endpoint bijection retains its stated hypothesis $a > 0$. $\square$

The source is `formal_shaik_reflection/Reflection.lean`. Its 31 named statements were compiled independently with Lean 4.15.0 and only the pinned `Init` dependency closure. Their axiom reports contain at most `propext` and `Quot.sound`, with no admitted proofs or classical choice. The source, object, compiler, dependency and output hashes are checked by `verify_certificate.py`. The two inverse maps are executable. This certifies the finite mechanism used above; the factorial, real-variable entropy and natural-density arguments remain written proofs. Shaik's supplementary sharp-barrier module already contains the recursive maximal comparison [3]; the reflection principle is not claimed as a new result.

The moving low parameters in Shaik's optional all-prefix route are

$$\eta_L = g - \frac{K_0}{L}, \quad r_L = 1 - \frac{K_0}{2L}, \quad \alpha_L = 1 - \frac{K_1}{L}, \quad g = 1 - a_0.$$

Here α_L is a fraction of the tolerance allocated to a Boolean barrier; it is not the Collatz threshold ratio r_L . The files `MovingLowParameters.lean` and `Alternates/AllPrefix/MovingLowParameters.lean` distinguish these roles. The latter retains the exact error

$$\frac{g}{\lambda} - \frac{\alpha_L \eta_L}{\lambda} = \frac{1}{\lambda} \left\{ \frac{K_0 + K_1 g}{L} - \frac{K_0 K_1}{L^2} \right\}.$$

The complete consumer `Alternates/AllPrefix/Implementation/MovingLowDensity.lean` already proves a square-root prefactor and an endpoint-rate loss $O(1/L)$. Its all- $q \geq L$ startup assumes $K_1 g \geq 8$. There is no need for that auxiliary reserve in the following estimate of the same envelope and landing set.

Proposition 113 (Explicit moving low density). *Fix $K_0 > 0$, put $u_* = g/\lambda$, $b_* = \mathcal{B}(u_*)$, and use N_g, K_g from Proposition 109. Define*

$$D_0 = \frac{K_0}{\lambda} \mathcal{B}'(u_*), \quad C_{\text{lo}} = \frac{K_g}{g}, \quad L_0 = \max \left\{ 2, \left\lceil \frac{2K_0}{g} \right\rceil, N_g + 1, \left\lceil \frac{4}{g} \right\rceil + 1 \right\}.$$

For every integer $L \geq L_0$ and every integer $q \geq L$, with the actual parent rank $m = q - 1$,

$$\frac{|\text{landingBad}(q, \eta_L)|}{2^q} \leq 2^{-q} + \frac{C_{\text{lo}}}{\sqrt{q-1}} \exp \left[- \left(b_* - \frac{D_0}{L} \right) (q-1) \right]. \quad (210)$$

No K_1 occurs in its hypotheses or constants. If $m \leq cL$, its second term is at most $C_{\text{lo}} e^{cD_0} m^{-1/2} e^{-b_ m}$.*

Proof. The inequalities $0 < g < a_0$ follow from $1 < \lambda < 2$. For $L \geq L_0$, $g/2 \leq \eta_L < g$, $m \geq N_g$, and $\eta_L m \geq 2$. The uniform finite estimate, used at $\tau = g$, therefore gives

$$\frac{|I_m \setminus W_{\eta_L}|}{2^m} \leq \frac{2K_g}{g\sqrt{m}} e^{-m\mathcal{B}(\eta_L/\lambda)}.$$

This is a statement about the original W_{η_L} . Its proof chooses the auxiliary fraction $1 - 1/(\eta_L m)$ separately at each actual parent rank; the uniform proposition has already verified its affine-correction and lattice conditions. It does not change the tolerance or the orbit. Since $\mathcal{B}'(u) = \log((1/2 + u)/(1/2 - u))$ is increasing,

$$0 \leq b_* - \mathcal{B}(\eta_L/\lambda) = \int_{\eta_L/\lambda}^{g/\lambda} \mathcal{B}'(v) dv \leq \frac{D_0}{L}.$$

Finally the exact landing inequality is

$$|\text{landingBad}(q, \eta_L)| \leq |I_{q-1} \setminus W_{\eta_L}| + 1.$$

Dividing by 2^q , not 2^{q-1} , proves (210). The last assertion follows from $mD_0/L \leq cD_0$. Both the parent-rank shift and the isolated upper endpoint 2^{-q} have been retained. $\square$

Propagation into the existing moving profile. Write $A_* = 1 + 6/r_*$, where the positive rational r_* is the source's common lower ratio. Retain the source moving parameter package, its low StageSetup, feasible times and separated failure envelope. Take $L \leq S < M$ after their original startup and after

$$L \geq L_1 := \max\{L_0, \lceil 2D_0/b_* \rceil\}.$$

Retain the rank-transport restriction $(q+2)/(r_* 2^q) \leq 1/3$ for $L \leq q < M$. Let H be an upper bound for the cardinalities of the actual feasible-time sets; the source gives its explicit square-root bound. The exact per-rank first-bad estimate in `MovingSharpProfile.lean` now gives the low contribution

$$HA_* \sum_{q=L}^S (q+1) \left\{ 2^{-q} + \frac{C_{\text{lo}}}{\sqrt{q-1}} e^{-(b_* - D_0/L)(q-1)} \right\}. \quad (211)$$

All summands are attached to the same source first-bad sets. Put $z_0 = e^{-b_*/2}$. A fully explicit upper bound for (211) is

$$HA_* \left\{ 2(L+2)2^{-L} + \frac{3C_{\text{lo}}e^{b_*+D_0}}{(1-z_0)^2} \sqrt{L} e^{-b_*L} \right\}. \quad (212)$$

Indeed the infinite dyadic sum is $2(L+2)2^{-L}$. For $q = L+j$, $q \geq 2$ implies $(q+1)/\sqrt{q-1} \leq 3\sqrt{q-1} \leq 3\sqrt{L}(j+1)$. Also $b_* - D_0/L \geq b_*/2$, so the remaining series is bounded by

$$3\sqrt{L} e^{-(b_* - D_0/L)(L-1)} \sum_{j=0}^{\infty} (j+1) z_0^j \leq \frac{3e^{b_*+D_0}}{(1-z_0)^2} \sqrt{L} e^{-b_*L}.$$

This proves the stated constant rather than hiding it in an eventual rate comparison.

In `MovingEndpointProfile.lean`, retain its initial/high density bound d_{hi} . Adding its actual high-rank union yields the full separated-envelope bound

$$d_{\text{hi}}\{1 + HA_*(M+1)^2\} + HA_* \left\{ 2(L+2)2^{-L} + \frac{3C_{\text{lo}}e^{b_*+D_0}}{(1-z_0)^2} \sqrt{L} e^{-b_*L} \right\}.$$

The factor M^2 belongs to this all-rank union and is not replaced by the M from a different, deterministic-rank construction. The later Proposition 116 proves the needed empty-fibre restriction

Proof. The source `Pullback.lean` defines the target by $Y_M = 2^{\lfloor r_L M \rfloor}$ and requires its terminal budget at every $M \geq M_0$. By `Parameters.lean`, `central_terminal_identity`, that field is exactly

$$(a_0 + \eta_L)M + 1 + \eta_L \leq \lfloor r_L M \rfloor. \quad (214)$$

Once $r_L > 0$, the source's natural-number floor is the usual integer floor. Write $t = K_0 M / (2L)$. Since M is an integer, $\lfloor M - t \rfloor = M - \lceil t \rceil$. Substituting $a_0 + \eta_L = 1 - K_0/L$, (214) becomes the precise scalar condition

$$F(t) := 2t - \lceil t \rceil \geq 1 + g - K_0/L. \quad (215)$$

In particular the discontinuity immediately after an integer cannot be discarded.

Suppose $K_0 \geq 4 + g$. For every $M \geq L$, $t \geq K_0/2 \geq 2 + g/2$. If $t \leq 3$, then $F(t) \geq 2t - 3 \geq 1 + g$. If $t > 3$, then $\lceil t \rceil < t + 1$, so $F(t) > t - 1 > 2 > 1 + g$; here $0 < g < 1$. Thus (215) holds, including the closed parameter endpoint $K_0 = 4 + g$.

It remains to verify every other field, not just the terminal one. For $L \geq L_{\text{st}}$, $g/2 \leq \eta_L < g < 1$ and $1/2 < r_L < 1$. At $M \geq L \geq 8$,

$$4 \leq \lfloor M/2 \rfloor \leq \lfloor r_L M \rfloor < M,$$

so $1 < Y_M < 2^M$. If $n = \lfloor M/2 \rfloor \geq 4$, then $M \leq 2n + 1$ and $3(2n + 1) \leq 2^{n+1}$. The last inequality holds at $n = 4$; increasing n adds 6 on the left and doubles the right, so induction proves it. Consequently $3M \leq 2Y_M$, which is precisely the rational horizon restriction $M/(2Y_M) \leq 1/3$. All fields of the original StageSetup are proved with $M_0 = L$.

For necessity, first take $L > K_0/g$, so that $\eta_L > 0$. If $0 < K_0 \leq 2$, choose $M = L$. Then $F(K_0/2) = K_0 - 1 \leq 1 < 1 + \eta_L$, and the budget fails. If $2 < K_0 \leq 4$, take

$$M = \lfloor 4L/K_0 \rfloor + 1, \quad L > 2K_0/g.$$

This is an integer $M \geq L$, and $2 < t \leq 2 + K_0/(2L) < 3$, so $\lceil t \rceil = 3$. Therefore

$$F(t) \leq 1 + K_0/L < 1 + g - K_0/L.$$

Finally, if $4 < K_0 < 4 + g$, choose $M = L$ and

$$L > \max\{K_0/g, K_0/(4 + g - K_0)\}.$$

Now $2 < K_0/2 < 3$, so $F(K_0/2) = K_0 - 3 < 1 + g - K_0/L$. Each case gives a failure at a parent $M \geq L$. A package with $M_0 \leq L$ would have to satisfy its terminal field at that same M , proving the claimed impossibility for this exact interface. $\square$

The density producer and the stage now have a common explicit startup. For $K_0 \geq 4 + g$ and a fixed rational $0 < r_* < 1$, define

$$L_{\text{all}} = \max \left\{ L_0, L_{\text{st}}, \lceil 2D_0/b_* \rceil, \left\lceil \frac{K_0}{2(1 - r_*)} \right\rceil, \left\lceil \frac{9}{2r_*} \right\rceil \right\}. \quad (216)$$

For every $L \geq L_{\text{all}}$, the stage above, the landing bound (210), and its explicit summed profile (212) are simultaneously valid. Indeed $r_* \leq r_L$ by the fourth entry. For $q \geq L \geq 8$, $2^q \geq q^2$ (induction from $q = 4$), whence

$$\frac{q + 2}{r_* 2^q} \leq \frac{3}{2r_* q} \leq \frac{1}{3}.$$

This verifies the low-rank smallness condition at every actual landing rank. Retain the source high data, $r_* \leq r_{\text{hi}}$, the same feasible-time bound, and the original $L \leq S < M$ when using the separated-envelope formula. The remaining high startup conditions are unaffected.

For a fixed K_0 , this is the original pair (r_L, η_L) and its literal stopped map. Choosing a different K_0 chooses a different member of that family; identity of those two orbit constructions is not asserted. The author's public `MovingEndpointParameterPackage` still encodes $K_0 > 6$, and its constructor chooses $K_0 = 7$ and $K_1 = 8/g$. The proposition proves a sharper mathematical `StageSetup` constructor, not a changed type in the author's repository or a replay of that Lean package. In particular it does not alter the canonical timeout theorem's critical target. The scalar threshold and all other fields above have complete written proofs; the separate 31-statement reflection certificate does not formalize them.

parent parameter t	$F(t) = 2t - \lceil t \rceil$	comparison with $1 + g$
2	$t = 2 : F(t) = 2$	$2 > 1 + g$
$2 + \varepsilon, 0 < \varepsilon < g/2$	$1 + 2\varepsilon$	$1 + 2\varepsilon < 1 + g$
$2 + g/2 \leq t \leq 3$	$2t - 3$	$2t - 3 \geq 1 + g$
$t > 3$	$F(t) > t - 1$	$F(t) > 2 > 1 + g$

The second row explains why checking only $M = L$ is insufficient: a later parent can lie just to the right of the downward jump at $t = 2$. The allowed starting point is $K_0/2 \geq 2 + g/2$.

Figure 19: The exact floor mechanism behind Proposition 114. The finite budget uses $1 + g - K_0/L$; $1 + g$ is its limiting level and a sufficient uniform upper comparison. The original scalar interface and moving family are due to Shaik [3].

28.11 An exact least startup, including the subthreshold family

The threshold just proved concerns the requirement $M_0 \leq L$. For smaller K_0 , the same parameters still define a valid stage after a larger startup. The exact amount can be calculated, rather than inferred from an asymptotic existence theorem.

Proposition 115 (Least startup for the original moving stage). *Fix $K_0 > 0$ and an integer $L > 2K_0/g$. Keep the original $\eta_L = g - K_0/L$ and $r_L = 1 - K_0/(2L)$. Among all `StageSetup`(r_L, η_L), the smallest possible startup is exactly*

$$Z_L = \left\lceil \frac{(4 + g)L}{K_0} - 1 \right\rceil. \quad (217)$$

Every integer $M \geq Z_L$ satisfies all original stage conditions, whereas the terminal-budget field fails at $M = Z_L - 1$. In particular, writing $\rho = (4 + g)/K_0$,

$$\rho L - 1 \leq Z_L < \rho L, \quad \frac{1}{2} 2^{\rho L} \leq 2^{Z_L} < 2^{\rho L}. \quad (218)$$

The estimate is valid with its displayed integer rounding, including when ρL is itself an integer.

Proof. Put $c = K_0/(2L)$, $d = 1 + g - 2c = 1 + \eta_L$, and $B = (3 + d)/(2c) = (4 + g)L/K_0 - 1$. The hypothesis gives $0 < c < g/4$ and $1 < d < 2$. Moreover

$$B - \frac{2}{c} = \frac{gL}{K_0} - 1 > 1, \quad B > \frac{8}{g} + 1 > 9.$$

Thus $Z_L = \lceil B \rceil \geq 10$, and the integer $M_- = Z_L - 1$ satisfies

$$\frac{2}{c} < B - 1 \leq M_- < B.$$

Consequently $2 < cM_- < (3+d)/2 < 5/2$. On this interval $\lceil cM_- \rceil = 3$, so $F(cM_-) = 2cM_- - 3 < d$: the exact terminal field fails.

On the other hand, $M \geq Z_L$ implies $cM \geq cB = (3+d)/2 > 2$. If $cM \leq 3$, then $F(cM) = 2cM - 3 \geq d$. If $cM > 3$, then $F(cM) > cM - 1 > 2 > d$. This proves the terminal field at every later parent, without assuming that F is monotone across its downward jumps.

The remaining fields follow explicitly as before: $0 < \eta_L < g < 1$, $1/2 < r_L < 1$, and $M \geq Z_L \geq 10$. Hence $1 < 2^{\lfloor r_L M \rfloor} < 2^M$, and with $n = \lfloor M/2 \rfloor \geq 5$ the inequality $3M \leq 3(2n+1) \leq 2^{n+1} \leq 2^{1+\lfloor r_L M \rfloor}$ proves the rational horizon condition. We have constructed the original stage with $M_0 = Z_L$. Any smaller startup would include M_- , where its required terminal inequality is false, proving minimality. Finally $B \leq \lceil B \rceil < B+1$, which gives both comparisons in (218). $\square$

How the two startup results fit together. For every $L > 2K_0/g$, the condition $M_0 \leq L$ is now equivalent to the exact finite test

$$K_0 \geq (4+g) \frac{L}{L+1}. \quad (219)$$

Indeed $Z_L \leq L$ is equivalent to $\rho L - 1 \leq L$, because L is an integer. Taking arbitrarily large L recovers Proposition 114, including its closed endpoint. At $K_0 = 4+g$, the least startup is $L-1$; choosing L , as in that proposition, is also valid. For $K_0 < 4+g$, the exact excess is $Z_L - L = \lceil (\rho-1)L - 1 \rceil$. Thus a valid stage still exists, but it cannot satisfy startup at most L once this excess is positive.

What is retained when the startup is changed. The parameters, target formula and full-stage rule are unchanged. Here is the exact comparison between the startup-dependent maps. Let p_a, p_b be two valid source packages with the same (r, η) and startups $a \leq b$, and let $A_{a,b} = \{n \in \mathbb{N}_{>0} : 2^a \leq n < 2^b\}$. Write W_η for the original initial-window set and

$$k(n) = \text{boundedFirstPassage}(2^{\lfloor rm \rfloor}, m, n), \quad m = \lfloor \log_2 n \rfloor.$$

Direct substitution in the source's `stageLength`, `extendedWindow` and `firstPassagePullback` gives

	<code>stageMap</code> _{p_a} (n)	<code>stageMap</code> _{p_b} (n)	
$1 \leq n < 2^a$	n	n	(220)
$n \in A_{a,b}$	$\text{orbit}_{k(n)}(n)$	n	
$n \geq 2^b$	$\text{orbit}_{k(n)}(n)$	$\text{orbit}_{k(n)}(n)$.	

For $n \in A_{a,b} \cap W_\eta$, the original terminal witness and first-passage property give $1 \leq k(n) \leq m$ and $\text{orbit}_{k(n)}(n) \leq 2^{\lfloor rm \rfloor} < n$. Thus there is an explicit strict difference at every such retained annular start; no such conclusion is imposed on other annular starts.

More precisely, for every target set $S \subseteq \mathbb{N}_{>0}$, put $P_a(S) = \text{firstPassagePullback}_{p_a}(S)$ and similarly for b . Then

$$P_a(S) \triangle P_b(S) = \{n \in A_{a,b} : [n \in S] \text{ xor } [n \in W_\eta \text{ and } \text{orbit}_{k(n)}(n) \in S]\}. \quad (221)$$

Indeed the two maps and window memberships agree outside the annulus. Inside it, the b -window contains every start and its map is the identity, while the a -window retains exactly W_η . This proves the displayed set equality and, for every real $X \geq 1$, the bound

$$|\#(P_a(S) \cap [1, X]) - \#(P_b(S) \cap [1, X])| \leq \max\{0, \min(\lfloor X \rfloor + 1, 2^b) - 2^a\} \leq 2^b - 2^a.$$

This is a proved finite-annulus relation for one-stage pullbacks, not global identity of the two stopped maps or of repeated runs. It also shows exactly why the density producer (210) is unaffected: its underlying W_η does not involve the stage startup. To use the low-profile transport with a new terminal rank $J = \max\{L, Z_L\}$, one must use that literal cutoff, its corresponding stopped run and the original feasible-time and high conditions. Merely relabeling the earlier terminal rank would lose the original target. Formula (218) quantifies the dyadic threshold cost 2^{Z_L} exactly; it is not by itself a new natural-density endpoint theorem.

parent	exact position	terminal field
$Z_L - 1$	$2 < c(Z_L - 1) < (3 + d)/2$	$F(c(Z_L - 1)) < d$
$M \geq Z_L, cM \leq 3$	$cM \geq (3 + d)/2$	$F(cM) \geq d$
$M \geq Z_L, cM > 3$	$\lceil cM \rceil < cM + 1$	$F(cM) > 2 > d$

Here $c = K_0/(2L)$, $d = 1 + g - K_0/L$. The first row forces the minimum; the other rows prove every later rank, not just the next tested value.

Figure 20: The last failed parent and the exact repair, proved in Proposition 115. The stage interface, target and moving parameters are those of Shaik [3], `Pullback.lean`, `Parameters.lean` and `MovingLowSetup.lean`.

28.12 Reachable ranks and exact time supports in the moving construction

The optional moving construction has more structure than its final all-rank estimate uses. The following comparison uses the actual `MovingRecertificationRun` of Shaik [3], not a replacement orbit procedure. The source modules `Alternates/AllPrefix/Implementation/MovingTimeSupport.lean`, `MovingFirstBad.lean`, `MovingExecution.lean`, `MovingProfile.lean` and `MovingOrbitCeiling.lean` were read together. The full-path, first-passage and duration estimates below are credited to those sources; the exact rank restriction and its finite support/profile consequences are made explicit here.

Fix the source high data P , write $r = P.r_{\text{hi}}$, and fix a valid low package p with $0 < r_{\text{lo}} < 1$ and $0 < t_{\text{lo}} < a_0$. Assume

$$2 \leq L \leq S < M, \quad \max\{P.p_{\text{hi}}.M_0, p.M_0\} \leq L.$$

The moving family of Propositions 114 and 115 supplies such a low package on its stated domain. Define a finite deterministic list by

$$\begin{aligned} m_0 &= M, \quad t_0 = t_{M,M}, \quad q_0 = \lfloor rM \rfloor, \\ m_{j+1} &= q_j - 1, \quad (t_{j+1}, q_{j+1}) = \begin{cases} (t_{M, m_{j+1}}, \lfloor r m_{j+1} \rfloor), & m_{j+1} \geq S, \\ (t_{\text{lo}}, \lfloor r_{\text{lo}} m_{j+1} \rfloor), & m_{j+1} < S, \end{cases} \end{aligned} \quad (222)$$

continuing only while $q_j > L$, and retaining the first terminal entry $q_J \leq L$. Every active parent is at least L ; the original stage fields imply $1 \leq q_j < m_j$. In particular the list terminates and its

entries are distinct. Set $\mathcal{Q} = \{q_j : q_j \geq L\}$. This includes L if that rank occurs: the original failure envelope includes it, even though the execution theorem already stops there.

Proposition 116 (Exact rank restriction and integer time enclosure). *For each $q \in [L, M - 1]$, every original moving run ending at q has $q = q_j$ for a unique index j , and follows the parent and tolerance list above up to that index. Consequently its original feasible-time set and first-bad source set are empty when $q \notin \mathcal{Q}$.*

At $q = q_j$, put

$$E_j = \sum_{i=0}^j \{t_i(m_i + 1) + 3\}, \quad d_j = M + 1 - q_j,$$

and let C_{cut} be the source's original *movingFeasibleTimeCutoff*. Define integers

$$A_j = \max \left\{ 1, \left\lceil \frac{d_j - E_j}{g} \right\rceil \right\}, \quad B_j = \min \left\{ C_{\text{cut}}, \left\lfloor \frac{d_j + E_j}{g} \right\rfloor \right\}, \quad H_j = \max\{0, B_j - A_j + 1\}. \quad (223)$$

Every original feasible time at q_j belongs to the integer interval $[A_j, B_j]$; hence its cardinality is at most H_j . Moreover, for the source's unchanged moving potential $V(q)$,

$$E_j + V(q_j) \leq V(M + 1). \quad (224)$$

Thus the enclosure retains the remaining potential as well as the exact finite prefix cost.

Proof. The first constructor fixes q_0 . At every subsequent constructor, the source's `certified_endpoint_shell_eq` applies: the certified landing has shell index exactly $q_j - 1$. For the high constructor its tolerance is less than a_0 ; for the low constructor this is the assumed $t_{l_0} < a_0$. The high/low test is therefore determined by $q_j - 1$, and the next target is the corresponding literal floor in (222). Induction on constructors proves that every run is a prefix of this list until its rank is at most L . A run that continues further has strictly smaller ranks and cannot return to $[L, M - 1]$. Strict descent proves the uniqueness of j and emptiness off $\mathcal{Q}$.

Let h_i be the actual duration of block i . The source `certified_firstPassage_duration_error` gives

$$|gh_i - (m_i - q_i)| \leq t_i m_i + t_i + 2.$$

Since $m_i = q_{i-1} - 1$ after the first block, $\sum_{i=0}^j (m_i - q_i) = M - q_j - j$. For the actual cumulative time $h = \sum_{i=0}^j h_i$, it follows that

$$|gh - (M + 1 - q_j)| \leq \sum_{i=0}^j (t_i m_i + t_i + 2) + (j + 1) = E_j.$$

The $j + 1$ is the exact total one-rank offset, not a removed boundary term. The source already proves $h > 0$, and its feasible-time definition requires $h \leq C_{\text{cut}}$. Intersecting these restrictions with the displayed real interval proves (223), including the case of an empty integer interval. Finally, apply `movingTimePotential_step` at every deterministic step. It pays precisely $t_i(m_i + 1) + 3$. The resulting finite sum telescopes to (224). $\square$

Retaining the directed corridor and every integer endpoint. The symmetric enclosure above is valid, but the source proves more before taking absolute values. Write $p_j = j + 1$, $U_j = \sum_{i=0}^j t_i m_i$, $T_j = \sum_{i=0}^j t_i(m_i + 1)$, and $c_j = M - q_j - j$. Then every original run satisfies

$$c_j - U_j \leq gh < c_j + T_j + 2p_j. \quad (225)$$

Thus define

$$A_j^{\text{dir}} = \max\{1, \lceil (c_j - U_j)/g \rceil\}, \quad B_j^{\text{dir}} = \min\{C_{\text{cut}}, \lceil (c_j + T_j + 2p_j)/g \rceil - 1\}.$$

The ceiling minus one retains the strict upper endpoint even when its argument is an integer.

There is a further exact integer improvement. For each block put

$$\begin{aligned} a_i &= \max\{1, \lceil ((1 - t_i)m_i - q_i)/g \rceil\}, \\ b_i &= \min\{m_i, \lceil ((1 + t_i)(m_i + 1) - q_i + 1)/g \rceil - 1\}. \end{aligned}$$

Set $A_j^{\text{int}} = \sum_{i=0}^j a_i$ and $B_j^{\text{int}} = \min\{C_{\text{cut}}, \sum_{i=0}^j b_i\}$. Define

$$H_j^{\text{int}} = \begin{cases} 0, & \text{some } a_i > b_i, \ i \leq j, \\ \max\{0, B_j^{\text{int}} - A_j^{\text{int}} + 1\}, & \text{otherwise.} \end{cases} \quad (226)$$

The original feasible-time cardinality is at most H_j^{int} . When no block interval is empty, these enclosures are nested:

$$[A_j^{\text{int}}, B_j^{\text{int}}] \cap \mathbb{Z} \subseteq [A_j^{\text{dir}}, B_j^{\text{dir}}] \cap \mathbb{Z} \subseteq [A_j, B_j] \cap \mathbb{Z}.$$

In particular $H_j^{\text{int}} \leq H_j$, including the empty cases.

Proof. Shaik's `ShrinkingBarrierCore.lean`, `certified_firstPassage_duration_corridor`, gives

$$(1 - t_i)m_i - q_i \leq gh_i < (1 + t_i)(m_i + 1) - q_i + 1.$$

The stage-length theorems also give $1 \leq h_i \leq m_i$. Adding the directed inequalities, using $\sum_i (m_i - q_i) = M - q_j - j$, proves (225). Dividing each local inequality by $g > 0$ and rounding with its actual strictness gives $a_i \leq h_i \leq b_i$. An empty local interval therefore excludes every original run with that prefix.

For nonempty local intervals, the addition map

$$\prod_{i=0}^j ([a_i, b_i] \cap \mathbb{Z}) \longrightarrow [\sum_i a_i, \sum_i b_i] \cap \mathbb{Z}, \quad (h_0, \dots, h_j) \longmapsto \sum_i h_i$$

is onto. To construct a preimage of h , start with all a_i , then distribute the integer excess $h - \sum_i a_i$ in order, placing at coordinate i the smaller of the remaining excess and $b_i - a_i$. The excess is exhausted because it is at most $\sum_i (b_i - a_i)$. This proves that the displayed blockwise interval is the exact image of the independent scalar corridors. Clipping at C_{cut} gives (226).

Summing the lower inequalities and the strict upper inequalities for any tuple in that product proves its containment in the directed corridor. To compare with the earlier symmetric interval, use $d_j = c_j + p_j$ and $E_j = T_j + 3p_j$. The directed real lower endpoint exceeds $d_j - E_j$ by $\sum_i t_i + 2p_j$, while its real upper endpoint is $d_j + E_j - 2p_j$. Positivity of t_i , the strict upper rounding, and the common cutoff prove the second containment. The construction forgets correlations between block durations: its surjectivity is a statement about the product of scalar intervals, not about which duration tuples are realized by Collatz orbits. $\square$

The same failure envelope, with an exact finite profile. Let F_q denote the source movingFirstBadSourcesAtRank, let $\mathcal{E}$ be its original separated failure envelope, and let I_{bad} be its initial high failure set. Choose the same positive rational $r_* \leq \min(r, r_{\text{lo}})$, and retain the source smallness condition $(q+2)/(r_*2^q) \leq 1/3$. Writing

$$w(q) = 1 + \frac{3(q+2)}{r_*}, \quad \delta_q = \frac{|\text{landingBad}(q, \text{movingTargetTolerance}(q))|}{2^q},$$

the source's exact rankwise transport, restricted only by proved empty sets, gives

$$\frac{|\mathcal{E}|}{2^M} \leq \frac{|I_{\text{bad}}|}{2^M} + \sum_{j: q_j \geq L} H_j w(q_j) \delta_{q_j}. \quad (227)$$

Indeed the all-rank union equals the union over $\mathcal{Q}$; its sets have not changed. The map on starts, elapsed times and marked endpoints is the identity. Only empty index fibres are removed. The bound then follows from the original transport theorem and Proposition 116. The same estimate holds with H_j^{int} in place of H_j , by (226); this is a bound for the same feasible-time sets. It also holds with the smaller of this bound and any other proved common feasible-time bound. This is not a claim that every integer in the enclosing interval is realized, or that the finite union is disjoint.

For the moving low parameters on the domain of Proposition 113, each low summand in (227) has the fully explicit bound

$$H_j w(q_j) \left\{ 2^{-q_j} + C_{\text{lo}}(q_j - 1)^{-1/2} e^{-(b_* - D_0/L)(q_j - 1)} \right\}.$$

The actual parent $q_j - 1$, the upper landing endpoint 2^{-q_j} , and both integer cutoffs are retained. In the underlying transport estimate one may instead use any proved common upper bound H for the actual time sets (or the smaller of H and H_j). Extending only the nonnegative low sum then recovers the bound proved in (212).

Corollary 117 (Linear high-rank charge for the original moving envelope). *Use the high-phase finite domain and Δ_M of Corollary 111. Let H be the source's common feasible-time bound, or any proved common bound for the same feasible-time sets. The initial-plus-high part of the original moving envelope has proportion at most*

$$\Delta_M + H \left[\left(1 + \frac{3(S+4)}{r_*} \right) 2^{-S} + \frac{\Delta_M}{2} C_{\text{hi}}(S) M \right], \quad C_{\text{hi}}(S) = \frac{r}{1-r} \left(\frac{3}{r_*} + \frac{1+6/r_*}{S+1} \right). \quad (228)$$

In particular the earlier $M^2 \Delta_M$ payment can be replaced by a linear one for this very envelope, not merely for a different selective construction.

For $S = \lceil C_{\text{sw}} \log(M+2) \rceil$ and the source's $H = O(\sqrt{(M+2) \log(M+2)})$, its high contribution is

$$O \left((M+2)^{3/2-Q} + (M+2)^{1/2-C_{\text{sw}} \log 2} (\log(M+2))^{3/2} \right), \quad Q = \frac{2D_h^2}{\lambda^2}. \quad (229)$$

For every fixed $e > 0$, $O(M^{-e})$ follows from

$$Q \geq e + \frac{3}{2}, \quad C_{\text{sw}} \log 2 > e + \frac{1}{2}, \quad (230)$$

with all cap, startup, switch and stage conditions retained. The first margin is closed: its square-root logarithm has cancelled.

Proof. Until the switch, $q_0 \leq rM$ and $q_{j+1} = \lfloor r(q_j - 1) \rfloor \leq rq_j$. All entries above S belong to this high prefix. Therefore

$$\sum_{j:q_j>S} q_j \leq \frac{rM}{1-r}, \quad \#\{j : q_j > S\} \leq \frac{1}{S+1} \frac{rM}{1-r}.$$

It follows, without replacing a count by an independent full-rank union, that $\sum_{j:q_j>S} w(q_j) \leq C_{\text{hi}}(S)M$. The source initial high set has proportion at most Δ_M , and its high landing band has proportion $2^{-q} + \Delta_M/2$, by the already proved uniform high seed. For the endpoint term only, extend the positive sum to all integers $q > S$. The geometric identities give

$$\sum_{q=S+1}^{\infty} 2^{-q} = 2^{-S}, \quad \sum_{q=S+1}^{\infty} q2^{-q} = (S+2)2^{-S},$$

and thus

$$\sum_{q=S+1}^{\infty} w(q)2^{-q} = \left(1 + \frac{3(S+4)}{r_*}\right) 2^{-S}.$$

Insert these in the exact rankwise transport to obtain (228). Write $V = \log(M+2)$. Since $\Delta_M = K_\tau(D_h\sqrt{V})^{-1}(M+2)^{-Q}$, the product $HM\Delta_M$ is $O((M+2)^{3/2-Q})$ with no logarithmic factor. The other product is $O((M+2)^{1/2-C_{\text{sw}}\log^2 V^{3/2}})$. The initial Δ_M is bounded by the first term for large M . This proves (229), and a strict second margin absorbs the remaining $V^{3/2}$; the first requires no such absorption and admits equality. $\square$

Propagation to the source endpoint and the earlier comparisons. The displayed M^2 estimate in the author's `MovingProfile.lean` and `MovingEndpointProfile.lean` remains correct. Proposition 116 proves the previously required restriction to reachable ranks, so (228) now sharpens that same union. The earlier warning after (209) is discharged here by an explicit source-constructor proof, not by assuming a generated distribution. The same cancellation also closes the first high-phase margin in Corollary 111 for its already proved deterministic high sequence.

For the complete moving envelope, add the low contribution in (227), or its explicit bound (212), to (228). No low-stage set, K_0 , terminal cutoff, recertification condition or orbit witness is changed. The source's terminal-or-failure proof therefore applies to the same starts, and its clock and height estimates apply to the same paths. In particular this sharper high charge does not lower the low-phase critical target A_{FP} . It refines the high-parameter requirements of the written estimate; the author's bundled parameter type with its stronger sufficient inequalities has not been edited or independently replayed.

$$M \mapsto q_0 = \lfloor rM \rfloor \mapsto m_1 = q_0 - 1 \mapsto q_1 \mapsto \cdots \mapsto q_J \leq L.$$

At each certified landing the parent is exactly $q_j - 1$. All source runs share these ranks; durations remain variable. Their prefix error is $E_j = \sum_{i \leq j} \{t_i(m_i + 1) + 3\}$, and their integer times lie in the clipped interval $[A_j, B_j]$. Ranks not on this list have empty first-bad source sets.

Figure 21: The exact rank/time separation behind Proposition 116 and Corollary 117. The run and certified landing-shell mechanism are due to Shaik [3]; source modules are identified above.

28.13 The all-prefix endpoint with the refined parameter domain

The preceding estimates concern Shaik's literal separated failure envelope. We now carry them through its execution and density assembly. This is a refinement of the sufficient parameters of the all-prefix construction, not a new claim of natural-density almost-boundedness. The source already proves the moving endpoint. In the pinned commit, the relevant consumers are `Alternates/AllPrefix/Implementation/MovingEndpointProfile.lean`, `Alternates/AllPrefix/Witness.lean`, and `MovingEndpointAssembly.lean` [3]. The distinction from the earlier timeout reconstruction is substantive: here every continued low block is recertified by its original all-prefix envelope. No timeout-good set is substituted.

Write $\ell = \log 2$, $\lambda = \log_2 3$, $a_0 = \lambda/2$, $g = 1 - a_0$, and retain

$$b_* = \mathcal{B}(g/\lambda), \quad \kappa = b_*/\ell, \quad A_{\text{FP}} = \frac{1}{2\kappa}.$$

All logarithms not explicitly subscripted are natural.

Theorem 118 (Refined all-prefix endpoint, with one retained orbit). *Fix $A_{\max} > 0$, $e > 0$, $\beta > 0$, and $c > 1/(g\ell)$. Choose a rational $a_0 < r < 1$, a real*

$$0 < \tau < \min\{r - a_0, a_0, \beta/2\},$$

and fixed K_0, D_h, C_{sw} satisfying

$$K_0 \geq 4 + g, \quad Q := 2D_h^2/\lambda^2 \geq e + \frac{3}{2}, \quad C_{\text{sw}} > \max\left\{\frac{A_{\max}}{\ell}, (D_h/\tau)^2, \frac{e + 1/2}{\ell}\right\}. \quad (231)$$

The choice $D_h > 0$ is understood. There is a fixed source run datum P with high ratio r , high cap τ , and coefficient D_h . For any sequence $A : \mathbb{N} \rightarrow \mathbb{R}$ eventually bounded above by $A_{\max}$, put

$$\begin{aligned} x_M &= M + 2, \quad V_M = \log(M + 2), \quad W_M = \log(M + 3), \\ L_M &= \lceil A(M) \log_2(M + 2) \rceil, \quad S_M = \lceil C_{\text{sw}} V_M \rceil, \\ \Delta_M^{\text{buf}} &= \kappa L_M - \frac{1}{2} \log_2(M + 2) - \log_2 \log(M + 3). \end{aligned} \quad (232)$$

Assume $\Delta_M^{\text{buf}} \rightarrow +\infty$. Then the following hold.

For every sufficiently large M , take the original low parameters

$$r_{L_M} = 1 - \frac{K_0}{2L_M}, \quad \eta_{L_M} = g - \frac{K_0}{L_M}$$

and the StageSetup with startup exactly L_M constructed above. Let $E_M \subset I_M = [2^M, 2^{M+1}) \cap \mathbb{N}$ be precisely the source movingSeparatedFailureEnvelope(P, p_{L_M}, M, L_M, S_M). There are fixed, explicitly specified positive constants C_{low} and C_{high} , independent of M and of the sequence A , such that

$$\frac{|E_M|}{2^M} \leq \delta_M := C_{\text{low}} 2^{-\Delta_M^{\text{buf}}} + C_{\text{high}} (M + 2)^{-e}. \quad (233)$$

The eventual shell cutoff may depend on the tail of A .

Every $n \in I_M \setminus E_M$ has a single shortcut path, ending at $y = T^k(n)$, with

$$\begin{aligned} y &\leq 2^{L_M} < 2(M + 2)^{A(M)} \leq C_{\text{tar}} (\log n)^{A(M)}, \\ k &\leq \frac{M + 1 + \mathcal{V}_M(M + 1)}{g} < c \log n, \\ T^j(n) &\leq n^{1+\beta/2} \quad (0 \leq j \leq k), \quad C_{\text{tar}} = 2(1/\ell + 2)^{A_{\max}}, \\ \mathcal{V}_M(M + 1) &= \frac{D_h + \tau + 3}{1 - \sqrt{r}} \sqrt{V_M(M + 1)} + (S_M + 4)S_M. \end{aligned} \quad (234)$$

All statements use that same k and path.

Set $G_M = I_M \setminus E_M$ after one common cutoff M_0 , and $G_M = \emptyset$ before it. Then $G = \bigcup_M G_M$ has natural density one. No monotonicity of A or of δ_M is required. For every integer $X \geq 2^{M_0}$, with $N = \lfloor \log_2 X \rfloor$,

$$\begin{aligned} \#([1, X] \setminus G) &\leq 2^{M_0} - 1 \\ &+ \sum_{M=M_0}^{N-1} \min\{2^M, \lfloor 2^M \delta_M \rfloor\} \\ &+ \min\{X - 2^N + 1, \lfloor 2^N \delta_N \rfloor\}. \end{aligned} \tag{235}$$

For $X < 2^{M_0}$ the count is X .

Proof. Parameters and startups. The domain is nonempty: choose a dyadic rational in $(a_0, 1)$, then a positive τ in the stated interval, choose $D_h = \lambda\sqrt{(e+3/2)/2}$, and choose C_{sw} strictly larger than the three displayed numbers. The dyadic choice is explicit: choose j with $2^{-j} < 1 - a_0$, increasing it if necessary, and take $(\lfloor 2^j a_0 \rfloor + 1)2^{-j} < 1$. Construct $p_{\text{hi}} = \text{StageSetup}(r, \tau)$ using (142). Its proof only needs $r - a_0 - \tau > 0$, not the geometric clock inequality of the older parameter package. To fill every field of `ShrinkingBarrierRunData`, take its auxiliary fixed low ratio, cap and package also equal to r, τ, p_{hi} , and its rational lower ratio equal to r . Its moving low stage is the separate p_{L_M} , not that auxiliary field. Alternatively retain the auxiliary fixed low fields of an existing valid run datum with these same high fields: none of the moving estimates here uses its auxiliary low package.

Fix any rational $0 < r_* \leq r$; it is this common transport lower ratio that appears in the constants below. For large L , $r_* \leq r_L$, even if $r_* = r$. Proposition 114 and (216) give every low-stage and transport field with startup L , and the explicit low-density bound, without a K_1 parameter. The buffer hypothesis implies $L_M \rightarrow \infty$ and eventually $A(M) > 0$. Since

$$L_M \leq A_{\text{max}} V_M / \ell + 1, \quad S_M \geq C_{\text{sw}} V_M, \quad (C_{\text{sw}} - A_{\text{max}} / \ell) V_M \rightarrow \infty,$$

eventually $2 \leq L_M < S_M < M$, $p_{\text{hi}} \cdot M_0 \leq L_M$, and all common startup conditions hold. The high-domain conditions of Corollary 111 hold once

$$V_M \geq \max\{N_\tau / C_{\text{sw}}, 2 / (D_h \sqrt{C_{\text{sw}}})\}.$$

These are finite explicit tests; a uniform effective cutoff for an arbitrary input sequence would additionally require a modulus for its asserted tail properties.

One time-support constant. Put

$$H_0 = 3 + \frac{2}{g} \left\{ \frac{D_h + \tau + 3}{1 - \sqrt{r}} + (C_{\text{sw}} + 5)^2 \right\}.$$

The source bound `eventually_movingFeasibleTimes_card_lt_sqrt` uses only the run datum and $C_{\text{sw}} \geq 0$, uniformly over the real moving low packages. It gives $H = H_0 \sqrt{x_M V_M}$. For clarity its scalar estimate follows directly from $\mathcal{V}_M(M+1)$: when $V_M \geq 1$, $S_M + 4 \leq (C_{\text{sw}} + 5)V_M$, and eventually $V_M^2 \leq \sqrt{x_M V_M}$. Thus the potential is at most $\{(D_h + \tau + 3)/(1 - \sqrt{r}) + (C_{\text{sw}} + 5)^2\} \sqrt{x_M V_M}$. An interval of real width $2\mathcal{V}_M(M+1)/g$ has at most one plus that width integer points; the source's additional ceiling is covered by the stated H_0 . The sharper directed blockwise supports remain available, but are not needed for this uniform bound.

Low contribution with its constant. Use C_{lo}, D_0, L_0 from Proposition 113, and set

$$\begin{aligned} d &= \ell - b_* > 0, & C_d &= 2 + (2d \exp(1))^{-1/2}, \\ C_s &= \frac{3C_{\text{lo}} \exp(b_* + D_0)}{(1 - \exp(-b_*/2))^2}, \\ C_{\text{low}} &= H_0(1 + 6/r_*)(2C_d + C_s)\sqrt{A_{\text{max}}/\ell + 1}. \end{aligned}$$

For $L \geq 1$,

$$(L + 2)2^{-L} \leq C_d \sqrt{L} e^{-b_* L},$$

because the quotient is $(\sqrt{L} + 2/\sqrt{L})e^{-dL}$, and $\sup_{t>0} \sqrt{t} e^{-dt} = (2d \exp(1))^{-1/2}$. Since $V_M \leq W_M$, $W_M \geq 1$, and $L_M \leq (A_{\text{max}}/\ell + 1)W_M$, the explicit low series (212) is at most

$$C_{\text{low}} \sqrt{x_M} W_M e^{-b_* L_M} = C_{\text{low}} 2^{-\Delta_M^{\text{buf}}}.$$

The last equality is exact, retaining the $\log(M + 3)$ in the buffer.

Initial and high contribution with its constant. Put

$$\begin{aligned} \mu &= C_{\text{sw}} \ell - e - \frac{1}{2} > 0, & \chi_0 &= \frac{r}{1 - r} (1 + 9/r_*), \\ B_{\text{sw}} &= 1 + (3C_{\text{sw}} + 15)/r_*, \\ C_{\text{high}} &= \frac{K_\tau}{D_h} \left(1 + \frac{H_0 \chi_0}{2} \right) + H_0 B_{\text{sw}} \left(\frac{3}{2\mu \exp(1)} \right)^{3/2}. \end{aligned}$$

The initial seed is $d_M^{\text{hi}} = K_\tau (D_h \sqrt{V_M})^{-1} x_M^{-Q}$; this is the preceding high seed, not the buffer. In (228), $C_{\text{hi}}(S_M) \leq \chi_0$ and, for $V_M \geq 1$, $1 + 3(S_M + 4)/r_* \leq B_{\text{sw}} V_M$. The initial term is at most $(K_\tau/D_h) x_M^{-e}$. The high weighted seed is at most

$$\frac{H_0 \chi_0 K_\tau}{2D_h} M x_M^{1/2-Q} \leq \frac{H_0 \chi_0 K_\tau}{2D_h} x_M^{-e}.$$

This includes $Q = e + 3/2$ exactly. The dyadic endpoint term is at most

$$H_0 B_{\text{sw}} x_M^{-e} V_M^{3/2} e^{-\mu V_M} \leq H_0 B_{\text{sw}} \left(\frac{3}{2\mu \exp(1)} \right)^{3/2} x_M^{-e}.$$

The maximum follows by differentiating $t^{3/2} e^{-\mu t}$, whose maximum is at $t = 3/(2\mu)$. Adding the low contribution proves (233) on the original failure set.

The actual witness and its height. A source outside E_M passes the original initial certificate. At a continued threshold $q > L_M$, failure of its next certificate is exactly a source first-bad membership and hence is excluded. A successful certificate places the current value in shell $q - 1 \geq L_M$; both startups apply. The next threshold is strictly smaller. Induction on q therefore ends at $q \leq L_M$ on the same path, precisely the source proof `movingSource_terminal_or_failure`. Its retained-potential inequality gives

$$|gk - (M + 1 - q)| + \mathcal{V}_M(q) \leq \mathcal{V}_M(M + 1).$$

Nonnegativity of q and of the potential proves the time bound in (234). The displayed formula for the source potential is $O(\sqrt{M V_M}) + O(V_M^2) = o(M)$. Since $c g \ell > 1$, eventually $k < c M \ell \leq c \log n$.

The high block ceilings are bounded by their starting values to the power $1 + \tau$, and successive checkpoint values decrease. Every low block starts below 2^{S_M} ; its certified ceiling is at most $2^{S_M(1+\eta_{L_M})}$. Because $S_M = O(V_M)$, eventually $S_M(1 + \eta_{L_M}) \leq (\beta/2)M$. The source orbit-ceiling argument, or concatenation of these block ceilings, gives the stated height on every prefix of this same path. For the target, $L_M < A(M) \log_2(M+2) + 1$. For $n \in I_M$, $M \leq \log n/\ell$, and eventually $\log n \geq 1$; thus $M + 2 \leq (1/\ell + 2) \log n$. Eventual $0 < A(M) \leq A_{\max}$ proves the uniform C_{tar} .

Global count, including an incomplete last shell. The shells are disjoint. The bad count in a full valid shell is an integer bounded both by 2^M and by $2^M \delta_M$. In the last shell it is also bounded by its actual length $X - 2^N + 1$. Counting all earlier startup integers as bad proves (235). For any $M_0 \leq K \leq N$, it also gives

$$\#([1, X] \setminus G) \leq 2^K - 1 + 2X \sup_{K \leq M \leq N} \delta_M. \quad (236)$$

Indeed the full lengths from K through N sum to $2^{N+1} - 2^K \leq 2X$. Since $\delta_M \rightarrow 0$, first fix K after which all these errors are less than any prescribed positive number, then let $X \rightarrow \infty$, and finally send that number to zero. This proves natural density one without exchanging the two limits. $\square$

Exact comparison with the source's packaged good set. For fixed P, K_0, L, S , any two valid low packages with these same (r_L, η_L) and startups at most L give identical E_M , and identical stopped runs up to their first threshold at most L . In a run ending at $q \geq L$, every preceding continued threshold exceeds L ; its certified parent is at least L . The one-stage maps therefore agree by (220). Induction identifies the actual start, parity path, elapsed time, and next threshold, not just their estimates. The initial bad set and every rankwise landing predicate also agree, proving equality of the envelopes. Continuing below the terminal threshold is not part of this assertion. Thus, whenever the original `MovingEndpointParameterPackage` is given, intersecting its `movingEndpointGood` with I_M agrees eventually with our G_M . This is the identity bijection on starts and on those finite path records, with identity inverse and singleton fibres. For a new choice $4 + g \leq K_0 \leq 6$, instead, we have constructed a different member of the same generic family. The author's stronger bundled type has not been changed or claimed to accept that input.

Corollary 119 (Three clocks on the same moving good set). *On the good set of Theorem 118, the shortcut, ordinary, and (for odd sources) Syracuse clocks have respective upper bounds*

$$\begin{aligned} & \frac{2 \log n}{\log(4/3)} + O(\sqrt{\log n \log \log n}), \\ & \frac{3 \log n}{\log(4/3)} + O(\sqrt{\log n \log \log n}), \\ & \frac{\log n}{\log(4/3)} + O(\sqrt{\log n \log \log n}). \end{aligned}$$

The target is the one in (234); all three prefixes have height at most $n^{1+\beta}$, eventually. The constants depend on the fixed parameters, not on the values of the bounded moving sequence. The same bad-count estimate bounds exceptions to these simultaneous conclusions. On odd sources it bounds a count of integers, with no change of measure asserted.

Proof. Let s be the number of odd inputs in the actual k -step shortcut path. The exact integer affine identity (130), with its nonnegative numerator, gives

$$s \leq \frac{k + \log_2(y/n)}{\lambda} \leq \frac{k + L_M - M}{\lambda}.$$

As $L_M = O(V_M)$ and the potential is $O(\sqrt{MV_M})$, $k \leq M/g + O(\sqrt{MV_M})$ and $s \leq M/(2g) + O(\sqrt{MV_M})$. Expanding odd shortcut edges gives exactly $k + s$ ordinary steps, with inverse given by cutting the inserted edges, as in Proposition 83. The inserted heights are at most twice the following shortcut heights; hence $2n^{1+\beta/2} \leq n^{1+\beta}$ eventually. For odd n , truncate at the first shortcut crossing of 2^{L_M} , finish its even tail, and cut at odd states. This gives a Syracuse landing no higher than that crossing with at most s returns; the added tail only halves values. This is the exact clock map of Proposition 77, not a probabilistic parity assumption. Finally $2g\ell = \log(4/3)$. $\square$

Scope of the refinement. The critical target and moving buffer are unchanged. For fixed $A > A_{\text{FP}}$, $\delta = \kappa A - 1/2$, choosing $e > \delta$ in the theorem recovers the endpoint error $O(X(\log X)^{-\delta} \log \log X)$ by the shell summation already proved in Corollary 82, now for this recertified all-prefix good set. The timeout route and this route have not been identified. The refinements are the closed high margin, the explicit constants, the exact low-startup domain, removal of the auxiliary density reserve, and their complete same-witness assembly. They are written proofs based on the cited source constructions. The existing finite Lean certificates do not certify this analytic theorem or the entire Shaik package.

unchanged source subset	proved proportion bound
initial and high first-bad sources	$C_{\text{high}}(M+2)^{-e}$
low first-bad sources	$C_{\text{low}}2^{-\Delta_M^{\text{buf}}}$
Outside their original union: one path $n \mapsto T^k(n) \leq 2^{L_M}$. Its potential bounds k ; its certificates bound every prefix height. Expand odd edges to obtain $k + s$ ordinary steps; for odd sources, complete the terminal halving tail and cut at odd states. The last shell contributes at most $\min\{X - 2^N + 1, \lfloor 2^N \delta_N \rfloor\}$, not a full unqualified top-shell proportion.	

Figure 22: The full all-prefix assembly. Theorem 118 gives the exact constants, cutoffs and count; Corollary 119 proves the clock maps. The moving run, first-bad construction, potential and orbit ceilings are Shaik's [3].

28.14 A kernel-checked dyadic assembly and clock transfer

The counting passage in Theorem 118 now has a separate Lean certificate. Its objects are literal counts of positive integers, not an abstract sequence assumed to model those counts. The real shell-error estimate remains the written analytic input proved above; the certificate verifies the discrete consumer for every admissible shell budget.

Proposition 120 (Exact dyadic count certificate). *For $f : \mathbb{N} \rightarrow \mathbb{N}$, define*

$$C_f(a, t) = \sum_{i=0}^{t-1} f(a+i), \quad P_f(X) = C_f(1, X), \quad B_f(m) = C_f(2^m, 2^m).$$

For integers $K, J \geq 0$, put $N = K + J$. Whenever $2^N \leq X < 2^{N+1}$, $t = X + 1 - 2^N$ lies in $[1, 2^N]$, and

$$P_f(X) = P_f(2^K - 1) + \sum_{j=0}^{J-1} B_f(K+j) + C_f(2^N, t). \quad (237)$$

This identity requires no bound on the nonnegative weights f . If $f(n) \leq 1$ for every n , and $b : \mathbb{N} \rightarrow \mathbb{N}$ satisfies $B_f(K+j) \leq b(K+j)$ for every $0 \leq j \leq J$, then

$$P_f(X) \leq 2^K - 1 + \sum_{j=0}^{J-1} \min\{2^{K+j}, b(K+j)\} + \min\{X+1-2^N, b(N)\}. \quad (238)$$

More generally, if $D, R \in \mathbb{N}$ and $DB_f(K+j) \leq R2^{K+j}$ throughout this range, then

$$DP_f(X) \leq D(2^K - 1) + 2XR. \quad (239)$$

No monotonicity or positivity of D is needed for this integer inequality. Division to obtain a rational density bound uses $D > 0$.

For a decidable predicate P on $\mathbb{N}$, the indicator $f(n) = \mathbf{1}_{\neg P(n)}$ makes $P_f(X)$ exactly the number of missing P -points in $[1, X]$. The certificate proves this equality and both displayed count bounds for that literal count.

Finally, let $H, S, Y, B : \mathbb{N} \rightarrow \mathbb{N}$, and let $P(n)$ be the predicate that there exists one shortcut time $h \leq H(n)$ with at most $S(n)$ odd inputs, landing at most $Y(n)$, and all its prefix values at most $B(n)$. If its shell missing counts satisfy the hypotheses of (238), then that same right-hand side bounds missing ordinary Collatz witnesses with time at most $H(n) + S(n)$, landing at most $Y(n)$, and height at most $2B(n)$. More generally the input predicate can be any decidable retained set G : it suffices that $G(n)$ supplies that shortcut witness for $n = 2^{K+j} + i$, $0 \leq j \leq J$, $0 \leq i < 2^{K+j}$. The same capped and scaled bounds then hold for the missing ordinary witnesses, using the shell budgets of $\neg G$. No implication is required below 2^K , and the startup prefix is charged just once.

Proof. Splitting a consecutive sum of length $l+t$ after l terms gives $C_f(a, l+t) = C_f(a, l) + C_f(a+l, t)$, by reindexing $i = l+j$. Induction on J , using $2^{m+1} = 2^m + 2^m$, partitions the integers from 1 through $2^N - 1$ into the startup prefix and the indicated complete shells. Splitting the remaining t terms proves (237). Nonnegativity makes a partial-shell count no larger than its full shell count. The bound $f \leq 1$ also bounds it by its length. Applying both bounds, and bounding the startup prefix by $2^K - 1$, proves (238).

For the scaled statement, the complete shells telescope as

$$D \sum_{j=0}^{J-1} B_f(K+j) + R2^K \leq R2^N.$$

Induction proves this directly from the assumed per-shell inequality, including $J = 0$. The partial shell costs at most $R2^N$. Since $2^N \leq X$, the two contributions together are at most $2RX$, proving (239) after the startup term is added.

Substitution of the zero-one indicator counts precisely $\{1, \dots, X\} \setminus P$; neither 0 nor $X+1$ is counted. For the ordinary witness, expand the actual odd shortcut edges. The previously certified exact clock and height maps give length $h+s \leq H(n)+S(n)$, the same endpoint, and height at most $2B(n)$. Thus a missing ordinary witness implies a missing shortcut witness on the identical start. Summing this indicator inequality proves the final assertion, with no extra exceptional set. For a retained set, the shell-restricted implication $G(n) \Rightarrow P(n)$ gives $\mathbf{1}_{\text{no ordinary witness}}(n) \leq \mathbf{1}_{\neg G}(n)$ on every counted shell. Sum this inequality on that shell, then use the already proved capped or scaled count bound directly. The uncontrolled startup was already bounded by its length; there is no second startup loss to add. $\square$

Precisely what the formal artifact establishes. The file `formal_shaik_dyadic/DyadicP.refix.lean` contains twenty-three declarations checked by Lean 4.15.0. The principal exports are `dyadic_prefix_partition`, `dyadic_prefix_bound`, `missing_scaled_tail_bound`, and `raw_dyadic_exception_bound`, together with `retained_raw_dyadic_bound` and `retained_raw_scaled_tail_bound` for a retained input set. They quantify over all the natural lengths, cutoffs, weights and budgets just stated. This is not an exhaustive check only up to a finite cutoff. The import is the pinned `JointWitness` certificate, which in turn uses the unchanged `ClockAndRanks` certificate and its pinned `Init` dependency closure. Every reported axiom set is contained in `{propext, Quot.sound}`; there is no admission or `Classical.choice`. The verification script checks the source, objects, compiler, imports, dependency pins, declarations, and guarded execution receipt.

For the preceding endpoint theorem, the explicit coordinate substitution is $K = M_0$, $J = N - M_0$, $f = \mathbf{1}_{\mathbb{N} \setminus G}$, and $b(m) = \lfloor 2^m \delta_m \rfloor$. It converts (238) literally into (235); no endpoints are lost. After a fixed K makes the analytic shell errors at most a positive rational R/D , (239) gives the rational version of (236). This substitution and the analytic construction of G are a written comparison with the generic formal interface, not a compiled instantiation of Shaik's real-parameter construction in this certificate. The passage $X \rightarrow \infty$, followed by $R/D \rightarrow 0$, is still the written real-analysis argument. The certificate does not prove the entropy estimates, select the real logarithmic parameters, or replay Shaik's entire formal package. Shaik's moving assembly is the source consumer being reconstructed [3]; no novelty is claimed for dyadic counting or the exact Collatz clock expansion.

28.15 The exact comparison of the two moving endpoint routes

The timeout and all-prefix constructions admit a sharper comparison than merely sharing a target scale. Their headline witness sets are identical at equal parameters, whereas their certificate-defined failure sets have a proved inclusion in a definite direction. These are different statements: a witness predicate records what an orbit does; a certificate-defined set also records the tests used to establish it.

Proposition 121 (Forgetting the low certificate). *Use the source's fixed shrinking high data P , and write*

$$r_\ell = 1 - \frac{K_0}{2L}, \quad q_\ell(m) = \lfloor r_\ell m \rfloor.$$

Let $p_\ell : \text{StageSetup}(r_\ell, t_\ell)$ with $t_\ell < a_0$, and suppose

$$2 \leq L \leq S < M, \quad P.p_{\text{hi}}.M_0 \leq L, \quad p_\ell.M_0 \leq L.$$

All floors here are the source's nonnegative integer floors; the stage hypotheses imply $0 < q_\ell(m) < m$ for $m \geq L$. Compare the literal source envelopes

$$\begin{aligned} E_A &= \text{movingSeparatedFailureEnvelope}(P, p_\ell, M, L, S), \\ E_T &= \text{timeoutSeparatedFailureEnvelope}(P.\text{toTimeoutHigh}, K_0, M, L, S). \end{aligned}$$

Then

$$E_T \subseteq E_A, \quad I_M \setminus E_A \subseteq I_M \setminus E_T. \tag{240}$$

The initial-failure pieces agree, and the high first-bad pieces agree at each $S < q < M$, not just after counting.

There is an exact map of numerical run records behind this inclusion. Restrict an all-prefix record to blocks whose continuing threshold is greater than L , retaining the first terminal block. Delete the low window certificates and replace each low stage by its first passage and parent-rank deadline. This gives a timeout record with the same source, parent ranks, threshold ranks, block durations, cumulative times and actual orbit values. Its image consists exactly of the corresponding timeout records, cut at their first $q \leq L$, whose low departure points satisfy W_{t_ℓ} . The dyadic-halving completion branch is not part of this image. Restoring those tests gives the inverse on this image. No inverse is claimed on the remaining timeout records.

For $n \notin E_A$, the two stopping procedures therefore have the same terminal threshold and elapsed time. The same is true of every intervening orbit value, odd-input count and ordinary-step expansion.

Proof. The high data are unchanged by $P \mapsto P.\text{toTimeoutHigh}$. Unfolding the two high tolerances gives the same minimum $\min\{\tau, D\sqrt{\log(M+2)/m}\}$ for $m > 0$, with the same zero-rank branch. The high targets and bounded first-passage maps coincide.

At a low departure $x \in I_m \cap W_{t_\ell}$, $m \geq L$, the StageSetup terminal inequality gives $T^m(x) \leq 2^{q_\ell(m)}$. Thus the least passage exists, has duration at most m , and is exactly the duration stored by the low stage. This is precisely $\text{LowTimeoutPassage}(K_0, L, m, x)$. The two target ranks agree by their floor definitions. This uses `terminal_witness`, `stageLength_le_shell` and `stageLength_isFirstPassage` in `Pullback.lean`; it does not assume eventual descent of an uncertified source.

Every reached threshold has the first-passage landing band $2^{q-1} < x \leq 2^q$. If its next certificate has tolerance below a_0 , the source's `certified_landing_mem_lower_shell` excludes $x = 2^q$; consequently the next parent is exactly $m = q - 1$. For a continued block, $q > L$, so $m \geq L$. These facts verify every rank and startup field of the timeout constructor, including $m \leq q \leq m + 1$, positive next target, and strict rank decrease. Induction on the recorded blocks proves the map. Conversely a timeout record with the stated low tests uses the same least passages; the all-prefix constructors restore it block by block. This proves the inverse assertion for numerical records, independently of choices of proof terms.

A record whose final threshold satisfies $q > S$ has no earlier low block: a low block's output rank is strictly below its parent $m < S$, and all subsequent threshold ranks decrease. Hence both high first-bad definitions quantify over exactly the same records. The initial piece is identical by equality of the high tolerances.

Now take a low first-timeout record ending at rank $p \in [L + 1, S]$. Inspect its low departure checkpoints in chronological order. If an earlier one first fails W_{t_ℓ} , all preceding blocks convert back to the all-prefix record just proved. That failed checkpoint is a literal all-prefix first-bad endpoint at some $q \in [p, S]$. If no earlier checkpoint fails, the whole prefix converts. Its final point cannot satisfy W_{t_ℓ} : the terminal inequality would then give a passage by time $p - 1$, contradicting this timeout. It is therefore an all-prefix first-bad endpoint at $q = p$. In either case the source belongs to E_A . This proves (240), together with the already matched initial and high pieces.

The upper dyadic endpoint is not silently identified with shell $q - 1$. If $x = 2^q$, $L < q \leq S$, it fails the all-prefix low test but the timeout procedure succeeds by exact halving. It cannot be a timeout: for the target $q_\ell(q - 1) > 0$, the halving time $q - q_\ell(q - 1)$ is at most $q - 1$. Moreover, the inductive timeout record type permits further low halving blocks at such endpoints even though the stopping procedure finishes there. All their later values are powers of two of strictly smaller rank; none produces a low timeout or returns to a high rank. Thus those extra record continuations do not add a failure overlooked in the argument.

Finally, on $I_M \setminus E_A$, every required low certificate succeeds. Both deterministic procedures use the same least passage at each step and stop at the same first reached $q \leq L$. All-prefix certification also rejects a bad landing at $q = L$, although its stopping procedure needs no next block there: that extra rejection only makes E_A larger. No equality of the two complete failure envelopes is inferred. $\square$

The exact remaining difference. Let I be the common initial piece, H the common union of high first-bad pieces, C the all-prefix low first-bad union over $L \leq q \leq S$, and D the timeout union over $L < p \leq S$. The proof gives $D \subseteq C$ by mapping a timeout history to its first missing low certificate, possibly at an earlier, larger rank. Hence

$$E_A = I \cup H \cup C, \quad E_T = I \cup H \cup D, \quad (I_M \setminus E_T) \setminus (I_M \setminus E_A) = C \setminus (I \cup H \cup D). \quad (241)$$

This formula retains the source's inclusion of $q = L$ in C . It neither asserts that the pieces are disjoint nor introduces a hypothesis that the two sampling laws are independent.

Corollary 122 (Identical witness predicates and transferred counts). *For every A, c, β, C, M , with no asymptotic restriction,*

$$\text{timeoutEndpointWitnessGood}(A, c, \beta, C, M) = \text{movingEndpointWitnessGood}(A, c, \beta, C, M). \quad (242)$$

Both sides are literally

$$\left\{ n : \exists k, \quad k < c \log n, \quad T^k(n) < C(\log n)^{A(M)}, \quad T^j(n) \leq n^{1+\beta} \quad (0 \leq j \leq k) \right\}.$$

Their original existentially chosen constants need not be equal. Taking the larger target constant makes both source constructions map to this same literal predicate.

At matching finite parameters of Proposition 121, every proved bound $\#E_A \leq b_A(M)$ is also a bound for $\#E_T$. If a timeout estimate $\#E_T \leq b_T(M)$ is available, the exact integer budget is

$$b(M) = \min\{2^M, \lfloor b_A(M) \rfloor, \lfloor b_T(M) \rfloor\}.$$

For the original parameter package, the source's timeout witness theorem therefore has both count estimates without changing its orbit, clock or height clauses. In particular, the refined profile of Theorem 118 transfers to the original timeout envelope wherever the matching stage hypotheses hold. For $N = \lfloor \log_2 X \rfloor \geq M_0$, its assembled exception count is at most

$$2^{M_0} - 1 + \sum_{m=M_0}^{N-1} b(m) + \min\{X - 2^N + 1, b(N)\}.$$

The same bound applies to missing literal witnesses and, after the exact retained-set transfer of Proposition 120, to the corresponding ordinary-clock witnesses. No new startup charge or additional exceptional set is needed.

Proof. Unfold the two displayed source definitions in `TimeoutEndpointNaturalDensity.lean` and `Alternates/AllPrefix/NaturalDensity.lean`. Their bound variables, strict inequalities, power, and universal prefix clause coincide exactly. Increasing C preserves the landing inequality because $(\log n)^{A(M)} \geq 0$. The source witness proofs actually pass first through $T^k(n) \leq 2^{L_M}$, retaining one witness for all clauses.

The envelope count transfer follows from the proved inclusion, not from an identification of the certificates. Taking the smaller of two valid bounds does not require independence. Counts are integers, which justifies the floors, and cannot exceed the shell length. The last displayed formula is the exact dyadic assembly already proved and certified. The inclusion used to bound missing witnesses is in the correct direction: a source outside the timeout envelope has a witness, so a missing witness belongs to that envelope. Apply the same-start ordinary-clock map to that witness. $\square$

Source and scope. These statements compare Idris Ali Shaik’s actual pinned source [3], revision ef341084: `TimeoutRun`, `TimeoutFirstBad`, `TimeoutEnvelope`, `TimeoutExecution`, `TimeoutEndpointWitness`, `TimeoutEndpointNaturalDensity`, and the corresponding `Alternates/AllPrefix` modules. The run comparison is a complete written induction, not a replay of the source’s analytic Lean closure. The common witness equality is obtained by unfolding original definitions. The weaker sufficient scalar domain is a result about the original envelopes and generic stage interface; it does not redefine the author’s stronger bundled parameter type. On the all-prefix-good sources, equality of the whole traces transfers every refined clock and height estimate pointwise. On the larger timeout-good set, the corollary uses the timeout witness bounds: an improved count alone does not strengthen a pointwise clock. The independently checked count certificate supplies the integer consumer, not the analytic estimates.

Figure 23: The exact comparison retains numerical runs and distinguishes their certificate-defined failure sets from their identical headline witness predicates. Schematic layout, not a geometric model of the sets. Proof: Proposition 121 and Corollary 122. Original constructions: Idris Ali Shaik [3]. Reproducible source: `draw_shaik_route_comparison.py`.

28.16 Two-sided height control inside completed timeout blocks

A completed low block carries more information than a time limit: its endpoint is already below a prescribed threshold. Forward growth and backward recovery from that endpoint can therefore be used simultaneously. This improves the finite height estimate without adding a low window test, changing a run, or discarding any timeout-good source.

Proposition 123 (A completed-block height bound). *Let $T(x) = x/2$ for even x , and $T(x) = (3x + 1)/2$ for odd x . For positive integers x , let $y = T^h(x)$. At every $0 \leq v \leq h$,*

$$T^v(x) \leq \min \left\{ \left\lfloor \frac{3^v(x+1)}{2^v} \right\rfloor - 1, 2^{h-v}y \right\}. \quad (243)$$

Now suppose

$$2^m \leq x < 2^{m+1}, \quad h \leq m, \quad y \leq 2^q, \quad 1 \leq q < m.$$

For integers $0 \leq v \leq m$, put

$$A_m(v) = 3^v 2^{m+1-v} - 1, \quad D_{m,q}(v) = 2^{m+q-v}, \quad B(m, q) = \max_{0 \leq v \leq m} \min\{A_m(v), D_{m,q}(v)\}.$$

Then every prefix of this completed block is at most $B(m, q)$. This bound can be evaluated by one integer crossing. Define

$$a = \max\{0 \leq v \leq m : 2^{m+1}3^v - 2^v \leq 2^{m+q}\}. \quad (244)$$

The set is nonempty, $a < m$, and

$$B(m, q) = \max\{3^a 2^{m+1-a} - 1, 2^{m+q-a-1}\}. \quad (245)$$

In particular, with

$$\alpha = 1 - \frac{\log 2}{\log 3}, \quad \kappa = 1 + \alpha,$$

one has the strict bound

$$B(m, q) < 2^{m+1+\alpha(q-1)}. \quad (246)$$

The maximum defining B is the exact maximum of these two scalar bounds. It is not asserted to be the maximum attained by Collatz blocks.

Proof. For each nonnegative integer z , direct examination of its parity gives

$$2(T(z) + 1) \leq 3(z + 1), \quad z \leq 2T(z).$$

For odd z , the first is equality; for even z , it follows from $z + 2 \leq 3z + 3$. The second is equality for even z , and follows from $z \leq 3z + 1$ for odd z . Iterating the first inequality gives $2^v(T^v(x) + 1) \leq 3^v(x + 1)$. Iterating the second over the last $h - v$ steps gives $T^v(x) \leq 2^{h-v}T^h(x)$. Integrality proves (243), including $v = 0, h$.

Since $x + 1 \leq 2^{m+1}$, $h \leq m$, and $y \leq 2^q$, the two bounds are respectively at most $A_m(v)$ and $D_{m,q}(v)$ for $v \leq h$. Taking the maximum over the larger interval $0 \leq v \leq m$ is legitimate; no orbit segment is invented at the added times.

The sequence A_m is strictly increasing, and $D_{m,q}$ is strictly decreasing. At zero, $A_m(0) = 2^{m+1} - 1 \leq 2^{m+q} = D_{m,q}(0)$. At m , $A_m(m) = 2 \cdot 3^m - 1 > 2^q = D_{m,q}(m)$. After multiplication by $2^v > 0$, their comparison is exactly the inequality in (244). Thus $0 \leq a < m$; the minimum is $A_m(v)$ up to a , and $D_{m,q}(v)$ from $a + 1$ onward. This proves (245).

For the real upper bound let $b = \log_2 3 > 1$ and $c = (q - 1)/b$. If $v \leq c$, then

$$A_m(v) < 2^{m+1+(b-1)v} \leq 2^{m+1+(b-1)c}.$$

If $v > c$, then

$$D_{m,q}(v) = 2^{m+q-v} < 2^{m+q-c} = 2^{m+1+(b-1)c}.$$

As $(b - 1)c = \alpha(q - 1)$, every term in the finite maximum satisfies the stated strict inequality. This also covers $q = 1$, where $c = 0$. $\square$

Corollary 124 (The whole timeout run, on its original good set). *Let $S \geq 3$, and define the explicit integer*

$$B_S = B(S - 1, S - 2).$$

Every completed low block of the original timeout run has all its values at most B_S . In fact

$$B_S < 2^{\kappa S - 3\alpha}. \quad (247)$$

For any original timeout run starting at $n \in I_M$, every shortcut prefix, including a final dyadic-halving completion, is bounded by

$$\max\{n^{1+\tau}, B_S\}. \quad (248)$$

Here $\tau > 0$ is the original high tolerance cap. In particular, for $\beta \geq \tau$, the finite sufficient condition

$$B_S \leq 2^{(1+\beta)M}$$

gives the original ceiling $n^{1+\beta}$. It suffices that $\kappa S - 3\alpha \leq (1 + \beta)M$. If $\kappa S - 3\alpha \leq M$, every low-block value is already below n ; then only the high blocks determine the power ceiling.

For $S = \lceil C_{\text{sw}} \log(M + 2) \rceil$, fixed $C_{\text{sw}} > 0$, the last condition holds for all sufficiently large M . Thus this height argument needs neither the source's low margin $2S \leq \beta M$, nor a low all-prefix certificate, nor the density parameters K_1, Q, e . Their roles elsewhere in the endpoint and counting arguments are unchanged.

Proof. The literal low constructor supplies $h \leq m < S$, $x \in I_m$, and a positive target $q < m$, with $T^h(x) \leq 2^q$. Thus $m \geq 2$. For $v \leq m$, increasing m to $S - 1$ and q to $S - 2$ increases both scalar bounds. Enlarging their time interval to $0 \leq v \leq S - 1$ proves $B(m, q) \leq B_S$. Equation (246) gives

$$B_S < 2^{S+\alpha(S-3)} = 2^{\kappa S - 3\alpha}.$$

The first high block has, by its original window test, height at most $n^{1+\eta} \leq n^{1+\tau}$. Each later high block starts at a previous first-passage endpoint $x < n$, as proved in the source's `endpoint_1 t_start`; its height is at most $x^{1+\eta} \leq n^{1+\tau}$, since $0 < \eta \leq \tau$. Every low block has the bound just proved, with no window test used. The constructors concatenate the same actual orbit, so these estimates cover every prefix by induction on blocks. If the execution finishes by halving a dyadic endpoint, that tail decreases from an endpoint less than n . Since $n \geq 1$ and $\tau > 0$, it also satisfies (248).

Finally $n \geq 2^M$, so the displayed finite condition and $\beta \geq \tau$ bound both terms of the maximum by $n^{1+\beta}$. The logarithmic-switch assertion follows from $S \leq C_{\text{sw}} \log(M + 2) + 1$ and $\log(M + 2)/M \rightarrow 0$. $\square$

An exact example. For $x = 1887$, $m = 10$, $q = 9$, the first passage below 2^9 has $h = 10$, $y = 448$, and path

$$1887, 2831, 4247, 6371, 9557, 14336, 7168, 3584, 1792, 896, 448.$$

The forward bound is attained through time five and the backward bound from time five onward. Thus the minimum in (243) is the exact orbit at every time in this example. Its peak is 14336, below $B(10, 9) = 15551$. The five odd inputs give ordinary duration 15 and peak 28672. This example illustrates the mechanism; the preceding proof, not this single trajectory, establishes the universal bound.

Figure 24: Actual integer orbit and the two proved endpoint bounds, on a logarithmic vertical axis. For this example their minimum is attained at every displayed time. The dotted line is the larger bound using only the source shell, target rank and deadline. Proof: Proposition 123. The timeout-block application is to Shaik's construction [3]. Source: `draw_shaik_completed_height.py`.

Effect on the assembled statements. The map here is the identity on the original source integer and the complete numerical run. Its endpoint, duration, odd-input count, first passage and failure-envelope membership do not change. In Corollary 122, the timeout witness therefore has this stronger finite height estimate on *all* timeout-good sources, not only on the all-prefix-good subset. All density estimates and the exact partial-shell counts remain valid on the same sets. Expansion to ordinary Collatz time keeps duration $h + s_h(n)$ and the same endpoint; it bounds

ordinary intermediate values by twice (248), because an inserted odd step is $3z + 1 = 2T(z)$. That factor is retained, not absorbed without a margin. The original asymptotic headline power ceiling is unchanged, but its finite startup condition is weakened.

Attribution and verification. The original block types, endpoint decrease and high-stage height estimate are Idris Ali Shaik’s, revision `ef341084: TimeoutRun.lean, TimeoutOrbitCeiling.lean, OrbitCeiling.lean, and TimeoutEndpointWitness.lean` [3]. The elementary forward and backward inequalities above are proved directly. No priority claim is made for them. The improvement is their explicit two-sided use in these same source blocks and its propagation without changing the retained set. The universal statements here have written proofs; the accompanying integer regressions do not certify the infinite analytic package.

28.17 A kernel-checked certificate for the completed-block bounds

Proposition 125 (Discrete completed-height certificate). *The natural-number part of Proposition 123 has a matching twenty-nine-declaration Lean certificate. It proves (243), the exact shell maximum $B(m, q)$, the characterization of a as the greatest integer satisfying (244), the two-candidate equality (245), and monotonicity of B in both shell and target ranks. In particular, a completed block with $x < 2^{m+1}$, $h \leq m < S$, $T^h(x) \leq 2^q$, and $q < m$ satisfies*

$$T^v(x) \leq B(S - 1, S - 2) \quad (0 \leq v \leq h).$$

The imported exact clock map gives the same endpoint at ordinary time $h + s_h(x)$, with every ordinary prefix bounded by $2B(m, q)$. All these statements quantify over their full natural-number domains; they are not proofs by testing a bounded list of trajectories.

Proof and precise formal correspondence. The module `CompletedHeight.lean` imports the independently pinned `JointWitness` and `ClockAndRanks` modules. The imported `shortcut`, `raw`, `iterate` and `oddCount` are the same literal maps and recursive orbit used in the earlier clock certificate. Induction on time identifies this recursion with Shaik’s `orbit_zero` and `orbit_succ`; no change of map or time convention is made.

The declarations `forward_scaled` and `prefix_backward` establish the two multiplied integer inequalities. `prefix_forward_floor` uses natural division to obtain exactly the displayed floor, and `two_sided_height` takes the minimum. The definitions `forward`, `backward` and `maxThrough` are respectively $A_m(v)$, $D_{m,q}(v)$ and the inclusive maximum over $0, \dots, m$. The shell theorem even needs only $x < 2^{m+1}$, not the shell’s lower bound; its specialization to the original shell is immediate.

The finite recursive search `lastGood` has a proved greatest-index specification. `crossing_integer_iff` identifies its predicate with the integer inequality in (244), retaining the subtracted 2^v . The declarations `crossing_greatest` and `two_candidate_budget` prove the exact maximum and its two-term evaluation, under $1 \leq q < m$. Finally, `budget_mono` and `completed_uniform_height` give the switch bound, and `completed_raw_height` and `completed_raw_endpoint` apply the already certified ordinary clock correspondence.

Lean 4.15.0 checked the saved module with pinned imports. Its reported axiom sets are contained in `{propext, Quot.sound}`; there are no admitted statements. Source, object, compiler, dependencies, output, resource receipts and the three failed implementation attempts are retained with hashes in `formal_shaik_height/`. The successful single worker used 184229888 bytes at its measured peak, below the watched 2 GiB tree cap.

The real-exponent inequality (246), the source’s high-window bound, concatenation through its complete timeout run type, and the density estimates retain their written/source proofs. They are not claimed as additional exports of this certificate. Thus the certificate checks the exact discrete bounds consumed by Corollary 124, not an entire analytic package. The original run and high-stage mechanisms remain credited to Idris Ali Shaik [3]. $\square$

28.18 Exact low-rank schedules and finite timeout clocks

The leading timeout clocks were already obtained in Corollary 82. The remaining low-rank reserve can be made considerably more precise at finite parameters. This refinement uses Shaik’s actual timeout run and terminal completion, not the smaller all-prefix-good set: `TimeoutRun.lean`, `TimeoutTimeSupport.lean`, `TimeoutFirstBad.lean`, and `TimeoutEndpointWitness.lean`, revision `ef341084` of [3]. The source potential pays S^2 ; (125) already replaced that by a triangular sum. The prescribed floor map gives a still smaller, exactly computable reserve.

Proposition 126 (Low continuation, with its dyadic branch). *Fix integers $2 \leq L < S$, $K_0 > 0$, and $r_L = 1 - K_0/(2L) \in (0, 1)$. Write*

$$f(m) = \lfloor r_L m \rfloor, \quad Q(p) = f(p - 1),$$

and assume $1 \leq Q(p) < p - 1$ for $L < p \leq S$. These are the literal target and admissibility conditions of the timeout construction. Define finite sets of integers and integers recursively, using strictly smaller arguments:

$$\begin{aligned} \mathcal{D}(p) &= \{0\}, & F(p) &= 0 & (p \leq L), \\ \mathcal{D}(p) &= \{p - L\} \cup ([p - Q(p), p - 1]_{\mathbb{Z}} + \mathcal{D}(Q(p))), & (L < p \leq S), \\ F(p) &= p - 1 + F(Q(p)) & (L < p \leq S). \end{aligned} \quad (249)$$

Here $A + B = \{a + b : a \in A, b \in B\}$, not a disjoint union. Start at a checkpoint $x \in (2^{p-1}, 2^p]$, $1 \leq p \leq S$. Execute the original low procedure, stopping upon a threshold at most L , or completing a dyadic checkpoint 2^u , $u > L$, by $u - L$ halvings. For every successful continuation its total duration d satisfies

$$d \in \mathcal{D}(p), \quad p - L \leq d \leq F(p) \quad (p > L).$$

For $L < p \leq S$, moreover,

$$\begin{aligned} \min \mathcal{D}(p) &= p - L, & \max \mathcal{D}(p) &= F(p), \\ F(p) &\leq \frac{p(p-1) - L(L-1)}{2}, \\ F(p) &< \frac{p-1}{1-r_L} = \frac{2L(p-1)}{K_0}. \end{aligned} \quad (250)$$

The minimum and maximum are exact for the stated scalar construction. Neither equality of $\mathcal{D}(p)$ with actual orbit durations nor attainment of $F(p)$ by an orbit is asserted.

Proof. For positive integers z , $T(z) \geq z/2$. Thus $T^h(x) \geq x/2^h$. At a non-dyadic checkpoint $2^{p-1} < x < 2^p$, the parent is exactly $p - 1$. If its passage to $2^{Q(p)}$ succeeds, its deadline gives $h \leq p - 1$, whereas

$$2^{Q(p)} \geq T^h(x) \geq x/2^h > 2^{p-1-h}$$

forces the integer inequality $h \geq p - Q(p)$. Its landing is in $(2^{Q(p)-1}, 2^{Q(p)}]$. At the dyadic checkpoint $x = 2^p$, the specified terminal completion has exactly $p - L$ steps. Induction proves the membership assertion. It also explains why an upper endpoint is a separate branch, not another non-dyadic parent of rank $p - 1$.

All the scalar intervals are nonempty, since $Q(p) \geq 1$. Their recursive maximum is $p - 1 + F(Q(p))$, which is at least $p - L$, since $L \geq 2$. If $Q(p) > L$, their recursive minimum is $p - Q(p) + Q(p) - L = p - L$; if $Q(p) \leq L$, it is $p - Q(p) \geq p - L$. The dyadic branch attains the scalar minimum. This proves both extrema, including the terminal overshoot.

For the upper bounds, put $p_0 = p$, $p_{i+1} = Q(p_i)$ while $p_i > L$, and $m_i = p_i - 1$. The finite recursion gives $F(p) = \sum_{i:p_i > L} m_i$. These are distinct decreasing integers in $[L, p - 1]$, proving the triangular bound. Successive nonterminal parents satisfy $m_{i+1} = \lfloor r_L m_i \rfloor - 1 \leq r_L m_i$. If there are $t \geq 1$ such parents, then

$$F(p) \leq (p - 1) \sum_{i=0}^{t-1} r_L^i < (p - 1)/(1 - r_L).$$

The strict inequality uses $0 < r_L < 1$ and finiteness of t . $\square$

Proposition 127 (Only one low schedule can lead to a first timeout). *Use Shaik's TimeoutHighRunData, with $M \geq S$ and $p_{\text{Hi}} \cdot M_0 \leq L$, and the low hypotheses of the preceding proposition. Its non-dyadic high thresholds follow $m_0 = M$, $q_i = \lfloor r_{\text{Hi}} m_i \rfloor$, $m_{i+1} = q_i - 1$. Let $p_0 \leq S$ be the first threshold entering the low region, and suppose $p_0 > L$. Let $[a, b]_{\mathbb{Z}}$ be any proved integer interval containing its high arrival times, with $a \leq b$. For example, at each high block use*

$$a_i = \max\left(1, \left\lceil \frac{(1 - t_i)m_i - q_i}{g} \right\rceil\right), \quad b_i = \min\left(m_i, \left\lceil \frac{(1 + t_i)(m_i + 1) - q_i + 1}{g} \right\rceil - 1\right),$$

and $a = \sum a_i$, $b = \sum b_i$; this retains the closed lower and strict upper endpoints of (124). An empty block interval means that no such high continuation exists.

Continue $p_{j+1} = Q(p_j)$ until the first $p_t \leq L$. Only $p_0, \dots, p_{t-1}$ can support the source's low first-timeout sets. For $0 \leq j < t$ put

$$V_j = \sum_{i < j} (p_i - 1), \quad I_j = [a + p_0 - p_j, b + V_j]_{\mathbb{Z}}, \quad W_j = b - a + 1 + \sum_{i < j} (p_{i+1} - 1).$$

Every arrival time in timeoutFirstBadSourcesAtRank at p_j belongs to I_j , and $\#I_j = W_j$. The other low first-timeout sets, for $L < p \leq S$, are empty. This is not an assertion that the source's larger timeoutFeasibleTimes is empty off that list.

Fix a rational $0 < r_* \leq \min(r_{\text{Hi}}, r_L)$. Retain the rankwise reverse-transport hypotheses $p_j < M$ and $(p_j + 2)/(r_* 2^{p_j}) \leq 1/3$. Writing $\mathcal{B}_p = \text{timeoutLandingBad}(K_0, L, p)$, one obtains the finite union estimate

$$\frac{\#\bigcup_{L < p \leq S} \text{timeoutFirstBadSourcesAtRank}(p)}{2^M} \leq \sum_{j < t} W_j \left(2^{p_j - M} + \frac{3(p_j + 2)}{r_*} \right) \frac{\#\mathcal{B}_{p_j}}{2^{p_j}}. \quad (251)$$

Each W_j may also be replaced by its minimum with any previously proved cardinality bound for the source's feasible-time set at p_j .

Proof. In a run ending at an actual timeout, no previous checkpoint was dyadic. Indeed, from 2^u every subsequent positive-threshold first passage is an exact halving passage, remains dyadic, and cannot time out. At a final putative timeout rank $p > L$, the target $Q(p) \geq 1$ is reached from 2^p in $p - Q(p) \leq p - 1$ steps, directly contradicting that timeout. No such history can leave the low region: reached thresholds strictly decrease, and the high constructor requires parent $m \geq S$ with previous threshold $m + 1$.

Consequently every high continuation in a first-timeout history has parent $q_i - 1$, and every low continuation has parent $p_i - 1$. The floor maps fix both lists. This reasoning applies to all the source's inductive records, including ones which were not explicitly stopped at an earlier terminal rank: strict decrease prevents their return to $p > L$. It proves the asserted empty first-timeout sets, without removing any source from the literal envelope.

Before the timeout at p_j , every low block succeeds, so its duration d_i is in $[p_i - p_{i+1}, p_i - 1]_{\mathbb{Z}}$. The sum of these independent scalar intervals is exactly $[p_0 - p_j, V_j]_{\mathbb{Z}}$. To prove this last equality constructively, start every coordinate at its lower endpoint. For a requested excess e between zero and the sum of the widths, add the lesser of e and the first width to the first coordinate, subtract that addition from e , and repeat. The remaining excess is zero after the last coordinate. This is a section of the sum map on scalar tuples, not a procedure for constructing a Collatz orbit. Applying it also to $[a, b]$ gives the exact scalar interval I_j . Its cardinality is

$$b + V_j - (a + p_0 - p_j) + 1 = b - a + 1 + \sum_{i < j} (p_{i+1} - 1) = W_j.$$

Restrict the source transport to the intersection of I_j with its original feasible times. The original source, landing, direct first passage and reverse-loss bound are unchanged. Apply `timeoutFirstBadSourcesAtRank_card_le`'s underlying finite-time transport estimate to this smaller support. In fact its unabsorbed version, `lossFiltered_arbitraryTarget_transport_atTimes` in `TimeSupportTransport.lean`, gives the exact factor $(1 + 3(p_j + 2)2^{M-p_j}/r_*)\#\mathcal{B}_{p_j}$. Counting the restricted times by W_j , or by the stated minimum, dividing by 2^M , and summing proves (251). Since $p_j < M$, replacing 2^{p_j-M} by 1 recovers the source's uniform factor $(1 + 3(p_j + 2)/r_*)\#\mathcal{B}_{p_j}/2^{p_j}$. The unabsorbed expression retains the smaller finite boundary term. Overlaps only make the union smaller; no disjointness assumption is needed. $\square$

Corollary 128 (Finite clocks and height on the unchanged successful paths). *In the preceding setting every timeout-good path which enters the low phase has its original completed witness $y \leq 2^L$ by integer shortcut time*

$$k \leq K := b + F(p_0).$$

No separate additional S -step halving reserve is needed: the dyadic terminal branch is already included in F . For $n \in I_M$ on this successful set, define

$$J(K, M, L) = \max\{s \in [0, K]_{\mathbb{Z}} : 3^s 2^M \leq 2^{K+L}\}. \quad (252)$$

The set is nonempty whenever such a witness exists. The same path has at most $J(K, M, L)$ odd inputs and at most $K + J(K, M, L)$ ordinary Collatz steps to y . For odd n , completing the terminal even tail and cutting at odd states gives at most $J(K, M, L)$ Syracuse returns. Its shortcut height is at most

$$H(n) = \max\{n^{1+\tau}, B(S-1, S-2)\},$$

and its ordinary height is at most $2H(n)$. The explicit density multiplier in (251) and these clocks therefore apply without a change of accepted sources.

Proof. Apply Proposition 126 at the actual high landing, allowing its dyadic branch. The high time is at most b . If s is the odd-input count, the exact affine iterate identity gives $3^s n \leq 2^k y$. Hence $3^s 2^M \leq 2^{k+L} \leq 2^{K+L}$ and $s \leq k \leq K$. This both proves nonemptiness in (252) and bounds s . Expansion and cutting are the exact path maps of Proposition 83; expansion takes $k + s$ ordinary steps. The height is Corollary 124, including its factor two for the inserted ordinary odd edges. The final even tail only decreases values. $\square$

The maps and their information loss. An actual path is sent to its high prefix, decreasing low ranks, block durations and terminal branch. Its scalar duration record forgets intermediate values and parity, so it has no asserted inverse to an orbit. On independent scalar duration tuples, the sum map has the explicit section in the proof above; its fibres are exactly the tuples in their respective intervals with the same sum. Decorating the actual path with its new budget changes neither the path nor its accepted source and has the identity inverse after forgetting that decoration. This is the comparison needed for the original failure-envelope and witness statements.

Take $L = 5$, $K_0 = 4$, $r_L = 3/5$, $p = 6$; thus $Q(6) = 3$.

branch at $(2^5, 2^6]$	duration	terminal value
$x = 64$	$6 - 5 = 1$	32
$32 < x < 64$	$h \in [6 - 3, 5]_{\mathbb{Z}}$	$T^h(x) \leq 8$

The scalar set is $\mathcal{D}(6) = \{1, 3, 4, 5\}$, not $[1, 5]_{\mathbb{Z}}$. An actual successful interior example is $37 \mapsto 56 \mapsto 28 \mapsto 14 \mapsto 7$, of duration 4. The scalar set contains 5, but this figure does not claim an orbit realizes every scalar possibility.

Figure 25: The upper-endpoint completion and the interior first passage are distinct branches of the same timeout procedure. The strict lower landing boundary produces the lower duration $p - Q(p)$. Complete proof: Proposition 126. The underlying run and terminal completion are Shaik’s [3]; exact integer checks and reproducible figure source are recorded in `check_shaik_low_schedule.py` and `draw_shaik_low_schedule.py`.

For a concrete finite comparison, take $L = 20$, $K_0 = 4$, $p_0 = S = 64$. Then $r_L = 9/10$, and the low threshold list is

$$64, 56, 49, 43, 37, 32, 27, 23, 19.$$

The exact scalar reserve is $F(64) = 63 + 55 + 48 + 42 + 36 + 31 + 26 + 22 = 323$. The triangular reserve in (125) is 1826; the source’s S^2 potential reserve is 4096. These are bounds on the low part, not on the preceding high clock.

What has and has not changed. The new recursion and directed supports refine finite reserves in the existing timeout proofs. They do not change the leading clock constants already proved above, and no new asymptotic density exponent follows merely from this refinement. At a logarithmic switch the triangular bound still gives $F(p_0) = O(\log^2 M)$, below the existing square-root clock error. The gain is an exact, usable finite budget, including its floor schedule, possible duration gaps, and smaller rankwise transport multiplier. The stronger original parameter bundle is not modified; the finite result uses only the displayed stage and transport hypotheses. These are complete written proofs with finite regressions, not new Lean exports or a replay of the whole analytic dependency closure.

28.19 Joining the drift prefix to the timed almost-bounded path

The high-rank construction and the timed fixed-target theorem describe the same deterministic orbit, but supply different information about it. The former controls its height and the time already spent; the latter bounds its total number of odd returns. Subtracting these times leaves only a sublinear number of returns after a polylogarithmic landing. This removes the extra $\log_2 n$ in the crude raw-clock conversion.

Here $d = \log(4/3)$, $g = 1 - \frac{1}{2} \log_2 3$, so $2g \log 2 = d$. The analytic inputs are the uniform timed count reconstructed from Allikvere's version-2 argument ([author source](#), section "A time budget and an explicit rate", labels `prop:timed`, `thm:quant`) and Tao's first-passage method, and Shaik's high-certificate transport [3], reconstructed above. Precisely, the timed input used here is

$$\#\{n \leq X : n \text{ odd}, \nexists k \leq K(n) : \text{Syr}^k(n) \leq B\} \leq C_c X (\log B)^{-c},$$

$$K(n) = d^{-1} \log n + C_K (\log(n+2))^{4/5},$$

for all $X, B \geq 2$, B integral, $0 < c < 1/17.232$. The absolute C_K is independent of B, c . This is the proved timed fixed-target count earlier in this manuscript, including its stronger kernel exponent and exact power-interval cover, not an additional hypothesis about landing distributions.

Lemma 129 (The high prefix has two-sided clocks). *For any $\eta, e > 0$, there are constants A, C, M_0 , a deterministic integer $p_M = O_{\eta,e}(\log(M+2))$, and sets $G_M \subseteq I_M$, for $M \geq M_0$, such that*

$$\#(I_M \setminus G_M) \leq C 2^M M^{-e}, \quad Y_M := 2^{p_M} \gg_{\eta,e} (M+2)^2.$$

Each $n \in G_M$ has a first shortcut passage to Y_M of length H , landing $y \in (Y_M/2, Y_M]$, and odd-step count s . With $E_M = \sqrt{M \log(M+2)}$, these satisfy

$$H = M/g + O_{\eta,e}(E_M), \quad s = M/(2g) + O_{\eta,e}(E_M), \quad H + s = 3M/(2g) + O_{\eta,e}(E_M). \quad (253)$$

The shortcut prefix stays below $n^{1+\eta}$. For odd n , appending the even tail of y ends at $z = \text{oddpair}(y) = \text{Syr}^s(n) \leq Y_M$, with at most p_M additional halvings and no additional odd step. Moreover, s is the first Syracuse passage time to Y_M . The expanded ordinary prefix stays below $2n^{1+\eta}$.

Proof. Choose $a_0 < r < 1$, $0 < \tau < \min(r - a_0, \eta, 1)$, then D and C_{sw} so that

$$D/\sqrt{C_{\text{sw}}} \leq \tau, \quad \min(c_0 D^2, C_{\text{sw}} \log 2) > e + 2, \quad r C_{\text{sw}} \log 2 > 2.$$

Set $S = \lceil C_{\text{sw}} \log(M+2) \rceil$. Use only the high phase of Theorem 81: reject the initial failed certificate and the first failed high certificate or upper endpoint at threshold rank greater than S . Stop at the first threshold rank at most S , including its upper endpoint. No low certificate, timeout or low-density estimate is used.

Before this stop the ranks are exactly $m_0 = M$, $q_i = \lfloor r m_i \rfloor$, $m_{i+1} = q_i - 1$. The final threshold $p_M = q_j$ is therefore independent of n and satisfies $\lfloor r S \rfloor \leq p_M \leq S$. The high-failure calculation (128) gives $O(2^M M^{-e})$ discarded sources. In detail the normalized charge is $O((1 + M E_M) M^{-\phi})$, where $\phi = \min(c_0 D^2, C_{\text{sw}} \log 2) > e + 2$; this is $O(M^{-e})$. Since $2^{\lfloor r S \rfloor} \geq (M+2)^{r C_{\text{sw}} \log 2 / 2}$, the asserted lower bound on Y_M follows.

The two sides of (124), with $p_M = O(\log M)$, $j = O(\log M)$, and $\sum_i t_{m_i} m_i = O(E_M)$, give $gH = M + O(E_M)$. Each earlier segment stays above its own target, which is greater than Y_M ; hence their concatenation is the direct first passage to Y_M . Proposition 74 gives

$$n = 2^H y P / 3^s, \quad 0 \leq 1 - P < H / (3 Y_M).$$

Thus $P \geq 1/2$ for large M . Since $M \leq \log_2 n < M + 1$ and $p_M - 1 < \log_2 y \leq p_M$,

$$(\log_2 3)s = H + \log_2(y/n) + \log_2 P = H - M + O(\log(M + 2)).$$

Using $1 - g = \frac{1}{2} \log_2 3$ proves all three clocks. The decreasing high checkpoints and their W_{t_m} bounds give the stated shortcut height. Expanding an odd shortcut edge inserts the height $3x + 1$, twice its following shortcut value, and inserts one extra ordinary step. This proves the ordinary height and clock.

For odd n , cutting the expanded path at odd states and appending the terminal halvings gives the first s actual Syracuse returns. Every earlier odd state is before the shortcut crossing and exceeds Y_M . The appended tail only decreases and has length $v_2(y) \leq p_M$. This proves the first-passage assertion, including the case that $y = Y_M$ and its odd part is 1. $\square$

Lemma 130 (Exact same-orbit continuation). *Let n be odd and let s be its finite first Syracuse passage to $Y \geq 1$, with $z = \text{Syr}^s(n) \leq Y$. Suppose its first Syracuse hit of the integer target $B \geq 1$ has time $\sigma \leq K$. If $B \geq Y$, then $\sigma \leq s$. If $B < Y$, then $s \leq \sigma$, and the remaining $r = \sigma - s$ returns satisfy*

$$0 \leq r \leq K - s, \quad \ell_C \leq 3r + \log_2 z, \quad \ell_T \leq 2r + \log_2 z, \quad \max(\text{ordinary tail}) \leq 2^{r+1}Y. \quad (254)$$

Here ℓ_C, ℓ_T are the actual ordinary and shortcut lengths from z to $\text{Syr}^\sigma(n)$; the estimates include $r = 0$.

Proof. Before s , every odd state exceeds Y . The two orderings of first passages follow immediately, including equality. For $z_i = \text{Syr}^i(z)$, put $a_i = v_2(3z_i + 1) \geq 1$, $v = \sum_{i < r} a_i$. Exact concatenation gives $\ell_C = r + v$, $\ell_T = v$, and

$$2^v z_r = z \prod_{i < r} (3 + 1/z_i) \leq 4^r z.$$

Since $z_r \geq 1$, $v \leq 2r + \log_2 z$. Also $z_i \leq 2^i z$, because $\text{Syr}(x) \leq 2x$. Every inserted ordinary odd peak is at most $4z_i \leq 2^{i+2}Y$; for $i < r$ this is at most $2^{r+1}Y$. The intervening halvings decrease. The empty tail has height $z \leq Y$.

This construction is a map from two marked prefixes of the same orbit to their common first- B prefix. On $B < Y$ it concatenates the retained first- Y prefix and the actual tail of length $\sigma - s$; on $B \geq Y$ it truncates the retained prefix. It neither chooses a new source at z nor transports a probability law to z . Retaining n, s, σ determines all these states; forgetting the unused suffix is not an invertible map on arbitrary path records. $\square$

Theorem 131 (Drift clocks and height for almost-bounded targets). *For every $\beta, e > 0$ there is $A_{\beta,e} > 0$ such that, putting*

$$R_j(n) = \frac{j}{d} \log n + A_{\beta,e} (\log(n + 2))^{4/5}, \quad j = 1, 2, 3,$$

the following holds. For every $X \geq 2$, integer $B \geq 2$, and $0 < c < 1/17.232$, all but at most

$$C_c X (\log B)^{-c} + C_{\beta,e} X (\log(X + 2))^{-e} \quad (255)$$

positive integers $n \leq X$ have one actual orbit prefix ending at an odd integer at most B . Its shortcut length is at most $R_2(n)$, its ordinary length at most $R_3(n)$, and its ordinary height (including its initial state) at most $n^{1+\beta}$. It contains at most $R_1(n)$ odd steps; for odd n this is its Syracuse return count. The constants $A_{\beta,e}, C_{\beta,e}$ do not depend on B, c, X . Consequently, for every $f(n) \rightarrow +\infty$, the same three clocks and height reach a value strictly below $f(n)$ for a set of natural density one. The clocks do not depend on f .

Proof. First restrict to odd n . Take the high-prefix lemma with $0 < \eta < \beta$, and intersect its good set with the successful set of the timed fixed-target count. Their complements are counted on the original integers n , so the union bound is legitimate without independence or a landing-distribution assumption. For $n \in I_M$ outside this union, write $\sigma \leq K(n)$ for the first odd hit of B , and s, z, Y_M for the high passage.

If $B \geq Y_M$, the desired prefix is a truncation of the high prefix completed to its odd landing. Its clocks and height already have the asserted bounds, since $E_M = o((\log(n+2))^{4/5})$. If $B < Y_M$, the two-sided odd clock gives

$$0 \leq r = \sigma - s \leq C_K(\log(n+2))^{4/5} + O_{\beta,e}(E_M) = O_{\beta,e}((\log(n+2))^{4/5}).$$

Here $s = d^{-1} \log n + O_{\beta,e}(E_M)$, not just an upper bound. The preceding lemma bounds the remaining ordinary and shortcut lengths by $O_{\beta,e}((\log(n+2))^{4/5})$, because $\log_2 z \leq p_M = O_{\beta,e}(\log \log(n+2))$. Adding them to (253), including the at most p_M terminal halvings, proves the three clocks. The high ordinary height is $2n^{1+\eta} \leq n^{1+\beta}$ for large n , while the tail height satisfies

$$\log(2^{r+1}Y_M) = O_{\beta,e}((\log(n+2))^{4/5}) = o(\log n).$$

It is therefore at most n for large n , uniformly in B . Finite startup inputs can be put in the high exceptional set.

The dyadic shell count for the high exceptions is $O_{\beta,e}(X(\log(X+2))^{-e})$: split shells at $\frac{1}{2} \log_2 X$, bound the lower part by $O(\sqrt{X})$, and sum the upper bounds geometrically. The timed exceptions have the first term of (255). This proves that estimate for odd sources.

For general $n = 2^a u$, u odd, prepend its a halvings. Since $j \log 2/d > 1$ for $j = 2, 3$,

$$a + (j/d) \log u \leq (j/d) \log n, \quad \log(u+2) \leq \log(n+2).$$

The odd-step count acquires no term a , and the height becomes at most $\max(n, u^{1+\beta}) \leq n^{1+\beta}$. The exact fibres $u \mapsto 2^a u$ give the sum of odd exception counts at cutoffs $X/2^a$. The target-dependent terms sum to at most $2C_c X(\log B)^{-c}$. For the high terms with $X/2^a \geq \sqrt{X}$ use $\log(X/2^a + 2) \geq \frac{1}{2} \log X$; the remaining cutoffs contribute at most $\sum_{X/2^a < \sqrt{X}} X/2^a \leq 2\sqrt{X}$. Absorb these and bounded X into $C_{\beta,e}$, and the factor two into C_c . No equality of odd and all-integer densities is used.

Finally set $b(X) = \inf_{n \geq \sqrt{X}} f(n)$ and $B(X) = \lceil b(X) \rceil - 1$. Eventually $B(X) \geq 2$, $B(X) \rightarrow \infty$, and $B(X) < f(n)$ for $n \geq \sqrt{X}$. The at most $\sqrt{X}$ smaller inputs, together with (255), have total proportion tending to zero. All constants were selected before B or f . $\square$

Relation to the literature and the earlier clocks. The coefficient $3/d$ is the ordinary drift coefficient, not a new prediction. Inselmann's [1, Theorems 1.9–1.10] trajectory results already give ordinary and Syracuse power envelopes through times $3d^{-1} \log n$ and $d^{-1} \log n$, respectively, on natural-density-one sets. His shortcut theorem uses $2d^{-1} \log n$. Those statements have fixed power tolerances $n^{\pm\epsilon}$. The proof above instead retains the explicit $O((\log n)^{4/5})$ clock error and joins the high prefix to the arbitrary-diverging-target conclusion, with the displayed quantitative fixed-target exception count and one common height-controlled path. It uses the written analytic reconstructions; it is not a Lean certification of either complete source package or a priority claim.

Allikvere's original valuation telescope gives $(3/d + 1/\log 2) \log n + O((\log n)^{4/5})$. That remains a valid bound. The improvement here is a different use of its timed witness, not a correction to that telescope or to Tao's theorem. Mazur's logarithmic-time theorem can be joined at the

same first passage, but its inspected leading odd-clock constant is larger than $1/d$. Subtracting s from that estimate leaves an $O(\log n)$ remainder, so this particular substitution does not give the subpower tail-height estimate just proved. This says what the displayed bounds imply; it does not assert different actual orbits or the impossibility of a sharper Mazur clock.

$$n \xrightarrow[\text{height} \leq 2n^{1+\eta}]{s=d^{-1} \log n + O(\sqrt{\log n \log \log n})} z \leq Y_M,$$

$$z \xrightarrow[\text{height} \leq 2^{r+1} Y_M]{r \leq K(n) - s = O((\log n)^{4/5})} \text{Syr}^\sigma(n) \leq B.$$

This is the case $B < Y_M$. For $B \geq Y_M$, truncate the first arrow at its first odd hit of B . The same initial integer labels both arrows; no new sampling occurs at z .

Figure 26: Joining two clocks on one orbit. The high-prefix construction is Shaik’s; the total odd clock comes from the reconstructed Allikvere–Tao argument. Subtracting a lower prefix bound from an upper total bound controls the actual remaining path. Proofs: Lemmas 129, 130 and Theorem 131. The diagram is schematic, not a numerical trajectory.

29 What has been checked at this stage

Theorem 131 joins the reconstructed timed count and high-prefix transport on the same orbit. It proves the three drift clocks with an explicit sublinear remainder, the ordinary height bound and the uniform fixed-target exception count. Its complete argument is written above; the finite path regressions do not certify the analytic inputs, and it is not a new Lean export. Proposition 126 and Proposition 127 give exact low-rank duration sets and first-timeout supports; Corollary 128 propagates these into finite three-clock and height bounds on unchanged paths. These new general written proofs are not Lean exports. Proposition 123 and Corollary 124 give written two-sided height bounds on the unchanged timeout-good set. Their discrete completed-block bounds now have the matching certificate in Proposition 125; the real exponents, high-stage bounds and whole-run analytic assembly remain written/source results. Proposition 121 gives a written exact comparison of the original failure envelopes and numerical run records; Corollary 122 identifies the literal witness predicates and propagates the count bounds. This comparison has not been independently compiled as a source-package Lean theorem. Proposition 120 has a matching twenty-three-declaration Lean certificate for the discrete count and ordinary-clock consumer. Its scope is separate from the real analytic endpoint theorem and the full source dependency closure. Theorem 118 assembles the refined all-prefix parameter domain, explicit constants, original failure envelope, same-witness clock and height, and exact arbitrary-prefix count. Corollary 119 transports its actual paths to the other two clocks. These analytic statements have written proofs, not new Lean certificates. Proposition 116 proves the exact rank restriction and time enclosure; Corollary 117 propagates it to the unchanged moving envelope. These are written proofs.

Proposition 115 gives the least possible startup and an explicit subthreshold repair, with the finite threshold (219). These remain written proofs.

Proposition 114 proves the necessary and sufficient closed startup threshold for the original scalar interface, and (216) propagates it into the low profile. These are written proofs, not further Lean exports.

Proposition 112 has a new independently compiled 31-statement finite Lean certificate, including

literal counting semantics. Proposition 113 and its subsequent profile calculation are complete written analytic refinements, not Lean exports of the real-variable estimates.

The uniform square-root estimate and its exact reflection map are proved in Proposition 109 and Lemma 108. Corollaries 110 and 111 propagate it through the existing fixed and shrinking constructions. These are written analytic proofs, not new Lean exports.

Proposition 105 retains the full fixed-tolerance entropy rate by an explicit rank-dependent auxiliary barrier. Corollary 107 carries it through the original pullbacks and fixed-barrier descent, without altering their sets or clocks. These real-variable results have written proofs, not new Lean exports.

Proposition 103 and Corollary 104 retain the actual stopping map, remove the extra graded-clock startup payment, and admit the closed fixed-depth exponent with one simultaneous exceptional-count estimate. These are complete written proofs; no new analytic Lean export is claimed.

The sharp shifted-shell assembly, its integer linear boundary, and the unclipped same-schedule application are proved in Propositions 100–101 and Corollary 102. Their analytic bounds are written proofs, not new Lean exports.

Proposition 98 now has a separate nine-statement Lean certificate for its exact integer witness/count mechanism. Proposition 99 is a complete written endpoint calculation for the explicitly shifted source schedule; its real-variable estimates are not included in that certificate.

Proposition 96 retains the endpoint exponent on the original fixed-depth bootstrap sets, with its logarithmic factor and explicit prefactor recurrence. Proposition 97 uses their absorbing startup branch to remove the coarse intermediate target and clock losses. These are written deductions from the source transport and deterministic dynamics, not new Lean exports.

Proposition 94 reconstructs the full stretched-logarithmic exceptional power, with the same target, clock and height witness. Proposition 95 retains an explicit power-saving count for the graded fixed-power clock and carries the witness through all three clocks. The source already contains a positive power-saving antecedent for its graded theorem. These written deductions are not new Lean exports.

Corollary 90 retains the two boundary iterated-logarithmic exceptional powers of Shaik’s critical schedules. Proposition 91 reconstructs the arbitrary diverging final factor and preserves its shifted quantitative profile. Both use the exact ceiling and the literal simultaneous-witness set. Their real-variable proofs are written, not newly kernel checked.

Proposition 88 constructs every prescribed strict rate in Shaik’s first fixed-target corollary, including explicit stage startups. Its proof follows the exact rate through the source package to the same-witness global exceptional count. This is written source-level reconstruction, not an additional Lean certificate.

Proposition 84 adds an executable inverse, exact integer fibres and an independently checked binary-shift identity. Corollary 85 gives the full written counting and conditioned-image comparison. Proposition 86 now adds general Lean proofs of weighted reindexing and the complete-shell binomial count. Proposition 87 retains every incoming logarithmic exceptional exponent under a finite startup change; its exact finite-count inequality is also checked by Lean, while its real-variable absorption argument is given in full above.

Proposition 83 now has a scoped, independently compiled Lean certificate for its exact integer and path statements. The complete analytic density argument is not thereby formalized.

Section 28 now reconstructs the uniform prefix bound, integer timeout tail, deterministic time supports and full critical shell estimate. It retains the endpoint rate and proves additive-error shortcut, raw and odd-return clocks. This is a written reconstruction, not a replay of the complete Lean dependency closure.

Section 27 reconstructs selected deterministic proofs in Shaik’s manuscript and pinned Lean source, with exact product defects, smaller loss and fibre constants, and a proved passage-clock correspondence. Its full argument is also in cumulative chapter 01n. The rational checks cover actual finite paths and exact interval counts; they do not verify the whole natural-density package or its formal build.

Section 25 completes the four comparisons for actual first passage, the exact whole-source mixtures and the nonpassage completion. Its four written results retain all original band normalizers, targets and boundary terms; the independent finite regressions are in `check_mazur_genuine.py`. Section 26 now gives the exact totalizer loss, real-source comparison, time-carrying recurrence and fixed-target timed density estimates, with their complete written proofs. The finite checks in `check_mazur_real_timed.py` test the algebra and path implications; they do not certify infinite claims. No Lean replay is claimed.

Section 24 now reconstructs the original weighted endpoint fibres, sums the terminal factor inside the two one-sided tail estimates, and completes both full reference profiles. The total coefficient is bounded by $2 + 3B^{-4997/600000}$; the exterior majorant is $(4 + o_C(1))N^{-64C^2/(\log 2)^3}$. The exact finite kernel estimate retains its original mass, signed denominator corrections and separate interior/exterior charges. The genuine-passage assembly is now proved above; the next source work is the original clock/iteration consumers and connected Shaik comparison. Exact finite regressions do not certify the asymptotic estimates or constitute a Lean replay.

Section 23 now proves the raw-mass asymptotic from a finite Robbins remainder, exact Gaussian integral, strict floor correction and both explicit tails. It evaluates the two interior physical profiles using Rhin’s published input and an exact time translation. The Gaussian comparison sharpens the logarithmic factor in the component error. The exterior coefficient tail and full common-profile assembly follow in Section 24. No Lean replay is claimed.

Section 22 reconstructs the shifted supports, signed adjacent correction, two time recurrences and exact padded Abel formula. Its finite weighted discrepancy bound uses the actual empirical discrepancies, not a tested substitute for an asymptotic theorem. The raw-mass and interior calculations are completed in the next section; the exterior coefficient tail is proved subsequently. The source’s constant-20 variation already gives 40, rather than its later relaxed 64, after equal padding; this changes no exponent.

Section 21 gives complete finite proofs of the two raw phase means, exact signed band and kernel corrections, the all-endpoint mass envelope, and sharp positive exterior centering. Its checker preserves distinct tube supports and strict phase endpoints. The interior weighted-rotation estimate now has a written proof in Section 23. The exterior coefficient-tail estimate is not certified by these finite identities or bounded tests.

Section 18 proves the exact mass/test maps, conditioned prefix mixture, central likelihood comparison and its terminal-constant consequence. Its two tail estimates are reconstructed in Corollary 41; the assembled endpoint-test caps are proved more sharply in Proposition 65, with the exact native coefficients and both band denominators retained. The exact rational suite `check_mazur_descent.py` also checks complete propagation of this section into the cumulative chapter, residue collisions, reversal, adjacent ratios, sharp cancellation, signed tails and directed exponential comparisons. It does not replace the written analytic proofs or a formal replay.

Section 19 supplies complete written proofs of the fixed-total concentration and both tail inputs, and identifies the original terminal union with the genuine first-passage event on the full-prefix good set. Its scalar guards are proved eventually for the unchanged schedules. Section 20 now also bounds the good set’s complement under each actual band measure and sharpens the maximal-prefix tail. Evaluation of the common reference profile is now proved in Section 24.

Section 17 supplies the written finite reconstruction, with the boundary and time hypotheses retained. Its sharp absorption lemma is proved for every finite nonnegative comparison satisfying its explicit inequalities. The resulting scalar refinement preserves the source’s analytic inputs; it is not a certificate of those inputs. The independent checker `check_mazur_finite.py` tests affine inverses, actual valuation words, short-window injectivity, both boundary conventions, two-row sums, and sharp-absorption equality cases with rational arithmetic. Its scope and source hashes are recorded in `mazur_finite_checks.json`.

Propositions 27 and 28 are independent written proofs for the literal scalar bounds in Mazur’s source. Relevant manuscript formulas were checked on rendered pages 3, 7, 8 and 9. The selected Lean statements and rate proofs were read; the publisher’s build and axiom records remain attributed evidence, not a local replay of its complete formalization.

The full 2,216-line Allikvere source has received a first content reading. The arguments reconstructed here have narrower, explicit scopes: Proposition 1 uses Tao’s positive-majorant theorem; Propositions 2 and 3 are elementary exact statements; and (10) extends the parameter range of an existing local proof without changing its estimates. Proposition 4, Corollaries 5 and 7, and Lemma 6 now give the complete written comparison through the conditioned product formula, using Tao’s cited positive-majorant theorem. None depends on numerical evidence for convergence of Collatz orbits.

The accompanying exact finite checks compare residue histograms, valuation cylinders, first-passage intervals and the binomial identity used in the uniform fibre calculation. They test the formulas and their implementation, not the asymptotic Fourier theorem. The written reconstruction now reaches the natural-density and timed conclusions in Corollaries 25 and 26. Its dependency chain uses Tao’s positive-majorant theorem and Rhin’s published Diophantine input, rather than independently reproving those external results. The uniform first-passage assembly is proved in Corollary 19, with its actual-input law and full mass error retained. No formal Lean certificate of the complete analytic theorem is claimed. The leading constant and its uniform error are now proved in Proposition 15, using the primary Rhin and Kuipers–Niederreiter statements at the cited locators. The full-band residue discrepancy, absolute window replacement and uniform kernel upper bound are proved above, with the window step independent of Rhin’s estimate. The written arguments above are not formal Lean proofs.

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